



S6D0164X01

Rev. 1.00

MOBILE DISPLAY DRIVER IC

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Revision History

Ver.	Date	History
1.00	2007-10-16	- Added note.2 in Table 2.S6D0164 Pad Dimensions - Modified Figure 14. External power off sequence - Divided "MTP Initialization, Erase and program" into "a. Using VCI for MTP" and "b. Using VCI1 for MTP" - Modified Figure 95. Application Circuit Example - deleted schottky diode except schottky linked VGL-VSS - Modified unused bits in 8.3.30. MTP Data Write (R82h) and 8.3.31. Panel ID (R93h) - "0" → don't care - Added values of TBD in Table 12.DC Characteristics For LCD Driver Outputs - Added 17.3. EXTERNAL COMPONENT - Correct some misspellings
0.40	2007-08-28	- Added current consumption value in normal operation - Added note for current consumption measurement condition - Modified note for current consumption measurement condition - Modified MTP Initialization, erase and program flow - Modified MTP_MODE and MTP_EX parameters description in 8.3.29. MTP Control (R81h) - Modified 6.5. External Power On/Off Sequence - Modified MTP_DOUT parameter description in 8.3.30. MTP Data Write (R82h) - Correct some misspellings - Modified Table 31. Oscillation Frequency - Added additional description of INV1-0/FLD parameter in 8.3.5 LCD-Driving-Waveform Control (R02h) - Modified AC polarity in Figure 21. Interlace drive and output waveform (Two Line Mixed Inversion) - Added values of TBD in Table 12.DC Characteristics For LCD Driver Outputs - Added values of TBD in SAP parameter table of 8.3.12. Power Control 1 (R10h) - Added 8.3.31. Panel ID (R93h)
0.30	2007-08-03	- Added note of SM parameter in R01h command - Note. When SM = 1, NL setting is disable. - Modified FP and BP parameter description in R08h command - Deleted AB2A parameter in R10h command - Modified Note3 of VCM parameter in R14h - Modified Figure 26. Vertical Scroll Display - Modified MTP_SEL parameter description in R81h command - Added Note. of MTP_DIN parameter in R82h - Modified Figure 29, 30, 31 and 32. - Changed reset value of MTP_SEL parameter in 9.RESET FUNCTION - Modified Figure 62. Bit Assignment of GRAM Data on 80-8bit CPU Interface - Modified Figure 83. Structure of Resistor Ladder Network 2 - VINP1/VINN1, VINP6/VINN6 : 3R → 9R - Modified Figure 88. Setup Procedure For The 8-Color Mode - Modified Table 10. Absolute Maximum Rating - Modified Figure 2.The functional block diagram and signal pads of S6D0164. - deleted MTPD/MTPG in MTP box - Modified Table 2.S6D0164 Pad Dimensions - Modified Table 8.Miscellaneous pad description - Modified D1-0 parameter table in R07h command - Modified Note of Table 30.Display State and Interface - Modified Note of VC parameter in R11h command - Modified Figure 29.Flow of MTP Load / Read - Modified reset value of TEST_KEY parameter in 9.RESET FUNCTION - Modified Figure 36.Power-Up Pattern Diagram - Modified Table 42.Gradient Adjustment (1) - Modified Table 11.DC Characteristics - Modified Table 12.DC Characteristics For LCD Driver Outputs - Modified 6.5. External Power On/Off Sequence - Deleted Note of AB_VCI1 parameter in R11h command - Correct some misspellings in 9.3.19. Software Reset (R28h) - zero → high - Modified Figure 35.Configuration of the Internal Power-Supply Circuit - Modified Figure 74.16-/18-bit RGB Interface Timing and Figure 76.6-bit RGB Interface Timing - Changed form and representation on the whole.
0.20	2007-03-16	- Modified BLOCK DIAGRAM, VCI → VDD3 (p.7) - Distance from chip edge to output bump center on figure 3 is revised.(p.9) - Distance K on figure 4 is revised.(325um → 365um, p.10) - Modified FLM and CL1 pin descriptions (p.23) - Added CONTACT pin description (p.23) - Modified Index register range, ID6-0 → ID7-0 (p.30) - Added NOTE for 3-field interlaced function (p.35) - Added Figure 12. Two Line Mixed Inversion (p.37) - Changed parameter name (p.41) - TEMON → FLM_MON - Modified Start Oscillation(R0Fh) (p.48)

		- Added AB_VCI1 parameter (p.50) - Modified VCOMG description (p.56) - GVD6-0, VCM6-0, VML6-0 table full range description (p.54~58) - Modified Note3. (p.57) - Revised numerical expression for VCOM amplitude (p.59) - Added MTP_DOUT register (p.73) - Modified Figure 20, 21, 22 and 23. (p.74, p.75) - Modified Figure 26. (p.78) - Deleted contents related with DIV (p.126) - Corrected parameter that is not used from "don't care" to "0" - Correct some misspellings
0.10	2007-01-24	- Modified Table 23 (p.35) - Modified value equivalent to SAP register (p.48) - Modified Figure 11 (p.36) - Modified Figure 12 (p.56) - Added AC CHARACTERISTICS
	2007-02-12	- Modified Figure 2 : Added EN_EXCLK, EXCLK (p.13) 3-pin SPI → SPI - RVDD pin : P(Power) → O(Output) (p.19) - VDD pin : P(Power) → I(Input) (p.19) - Modified RW_WRB, E_RDB and ENABLE description (p.21,22) - Modified Table 14 : Added EN_EXCLK, EXCLK (p.23) - Modified Table 15 : TEST_CSN[1:0] → TEST_DA (p.23) - Deleted VCIR_VSSB register (p.28) - Added MDT of the 68mode Interface (p.37) - Modified value equivalent to VCIR register (p.56) - Modified Table 33 (p.79) - Modified Figure 79 (p.122) - Changed pin name : GVDDO pin → GVDD pin

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1. Introduction

1.1. Purpose of this document

This document is to provide an electrical and mechanical reference specification of S6D0164 driver IC developed by Samsung. This would be a reference design guide for display panel module makers so that the display module subassembly can be designed properly.

1.2. Product options

S6D0164 is a single-chip CMOS LCD controller/driver for color TFT-LCD displays supporting the panel resolution of 220 gates and 176xRGB columns. It contains a 260k-bit (176 x 220 x 18-bit) display graphic RAM and a full set of control functions. S6D0164 comes with a multiple options based on customer's need, where some of functions are slightly different. The description of each option is described in Table 1. For example, S6D0164-X01 version supports various types of peripheral interface such as 80-series MCU interface (8-/9-/16-/18-bit data), 68-Series MCU interface (8-/9-/16-/18-bits data), and serial interface.

Table 1. List of S6D0164 options.

Options	Remarks
-X01	Reference design of S6D0164 which supports various Host interfaces

2. Features

- **176-RGB x 220-dot Resolution, 528ch Source Driver & 220ch Gate Driver**
- **Various color-display control functions**
 - 262,144 colors can be displayed at the same time (including gamma adjust)
 - 65,536 colors, 8 colors can be displayed
- **MPU/SPI/RGB Interfaces**
 - 18-/16-/9-/8-bit high-speed parallel bus interfaces including MDT (Multiple Data Transfer) mode.
 - Serial peripheral interface(SPI)
 - 18-/16-/6-bit RGB interfaces for motion picture display.
- **Various Graphic Operations**
 - Window-Addressing Function to display motion picture independently of still image display
 - Image Rotating / Mirroring Function
- **Built-in GRAM : 176 x 18 x 220 = 696,960 bits**
- **Alternating functions for TFT-LCD common-electrode power**
- **Low-power operation supports**
 - Power-save functions (standby mode, deep-standby mode)
 - Partial display in any position
 - Voltage step-up circuit up to 7 times for generating driving voltage
 - Charge-recyclizing function for the switching performance of step-up circuits and operational amplifiers
- **An Internal oscillator and external hardware reset**
- **Internal power supply circuit**
 - Voltage Step-up circuit: five to seven times, positive-polarity inversion
- **Operating voltage**
 - 1.65V–3.3V for I/O supply voltage range for VDD3
 - 2.5V–3.3V for Analog supply voltage range for VCI
- **Voltage supplies generated internally**
 - 2.5V to 5.0V Source output voltage range for GVDD to AVSS
 - 4.5V to 6.0V (See Note 1) Power supply for driver circuit range for AVDD to AVSS
 - Max 6.0V Common electrode output voltage range for VCOM
 - Min 11.25V to 16.5V (See Note 2) Positive Gate output voltage range for VGH to AVSS
 - -13.75 V to -6.75V (See Note 2) Negative Gate output voltage range for VGL to AVSS

Note.

1. AVDD Min: When VCI1 = 2.25V, AVDD Max: When VCI1=3.0V
2. |VGH| & |VGL| Min : When VCI1 = 2.25V, |VGL| Max : When VCI1=2.75V, |VGH - VGL| Max : 30.0V
3. |VGH| max. should be lower than or equal to 16.5V in normal operating condition, regardless of VCI1 & BT settings.

3. Display Module Block Diagram

3.1. Signal flow of the display module and its relationship

Figure 1 shows the block diagram of a mobile display panel module and related interface signals required by set makers and module makers. Level I interface signals are usually required by a set maker who would then request the display module such a function, and Level II interface signals are required by a display module maker for its own purpose.

There are also Level III signals which is for the internal use only for the driver IC itself. Level III signals are not intended to be released to the customers since it is designed for a specific manufacturing purpose, and it may alter the functions of IC if it is misused.

This reference specification only provide a guideline to Level I and II interface signals since some of specifications related to Level I and II need a margin on IC side and would not be necessarily the same as the one in Level I and II specification even though both uses the same interface signals. This is mainly due to the parasitic and design requirements within the flexible PCB used by a display module maker. IC specification will offer related information among Level I/II on how each interface signal relates each other.

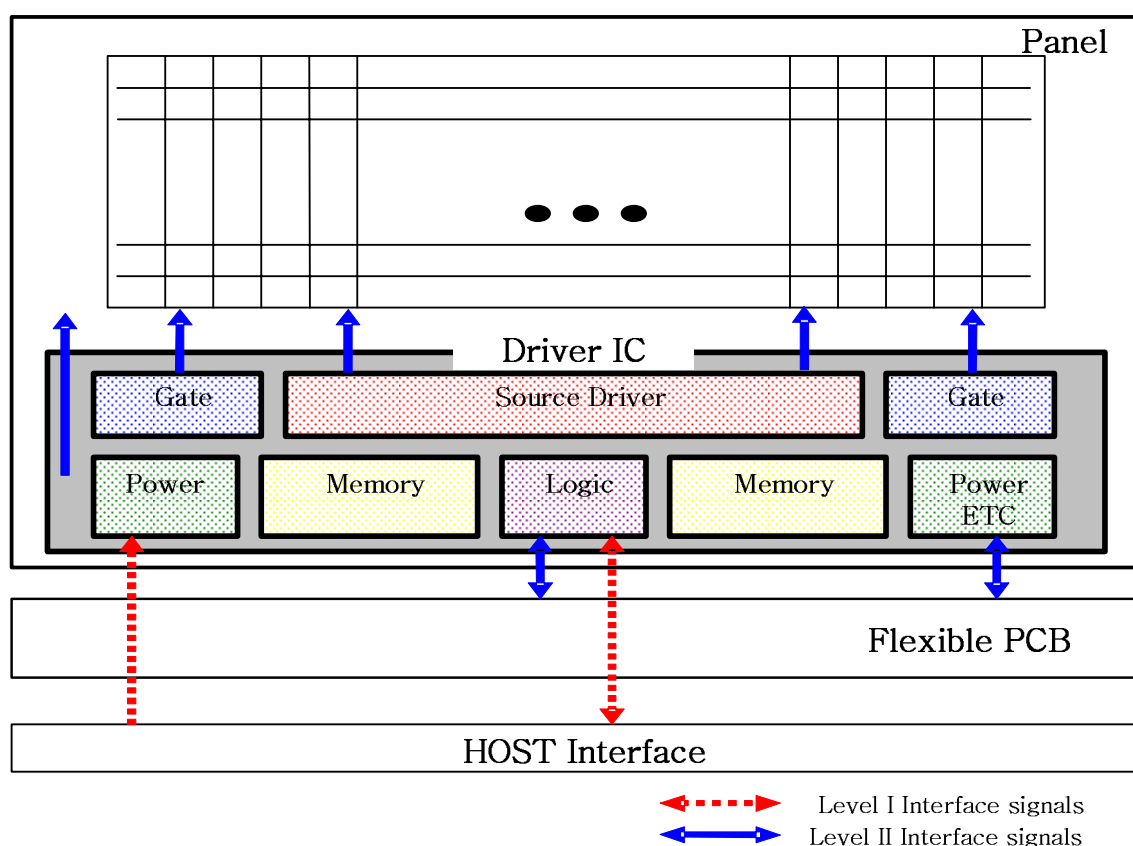


Figure 1. The interface signal flow of a mobile display panel module

3.2. Function Block Diagram and Signal Pads of S6D0164

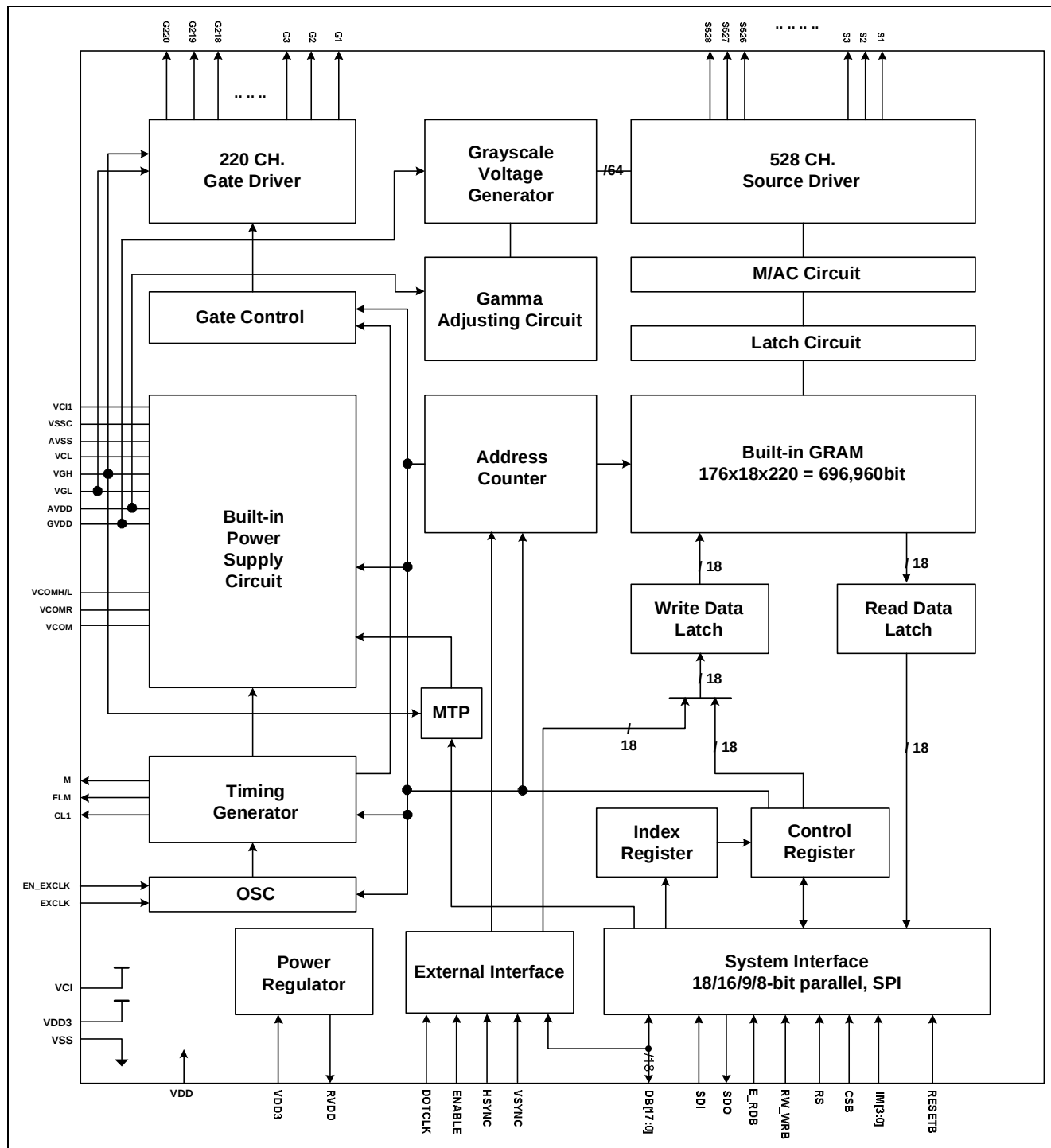


Figure 2. The functional block diagram and signal pads of S6D0164.

4. Chip Information

4.1. PAD Configuration

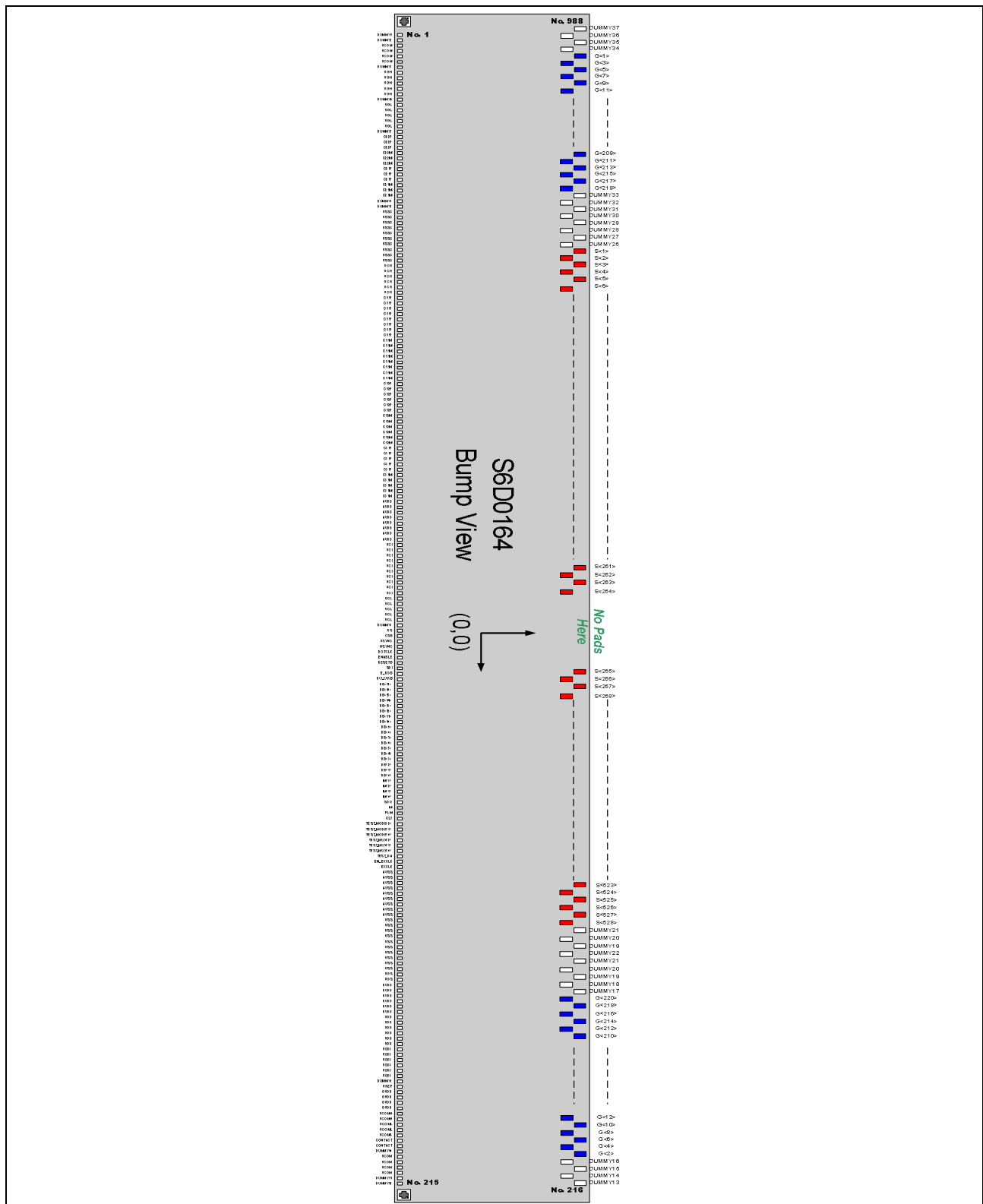


Figure 3. Chip Layout

4.2. Bump PAD Information

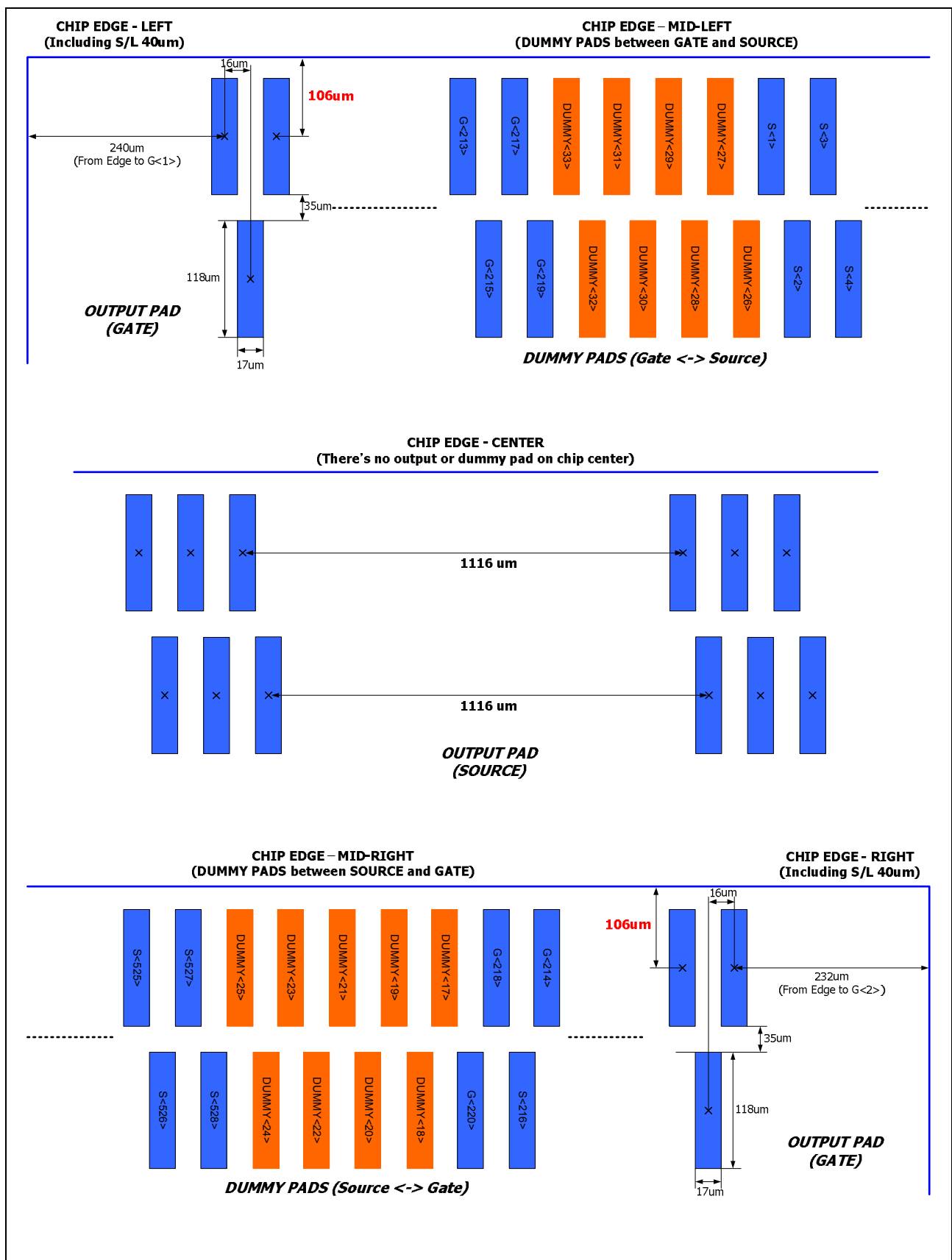


Figure 4. Pad Configuration – Output

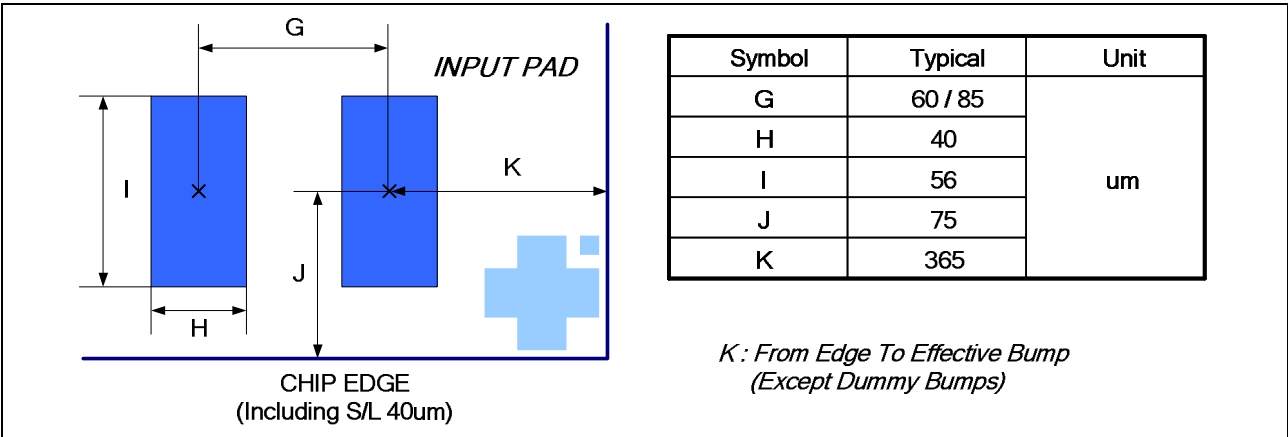


Figure 5. Pad Configuration - Input

Table 2. S6D0164 Pad Dimensions

Items	Pad name.	Size		Unit
		X	Y	
Chip Size ¹⁾	-	13,880	820	um
Bumped Pad Size	Input Pads (60um / 85um pitch)	40±2	56±2	
	Output Pads (16um pitch)	17±2	118±2	
Bumped Pad Height	In Wafer	15 (Typ.) ± 2		
	In Chip	Under 1		

- Note.
1. Scribe lane is included in this chip size (Scribe lane: 80um)
 2. wafer thickness :
- S6D0164●●●●-●●●●8 : 300±10 um

4.3. ALIGN KEY CONFIGURATION AND COORDINATES

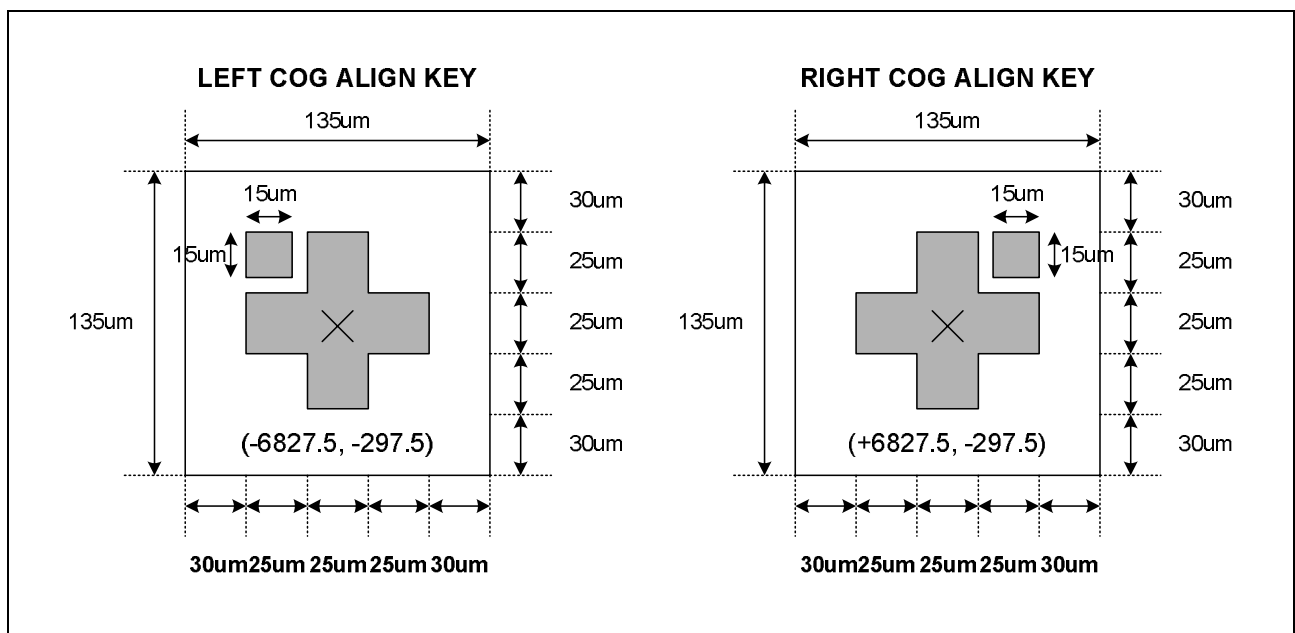


Figure 6. COG Align Keys

5. IC Pad Description

5.1. Pads for Power Supplies

Table 3. Pads for power supplies.

Symbol	I/O	Description
RVDD	O	Voltage regulator output for VDD. Connect this to VDD pad for supplying power. Connect a capacitor for stabilization.
VDD	I	Power supply for memory and internal logic circuit. Connect this pad to RVDD pad. Do not apply any external power to this pad over 1.5V.
VDD3	P	Power supply for I/O blocks.
VCI	P	Power supply for analog and voltage booster block.
AVSS	P	GND for analog circuits
VSSC	P	GND for booster circuits.
VSS	P	GND for logic circuits.
AVDD	O	Internally generated voltage output pad for source driver block. Output voltage of 1 st booster circuit (=2 x VCI1) Input voltage to 2 nd booster circuit. This pad needs an external bypass capacitor.
VCI1	O	Reference input voltage for 1 st booster circuit. Connect a capacitor for stabilization. Note. VCI1 cannot exceed 3V
VGH	O	Positive power output of the 2 nd booster circuit. Gate "ON" level voltage. Connect a capacitor for storage function.
VGL	O	Negative power output of the 2 nd booster circuit. Gate "OFF" level voltage. Connect a capacitor for storage function.
VCL	O	3 rd booster output voltage. Power supply for generating VCOML block. Connect a capacitor for storage function.
VGS	I	Reference voltage input for grayscale voltage generator. Connect an external resistor or to system ground.
VREF	I/O	Reference voltage for generating GVDD voltage.
GVDD	O	Reference voltage input for grayscale voltage generator. Reference voltage input for VCOMH / VCOML voltage generator. An internal register can be used to adjust the GVDD voltage. Connect a capacitor for stabilization.
VCOMH	O	High level output voltage of VCOM. An internal register can be used to adjust the VCOMH voltage. Connect a capacitor for stabilization.
VCOML	O	Low level output voltage of VCOM. An internal register can be used to adjust the difference voltage between VCOMH and VCOML. Connect a capacitor for stabilization.

Table 4. Power supply pad description (continued)

Symbol	I/O	Description
VCOM	O	Power supply pad for the TFT- display common electrode. The alternating cycle can be set by the "M" pad. Charge recycling method is used with VCI voltage. Connect this pad to the TFT-display common electrode
VCOMR	I/O	Reference voltage input pad for VCOMH. When VCOMH voltage is adjusted externally, halt the internal adjuster of VCOMH by setting the register and insert a variable resistor between GVDD and VSS. When VCOMH is not adjusted externally, leave this pad open and adjust VCOMH by setting the internal register.
C11P, C11M C12P, C12M	-	Connect the charge-pumping capacitor for generating AVDD level.
C21P, C21M C22P, C22M	-	Connect the charge-pumping capacitor for generating VGH, VGL level.
C31P, C31M	-	Connect the charge-pumping capacitor for generating VCL level.

5.2. Signal pads for Logic interface

Table 5. Signal pads for Interface Logic

Symbol	I/O	Description																																																				
IM[3:0] / ID	I	Selects the interface mode.																																																				
		<table><tr><th>IM[3:0]</th><th>Description</th></tr><tr><td>0000</td><td>68-Series 16-bit MPU interface</td></tr><tr><td>0001</td><td>68-Series 8-bit MPU interface</td></tr><tr><td>0010</td><td>80-Series 16-bit MPU interface</td></tr><tr><td>0011</td><td>80-Series 8-bit MPU interface</td></tr><tr><td>010x</td><td>Serial peripheral interface (SPI) IM[0] = ID</td></tr><tr><td>011x</td><td>Reserved</td></tr><tr><td>1000</td><td>68-Series 18-bit MPU interface</td></tr><tr><td>1001</td><td>68-Series 9-bit MPU interface</td></tr><tr><td>1010</td><td>80-Series 18-bit MPU interface</td></tr><tr><td>1011</td><td>80-Series 9-bit MPU interface</td></tr><tr><td>11xx</td><td>Reserved</td></tr></table>	IM[3:0]	Description	0000	68-Series 16-bit MPU interface	0001	68-Series 8-bit MPU interface	0010	80-Series 16-bit MPU interface	0011	80-Series 8-bit MPU interface	010x	Serial peripheral interface (SPI) IM[0] = ID	011x	Reserved	1000	68-Series 18-bit MPU interface	1001	68-Series 9-bit MPU interface	1010	80-Series 18-bit MPU interface	1011	80-Series 9-bit MPU interface	11xx	Reserved																												
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		1010	80-Series 18-bit MPU interface																																																			
		1011	80-Series 9-bit MPU interface																																																			
		11xx	Reserved																																																			
RESETB	I	Reset pad. Initializes the IC when it is low. Must be reset after power-on.																																																				
CSB	I	Chip Select. - Low: IC is selected and can be accessed. - High: IC is not selected and cannot be accessed. Must be connected to VDD3 level when not in use.																																																				
RS	I	Register Select. - Low : Index / status register - High : Control register Must be connected to VDD3 or VSS level when SPI mode.																																																				
RW_WRB / SCL	I	In 68-Series mode, this is used to select the read or write operation. (RW) In 80-system mode, this is used as a write strobe signal (WRB). In SPI mode, it is used as a synchronous clock (SCL).																																																				
E_RDB	I	In 68-system mode, this is used as write/read enable strobe (E). In 80-system mode, this is used as a read strobe signal. (RDB). Must be connected to VDD3 or VSS level when SPI mode.																																																				
DB[17:0] [Note]	I/O	Data Bus.																																																				
		<table><tr><th colspan="2">Interface Mode</th><th colspan="2">Description</th></tr><tr><th>IM[3:0]</th><th></th><th>Index</th><th>Data</th></tr><tr><td>0000</td><td>68-Series16-bit MPU interface</td><td>DB[8:1]</td><td>DB[17:10], DB[8:1]</td></tr><tr><td>0001</td><td>68-Series 8-bit MPU interface</td><td>DB[17:10]</td><td>DB[17:10]</td></tr><tr><td>0010</td><td>80-Series 16-bit MPU interface</td><td>DB[8:1]</td><td>DB[17:10], DB[8:1]</td></tr><tr><td>0011</td><td>80-Series 8-bit MPU interface</td><td>DB[17:10]</td><td>DB[17:10]</td></tr><tr><td>010x</td><td>Serial peripheral interface (SPI)</td><td>-</td><td>-</td></tr><tr><td>011x</td><td>Reserved</td><td>-</td><td>-</td></tr><tr><td>1000</td><td>68-Series 18-bit MPU interface</td><td>DB[8:1]</td><td>DB[17:0]</td></tr><tr><td>1001</td><td>68-Series 9-bit MPU interface</td><td>DB[17:10]</td><td>DB[17:9]</td></tr><tr><td>1010</td><td>80-Series 18-bit MPU interface</td><td>DB[8:1]</td><td>DB[17:0]</td></tr><tr><td>1011</td><td>80-Series 9-bit MPU interface</td><td>DB[17:10]</td><td>DB[17:9]</td></tr><tr><td>11xx</td><td>Reserved</td><td>-</td><td>-</td></tr></table>	Interface Mode		Description		IM[3:0]		Index	Data	0000	68-Series16-bit MPU interface	DB[8:1]	DB[17:10], DB[8:1]	0001	68-Series 8-bit MPU interface	DB[17:10]	DB[17:10]	0010	80-Series 16-bit MPU interface	DB[8:1]	DB[17:10], DB[8:1]	0011	80-Series 8-bit MPU interface	DB[17:10]	DB[17:10]	010x	Serial peripheral interface (SPI)	-	-	011x	Reserved	-	-	1000	68-Series 18-bit MPU interface	DB[8:1]	DB[17:0]	1001	68-Series 9-bit MPU interface	DB[17:10]	DB[17:9]	1010	80-Series 18-bit MPU interface	DB[8:1]	DB[17:0]	1011	80-Series 9-bit MPU interface	DB[17:10]	DB[17:9]	11xx	Reserved	-	-
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		1011	80-Series 9-bit MPU interface	DB[17:10]	DB[17:9]																																																	
		11xx	Reserved	-	-																																																	
Must be connected to VDD3 or VSS level when not used.																																																						
SDI	I	Serial input data. Must be connected to VDD3 or VSS level when not in use.																																																				
SDO	O	Serial output data. Leave this pad open when not in use.																																																				

Note. When used as system interface.

Table 6. RGB interface pad description (Continued)

Symbol	I/O	Description
ENABLE	I	Data enable signal of RGB interface. When ENABLE is in active state, data on RGB bus is valid. But when this is not in active state, data on RGB bus is invalid. (For details, refer to the description of EPL register) Must be connected to VDD3 or VSS level when not used.
VSYNC	I	Synchronous signal of frame. (Active Low Pad) Must be connected to VDD3 or VSS level when not used.
HSYNC	I	Synchronization signal of a horizontal line. (Active Low Pad) Must be connected to VDD3 or VSS level when not used.
DOTCLK	I	Data Clock of RGB interface. Must be connected to VDD3 or VSS level when not used.
DB[17:0] [Note]	I	Used as an input data bus for RGB I/F. - 6-bit interface: DB[17:12] - 16-bit interface: {DB[17:13], & DB[11:1]} - 18-bit interface: DB[17:0] Must be connected to VDD3 or VSS level when not used.

Note. When used as RGB I/F

Table 7. Display pad description

Symbol	I/O	Description
S1 – S528	O	Source driver output pads. The SS bit can change the shift direction of the source signal. For example, if SS = 0, gray data of S1 is read from RAM address 0000h. If SS = 1, contents of RAM address 0000h is out from S528. S1, S4, S7, ... S(3n-2) : display Red (R) (SS = 0) S2, S5, S8, ... S(3n-1) : display Green (G) (SS = 0) S3, S6, S9, ... S(3n) : display Blue (B) (SS = 0)
G1 – G220	O	Gate driver output pads. The output of driving circuit is whether VGH or VGL VGH : gate-ON level VGL : gate-OFF level

Table 8. Miscellaneous pad description

Symbol	I/O	Description
M	O	Output pads used only for a test purpose at IC-side. In normal operation, leave this unconnected.
FLM	O	Tearing effect output pad to synchronize MCU to frame writing, activated by S/W command. When this is not activated, this pad should be low. If not in use, leave this unconnected.
CL1	O	Output pads used only for a test purpose at IC-side. In normal operation, leave this unconnected.
EN_EXCLK	I	Enable external clock input Connect this pad to VSS level if the pad is not in use.
EXCLK	I	External clock input pad. Connect this pad to VSS level if the pad is not in use.
CONTACT	-	Contact resistance measurement pad. In normal operation, leave this unconnected. These pads are at VSS level. When measuring an ohmic resistance of the contact, do not apply any power.
DUMMY1 ~ DUMMY12	-	Dummy pins located at the input side. There is no electrical potential at any dummy pins.
DUMMY13 ~ DUMMY37	-	Dummy pins located at the output side. There is no electrical potential at any dummy pins.

Table 9. Test pad description

Symbol	I/O	Description
TEST_MODE[2:0]	I	Input pads used only for a test purpose at IC-side. In normal operation, connect this pad to VSS.
TEST_MUX[2:0]	I	Input pads used only for test purpose at IC-side. In normal operation, connect this pad to VSS.
TEST_DA	I	Input pads used only for a test purpose at IC-side. In normal operation, connect this pad to VSS.

6. Electrical Specifications

This chapter describes the DC and AC electrical specification of the IC.

6.1. Absolute Maximum Ratings

Table 10. Absolute Maximum Rating

Item	Symbol	Rating	Unit
Supply voltage for logic block	VDD - VSS	-0.3 to +3.3	V
Supply voltage for I/O block	VDD3 - VSS	-0.3 to +5.0	V
Supply voltage for step-up circuit	VCI - VSS	-0.3 to +5.0	V
LCD Supply Voltage range	AVDD - VSS	-0.3 to +6.5	V
	VGH - VSS	-0.3 to +22.0	V
	VSS - VGL	-0.3 to +22.0	V
	VSS - VCL	-0.3 to +5.0	V
	VGH - VGL	-0.3 to +33	V
Input Voltage range	Vin	- 0.3 to VDD3 + 0.5	V
Operating temperature	Topr	-40 to +85	°C
Storage temperature	Tstg	-55 to +110	°C

Note.

1. Absolute maximum rating is the limit value. When the IC is exposed operating environment beyond this range, the IC is not guaranteed for the normal operations and may be damaged permanently, not be able to be recovered.
2. Operating temperature means the possible range of device-operating temperature. It does not mean that the performance of the driver IC would be guaranteed over this temperature range.
3. Absolute maximum rating is guaranteed only when our company's package used.

6.2. DC Characteristics

6.2.1. Basic Characteristics

Table 11. DC Characteristics

(VSS = 0V)

Characteristic	Symbol	CONDITION	MIN	TYP	MAX	Unit	Note
Operating voltage	VDD3		1.65	-	3.3	V	*1
LCD driving voltage	VGH		11.25	-	16.50	V	*2
	VGL		-13.75	-	-6.75	V	*2
	VCL		-3	-	-2.25	V	*3
	AVDD		4.5	-	6.0	V	*3
	GVDD		2.5	-	5.0	V	
Input high voltage	V _{IH}		0.7*VDD3	-	VDD3	V	*4
Input low voltage	V _{IL}		0	-	0.3*VDD3	V	*4
Output high voltage	V _{OH}	I _{OH} = -0.5mA	0.8*VDD3	-	VDD3	V	*5
Output low voltage	V _{OL}	I _{OL} = 0.5mA	0.0	-	0.2*VDD3	V	*5
Input leakage current	I _{IL}	VIN = VSS or VDD3	-1.0		1.0	uA	*4
Output leakage current	I _{OL}	VIN = VSS or VDD3	-3.0		3.0	uA	*5
Operating frequency	Fosc	Frame freq. = 60 Hz Display line = 220	0.9*TYP	227	1.1*TYP	kHz	*6
Internal reference power supply voltage	VCI		2.5	-	3.3	V	
1 st step-up input voltage	VCI1		1.35	-	3	V	
1 st step-up output efficiency	AVDD	ILOAD = 2 mA	90	95	100	%	
2 nd step-up output efficiency	VGH	ILOAD = 0.1 mA	90	95	100	%	
3 rd step-up output efficiency	VGL	ILOAD = 0.1 mA	90	95	100	%	
4 th step-up output efficiency	VCL	ILOAD = 0.3 mA	90	95	100	%	

Note.

1. VSS = 0V.
2. |VGH| & |VGL| Min : When VCI1 = 2.25V, |VGL| Max : When VCI1=2.75V, |VGH - VGL| Max : 30.0V
cf> |VGH| max. should be lower than or equal to 16.5V in normal operating condition, regardless of VCI1 & BT settings.
3. AVDD & |VCL| Min: When VCI1 = 2.25V, AVDD & |VCL| Max: When VCI1=3.0V
4. Applied pins; CSB, E_RDB, RW_WRB, RS, DB[17:0], RESETB, HSYNC, VSYNC, DOTCLK, ENABLE, SDI, EXCLK
5. Applied pins; DB[17:0], M, FLM, CL1, SDO
6. Target frame frequency : 60 Hz, Display line = 220, Back porch = 8, Front porch = 8, RTN2-0 = "000"
cf> Fosc can be observed indirectly by measuring CL1 pad. (Fosc / 16)

Table 12. DC Characteristics For LCD Driver Outputs

(VDD = 1.5V, VDD3 = 3.0V, VSS = 0V)

Characteristic	Symbol	Condition	MIN	TYP	MAX	Unit	Note
LCD Gate Driver Output "On" Resistance	Ronvgh	I(sink)=100uA	-	-	3	kΩ	-
	Ronvgl	I(source)=100uA	-	-	3	kΩ	-
LCD Source Driver Output "On" Resistance	Ronp	I(sink)=100uA	-	-	20	kΩ	-
	Ronn	I(source)=100uA	-	-	20	kΩ	-
LCD Binary Driver Output "On" Resistance	Ronpb	I(sink)=100uA	-	-	300	kΩ	-
	Ronnb	I(source)=100uA	-	-	300	kΩ	-
Output Voltage Deviation (Mean Value)	ΔV_o	$4.2V \leq V_{SO}$	-	-	± 55	mV	*1
		$0.8V < V_{SO} < 4.2V$	-	-	± 25	mV	*1
		$V_{SO} \leq 0.8V$	-	-	± 55	mV	*1
LCD Source Driver Delay	tSD	AVDD = 5.0V GVDD = 4.5V SAP = "0100"	-	-	20.9	us	*2
Current Consumption during Normal Operation	IVDD3	No load, Ta = 25 °C, VDD3=VCI=3V Frame(f)=60Hz	-	-	150	uA	*3
	IVCI		-	-	4.8	mA	*3
Current Consumption during Standby Mode	Istby_VDD3	VDD3= VCI = 3.0V Ta = 25°C	-	-	15.0	uA	*4
	Istby_VCI		-	-	5.0	uA	
Current Consumption during Deep Standby Mode	Idstby_VDD3	VDD3= VCI = 3.0V Ta = 25°C	-	-	2.0	uA	*4
	Idstby_VCI		-	-	5.0	uA	

Note.

1. VSO the output voltage of analog output pins S1 to S528
2. tSD : LCD Source driver delay
3. Target frame frequency = 60 Hz, Display line = 220, Back porch = 8, Front porch = 8, RTN3-0 = "0000"
VC3-0 = "1011", DC11/DC10/DC21/DC20/DC31/DC30 = "000000", BT2-0 = "000", GVD6-0 = "11111111"
VCM6-0 = "11111111", VML6-0 = "11111111", SAP3-0 = "0011", VCIR2-0 = "011"
4. Target frame frequency = 60 Hz, Display line = 220, Back porch = 8, Front porch = 8, RTN3-0 = "0000"
VC3-0 = "1011", DC11/DC10/DC21/DC20/DC31/DC30 = "000000", BT2-0 = "000", GVD6-0 = "11111111"
VCM6-0 = "11111111", VML6-0 = "11111111"

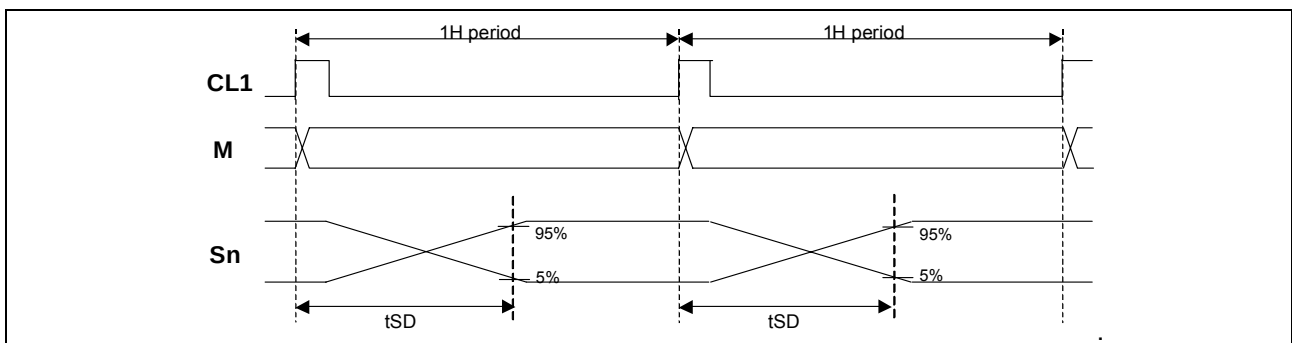


Figure 7. LCD source driver delay

6.3. AC characteristics

Table 13. Parallel Write Interface Characteristics (68 Mode)

(VDD = 1.5V, VDD3 = 1.65 to 3.3V, TA = -40 to +85oC)

Characteristic		Symbol	Specification		Unit
			Min.	Max.	
Cycle time	Write	tCYCW68	100	-	ns
	Read	tCYCR68	500	-	
Pulse rise / fall time		tR, tF	-	15	
Pulse width low	Write	tWLW68	33	-	
	Read	tWLR68	250	-	
Pulse width high	Write	tWHW68	33	-	
	Read	tWHR68	250	-	
RS,RW to CSB, E setup time		tAS68	10	-	
RS,RW to CSB, E hold time		tAH68	2	-	
CSB to E time		tCW68	15	-	
Write data setup time		tWDS68	60	-	
Write data hold time		tWDH68	15	-	
Read data delay time		tRDD68	-	200	
Read data hold time		tRDH68	5	-	

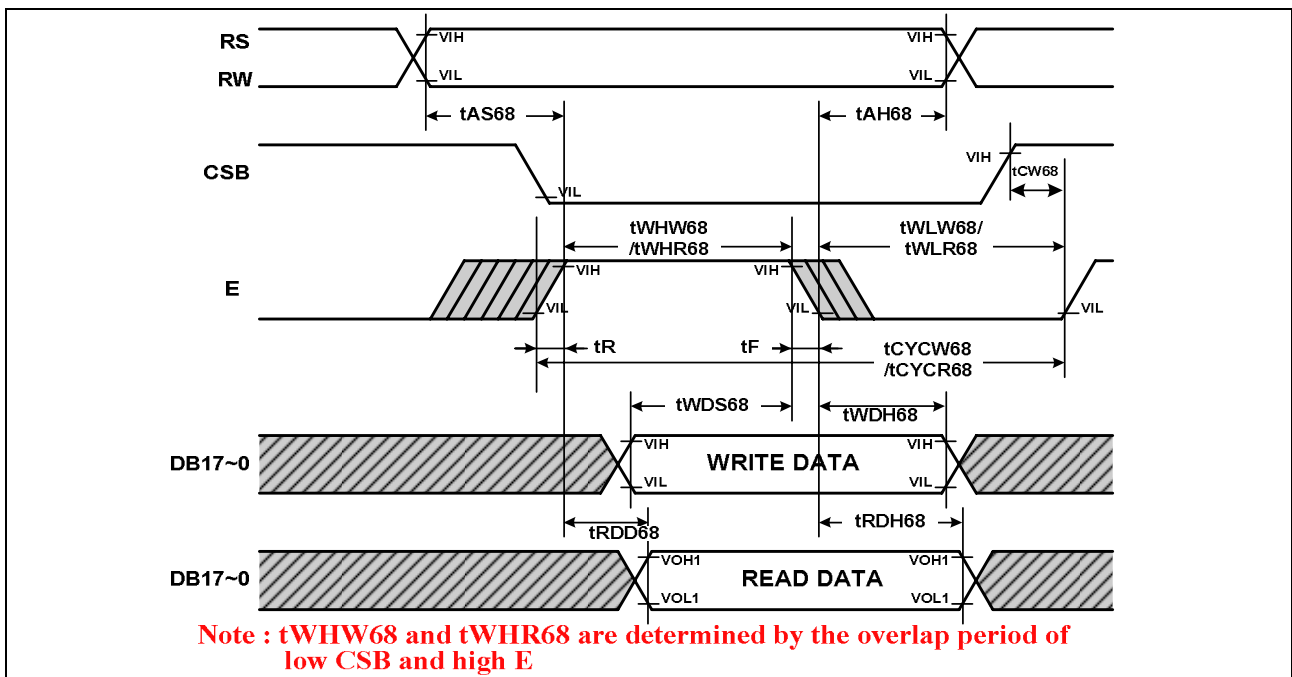

Figure 8. AC Characteristics (68 Mode)

Table 14. Parallel Write Interface Characteristics (80 Mode)

(VDD = 1.5V, VDD3 = 1.65 to 3.3V, TA = -40 to +85oC)

Characteristic		Symbol	Specification		Unit
			Min.	Max.	
Cycle time	Write	tCYCW80	100	-	ns
	Read	tCYCR80	500	-	
Pulse rise / fall time		tR, tF	-	15	
Pulse width low	Write	tWLW80	33	-	
	Read	tWLR80	250	-	
Pulse width high	Write	tWHW80	33	-	
	Read	tWHR80	250	-	
RS to CSB, WRB(RDB) setup time		tAS80	10	-	
RS to CSB, WRB(RDB) hold time		tAH80	2	-	
CSB to WRB(RDB) time		tCW80	15	-	
Write data setup time		tWDS80	20	-	
Write data hold time		tWDH80	10	-	
Read data delay time		tRDD80	-	200	
Read data hold time		tRDH80	10	-	

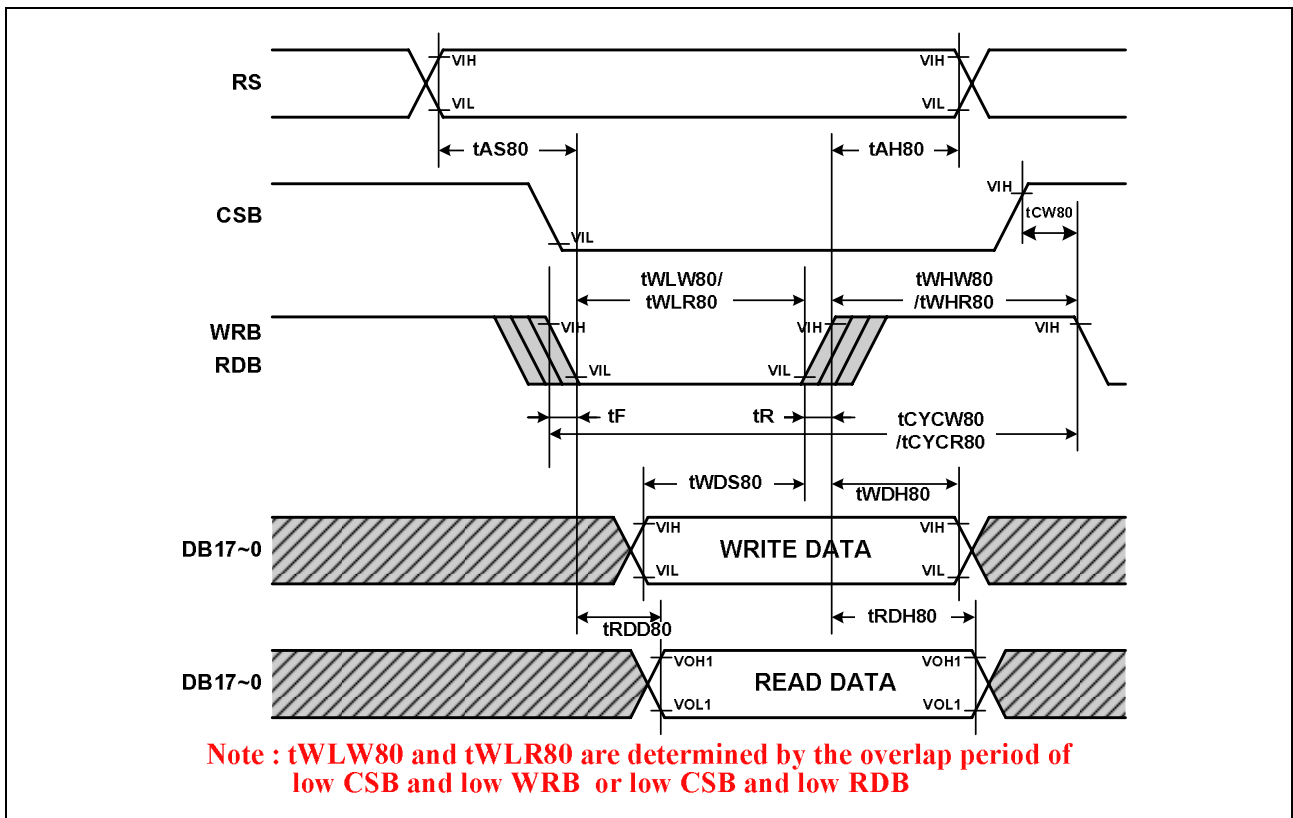


Figure 9. AC Characteristics (80 Mode)

Table 15. Clock Synchronized Serial Write Mode Characteristics

(VDD = 1.5V, VDD3 = 1.65 to 3.3V, TA = -40 to +85oC)

Characteristic	Symbol	specification		Unit
		Min.	Max.	
Serial clock write cycle time	tscyc	130	-	ns
Serial clock read cycle time	tscyc	250	-	ns
Serial clock rise / fall time	tR, tF	-	15	ns
Pulse width high for write	tSCHW	50	-	ns
Pulse width high for read	tSCHR	110	-	ns
Pulse width low for write	tSCLW	50	-	ns
Pulse width low for read	tSCLR	110	-	ns
Chip Select setup time	tCSS	20	-	ns
Chip Select hold time	tCSH	60	-	ns
Serial input data setup time	tSIDS	30	-	ns
Serial input data hold time	tSIDH	30	-	ns
Serial output data delay time	tSODD	-	130	ns
Serial output data hold time	tSODH	5	-	ns

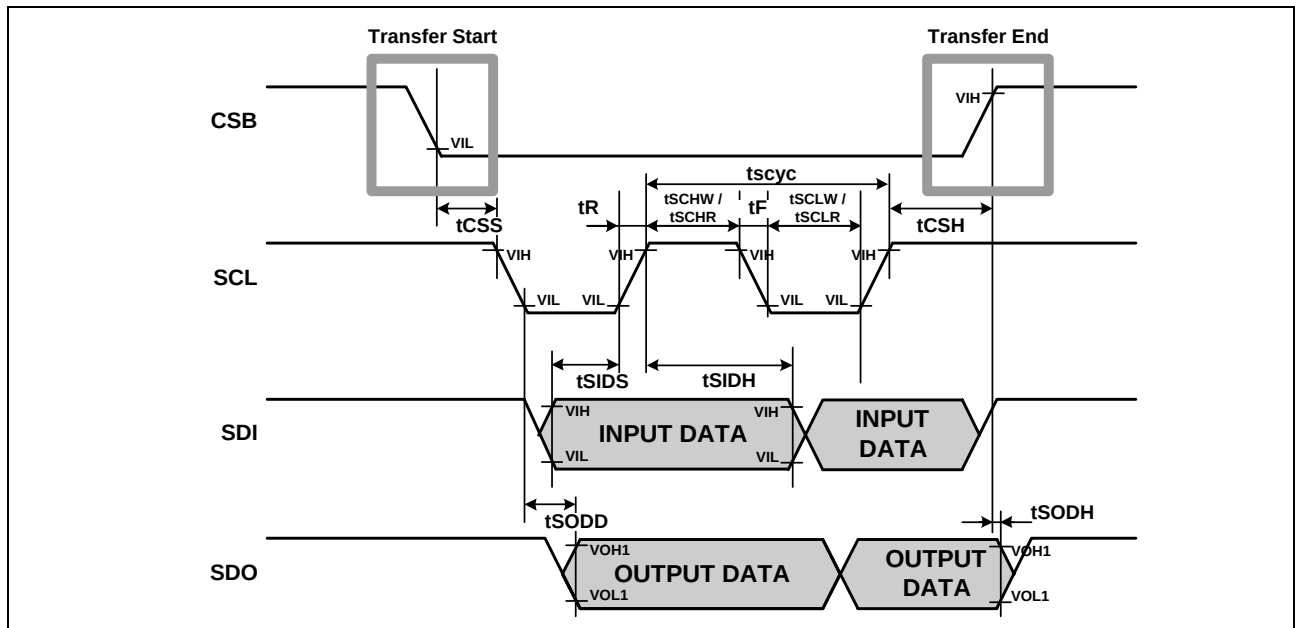


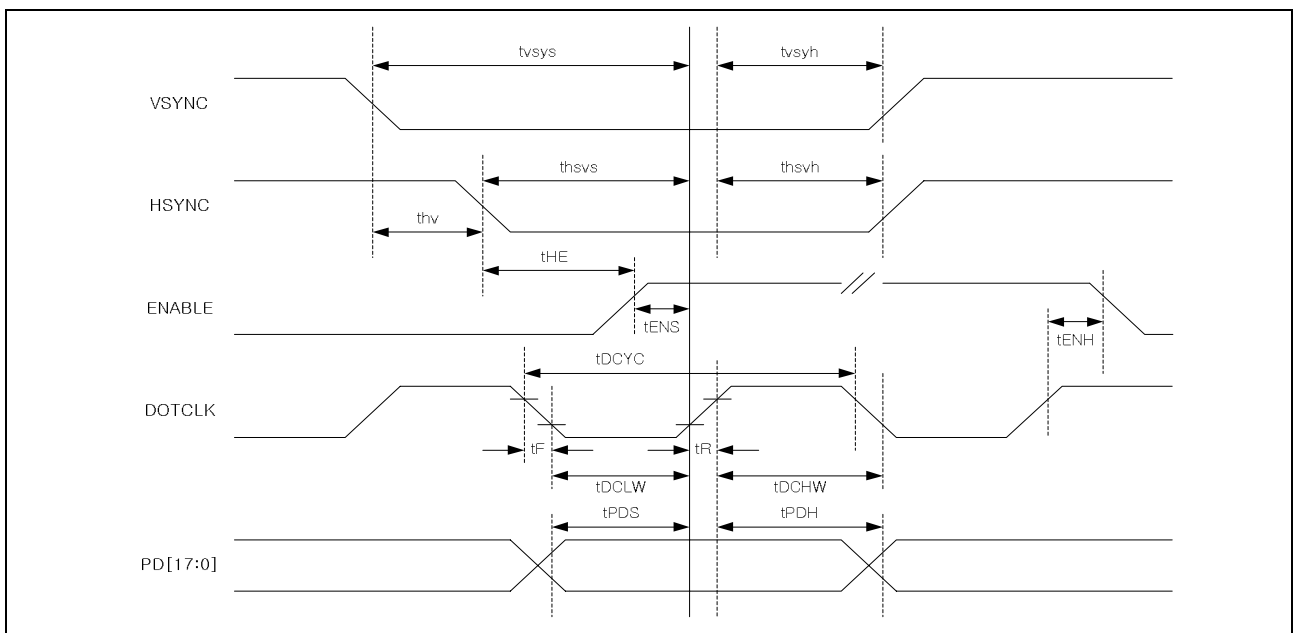
Figure 10. AC Characteristics (SPI Mode)

Table 16. RGB Data Interface Characteristics

(VDD = 1.5V, VDD3 = 1.65 to 3.3V, TA = -40 to +85oC)

Characteristic	Symbol	18/16bit RGB interface		6bit RGB interface		Unit
		Min.	Max.	Min.	Max.	
DOTCLK cycle time	tDCYC	100	-	100	-	ns
DOTCLK rise / fall time	tR, tF	-	15	-	15	
DOTCLK Pulse width high	tDCHW	40	-	40	-	
DOTCLK Pulse width low	tDCLW	40	-	40	-	
Vertical Sync Setup Time	tvsys	30	-	30	-	
Vertical Sync Hold Time	tvsyh	30	-	30	-	
Horizontal Sync Setup Time	thsys	30	-	30	-	
Horizontal Sync Hold Time	thsyh	30	-	30	-	
ENABLE setup time	tENS	30	-	30	-	
ENABLE hold time	tENH	20	-	20	-	
PD data setup time	tPDS	30	-	30	-	
PD data hold time	tPDH	20	-	20	-	
HSYNC-ENABLE Time	tHE	1	HBP	1	HBP	tDCYC
VSYNC-HSYNC Time	thv	1	175	1	527	

Note. HBP is Horizontal Back-porch.



(When VSPL=0, HSPL=0, DPL=0, EPL=1)

Figure 11. Figure 90 : AC Characteristics (RGB Interface Mode)

Note. When RGB interface mode, PD [17:0] corresponds to DB [17:0]

6.4. Reset Input Timing

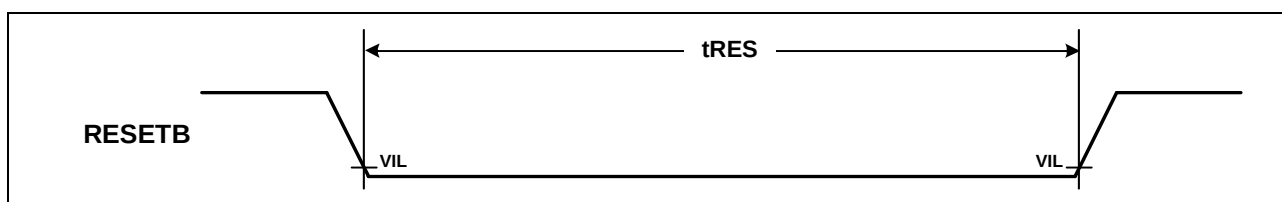


Figure 12. AC characteristics (RESET timing)

Note. Reset low pulse width shorter than 10us do not make reset. It means undesired short pulse such as glitch, bouncing noise or electrostatic discharge do not cause irregular system reset. Please refer to the table below.

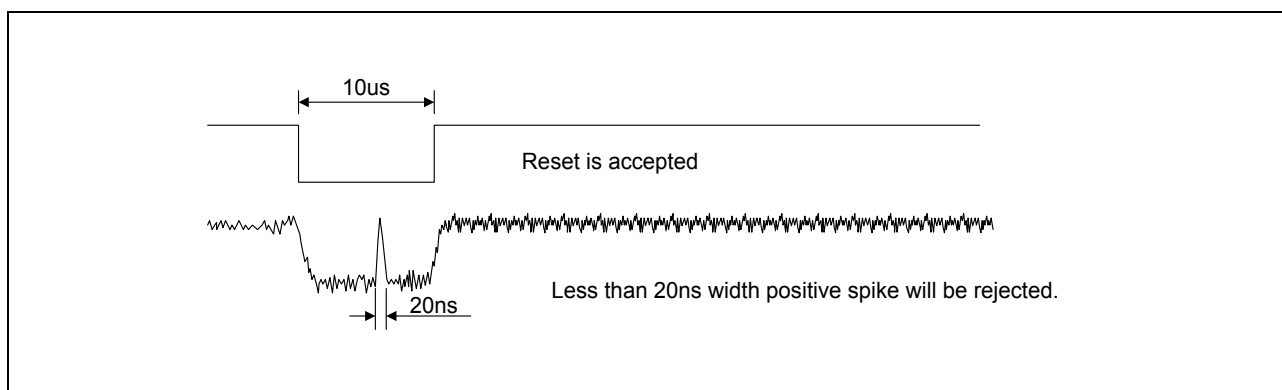
Parameter	Description	Min	Max	Unit
tRES	Reset low pulse width	10	-	us

Table 17. Reset Operation regarding tRES Pulse Width

tRES Pulse	Action
Shorter than 5 us	No reset
Longer than 10 us	Reset
Between 5 us and 10 us	Not determined

1. User may or may not use RESETB pin. In order to use it, user should satisfy the conditions described in the above tables. But when not wants to use RESETB, user may fix this pin to VDD3 level because internally generated POR (Power-On-Reset) is used.

2. Spike Rejection also applies during a valid reset pulse as shown below:



6.5. External Power On/Off Sequence

6.5.1. External Power On Sequence

VDD3 must be applied earlier than VCI or at least applied simultaneously with VCI. When regulator cap is 1uF, RESETB must be applied after VCI have been applied. The applied time gap between VCI and RESETB is minimum 1ms. As regulator cap becomes larger, this time gap must be increased. Otherwise function is not guaranteed.

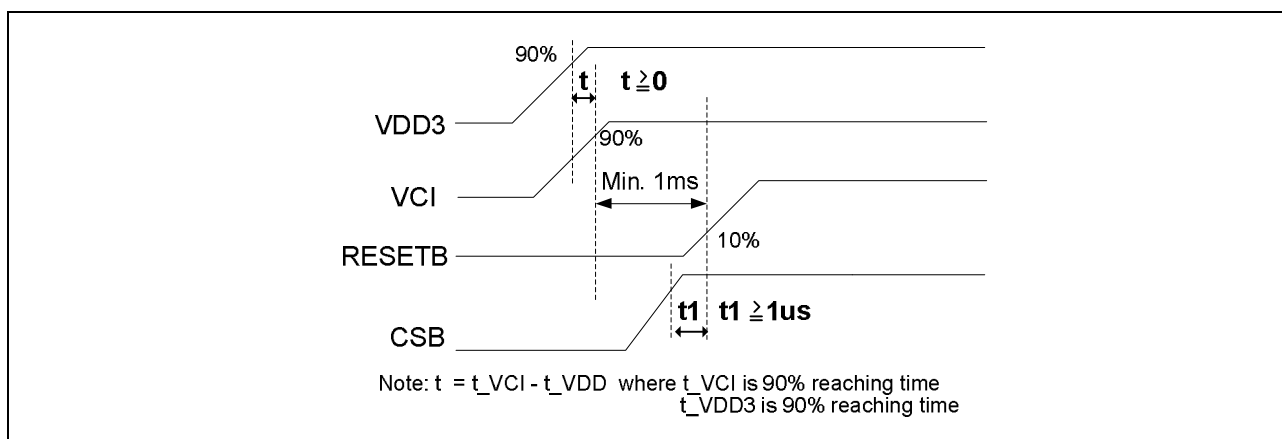


Figure 13. External power on sequence

6.5.2. External Power Off Sequence

VDD3 must be powered down later than VCI or at least powered down simultaneously with VCI. VCI must be powered down after RESETB have been powered down. The time gap of powered down between RESETB and VCI is minimum 1ms. Otherwise function is not guaranteed.

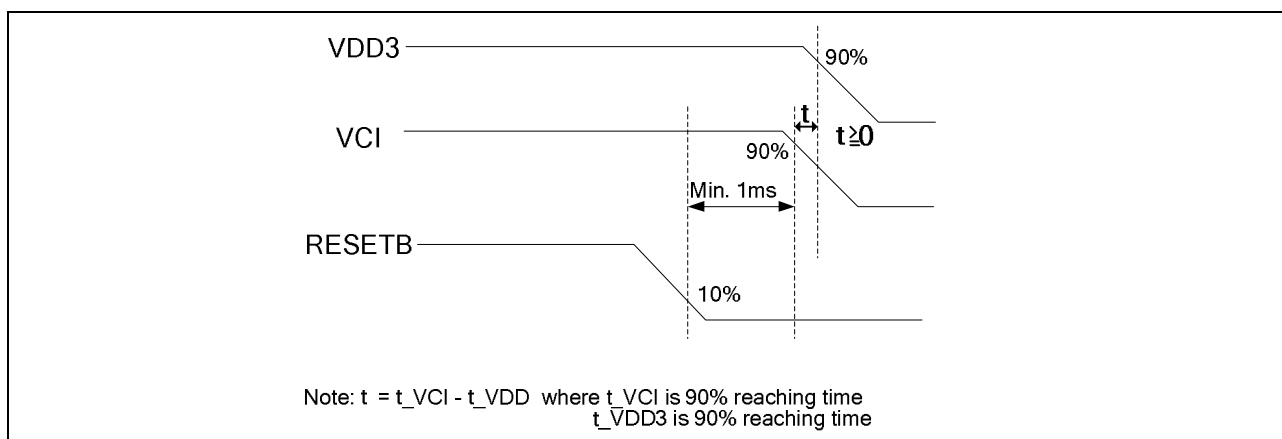


Figure 14. External power off sequence

7. FUNCTIONAL DESCRIPTION

7.1. SYSTEM INTERFACE

The IC has nine high-speed system interfaces: a 80-Series 18-/16-/9-/8-bit bus, a 68-Series 18-/16-/9-/8-bit and SPI(Serial Peripheral Interface).

The IC has three 18-bit registers: an index register (IR), a write data register (WDR), and a read data register (RDR). The IR stores index information for control register and GRAM. WDR temporarily stores data to be written into control register and GRAM. RDR temporarily stores data read from GRAM. Data written into the GRAM from MPU is initially written to WDR and then written to the GRAM automatically. Data is read through the RDR when reading from the GRAM, and the first read data is invalid and the second and the following data are valid.

Execution time for instruction, except oscillation start, is 0-clock cycle so that instructions can be written in succession.

Table 18. Register Selection (18-/16-/9-/8- Parallel Interface)

SYSTEM	RW_WRB	E_RDB	RS	Operations
68	0	1	0	Write index to IR
	1	1	0	Read internal status
	0	1	1	Write to control register and GRAM through WDR
	1	1	1	Read from GRAM through RDR
80	0	1	0	Write index to IR
	1	0	0	Read internal status
	0	1	1	Write to control register and GRAM through WDR
	1	0	1	Read from GRAM through RDR

Table 19. CSB signal (GRAM update control)

CSB	Operation
0	Data is written to GRAM, GRAM address is updated
1	Data is not written to GRAM, GRAM address is not updated

Table 20. Register Selection (Serial Peripheral Interface)

R/W bit	RS bit	Operation
0	0	Write index to IR
1	0	Read internal status
0	1	Write data to control register and GRAM through WDR
1	1	Read data from GRAM through RDR

7.2. RGB INTERFACE

The IC has RGB interface for the reproduction of motion picture display. When the RGB interface is used, the synchronization signals (VSYNC, HSYNC, and DOTCLK) are available for the display operation. The data for display (DB [17:0]) can be written according to the values of ENABLE and DOTCLK. This allows a flicker-free update of screen.

7.3. GRAPHICS RAM

The graphics RAM (GRAM) has 18-bits/pixel and stores the bit-pattern data of 176-RGB x 220 pixels.

The IC has an address counter for GRAM access. The address counter (AC) assigns addresses to the GRAM. When an address set instruction is performed, the address from system interface is sent to this AC. After writing into GRAM, AC is automatically increased (or decremented) by 1. When read data from GRAM, AC is not updated. Window Address Function allows data to be written only into the Window specified by designated control registers.

7.4. PANEL INTERFACE CONTROLLER

The Panel Interface Controller generates timing signals for TFT-LCD Driver and control signals for the operation of internal circuits such as source driver array and GRAM. GRAM read operations done by this Panel Interface Controller and GRAM write operations done through system interface are performed independently to avoid the interference between them.

7.5. GRAYSCALE VOLTAGE GENERATOR

The grayscale voltage circuit block generates a certain voltage level that is specified by the grayscale - adjusting resistor for LCD driver circuit. By use of the generator, 262,144 colors can be displayed. For details, see the Y-adjusting resistor section.

7.6. OSCILLATION CIRCUIT (OSC)

The IC can provide R-C oscillator without an external resistor. The appropriate oscillation frequency for controlling operating voltage, display size, and frame frequency, can be obtained by adjusting the register setting value [R0Fh]. Clock pulse can also be supplied externally. Since R-C oscillator is suspended during the standby mode, current consumption can be reduced. For details, see the Oscillation Circuit section.

7.7. SOURCE DRIVER ARRAY

The liquid crystal display source driver array consists of 528 drivers (S1 to S528).

Display pattern data is latched when 528-bit data is ready. Then the latched data enables the source drivers to output the expected voltage level. The SS bit can change the shift direction of 528-bit data by selecting an appropriate direction for the panel configuration.

7.8. GATE DRIVER ARRAY

The liquid crystal display gate driver array consists of 220 gate drivers (G1 to G220).

The VGH or VGL level is controlled by the signal from the gate control circuit. G1 and G220 are IC test pads for itself.

7.9. GRAM ADDRESS MAP

The image data stored in GRAM corresponds to real pixel on display as shown below.

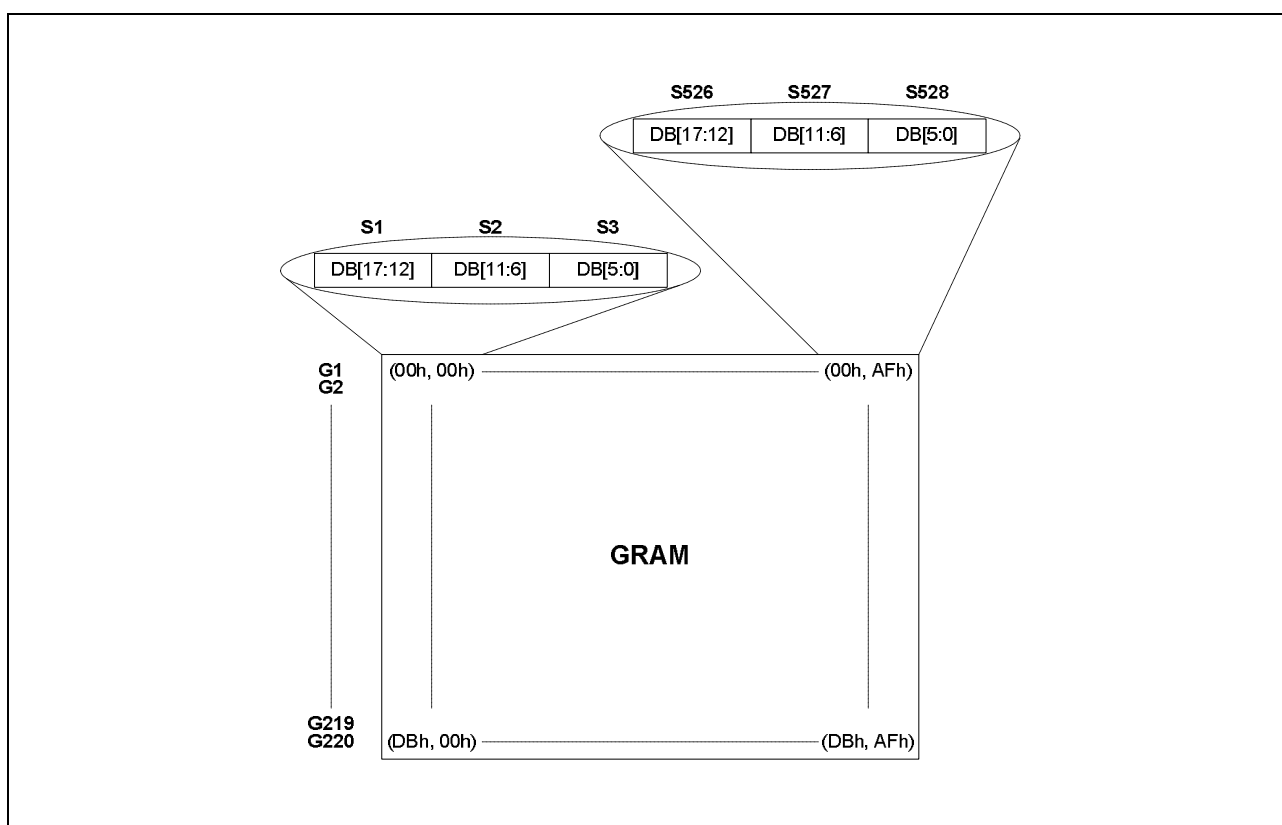


Figure 15. GRAM Address and Display Image

Note. The display condition of this figure is like this. SS = 0, BGR = 0, GS = 0.

8. Instruction Sets

8.1. Introduction

The IC uses 18-bit bus architecture. To execute an instruction to the IC, the control information from the external 18/16/9/8-bit data is stored in Index Register (IR) and Control Register (CR) as described later.

The internal operation of the IC is determined by the set of data sent from MCU. These data, which consists of the register selection signal (RS), the write/read signals (E/RWB for 68-Series, WRB/RDB for 80-Series), and the internal 16-bit data bus signals (IB15 to IB0), generate instructions.

There are eight categories of instructions that

- Specifies the index
- Reads the status
- Controls the display
- Controls power management
- Processes the graphics data
- Sets internal GRAM addresses
- Transfers data to and from the internal GRAM
- Sets grayscale level for the internal grayscale palette table

Normally, instructions writing data are used the most frequently. So, the automatic update of internal GRAM address after each data write can lighten the microcomputer's load. Because instructions are executed in 0 cycles, they can be written in succession.

The 16-bit instruction assignment varies with interface mode specified by IM. And the assignment for each interface mode is shown in SYSTEM INTERFACE section described later.

8.2. Instruction Set

Table 21. Instruction set I

Reg.No	R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0	Register Name / Description
IR	W	0	X	X	X	X	X	X	X	X	ID7	ID6	ID5	ID4	ID3	ID2	ID1	ID0	Index / Sets the index register value
SR	R	0	X	X	X	X	X	X	X	L8	L7	L6	L5	L4	L3	L2	L1	L0	Status read / Reads the internal status of the S6D0164
R00h	R	1	0	0	0	0	0	0	0	1	0	1	1	0	0	1	0	0	Device code read / Read 0164H
R01h	W	1	VSPL (0)	HSPL (0)	DPL (0)	EPL (0)	0	SM (0)	GS (0)	SS (0)	0	0	0	NL4 (1)	NL3 (1)	NL2 (1)	NL1 (0)	NL0 (0)	Driver output control(R01H) / VSPL: set polarity of VSYNC pin. HSPL: set polarity of HSYNC pin. DPL: set polarity of DOTCLK pin. EPL: set polarity of ENABLE pin. SM: gate driver division drive control GS: gate driver shift direction SS: source driver shift direction. NL4-0: number of driving lines.
R02h	W	1	0	0	0	0	0	0	INV1 (0)	INV0 (1)	0	0	0	0	0	0	0	FLD (0)	LCD-Driving-waveform control (R02H)/ INV1-0: Line/Frame inversion setting FLD: Interface Mode Control
R03h	W	1	0	0	0	BGR (0)	0	0	MDT1 (0)	MDT0 (0)	0	0	ID1 (1)	ID0 (1)	AM (0)	0	0	0	Entry mode(R03H) / BGR: RGB swap control MDT1-0: Multiple Data Transfer ID1-0: address counter Increment / Decrement control AM: horizontal / vertical RAM update
R07h	W	1	0	0	0	FLM_MON (0)	0	0	0	0	0	0	0	GON (0)	CL (0)	REV (0)	D1 (0)	D0 (0)	Display control (R07H) / FLM_MON: Enable Frame Flag Output(FLM) GON: gate on/off control CL: 8-color display mode enable REV: display area inversion drive D1-0: source output control
R08h	W	1	0	0	0	0	FP3 (1)	FP2 (0)	FP1 (0)	FP0 (0)	0	0	0	0	BP3 (1)	BP2 (0)	BP1 (0)	BP0 (0)	Blank period control 1 (R08H)/ FP3-0: Front porch setting BP3-0: Back porch setting
R0Bh	W	1	NO3 (0)	NO2 (0)	NO1 (0)	NO0 (1)	SDT3 (0)	SDT2 (0)	SDT1 (0)	SDT0 (1)	0	0	0	0	RTN3 (0)	RTN2 (0)	RTN1 (0)	RTN0 (0)	Frame cycle control (R0BH)/ NO3-0: specify the amount of non-overlap SDT3-0: set amount of source delay RTN3-0: set the 1-H period
R0Ch	W	1	0	0	0	0	0	0	0	RM (0)	0	0	0	DM (0)	0	0	RIM1 (0)	RIM0 (0)	External interface control(R0CH) / RM: specify the interface for RAM access DM: specify display operation mode RIM1-0: specify RGB-LUT mode
R0Fh	W	1	0	0	0	FOSC4 (0)	FOSC3 (0)	FOSC2 (1)	FOSC1 (0)	FOSC0 (1)	0	0	0	0	0	0	0	OSC ON (1)	Start oscillation(R0FH) / FOSC4-0: Adjust frequency of oscillator OSC0N: oscillator ON
R10h	W	1	0	0	0	0	SAP3 (0)	SAP2 (0)	SAP1 (1)	SAP0 (0)	0	0	0	0	0	0	DSTB (0)	STB (0)	Power control 1 (R10H) / SAP3-0: adjust amount of current for source driver amp DSTB: deep standby mode control STB: standby mode control
R11h	W	1	0	0	0	APON (0)	PON3 (0)	PON2 (0)	PON1 (0)	PON (0)	0	AB_VCI1 (0)	AON (0)	VC1_E N (0)	VC3 (0)	VC2 (0)	VC1 (0)	VC0 (0)	Power control 2 (R11H) / APON: auto power on control PON3: VCL booster control PON2: VGL booster control PON1: VGH booster circuit control PON: AVDD booster circuit control AB_VCI1: Set VCI1 output equal to VCI AON: operation start bit for the amplifier. VC1_EN: VCI1 amplifier control VC3-0: set VCI1 voltage
R12h	W	1	0	BT2 (0)	BT1 (0)	BT0 (0)	0	0	DC11 (0)	DC10 (0)	0	0	DC21 (0)	DC20 (0)	0	0	DC31 (0)	DC30 (0)	Power control 3 (R12H) / BT2-0: Adjust scale factor DC11-0: Adjust the frequency in stepup1 DC21-0: Adjust the frequency in stepup2 DC31-0: Adjust the frequency in stepup3
R13h	W	1	0	0	0	DCR_EX (0)	0	DCR2 (0)	DCR1 (0)	DCR0 (0)	0	GVD6 (0)	GVD5 (0)	GVD4 (0)	GVD3 (0)	GVD2 (0)	GVD1 (0)	GVD0 (0)	Power control 4 (R13H) / DCR_EX: Input signal selection. DCR2-0: Set clock cycle for step-up circuit. GVD6-0: set GVD0 voltage
R14h	W	1	VCOM G (1)	VCM6 (0)	VCM5 (0)	VCM4 (0)	VCM3 (0)	VCM2 (0)	VCM1 (0)	VCM0 (0)	VCMR (0)	VML6 (0)	VML5 (0)	VML4 (0)	VML3 (0)	VML2 (0)	VML1 (0)	VML0 (0)	Power control 5 (R14H) / VCOMG: VCOML -> GND VCMR: VCOMH control VCM6-0: set the VCOMH voltage VML6-0: set the VCOML Amplitude
R15h	W	1	0	0	0	0	0	0	0	0	0	VCIR2 (0)	VCIR1 (0)	VCIR0 (0)	0	0	0	0	VCI Recycling (R15H) / VCIR2-0: VCI Recycling period setting
R20h	W	1	0	0	0	0	0	0	0	0	AD7 (0)	AD6 (0)	AD5 (0)	AD4 (0)	AD3 (0)	AD2 (0)	AD1 (0)	AD0 (0)	RAM address register (R20H) / AD7-AD0
R21h	W	1	0	0	0	0	0	0	0	0	AD15 (0)	AD14 (0)	AD13 (0)	AD12 (0)	AD11 (0)	AD10 (0)	AD9 (0)	AD8 (0)	RAM address register (R21H) / AD15-AD8
R22h	W	1	WD17-0: Pin assignment varies according to the interface method																Write data to GRAM (R22H) / WD17-0: Input data for GRAM
	R	1	RD17-0: Pin assignment varies according to the interface method																Read data from GRAM (R22H) / RD17-0: Read data from GRAM
R28h	W	1	0	0	0	0	0	0	0	0	1	1	0	0	1	1	1	0	Software RESET(R28h) / 00CEh: Software RESET
R30h	W	1	0	0	0	0	0	0	0	0	0	0	0	SCN4 (0)	SCN3 (0)	SCN2 (0)	SCN1 (0)	SCN0 (0)	Gate Scan Position (R30H) / SCN4-0: scan starting position of gate
R31h	W	1	0	0	0	0	0	0	0	0	SEA7 (1)	SEA6 (1)	SEA5 (0)	SEA4 (1)	SEA3 (1)	SEA2 (0)	SEA1 (1)	SEA0 (1)	Vertical scroll control 1 (R31H,R32H) / SEA7-0: Scroll End Address SEA0-0: Scroll Start Address
R32h	W	1	0	0	0	0	0	0	0	0	SSA7 (0)	SSA6 (0)	SSA5 (0)	SSA4 (0)	SSA3 (0)	SSA2 (0)	SSA1 (0)	SSA0 (0)	Vertical scroll control 2(R33H) / SST7-0: Scroll Start and Step
R33h	W	1	0	0	0	0	0	0	0	0	SST7 (0)	SST6 (0)	SST5 (0)	SST4 (0)	SST3 (0)	SST2 (0)	SST1 (0)	SST0 (0)	Partial screen driving position (R34H,R35H) / SST17-10: screen start position SST7-10: screen end position
R34h	W	1	0	0	0	0	0	0	0	0	SE17 (1)	SE16 (1)	SE15 (0)	SE14 (1)	SE13 (1)	SE12 (0)	SE11 (1)	SE10 (1)	Horizontal window address (R36H,R37H) / HEA7-0: Horizontal window address end position HEA0-0: Horizontal window address start position
R35h	W	1	0	0	0	0	0	0	0	0	SS17 (0)	SS16 (0)	SS15 (0)	SS14 (0)	SS13 (0)	SS12 (0)	SS11 (0)	SS10 (0)	Vertical window Address (R38H,R39H) / VEA7-0: Vertical window address end position VEA0-0: Vertical window address start position
R36h	W	1	0	0	0	0	0	0	0	0	HEA7 (1)	HEA6 (0)	HEA5 (1)	HEA4 (0)	HEA3 (1)	HEA2 (1)	HEA1 (1)	HEA0 (1)	Horizontal window address (R36H,R37H) / HEA7-0: Horizontal window address end position HEA0-0: Horizontal window address start position
R37h	W	1	0	0	0	0	0	0	0	0	HSA7 (0)	HSA6 (0)	HSA5 (0)	HSA4 (0)	HSA3 (0)	HSA2 (0)	HSA1 (0)	HSA0 (0)	Vertical window Address (R38H,R39H) / VEA7-0: Vertical window address end position VEA0-0: Vertical window address start position
R38h	W	1	0	0	0	0	0	0	0	0	VEA7 (1)	VEA6 (1)	VEA5 (0)	VEA4 (1)	VEA3 (1)	VEA2 (0)	VEA1 (1)	VEA0 (1)	Vertical window Address (R38H,R39H) / VEA7-0: Vertical window address end position VEA0-0: Vertical window address start position
R39h	W	1	0	0	0	0	0	0	0	0	VSA7 (0)	VSA6 (0)	VSA5 (0)	VSA4 (0)	VSA3 (0)	VSA2 (0)	VSA1 (0)	VSA0 (0)	Gamma control 1 (R50H) / Adjust Gamma voltage
R50h	W	1	0	0	0	0	PKP 13 (0)	PKP 12 (0)	PKP 11 (0)	PKP 10 (0)	0	0	0	0	PKP 03 (0)	PKP 02 (0)	PKP 01 (0)	PKP 00 (0)	Gamma control 2 (R51H) / Adjust Gamma voltage
R51h	W	1	0	0	0	0	PKP 33 (0)	PKP 32 (0)	PKP 31 (0)	PKP 30 (0)	0	0	0	0	PKP 23 (0)	PKP 22 (0)	PKP 21 (0)	PKP 20 (0)	Gamma control 3 (R52H) / Adjust Gamma voltage
R52h	W	1	0	0	0	0	PKP 53 (0)	PKP 52 (0)	PKP 51 (0)	PKP 50 (0)	0	0	0	0	PKP 43 (0)	PKP 42 (0)	PKP 41 (0)	PKP 40 (0)	Gamma control 4 (R53H) / Adjust Gamma voltage
R53h	W	1	0	0	0	0	PRP 13 (0)	PRP 12 (0)	PRP 11 (0)	PRP 10 (0)	0	0	0	0	PRP 03 (0)	PRP 02 (0)	PRP 01 (0)	PRP 00 (0)	Gamma control 4 (R53H) / Adjust Gamma voltage

R54h	W	1	0	0	0	0	PKN 13 (0)	PKN 12 (0)	PKN 11 (0)	PKN 10 (0)	0	0	0	0	PKN 03 (0)	PKN 02 (0)	PKN 01 (0)	PKN 00 (0)	Gamma control 5 (R54H)/ Adjust Gamma voltage
R55h	W	1	0	0	0	0	PKN 33 (0)	PKN 32 (0)	PKN 31 (0)	PKN 30 (0)	0	0	0	0	PKN 23 (0)	PKN 22 (0)	PKN 21 (0)	PKN 20 (0)	Gamma control 6 (R55H)/ Adjust Gamma voltage
R56h	W	1	0	0	0	0	PKN 53 (0)	PKN 52 (0)	PKN 51 (0)	PKN 50 (0)	0	0	0	0	PKN 43 (0)	PKN 42 (0)	PKN 41 (0)	PKN 40 (0)	Gamma control 7 (R56H)/ Adjust Gamma voltage
R57h	W	1	0	0	0	0	PRN 13 (0)	PRN 12 (0)	PRN 11 (0)	PRN 10 (0)	0	0	0	0	PRN 03 (0)	PRN 02 (0)	PRN 01 (0)	PRN 00 (0)	Gamma control 8 (R57H)/ Adjust Gamma voltage
R58h	W	1	0	0	0	VRP 14 (0)	VRP 13 (0)	VRP 12 (0)	VRP 11 (0)	VRP 10 (0)	0	0	0	0	VRP 04 (0)	VRP 03 (0)	VRP 02 (0)	VRP 01 (0)	Gamma control 9 (R58H)/ Adjust Amplitude voltage
R59h	W	1	0	0	0	VRN 14 (0)	VRN 13 (0)	VRN 12 (0)	VRN 11 (0)	VRN 10 (0)	0	0	0	0	VRN 04 (0)	VRN 03 (0)	VRN 02 (0)	VRN 01 (0)	Gamma control 10 (R59H)/ Adjust Amplitude voltage

Table 22. Instruction table 2

Reg.No	R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0	Register Name / Description
R80h	W	1	0	0	0	0	0	0	0	0	TEST_ KEY7 (0)	TEST_ KEY6 (0)	TEST_ KEY5 (0)	TEST_ KEY4 (0)	TEST_ KEY3 (0)	TEST_ KEY2 (0)	TEST_ KEY1 (0)	TEST_ KEY0 (0)	MTP Test Key (R80H) Test Key to update MTP Value.
R81h	W	1	MTP_ MODE (0)	MTP_ EX (0)	0	MTP_ SEL (0)	0	0	0	MTP_ ERB (1)	0	0	0	MTP_ WRB (1)	0	0	0	MTP_ LOAD (0)	MTP Control Registers (R81H)
R82h	W	1	0	0	0	0	0	0	0	0	0	0	0	MTP_ _DIN4 (0)	MTP_ _DIN3 (0)	MTP_ _DIN2 (0)	MTP_ _DIN1 (0)	MTP_ _DIN0 (0)	MTP Data Write (R82H)
	R	1	X	X	X	X	X	X	X	X	X	X	MTP_ DOUT5 (0)	MTP_ DOUT4 (0)	MTP_ DOUT3 (0)	MTP_ DOUT2 (0)	MTP_ DOUT1 (0)	MTP_ DOUT0 (0)	MTP Data Read (R82H)
R93h	R	1	P_ID1	P_ID0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	Panel ID (R93h)

8.3. Description of Instructions

8.3.1. Index Register (IR)

The index instruction specifies the RAM control indexes (R00h to R93h). It sets the register number in the range of 00000000 to 11111111 in binary form.

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	0	0	0	0	0	0	0	0	0	ID7	ID6	ID5	ID4	ID3	ID2	ID1	ID0

8.3.2. Status Read

The status read instruction reads out the internal status of the IC.

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
R	0	0	0	0	0	0	0	0	L8	L7	L6	L5	L4	L3	L2	L1	L0

L8–0: Indicate the driving raster-row position in scan address of GRAM where the liquid crystal display is being driven.

8.3.3. System Control (R00h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
R	1	0	0	0	0	0	0	0	1	0	1	1	0	0	1	0	0

Read device code “0164h”.

8.3.4. Driver Output Control (R01h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	VSPL	HSPL	DPL	EPL	0	SM	GS	SS	0	0	0	NL4	NL3	NL2	NL1	NL0

VSPL: reverses the polarity of the VSYNC signal.

VSPL= "0": VSYNC is low active.

VSPL= "1": VSYNC is high active.

HSPL: reverses the polarity of the HSYNC signal.

HSPL= "0": HSYNC is low active.

HSPL= "1": HSYNC is high active.

DPL: reverses the polarity of the DOTCLK signal.

DPL= "0": Display data is fetched at DOTCLK's rising edge.

DPL= "1": Display data is fetched at DOTCLK's falling edge.

EPL: Set the polarity of ENABLE pin while using RGB interface.

EPL = "0": ENABLE ="Low" / write data of DB17-0

ENABLE ="High" / do not write data of DB17-0

EPL = "1": ENABLE ="High" / write data of DB17-0

ENABLE ="Low" / do not write data of DB17-0

Table 23. Relationship between EPL, ENABLE and RAM access

EPL	ENABLE	RAM write	RAM address
0	0	Valid	Updated
0	1	Invalid	Held
1	0	Invalid	Held
1	1	Valid	Updated

SM: Select the division drive method of the gate driver. When SM = 0, even/odd division is selected; SM = 1, upper/lower division drive is selected by NL4-0(Upper NL4-0/2 and Lower NL4-0/2). Various connections between TFT panel and the IC can be supported with the combination of SM and GS bit.

Note. When SM = 1, NL setting is disable.

GS: Set the order of Gate Clock generation. When GS = 0, G1 is output first and G220 is finally output. When GS = 1, G220 is output first and G1 is finally output (NL = 5'b11100). But in case of NL = 5'b01100, when GS = 0, G1 is output first and G96 is finally output, and when GS = 1, G96 is output first and G1 is finally output

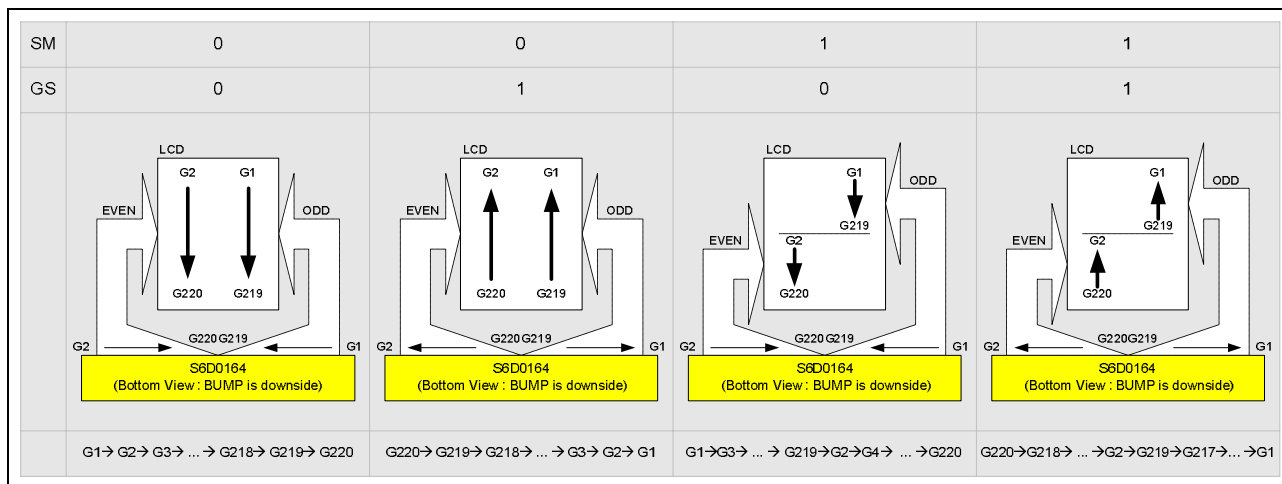


Figure 16. Gate Clock Generation order selection using GS and SM (NL = 5'b11100, SCN = 5'b00000)

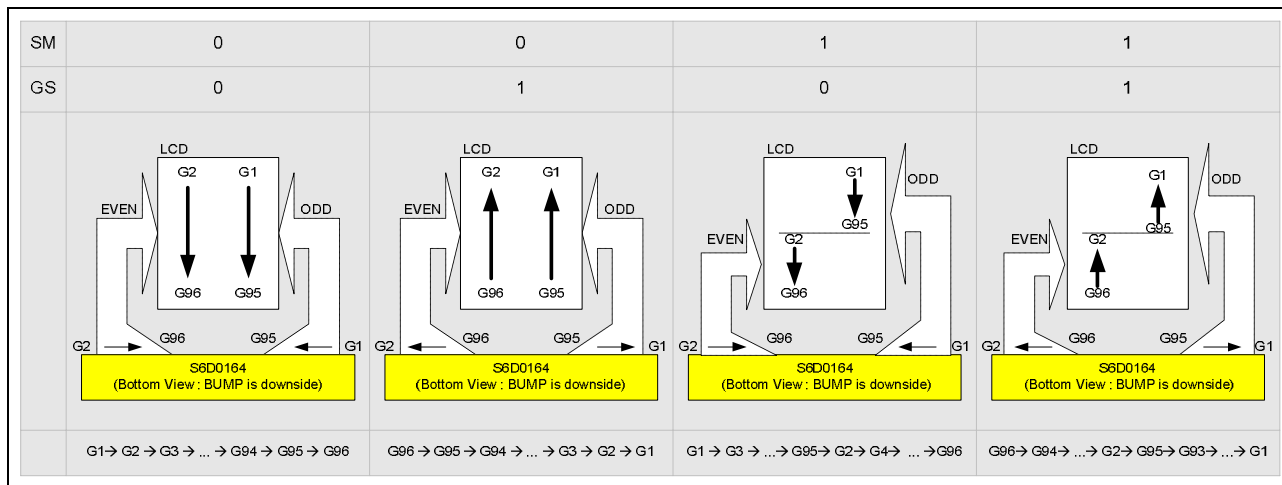


Figure 17. Gate Clock Generation order selection using GS and SM (NL = 5'b01100, SCN = 5'b00000)

SS

Select the direction of the source driver channel in pixel unit.

When user changes the value of SS, memory should be updated to apply the change.

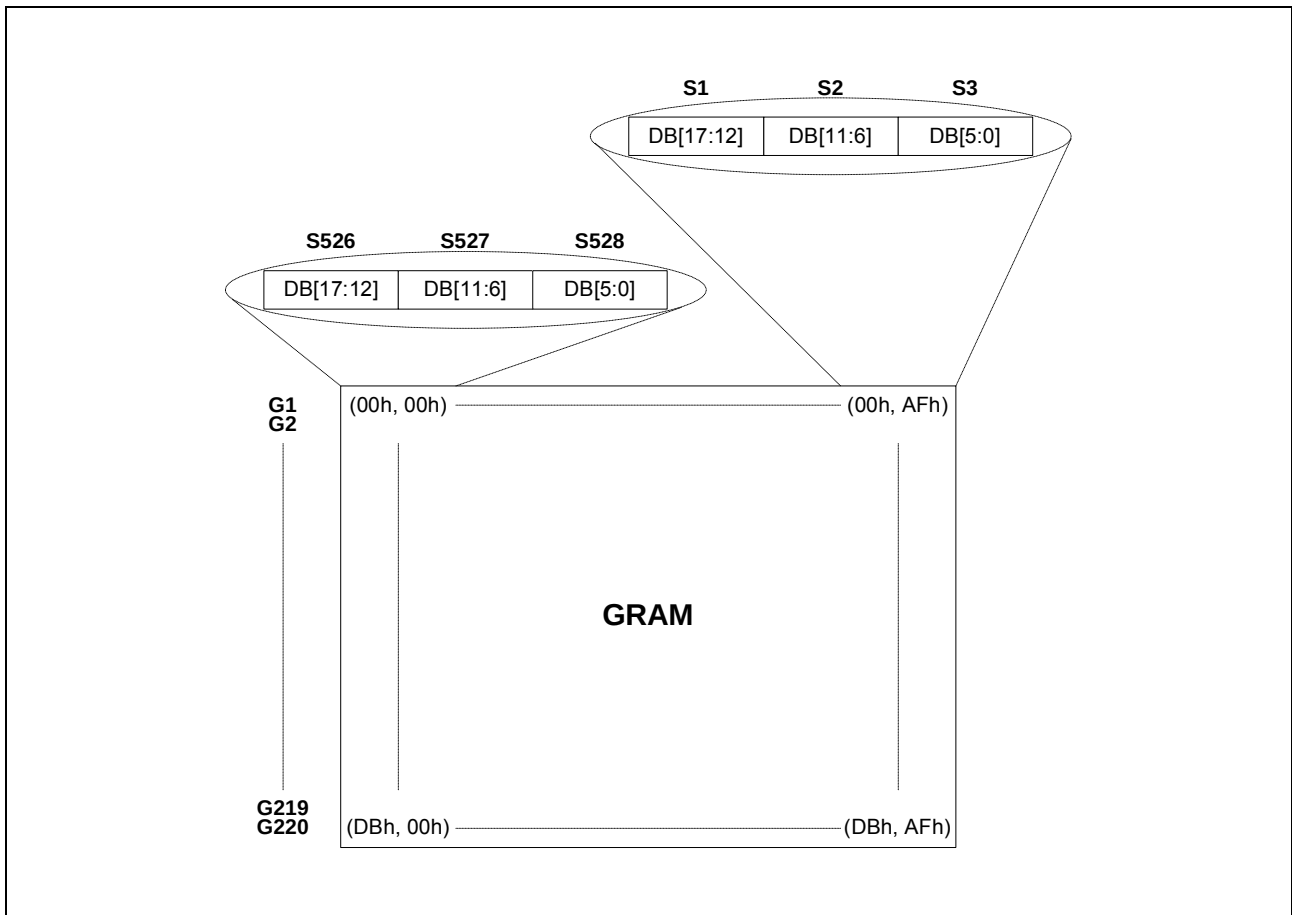


Figure 18. Image mirroring using SS register (SS = "1")

Note.

The display condition of this figure is like this.

SS = 1, BGR = 0, GS = 0.

NL

Specify the number of horizontal lines to be driven. The number of the lines can be adjusted in units of eight. GRAM address mapping is independent of this setting. The set value should be more than the panel size. Do not change setting of NL [4:0] in DISPLAY ON STATE.

Table 24. NL bit and Drive Duty (SCN = "00000")

NL[4:0]	Display Size	Drive Line	Gate Driver- Lines Used
00000	Reserved		
00001	528 X 8 dots	8	G1 to G8
00010	528 X 16 dots	16	G1 to G16
00011	528 X 24 dots	24	G1 to G24
00100	528 X 32 dots	32	G1 to G32
00101	528 X 40 dots	40	G1 to G40
00110	528 X 48 dots	48	G1 to G48
00111	528 X 56 dots	56	G1 to G56
01000	528 X 64 dots	64	G1 to G64
01001	528 X 72 dots	72	G1 to G72
01010	528 X 80 dots	80	G1 to G80
01011	528 X 88 dots	88	G1 to G88
01100	528 X 96 dots	96	G1 to G96
01101	528 X 104 dots	104	G1 to G104
01110	528 X 112 dots	112	G1 to G112
01111	528 X 120 dots	120	G1 to G120
10000	528 X 128 dots	128	G1 to G128
10001	528 X 136 dots	136	G1 to G136
10010	528 X 144 dots	144	G1 to G144
10011	528 X 152 dots	152	G1 to G152
10100	528 X 160 dots	160	G1 to G160
10101	528 X 168 dots	168	G1 to G168
10110	528 X 176 dots	176	G1 to G176
10111	528 X 184 dots	184	G1 to G184
11000	528 X 192 dots	192	G1 to G192
11001	528 X 200 dots	200	G1 to G200
11010	528 X 208 dots	208	G1 to G208
11011	528 X 216 dots	216	G1 to G216
11100	528 X 220 dots	220	G1 to G220
11101 11111	Setting Disable		

Note.

1. A FP (front porch) and BP (back porch) period will be inserted as blanking period (All gates output VGL level) before / after the driver scan through all of the scans.
2. When SM = 1, NL setting is disable.

8.3.5. LCD-Driving-Waveform Control (R02h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	INV1	INV0	0	0	0	0	0	0	0	FLD

INV1-0 / FLD

Set LCD inversion method as show below.

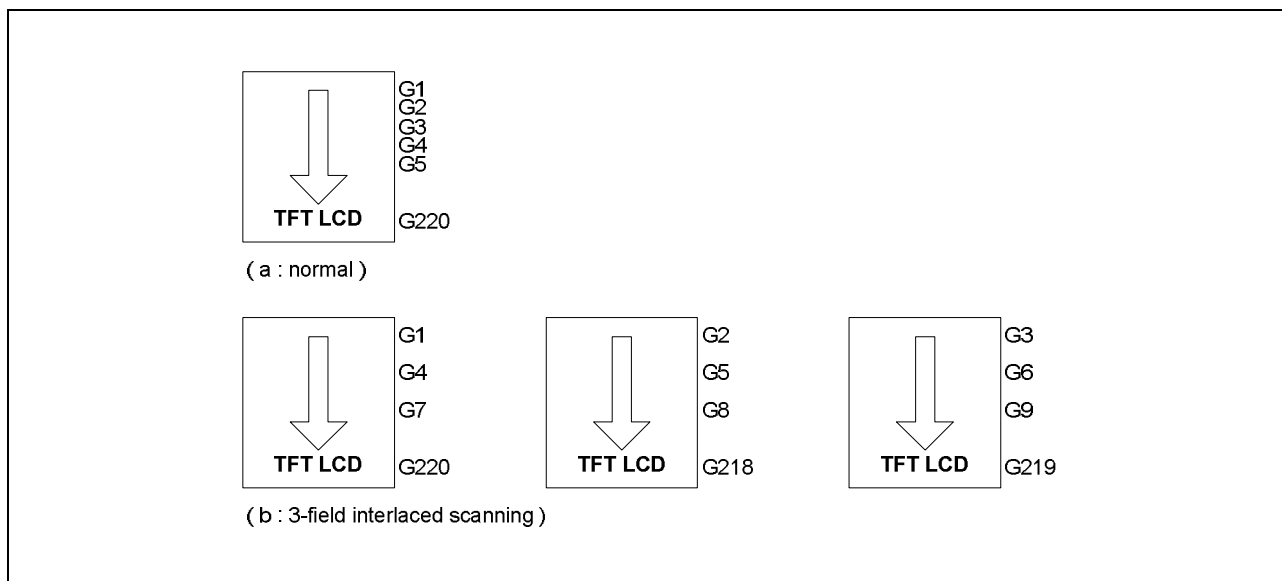
Enables or disables 3-field interlaced scanning function like below. When you want to save power consumption, you'd better enable 3 field interlaced scanning function. 3-field interlaced scanning function and Two Line Mixed Inversion scanning function are properly operated in FP=2, SM=0 and VCIR2-0=000.

Table 25. LCD inversion selection / Interlaced scanning method control

INV[1:0]	FLD	Description
00	0	Frame Inversion
	1	3 field interlace
01	0	Line Inversion
	1	Setting Disable
10	0	Two Line Mixed Inversion
	1	Setting Disable
11	0	No Inversion. Active with positive polarity (VCOM = Low)
	1	No Inversion. Active with negative polarity (VCOM = High)

Note.

1. The interlaced scanning method can be set in the 3-field interlace when both window addresses and display area is related with 176x220 resolution in normal mode e.g. window address setting, partial screen driving position, number of driving lines.
2. Employing 3-field interlaced scanning method could deteriorate the display quality.

**Figure 19. Interlaced scanning methods**

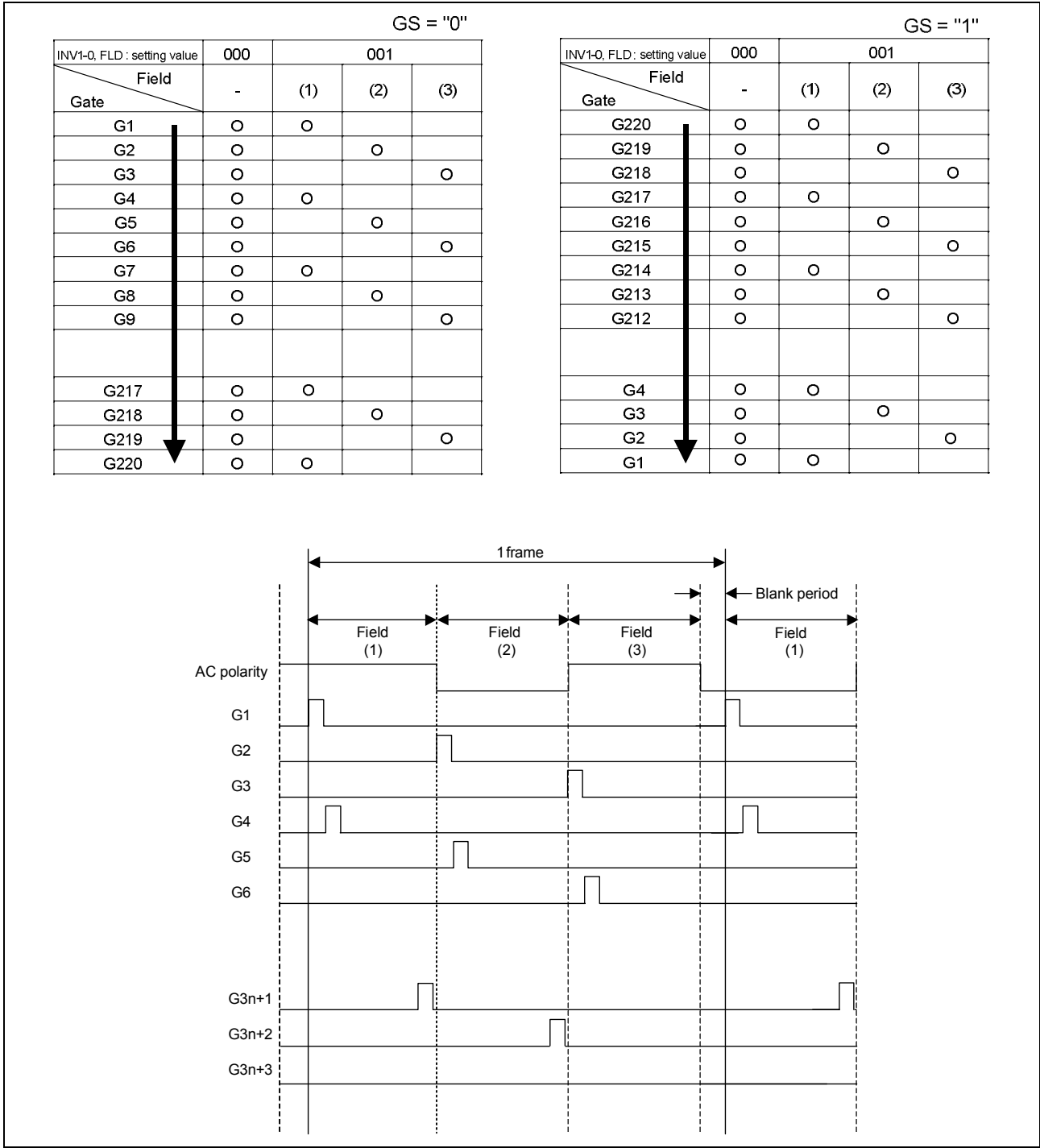


Figure 20. Interlace drive and output waveform (3 Field Interlace)

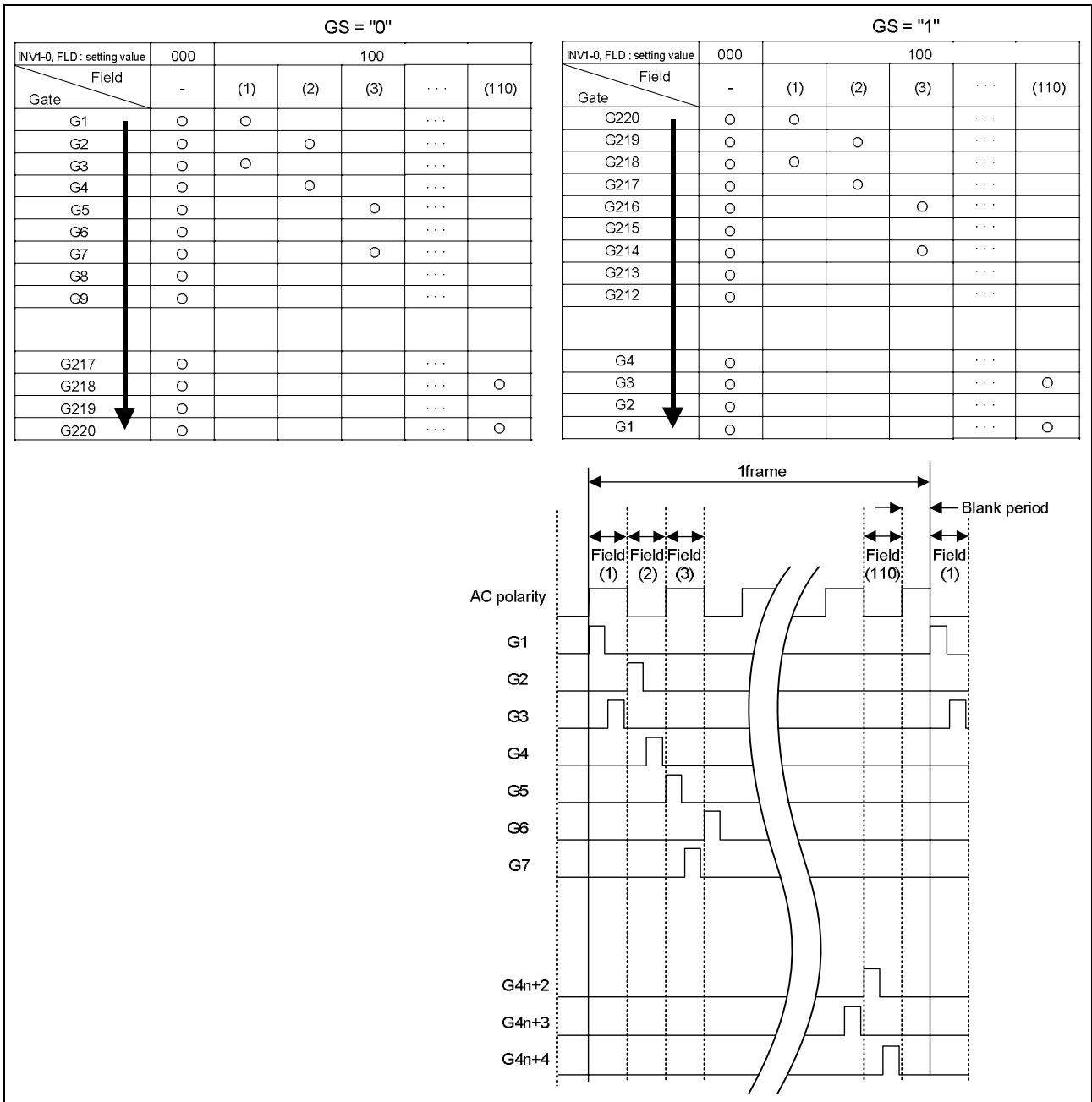


Figure 21. Interlace drive and output waveform (Two Line Mixed Inversion)

8.3.6. Entry Mode (R03h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	BGR	0	0	MDT1	MDT0	0	0	ID1	ID0	AM	0	0	0

BGR

When 18-bit data is written to GRAM through DB bus, RGB assignment can be changed.

- BGR = 0; {DB [17:12], DB [11:6], DB [5:0]} is assigned to {R, G, B}. Actually the analog value that corresponds to DB [17:12] is output firstly at source output

- BGR = 1; {DB [17:12], DB [11:6], DB [5:0]} is assigned to {B, G, R}. Actually the analog value that corresponds to DB [5:0] is output firstly at source output.

MDT1: This bit is active on the 68/80-system of 8-bit bus, and the data for 1-pixel is transported to the memory for 3 write cycles. This bit is on the 68/80-system of 16-bit bus, and the data for 1-pixel is transported to the memory for 2 write cycles. When the 68/80-system interface mode is not set in the 8-bit or 16-bit mode, set MDT1 bit to be "0".

MDT0: When 8-bit or 16-bit 68/80 interface mode and MDT1 bit = 1, MDT0 defines color depth for the IC.

Interface Mode	MDT1	MDT0	Write data to GRAM
68/80-system 8-bit	0	*	Note: n= lower 8 byte of address (0 to 175)
	1	0	Note: n= lower 8 byte of address (0 to 175)
	1	1	Note: n= lower 8 byte of address (0 to 175)

Interface Mode	MDT1	MDT0	Data Assignment
68/80-system 16-bit	0	1	1st Transmission2nd Transmission INPUT DATA DB 15DB 14DB 13DB 12DB 11DB 10DB 7DB 6DB 5DB 4DB 3DB 2DB 15DB 14DB 13DB 12DB 11DB 10 RGB Arrangement R1 5R1 4R1 3R1 2R1 1R1 0G1 5G1 4G1 3G1 2G1 1G1 0B1 5B1 4B1 3B1 2B1 1B1 0 Output S (3n + 1)S (3n + 2)S (3n + 3) Note: n= lower 8 byte of address (0 to 175)
			2nd Transmission3rd Transmission INPUT DATA DB 7DB 6DB 5DB 4DB 3DB 2DB 15DB 14DB 13DB 12DB 11DB 10DB 7DB 6DB 5DB 4DB 3DB 2 RGB Arrangement R2 5R2 4R2 3R2 2R2 1R2 0G2 5G2 4G2 3G2 2G2 1G2 0B2 5B2 4B2 3B2 2B2 1B2 0 Output S (3n + 4)S (3n + 5)S (3n + 6) Note: n= lower 8 byte of address (0 to 175)
	1	0	1st Transmission2nd Transmission INPUT DATA DB 17DB 16DB 15DB 14DB 13DB 12DB 11DB 10DB 8DB 7DB 6DB 5DB 4DB 3DB 2DB 1DB 17DB 16 RGB Arrangement R5R4R3R2R1R0G5G4G3G2G1G0B5B4B3B2B1B0 Output S (3n + 1)S (3n + 2)S (3n + 3) Note: n= lower 8 byte of address (0 to 175)
			1st Transmission2nd Transmission INPUT DATA DB 2DB 1DB 17DB 16DB 15DB 14DB 13DB 12DB 11DB 10DB 8DB 7DB 6DB 5DB 4DB 3DB 2DB 1 RGB Arrangement R5R4R3R2R1R0G5G4G3G2G1G0B5B4B3B2B1B0 Output S (3n + 1)S (3n + 2)S (3n + 3) Note: n= lower 8 byte of address (0 to 175)

8-bit (68/80-system), MDT10 = 10: 262k-color mode (3 times of 6-bit data transfer to GRAM)

8-bit (68/80-system), MDT10 = 11: 65k-color mode (5-bit, 6-bit, 5-bit data transfer to GRAM)

16-bit (68/80-system), MDT10 = 01: 262k-color mode (3 times of 2-pixel data transfer to GRAM)

16-bit (68/80-system), MDT10 = 10: 262k-color mode (16-bit, 2-bit data transfer to GRAM)

16-bit (68/80-system), MDT10 = 11: 262k-color mode (2-bit, 16-bit data transfer to GRAM)

ID

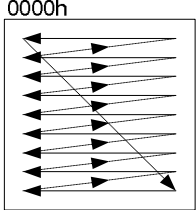
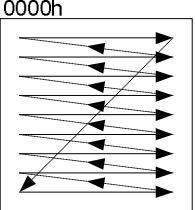
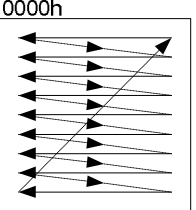
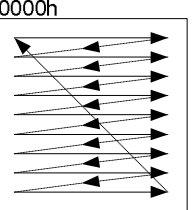
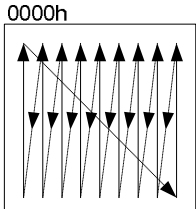
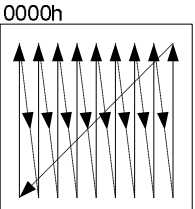
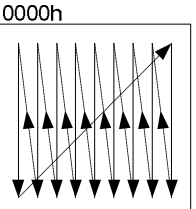
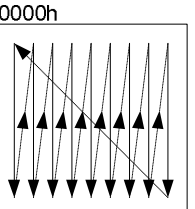
When ID [1], ID [0] = 1, the address counter (AC) is automatically increased by 1 after the data is written to the GRAM. When ID [1], ID [0] = 0, the AC is automatically decreased by 1 after the data is written to the GRAM.

The increment/decrement setting of the address counter using ID [1:0] is done independently for the horizontal address and vertical address.

AM

Set the automatic update method of the AC after the data is written to GRAM. When AM = "0", the data is continuously written in horizontally. When AM = "1", the data is continuously written vertically. When window addresses are specified, the GRAM in the window range can be written to according to the ID [1:0] and AM.

Table 26. Address Direction Setting

	ID[1:0] = "00" H: decrement V: decrement	ID[1:0] = "01" H: increment V: decrement	ID[1:0] = "10" H: decrement V: increment	ID[1:0] = "11" H: increment V: increment
AM="0" Horizontal Update				
AM="1" Vertical Update				

Note. When window addresses have been set, the GRAM can only be written within the window.

When AM or ID is set, the start address should be written accordingly prior to memory write.

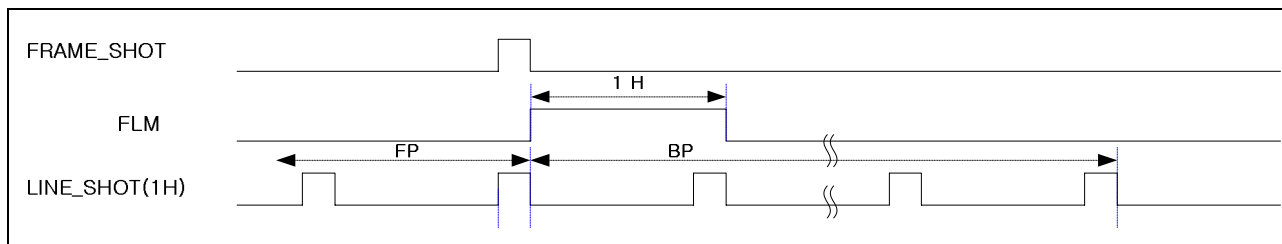
8.3.7. Display Control (R07h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	FLM_MON	0	0	0	0	0	0	0	GON	CL	REV	D1	D0

FLM_MON:

FLM_MON = 1, Enable the Frame flag output signal from the FLM signal line for preventing Tearing Effect.

FLM_MON = 0, Disable the Frame flag output signal from the FLM signal line for preventing Tearing Effect.



GON: Gate on/off control signal. All gate outputs are set to be gate off level when GON = 0.

When GON = 1, gate driver is working: G1 to G220 output is either VGH or VGL level. See the Instruction set up flow for further description on the display on/off flow.

GON	Gate Output
0	All gates off (VGL)
1	Gate on(VGH / VGL)

CL: CL = 1 selects 8-color display mode. For details, see the section on 8-COLOR DISPLAY MODE.

CL	Number of display colors
0	262,144 colors
1	8 colors

REV: Displays all character and graphics display sections with reversal when REV = 1. For details, see the Reversed Display Function section. Since the grayscale level can be reversed, display of the same data is enabled on normally white and normally black panels.

REV	GRAM Data	Display Area	
		Positive	Negative
0	6'b000000	V63	V0
	:	:	:
	6'b111111	V0	V63
1	6'b000000	V0	V63
	:	:	:
	6'b111111	V63	V0

D1-0: Display is on when D1 = 1 and off when D1 = 0. When off, the display data remains in the GRAM, and can be re-displayed instantly by setting D1 = 1. When D1 is 0, the display is off with the entire source outputs set to the VSS level. Because of this, the S6D0164 can control the charging current for the LCD with AC driving. Control the display on/off while control GON. For details, see the Instruction set up flow.

When D1-0 = 01, the internal display of the S6D0164 is performed although the display is off. When D1-0 = 00, the internal display operation halts and the display is off.

D1	D0	GON	Source output	Gate Output	VCOM Output	display
0	0	X	AVSS	AVSS	AVSS	off
0	1	0	AVSS	VGL	AVSS	off
		1	AVSS	Operate	AVSS	on
1	0	0	White on Normally White Panel Black on Normally Black Panel	VGL	Operate	off
		1	White on Normally White Panel Black on Normally Black Panel	Operate	Operate	on
1	1	0	Normal Display	VGL	Operate	off
		1	Normal Display	Operate	Operate	on

Note.

1. In standby mode or D1-0 = 00, gate outputs go to AVSS level
2. Writing from MCU to GRAM is independent of D1-0.
3. When source output is the same phase with VCOM, white screen is displayed at normally white LCD panel

8.3.8. Blank Period Control 1 (R08h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	FP3	FP2	FP1	FP0	0	0	0	0	BP3	BP2	BP1	BP0

The blanking period in the front and end of the display area can be defined using this register.

When N-raster-row is driving, a blank period is inserted after all screens are drawn. Front and Back porch can be adjusted using FP3-0 and BP3-0 bits (R08h).

FP3-0/BP3-0: Set the blanking periods (the front and back porch) which are placed at the beginning and the end of the data in. FP3-0 is for a front porch and BP3-0 is for a back porch. When front and back porches are set, the settings should meet the following conditions.

$BP + FP \leq 16$ raster-rows

$FP \geq 2$ raster-rows

$BP \geq 2$ raster-rows

When the external display interface is in use, the back porch (BP) will start on the falling edge of the VSYNC signal and the display operation will commence at the end of the back-porch period. The front porch (FP) will start when data for the number of raster-rows specified by the NL bits has been displayed. During the period between the completion of the front-porch period and the next VSYNC signal, the display will remain blank.

Table 27. Front/Back Porch

FP3 BP3	FP2 BP2	FP1 BP1	FP0 BP0	# of Raster Periods In the Front Porch # of Raster Periods In the Back Porch
0	0	0	0	Setting Disabled
0	0	0	1	Setting Disabled
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	Setting Disabled

8.3.9. Frame Cycle Control (R0Bh)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	NO3	NO2	NO1	NO0	SDT3	SDT2	SDT1	SDT0	0	0	0	0	RTN3	RTN2	RTN1	RTN0

NO3-0: Set amount of non-overlap for the gate output.

NO3	NO2	NO1	NO0	Amount of non-overlap		
				Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)	
					18-bit RGB	6-bit RGB
0	0	0	0	Setting disable	Setting disable	Setting disable
0	0	0	1	1 INCLK	8 dot clock	8*3 dot clock
0	0	1	0	2 INCLK	16 dot clock	16*3 dot clock
0	0	1	1	3 INCLK	24 dot clock	24*3 dot clock
0	1	0	0	4 INCLK	32 dot clock	32*3 dot clock
0	1	0	1	5 INCLK	40 dot clock	40*3 dot clock
0	1	1	0	6 INCLK	48 dot clock	48*3 dot clock
0	1	1	1	7 INCLK	56 dot clock	56*3 dot clock
1	0	0	0	8 INCLK	64 dot clock	64*3 dot clock
1	0	0	1	9 INCLK	72 dot clock	72*3 dot clock
1	0	1	0	10 INCLK	80 dot clock	80*3 dot clock
1	0	1	1	Setting disable	88 dot clock	88*3 dot clock
1	1	0	0	Setting disable	96 dot clock	96*3 dot clock
1	1	0	1	Setting disable	104 dot clock	104*3 dot clock
1	1	1	0	Setting disable	112dot clock	112*3 dot clock
1	1	1	1	Setting disable	120dot clock	120*3 dot clock

Note. The amount of non-overlap time is defined from starting time of 1H.

SDT3-0: Set delay amount from gate edge (end) to source output.

SDT3	SDT2	SDT1	SDT0	Delay amount of the source output		
				Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)	
					18-bit RGB	6-bit RGB
0	0	0	0	Setting disable	Setting disable	Setting disable
0	0	0	1	1 INCLK	8 dot clock	8*3 dot clock
0	0	1	0	2 INCLK	16 dot clock	16*3 dot clock
0	0	1	1	3 INCLK	24 dot clock	24*3 dot clock
0	1	0	0	4 INCLK	32 dot clock	32*3 dot clock
0	1	0	1	5 INCLK	40 dot clock	40*3 dot clock
0	1	1	0	6 INCLK	48 dot clock	48*3 dot clock
0	1	1	1	7 INCLK	56 dot clock	56*3 dot clock
1	0	0	0	8 INCLK	64 dot clock	64*3 dot clock
1	0	0	1	9 INCLK	72 dot clock	72*3 dot clock
1	0	1	0	10 INCLK	80 dot clock	80*3 dot clock
1	0	1	1	Setting disable	88 dot clock	88*3 dot clock
1	1	0	0	Setting disable	96 dot clock	96*3 dot clock
1	1	0	1	Setting disable	104 dot clock	104*3 dot clock
1	1	1	0	Setting disable	112dot clock	112*3 dot clock
1	1	1	1	Setting disable	120dot clock	120*3 dot clock

RTN3-0: Set the 1H period (1 raster-row).

RTN3	RTN2	RTN1	RTN0	1 Horizontal clock cycle (CL1)
0	0	0	0	16 INCLK
0	0	0	1	17 INCLK
0	0	1	0	18 INCLK
0	0	1	1	19 INCLK
0	1	0	0	20 INCLK
0	1	0	1	21 INCLK
0	1	1	0	22 INCLK
0	1	1	1	23 INCLK
1	0	0	0	24 INCLK
1	0	0	1	25 INCLK
1	0	1	0	26 INCLK
1	0	1	1	27 INCLK
1	1	0	0	28 INCLK
1	1	0	1	29 INCLK
1	1	1	0	30 INCLK
1	1	1	1	31 INCLK

8.3.10. External Display Interface Control (R0Ch)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	RM	0	0	0	DM	0	0	RIM1	RIM0

RM: Specify the interface for GRAM access as shown below. This register and DM register can be set independently. The display data can be written through System Interface by clearing this register while the RGB interface is used.

Table 28. RM and GRAM Access Interface

RM	GRAM Access Interface
0	System interface
1	RGB interface

DM: Specify the display operation mode. The interface can be set based on the bits of DM. This setting enables switching interface between internal operation and the external display interface.

Switching between two external display interfaces (RGB interface) should not be done.

DM	Display Operation Mode
0	Internal clock operation
1	RGB interface

RIM1-0: Specify the RGB interface mode when the RGB interface is used. Specifically, this setting specifies the mode when the bits of DM and RM are set to RGB interface. These bits should be set before display operation through the RGB interface and should not be set during operation.

RIM1	RIM0	RGB Interface Mode
0	0	18-bit RGB interface (one transfer/pixel)
0	1	16-bit RGB interface (one transfer /pixel)
1	0	6-bit RGB interface (three transfers /pixel)
1	1	Setting disabled

You should notice that some display functions, which will be described later, cannot be used according to the display mode shown below.

Table 29. Display Functions and Display Modes

Function	External Clock Operation Mode	Internal Clock Operation Mode
Partial Display	Cannot be used	Can be used
Scroll Function	Cannot be used	Can be used
Rotation	Cannot be used	Can be used
Mirroring	Cannot be used	Can be used
Window Function	Cannot be used	Can be used

Depending on the external display interface setting, various interfaces can be specified to match the display state. While displayed motion pictures (RGB interface), the data for display can be written in high-speed write mode, which achieves both low power consumption and high-speed access.

Table 30. Display State and Interface

Display State	Operation Mode	RAM Access (RM)	Display Operation Mode (DM)
Still Pictures	Internal Clock	System interface (RM=0)	Internal clock (DM=0)
Motion Pictures	RGB interface (1)	RGB interface (RM=1)	RGB interface (DM=1)
Rewrite still picture area while displaying motion pictures	RGB interface (2)	System interface (RM=0)	RGB interface (DM=1)

Note.

1. The instruction register can only be set through the system interface (SPI).
2. The RGB interface mode should not be set during operation.
3. For more information about the transition flow of each operation mode, see the Transition Sequences between Display Modes section.

Internal Clock Mode

All display operation is controlled by signals generated by the internal clock in internal clock mode. All inputs through the external display interface are invalid. The internal RAM can be accessed only via the system interface.

RGB Interface Mode (1)

The displays operations are controlled by the frame synchronization clock (VSYNC), raster-row synchronization signal (HSYNC), and dot clock (DOTCLK) in RGB interface mode. These signals should be supplied during display operation in this mode.

The display data is transferred to the internal RAM via DB17-0 for each pixel. Combining the function of the high-speed write mode and the window address enables display of both the motion picture area and the internal RAM area simultaneously. In this method, data is only transferred when the screen is updated, which reduces the amount of data transferred.

The periods of the front (FP), back (BP) porch, and the display are automatically generated in the S6D0164 by counting the raster-row synchronization signal (HSYNC) based on the frame synchronization signal (VSYNC).

RGB Interface Mode (2)

When RGB interface is in use, data can be written to RAM via the system interface. This write operation should be performed while data for display is not being transferred via RGB interface (ENABLE = active). Before the next data transfer for display via RGB interface, the setting above should be changed, and then the address and index (R22h) should be set.

8.3.11. Start Oscillation (R0Fh)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	FOSC4	FOSC3	FOSC2	FOSC1	FOSC0	0	0	0	0	0	0	0	OSC ON

FOSC4-0: Select the oscillation frequency of internal oscillator

Table 31. Oscillation Frequency

FOSC[4:0]	Oscillation Frequency	FOSC[4:0]	Oscillation Frequency
00000	X 0.84	10000	X 1.34
00001	X 0.86	10001	X 1.39
00010	X 0.88	10010	X 1.44
00011	X 0.90	10011	X 1.50
00100	X 0.92	10100	X 1.57
00101	X 0.95	10101	X 1.64
00110	X 0.97	10110	X 1.72
00111	X 1.00	10111	X 1.80
01000	X 1.03	11000	X 1.90
01001	X 1.06	11001	X 2.00
01010	X 1.09	11010	X 2.12
01011	X 1.13	11011	X 2.25
01100	X 1.16	11100	X 2.40
01101	X 1.20	11101	X 2.58
01110	X 1.24	11110	X 2.77
01111	X 1.29	11111	X 3.01

Note. If the default oscillation frequency is 227 KHz and register setting of FOSC [4:0] is 00111.

OSCON: This instruction starts the oscillator from the Halt State in the standby mode. After this instruction, wait at least 10 ms for oscillation to stabilize before giving the next instruction.

When OSCON = 1: OSC ON, OSCON = 0: OSC OFF

8.3.12. Power Control 1 (R10h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	SAP3	SAP2	SAP1	SAP0	0	0	0	0	0	0	DSTB	STB

SAP3-0: Adjust the slew-rate of the operational amplifier for the source driver. If higher SAP3-0 is set, LCD panel having higher resolution or higher frame frequency can be driven because the slew-rate of the operational amplifier is increased. But these bits must be set as adequate value because the amount of fixed current of the operational amplifier is also adjusted. During non-display, when SAP3-0 = "0000", operational amplifiers are turned off, so current consumption can be reduced.

SAP3	SAP2	SAP1	SAP0	Source Amp. Current Level	Slew rate[us/V]	Delay[us]
0	0	0	0	Amp. Stop	-	-
0	0	0	1	Setting Disable		
0	0	1	0	Setting Disable		
0	0	1	1	Slow 2	5.38	26.9
0	1	0	0	Medium Slow 1	4.18	20.9
0	1	0	1	Medium Slow 2	3.44	17.2
0	1	1	0	Medium Slow 3	2.88	14.4
0	1	1	1	Medium Slow 4	2.54	12.7
1	0	0	0	Medium Fast 1	2.40	12.0
1	0	0	1	Medium Fast 2	2.04	10.2
1	0	1	0	Medium Fast 3	1.96	9.8
1	0	1	1	Medium Fast 4	1.76	8.8
1	1	0	0	Fast1	1.66	8.3
1	1	0	1	Fast2	1.60	8.0
1	1	1	0	Fast3	1.52	7.6
1	1	1	1	Fast4 (the Fastest)	1.42	7.1

DSTB: When DSTB = 1, the S6D0164 enters the deep standby mode, where the power supply for the internal logic is turned off to save more power than the standby mode. Writing the GRAM data or setting any instructions are prohibited during the deep-standby mode and they must be reset after releasing from the deep standby mode.

For more information, see the Standby Mode section.

STB: When STB = 1, the S6D0164 enters the standby mode, where display operation completely stops, halting all the internal operations including the internal R-C oscillator. Further, no external clock pulses are supplied. For more information, see the Standby Mode section.

Outputs	Conditions
VCOM	AVSS
Gate	AVSS
Source	AVSS

8.3.13. Power Control 2 (R11h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	APON	PON3	PON2	PON1	PON	0	AB_VCI1	AON	VCI1_EN	VC3	VC2	VC1	VC0

APON: This is an automatic-boosting-operation-starting bit for the booster circuits. In case of APON=0, the automatic booster sequence circuit is stopped, but the booster circuits are independently operated by PON, PON1, PON2 and PON3 bits. In case of APON=1, booster circuits are automatically and sequentially operated. For further information about timing, please refer to the SET UP FLOW OF POWER SUPPLY.

PON3: This is an operation-starting bit for the booster circuit 3(VCL). In case of PON3 = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the PON3= 1, please refer to the SET UP FLOW OF POWER SUPPLY.

PON2: This is an operation-starting bit for the booster circuit 2(VGL). In case of PON2 = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the PON2= 1, please refer to the SET UP FLOW OF POWER SUPPLY.

PON1: This is an operation-starting bit for the booster circuit 2(VGH). In case of PON1 = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the PON1= 1, please refer to the SET UP FLOW OF POWER SUPPLY.

PON: This is an operation-starting bit for the booster circuit1. In case of PON = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the PON = 1, please refer to the SET UP FLOW OF POWER SUPPLY.

AB_VCI1: Set VCI1 output equal to VCI. VCI1 output is internally connected to VCI via switching circuit when AB_VCI1="H."

AON: This is an operation-starting bit for the Amplifier. In case of AON = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the AON= 1, please refer to the SET UP FLOW OF POWER SUPPLY.

VCI1_EN: Internal VCI1 generation amplifier operation control bit. When VCI1_EN=0, VCI1 voltage is not generated.

VC3-0: Set the VCI1 voltage. These bits set the VCI1 voltage up to 3V as the nominal output
(Upper limit value may depend on VCI voltage)

VC3	VC2	VC1	VC0	VCI1
0	0	0	0	1.35
0	0	0	1	1.75
0	0	1	0	2.07
0	0	1	1	2.16
0	1	0	0	2.25
0	1	0	1	2.34
0	1	1	0	2.43
0	1	1	1	2.52
1	0	0	0	2.58
1	0	0	1	2.64
1	0	1	0	2.70
1	0	1	1	2.76
1	1	0	0	2.82
1	1	0	1	2.88
1	1	1	0	2.94
1	1	1	1	3

Note. Do not set any higher VCI1 level than $V_{CI} - 0.15V$.

8.3.14. Power Control 3 (R12h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	BT2	BT1	BT0	0	0	DC11	DC10	0	0	DC21	DC20	0	0	DC31	DC30

BT2-0: The output factor of step-up is switched. Adjust scale factor of the step-up circuit by the voltage used. When the step-up operating frequency is high, the driving ability of the step-up circuit and the display quality become high, but the current consumption is increased. Adjust the frequency considering the display quality and the current consumption.

BT2	BT1	BT0	VGH	VGL	Notes*	
0	0	0	5 X Vci1	-3X Vci1	13.75V	-8.25V
0	0	1	5 X Vci1	-4X Vci1	13.75V	-11V
0	1	0	6 X Vci1	-3X Vci1	16.5V	-8.25V
0	1	1	6 X Vci1	-4X Vci1	16.5V	-11V
1	0	0	6 X Vci1	-5X Vci1	16.5V	-13.75V
1	0	1	7 X Vci1	-4X Vci1	19.25V	-11V
1	1	0	Setting disabled			
1	1	1	Setting disabled			

Note. The values in table above are example of nominal upper-limit by register setting when VCI1=2.75V.

DC11-0: The operating frequency in the step-up circuit1 is selected. When the step-up operating frequency is high, the driving ability of the step-up circuit and the display quality become high, but the current consumption is increased. Adjust the frequency considering the display quality and the current consumption.

DC11	DC10	Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)
		f(CL1) : f(DCCLK1)	f(DCCLK):f(DCCLK1)
0	0	1:4	1:1
0	1	1:2	1:0.5
1	0	1:1	1:0.25
1	1	Setting disabled	Setting disabled

Note.

1. DCCLK1 is pumping clock for step-up circuit1
2. f(CL1) is horizontal frequency (1 raster-row)

DC21-0: The operating frequency in the step-up circuit 2 is selected.

DC21	DC20	Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)
		f(CL1) : f(DCCLK2)	f(DCCLK) : f(DCCLK2)
0	0	1:2	1:0.5
0	1	1:1	1:0.25
1	0	1:0.5	1:0.125
1	1	1:0.25	1:0.0625

Note. DCCLK2 is pumping clock for step-up circuit2

DC31-0: The operating frequency in the step-up circuit 3 is selected.

DC31	DC30	Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)
		f(CL1) : f(DCCLK3)	f(DCCLK) : f(DCCLK3)
0	0	1:4	1:1
0	1	1:2	1:0.5
1	0	1:1	1:0.25
1	1	Setting disabled	Setting disabled

Note. DCCLK3 is pumping clock for step-up circuit3

8.3.15. Power Control 4 (R13h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	DCR_EX	0	DCR2	DCR1	DCR0	0	GVD6	GVD5	GVD4	GVD3	GVD2	GVD1	GVD0

DCR_EX: Input signal selection register for external interface mode. (0: internal operation clock, 1: DOTCLK) Set DCR_EX bit to 1 for DOTCLK to be DCCLK (clock cycle for step-up circuit) source when external interface mode is in use (DM=1).

DCR 2-0: Set clock cycle for step-up circuit in external interface mode. Please set DCR_EX bit to “1” and DCR1-0 value when external interface is in use. In this case, DOTCLK must be input periodically and continuously.

DCR2	DCR1	DCR0	Clock cycle for step-up circuits (DCCLK) in external interface mode	
			18-bit RGB	6-bit RGB
0	0	0	6 dot clock	18 dot clock
0	0	1	8 dot clock	24 dot clock
0	1	0	12 dot clock	36 dot clock
0	1	1	16 dot clock	48 dot clock
1	0	0	24 dot clock	72 dot clock
1	0	1	32 dot clock	96 dot clock
1	1	0	48 dot clock	144 dot clock
1	1	1	64 dot clock	192 dot clock

GVD6-0: Set the amplifying factor of the GVDD voltage (the voltage for the Gamma voltage). It allows ranging from 2.5V to 5.0V.

GVD6-0	GVDD Voltage	GVD6-0	GVDD Voltage
0000000	2.50V	1000000	3.76V
0000001	2.52V	1000001	3.78V
0000010	2.54V	1000010	3.80V
0000011	2.56V	1000011	3.82V
0000100	2.58V	1000100	3.84V
0000101	2.60V	1000101	3.86V
0000110	2.62V	1000110	3.88V
0000111	2.64V	1000111	3.90V
0001000	2.66V	1001000	3.92V
0001001	2.68V	1001001	3.94V
0001010	2.70V	1001010	3.96V
0001011	2.72V	1001011	3.98V
0001100	2.74V	1001100	4.00V
0001101	2.76V	1001101	4.02V
0001110	2.78V	1001110	4.04V
0001111	2.80V	1001111	4.06V
0010000	2.81V	1010000	4.07V
0010001	2.83V	1010001	4.09V
0010010	2.85V	1010010	4.11V
0010011	2.87V	1010011	4.13V
0010100	2.89V	1010100	4.15V

GVD6-0	GVDD Voltage	GVD6-0	GVDD Voltage
0010101	2.91V	1010101	4.17V
0010110	2.93V	1010110	4.19V
0010111	2.95V	1010111	4.21V
0011000	2.97V	1011000	4.23V
0011001	2.99V	1011001	4.25V
0011010	3.01V	1011010	4.27V
0011011	3.03V	1011011	4.29V
0011100	3.05V	1011100	4.31V
0011101	3.07V	1011101	4.33V
0011110	3.09V	1011110	4.35V
0011111	3.11V	1011111	4.37V
0100000	3.13V	1100000	4.39V
0100001	3.15V	1100001	4.41V
0100010	3.17V	1100010	4.43V
0100011	3.19V	1100011	4.45V
0100100	3.21V	1100100	4.47V
0100101	3.23V	1100101	4.49V
0100110	3.25V	1100110	4.51V
0100111	3.27V	1100111	4.53V
0101000	3.29V	1101000	4.55V
0101001	3.31V	1101001	4.57V
0101010	3.33V	1101010	4.59V
0101011	3.35V	1101011	4.61V
0101100	3.37V	1101100	4.63V
0101101	3.39V	1101101	4.65V
0101110	3.41V	1101110	4.67V
0101111	3.43V	1101111	4.69V
0110000	3.44V	1110000	4.70V
0110001	3.46V	1110001	4.72V
0110010	3.48V	1110010	4.74V
0110011	3.50V	1110011	4.76V
0110100	3.52V	1110100	4.78V
0110101	3.54V	1110101	4.80V
0110110	3.56V	1110110	4.82V
0110111	3.58V	1110111	4.84V
0111000	3.60V	1111000	4.86V
0111001	3.62V	1111001	4.88V
0111010	3.64V	1111010	4.90V
0111011	3.66V	1111011	4.92V
0111100	3.68V	1111100	4.94V
0111101	3.70V	1111101	4.96V
0111110	3.72V	1111110	4.98V
0111111	3.74V	1111111	5.00V

Note. Do not set any higher GVDD level than AVDD-0.3V

8.3.16. Power Control 5 (R14h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	VCOMG	VCM6	VCM5	VCM4	VCM3	VCM2	VCM1	VCM0	VCMR	VML6	VML5	VML4	VML3	VML2	VML1	VML0

VCOMG: When VCOMG = 1, low level of VCOM signal is to be fixed at AVSS. Therefore, the amplitude of VCOM signal is determined as $|VCOMH - AVSS|$ regardless of VML setting. In this case, VCOML pin should be connected to GND, because VCOML amp is off and VCOML output is floated.

When VCOMG=0, the amplitude of VCOM signal is determined as $|VCOMH - VCOML|$.

VCM6-0: Set the VCOMH voltage (a high level voltage at the Vcom alternating drive), these bits amplify the VcomH voltage from 0.4015 to 1.1000 times the GVDD voltage.

Table 32. VCOMH Setting

VCM6-0	VCOMH Output	VCM6-0	VCOMH Output
0000000	GVDD X 0.4015	1000000	GVDD X 0.7535
0000001	GVDD X 0.4070	1000001	GVDD X 0.7590
0000010	GVDD X 0.4125	1000010	GVDD X 0.7645
0000011	GVDD X 0.4180	1000011	GVDD X 0.7700
0000100	GVDD X 0.4235	1000100	GVDD X 0.7755
0000101	GVDD X 0.4290	1000101	GVDD X 0.7810
0000110	GVDD X 0.4345	1000110	GVDD X 0.7865
0000111	GVDD X 0.4400	1000111	GVDD X 0.7920
0001000	GVDD X 0.4455	1001000	GVDD X 0.7975
0001001	GVDD X 0.4510	1001001	GVDD X 0.8030
0001010	GVDD X 0.4565	1001010	GVDD X 0.8085
0001011	GVDD X 0.4620	1001011	GVDD X 0.8140
0001100	GVDD X 0.4675	1001100	GVDD X 0.8195
0001101	GVDD X 0.4730	1001101	GVDD X 0.8250
0001110	GVDD X 0.4785	1001110	GVDD X 0.8305
0001111	GVDD X 0.4840	1001111	GVDD X 0.8360
0010000	GVDD X 0.4895	1010000	GVDD X 0.8415
0010001	GVDD X 0.4950	1010001	GVDD X 0.8470
0010010	GVDD X 0.5005	1010010	GVDD X 0.8525
0010011	GVDD X 0.5060	1010011	GVDD X 0.8580
0010100	GVDD X 0.5115	1010100	GVDD X 0.8635
0010101	GVDD X 0.5170	1010101	GVDD X 0.8690
0010110	GVDD X 0.5225	1010110	GVDD X 0.8745
0010111	GVDD X 0.5280	1010111	GVDD X 0.8800
0011000	GVDD X 0.5335	1011000	GVDD X 0.8855
0011001	GVDD X 0.5390	1011001	GVDD X 0.8910
0011010	GVDD X 0.5445	1011010	GVDD X 0.8965
0011011	GVDD X 0.5500	1011011	GVDD X 0.9020
0011100	GVDD X 0.5555	1011100	GVDD X 0.9075
0011101	GVDD X 0.5610	1011101	GVDD X 0.9130
0011110	GVDD X 0.5665	1011110	GVDD X 0.9185
0011111	GVDD X 0.5720	1011111	GVDD X 0.9240

VCM6-0	VCOMH Output	VCM6-0	VCOMH Output
0100000	GVDD X 0.5775	1100000	GVDD X 0.9295
0100001	GVDD X 0.5830	1100001	GVDD X 0.9350
0100010	GVDD X 0.5885	1100010	GVDD X 0.9405
0100011	GVDD X 0.5940	1100011	GVDD X 0.9460
0100100	GVDD X 0.5995	1100100	GVDD X 0.9515
0100101	GVDD X 0.6050	1100101	GVDD X 0.9570
0100110	GVDD X 0.6105	1100110	GVDD X 0.9625
0100111	GVDD X 0.6160	1100111	GVDD X 0.9680
0101000	GVDD X 0.6215	1101000	GVDD X 0.9735
0101001	GVDD X 0.6270	1101001	GVDD X 0.9790
0101010	GVDD X 0.6325	1101010	GVDD X 0.9845
0101011	GVDD X 0.6380	1101011	GVDD X 0.9900
0101100	GVDD X 0.6435	1101100	GVDD X 0.9955
0101101	GVDD X 0.6490	1101101	GVDD X 1.0010
0101110	GVDD X 0.6545	1101110	GVDD X 1.0065
0101111	GVDD X 0.6600	1101111	GVDD X 1.0120
0110000	GVDD X 0.6655	1110000	GVDD X 1.0175
0110001	GVDD X 0.6710	1110001	GVDD X 1.0230
0110010	GVDD X 0.6765	1110010	GVDD X 1.0285
0110011	GVDD X 0.6820	1110011	GVDD X 1.0340
0110100	GVDD X 0.6875	1110100	GVDD X 1.0395
0110101	GVDD X 0.6930	1110101	GVDD X 1.0450
0110110	GVDD X 0.6985	1110110	GVDD X 1.0505
0110111	GVDD X 0.7040	1110111	GVDD X 1.0560
0111000	GVDD X 0.7095	1111000	GVDD X 1.0615
0111001	GVDD X 0.7150	1111001	GVDD X 1.0670
0111010	GVDD X 0.7205	1111010	GVDD X 1.0725
0111011	GVDD X 0.7260	1111011	GVDD X 1.0780
0111100	GVDD X 0.7315	1111100	GVDD X 1.0835
0111101	GVDD X 0.7370	1111101	GVDD X 1.0890
0111110	GVDD X 0.7425	1111110	GVDD X 1.0945
0111111	GVDD X 0.7480	1111111	GVDD X 1.1000

Note.

1. $V_{comH} = GVDD \times (0.4015 + 0.0055 \times VCM)$
2. When using VCI recycling function, VCOMH voltage should be higher than VCI.
3. TOTAL_VCM [6:0] is VCM [6:0] + VCM_OFFSET_MTP [4:0] when MTP_SEL=0 and is VCM [6:0] + VCM_OFFSET_REG [4:0] when MTP_SEL=1.

VCMR: If VCMR is LOW, VCOMH is adjusted by VCM6-0 register and VCOMR pin is used to monitor the input voltage of the amplifier which outputs the VCOMH voltage.

If VCMR is HIGH, VCM6-0 register is ignored and VCOMH voltage is adjusted by VCOMR voltage. VCOMR voltage is externally supplied. The relationship between VCOMH and VCOMR is given as $VCOMH=1.1 \times VCOMR$.

VML6-0: Set the alternating amplitudes of VCOM at the VCOM alternating drive. These bits amplify VCOM from 0.534 to 1.20 times the GVDD voltage. When the VCOM alternation is not driven, the settings become invalid.

Table 33. VCOM Amplitude Control

VML[6:0]	VCOM Amplitude	VML[6:0]	VCOM Amplitude
0000000	Setting disable	1000111	GVDD X 0.864
.....	Setting disable	1001000	GVDD X 0.870
0001111	Setting disable	1001001	GVDD X 0.876
0010000	GVDD X 0.534	1001010	GVDD X 0.882
0010001	GVDD X 0.540	1001011	GVDD X 0.888
0010010	GVDD X 0.546	1001100	GVDD X 0.894
0010011	GVDD X 0.552	1001101	GVDD X 0.900
0010100	GVDD X 0.558	1001110	GVDD X 0.906
0010101	GVDD X 0.564	1001111	GVDD X 0.912
0010110	GVDD X 0.570	1010000	GVDD X 0.918
0010111	GVDD X 0.576	1010001	GVDD X 0.924
0011000	GVDD X 0.582	1010010	GVDD X 0.930
0011001	GVDD X 0.588	1010011	GVDD X 0.936
0011010	GVDD X 0.594	1010100	GVDD X 0.942
0011011	GVDD X 0.600	1010101	GVDD X 0.948
0011100	GVDD X 0.606	1010110	GVDD X 0.954
0011101	GVDD X 0.612	1010111	GVDD X 0.960
0011110	GVDD X 0.618	1011000	GVDD X 0.966
0011111	GVDD X 0.624	1011001	GVDD X 0.972
0100000	GVDD X 0.630	1011010	GVDD X 0.978
0100001	GVDD X 0.636	1011011	GVDD X 0.984
0100010	GVDD X 0.642	1011100	GVDD X 0.990
0100011	GVDD X 0.648	1011101	GVDD X 0.996
0100100	GVDD X 0.654	1011110	GVDD X 1.002
0100101	GVDD X 0.660	1011111	GVDD X 1.008
0100110	GVDD X 0.666	1100000	GVDD X 1.014
0100111	GVDD X 0.672	1100001	GVDD X 1.020
0101000	GVDD X 0.678	1100010	GVDD X 1.026
0101001	GVDD X 0.684	1100011	GVDD X 1.032
0101010	GVDD X 0.690	1100100	GVDD X 1.038
0101011	GVDD X 0.696	1100101	GVDD X 1.044
0101100	GVDD X 0.702	1100110	GVDD X 1.050
0101101	GVDD X 0.708	1100111	GVDD X 1.056
0101110	GVDD X 0.714	1101000	GVDD X 1.062
0101111	GVDD X 0.720	1101001	GVDD X 1.068
0110000	GVDD X 0.726	1101010	GVDD X 1.074
0110001	GVDD X 0.732	1101011	GVDD X 1.080
0110010	GVDD X 0.738	1101100	GVDD X 1.086
0110011	GVDD X 0.744	1101101	GVDD X 1.092
0110100	GVDD X 0.750	1101110	GVDD X 1.098
0110101	GVDD X 0.756	1101111	GVDD X 1.104
0110110	GVDD X 0.762	1110000	GVDD X 1.110

VML[6:0]	VCOM Amplitude	VML[6:0]	VCOM Amplitude
0110111	GVDD X 0.768	1110001	GVDD X 1.116
0111000	GVDD X 0.774	1110010	GVDD X 1.122
0111001	GVDD X 0.780	1110011	GVDD X 1.128
0111010	GVDD X 0.786	1110100	GVDD X 1.134
0111011	GVDD X 0.792	1110101	GVDD X 1.140
0111100	GVDD X 0.798	1110110	GVDD X 1.146
0111101	GVDD X 0.804	1110111	GVDD X 1.152
0111110	GVDD X 0.810	1111000	GVDD X 1.158
0111111	GVDD X 0.816	1111001	GVDD X 1.164
1000000	GVDD X 0.822	1111010	GVDD X 1.170
1000001	GVDD X 0.828	1111011	GVDD X 1.176
1000010	GVDD X 0.834	1111100	GVDD X 1.182
1000011	GVDD X 0.840	1111101	GVDD X 1.188
1000100	GVDD X 0.846	1111110	GVDD X 1.194
1000101	GVDD X 0.852	1111111	GVDD X 1.200
1000110	GVDD X 0.858		

Note.

1. Adjust the settings between GVDD and VML6-0 so that the Vcom amplitudes are lower than 6.0 V.
2. VcomL voltage should be satisfied the following condition. : $0.0V > V_{comL} > V_{CL} + 0.5V$.
3. $V_{COM} \text{ amplitude} = GVDD \times (0.534 + 0.006 \times (V_{ML} - 15))$

8.3.17. VCI Recycling (R15h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	0	VCIR2	VCIR1	VCIR0	0	0	0	0

VCIR2-0: VCI recycling period is sustained for the number of clock cycle which is set on VCIR2-0.

VCIR2	VCIR1	VCIR0	Period of VCIR recycling			
			VCI recycling period. (Synchronized with OSC clock) * unit : clock cycle			RGB I/F operation (Synchronized with DOTCLK)
			Sn	Vcom1	Vcom2	
0	0	0	Off			Off
0	0	1	setting disable			Setting disable
0	1	0	2	1 / 2	2 / 1	16 dot clock
0	1	1	3	1.5 / 3	3 / 1.5	32 dot clock
1	0	0	4	2 / 4	4 / 2	48 dot clock
1	0	1	5	2.5 / 5	5 / 2.5	64 dot clock
1	1	0	6	3 / 6	6 / 3	80 dot clock
1	1	1	7	3.5 / 7	7 / 3.5	96 dot clock

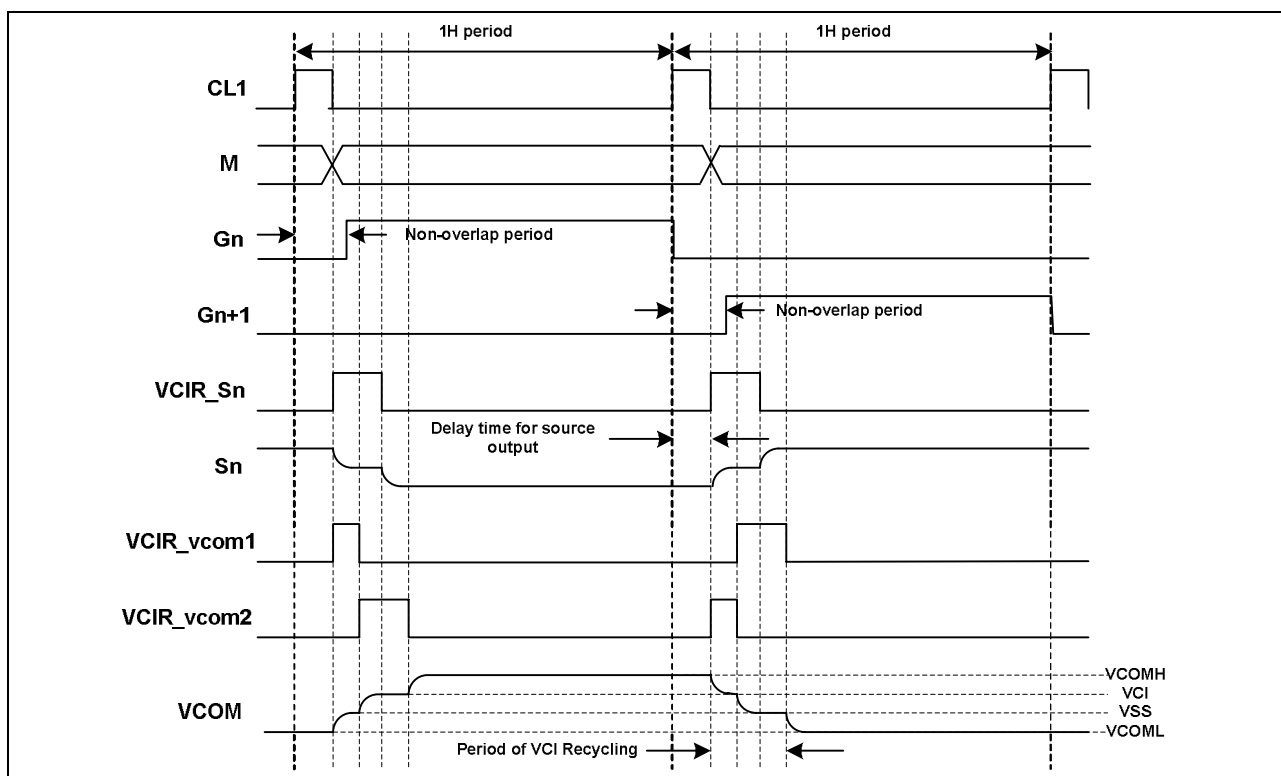


Figure 22. Set Delay From Gate Output To Source Output And VCIR Signal

8.3.18. GRAM Address Set (R20h,R21h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	AD7	AD6	AD5	AD4	AD3	AD2	AD1	AD0
W	1	0	0	0	0	0	0	0	0	AD15	AD14	AD13	AD12	AD11	AD10	AD9	AD8

AD

You can write initial GRAM address into internal Address Counter (AC). When GRAM data is transferred through System Interface or RGB Interface, the AC is automatically updated according to AM and ID. This allows consecutive write without re-setting address in AC. But when GRAM data is read, the AC is not automatically updated.

GRAM address setting is not allowed in Standby mode. Ensure that the address is set within the specified window area specified with VSA, VEA, HSA and HEA.

When RGB interface is used (RM="1") to access GRAM, AD [15:0] will be set in the address counter at the falling edge of the VSYNC signal. And when one uses System Interface to access GRAM (RM = "0"), AD [15:0] will be set upon the execution of an instruction.

Table 34. GRAM Address Range

AD[15:0]	GRAM setting
"0000H" to "00AF"H	Bitmap data for G1
"0100H" to "01AF"H	Bitmap data for G2
"0200H" to "02AF"H	Bitmap data for G3
"0300H" to "03AF"H	Bitmap data for G4
⋮	⋮
"D800H" to "D8AF"H	Bitmap data for G217
"D900H" to "D9AF"H	Bitmap data for G218
"DA00H" to "DAAF"H	Bitmap data for G219
"DB00H" to "DBAF"H	Bitmap data for G220

8.3.19. Write Data to GRAM (R22h)

R/W	RS	
W	1	RAM write data (WD17 ~ WD0). ※Interface mode controls the width of WD

WDR

Data on DB bus is expanded to 18-bits before being written to GRAM and the data determines grayscale level of S6D0164's source output. Please keep in mind that the expansion format varies with interface mode. GRAM cannot be accessed in Standby mode. When data is written to GRAM via system interface while another data is being written to through RGB interface, please make sure that the two write operations does not conflict.

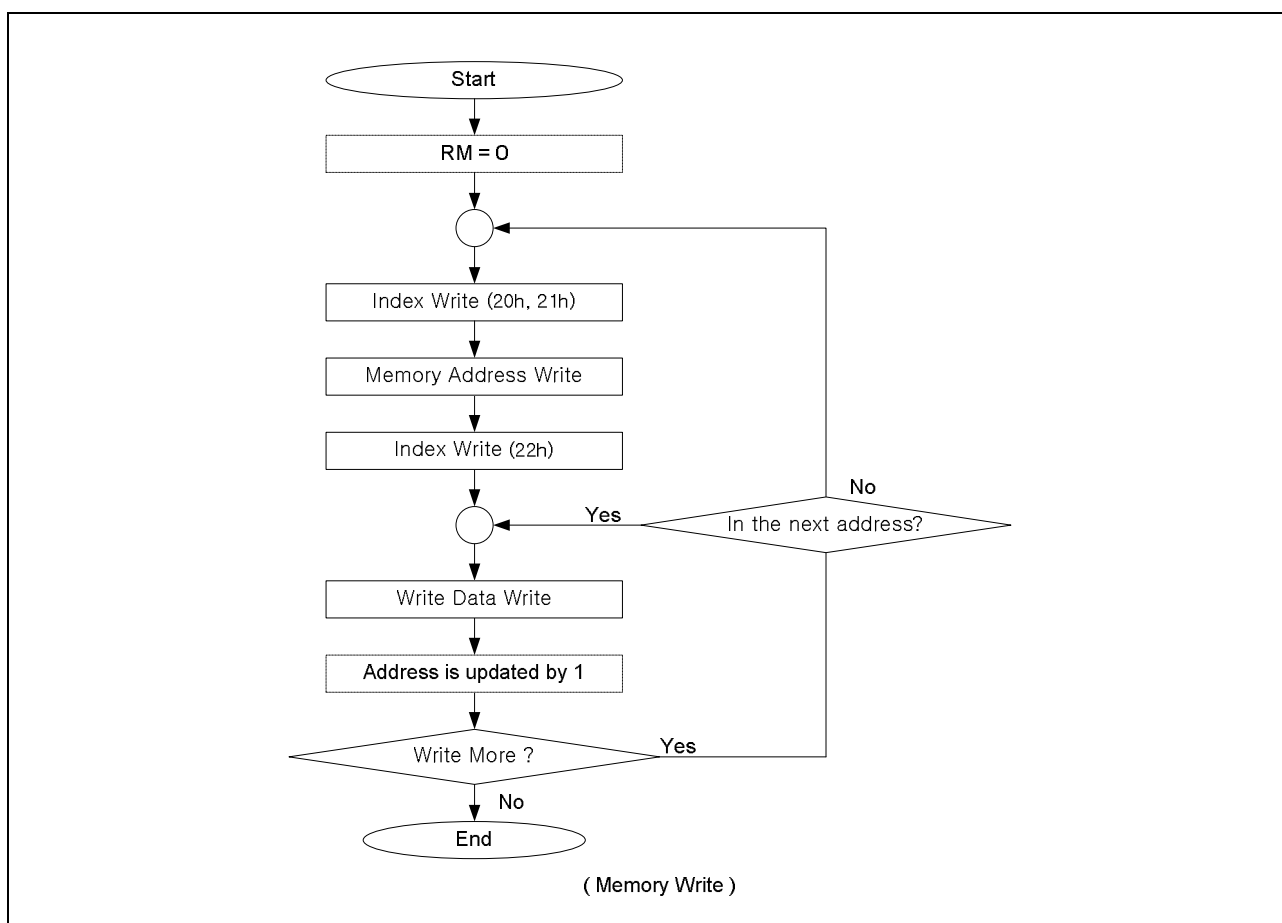


Figure 23. Memory Data Write Sequence

8.3.20. Read Data from GRAM (R22h)

R/W	RS	
R	1	RAM read data (RD17 ~ RD0). ※Interface mode controls the width of RD

RDR

You may read data from GRAM using this register. When you make read operations, you can get a proper data on the second read operation as shown below. The first word you get just after address setting may be invalid.

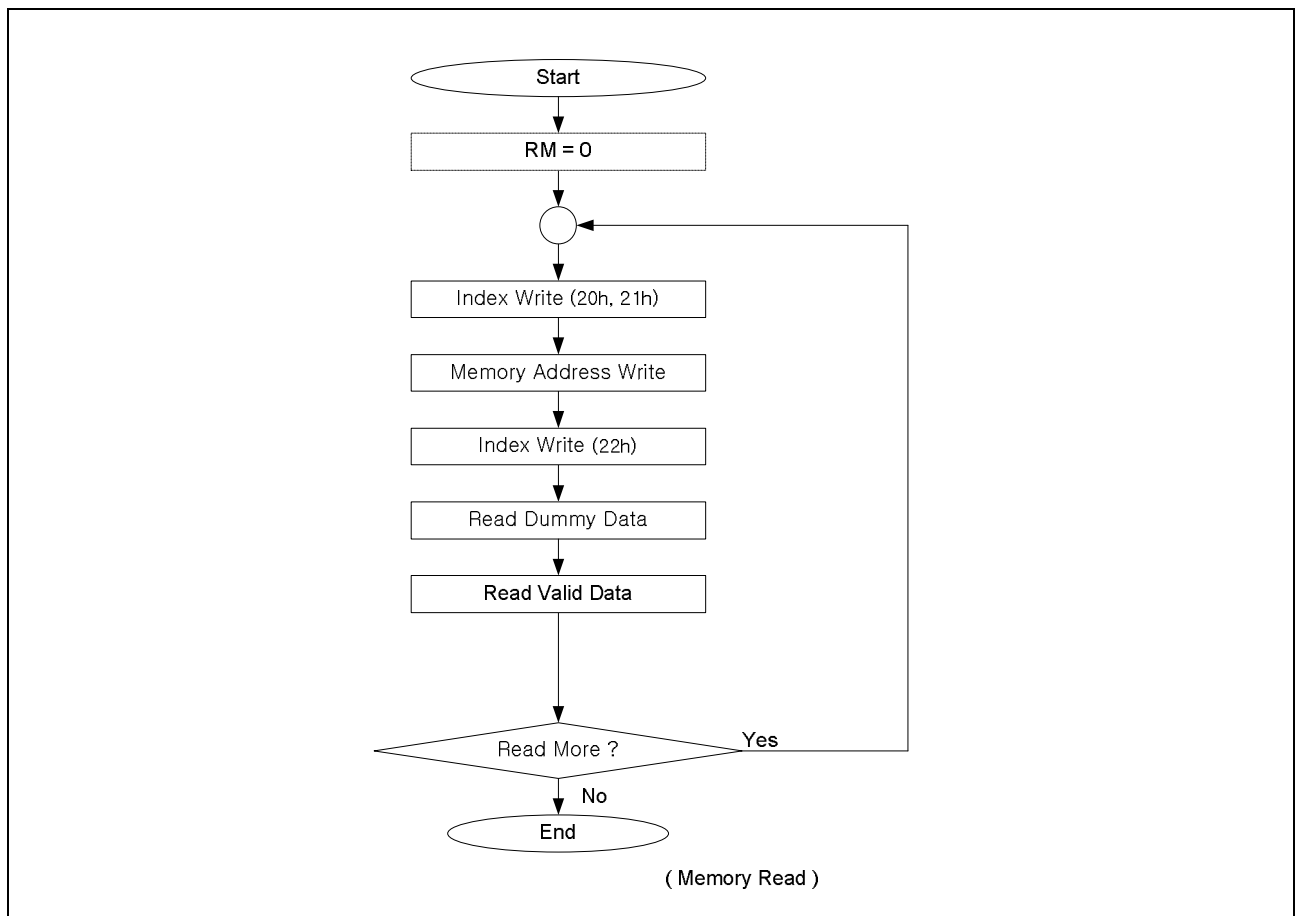


Figure 24. Memory Data Read Sequence

8.3.21. Software Reset (R28h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	1	1	0	0	1	1	1	0

When Software Reset parameter is 00CEh, it cause a software reset. This register automatically set to high after a Software Reset.

8.3.22. Gate Scan Position (R30h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	0	0	0	SCN4	SCN3	SCN2	SCN1	SCN0

SCN4-0: Set the scanning start position of the gate driver.

SCN4-0	Start Position	
	GS=0	GS=1
00000	G1	G220
00001	G9	G216
00010	G17	G208
---	---	---
11001	G201	G24
11010	G209	G16
11011	G217	G8

Note. Ensure that gate start position (SCN) + the number of LCD driver lines (NL) $\leq$ 220 when GS = 0, and that gate start position (SCN) - the number of LCD driver lines (NL) $\geq$ 0 when GS = 1

* Initial Value: SCN4-0 = 00000

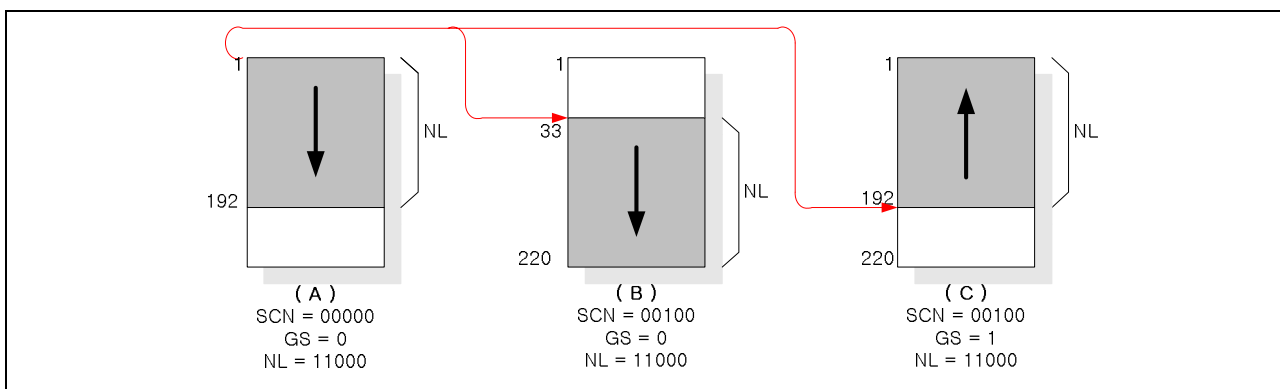


Figure 25. Gate Scan Position Control

8.3.23. Vertical Scroll Control 1 (R31h, R32h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	SEA 7	SEA 6	SEA 5	SEA 4	SEA 3	SEA 2	SEA 1	SEA 0
W	1	0	0	0	0	0	0	0	0	SSA 7	SSA 6	SSA 5	SSA 4	SSA 3	SSA 2	SSA 1	SSA 0

SSA7-0: Specify scroll start address at the scroll display for vertical smooth scrolling.

SSA7	SSA6	SSA5	SSA4	SSA3	SSA2	SSA1	SSA0	Scroll Start Address
0	0	0	0	0	0	0	0	0 raster-row
0	0	0	0	0	0	0	1	1 raster-row
0	0	0	0	0	0	1	0	2 raster-row
0	0	0	0	0	0	1	1	3 raster-row
0	0	0	0	0	1	0	0	4 raster-row
0	0	0	0	0	1	0	1	5 raster-row
: : : : : :								:
1	1	0	1	1	0	0	0	216 raster-row
1	1	0	1	1	0	0	1	217 raster-row
1	1	0	1	1	0	1	0	218 raster-row
1	1	0	1	1	0	1	1	219 raster-row

SEA7-0: Specify scroll end address at the scroll display for vertical smooth scrolling.

SEA7	SEA6	SEA5	SEA4	SEA3	SEA2	SEA1	SEA0	Scroll End Address
0	0	0	0	0	0	0	0	0 raster-row
0	0	0	0	0	0	0	1	1 raster-row
0	0	0	0	0	0	1	0	2 raster-row
0	0	0	0	0	0	1	1	3 raster-row
0	0	0	0	0	1	0	0	4 raster-row
0	0	0	0	0	1	0	1	5 raster-row
: : : : : :								:
1	1	0	1	1	0	0	0	216 raster-row
1	1	0	1	1	0	0	1	217 raster-row
1	1	0	1	1	0	1	0	218 raster-row
1	1	0	1	1	0	1	1	219 raster-row

Note.

1. Do not set any higher raster-row than 219 ("DB"H).
2. Set $SS17-10 \leq SSA7-0$, if set out of range, $SSA7-0 = SS17-10$.
3. Set $SE17-10 \geq SEA7-0$, if set out of range, $SEA7-0 = SE17-10$.

Vertical Scroll Control 2 (R33h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	SST7	SST6	SST5	SST4	SST3	SST2	SST1	SST0

SST8-0: Specify scroll start and step at the scroll display for vertical smooth scrolling. Any raster-row from the 1st to 220th can be scrolled for the number of the raster-row. After 219th raster-row is displayed, the display restarts from the first raster-row. When SST7-0 = 00000000, Vertical Scroll Function is disabled.

SST7	SST6	SST5	SST4	SST3	SST2	SST1	SST0	Scroll Step
0	0	0	0	0	0	0	0	0 raster-row
0	0	0	0	0	0	0	1	1 raster-row
0	0	0	0	0	0	1	0	2 raster-row
0	0	0	0	0	0	1	1	3 raster-row
0	0	0	0	0	1	0	0	4 raster-row
0	0	0	0	0	1	0	1	5 raster-row
: : : : : :								:
1	1	0	1	1	0	0	0	216 raster-row
1	1	0	1	1	0	0	1	217 raster-row
1	1	0	1	1	0	1	0	218 raster-row
1	1	0	1	1	0	1	1	219 raster-row

Note.

- Do not set any higher raster-row than 219 ("DB"H)
- Set $SS17-10 < SSA7-0 + SST7-0 \leq SEA7-0 \leq SE17-10$, if set out of range, Scroll function is disabled

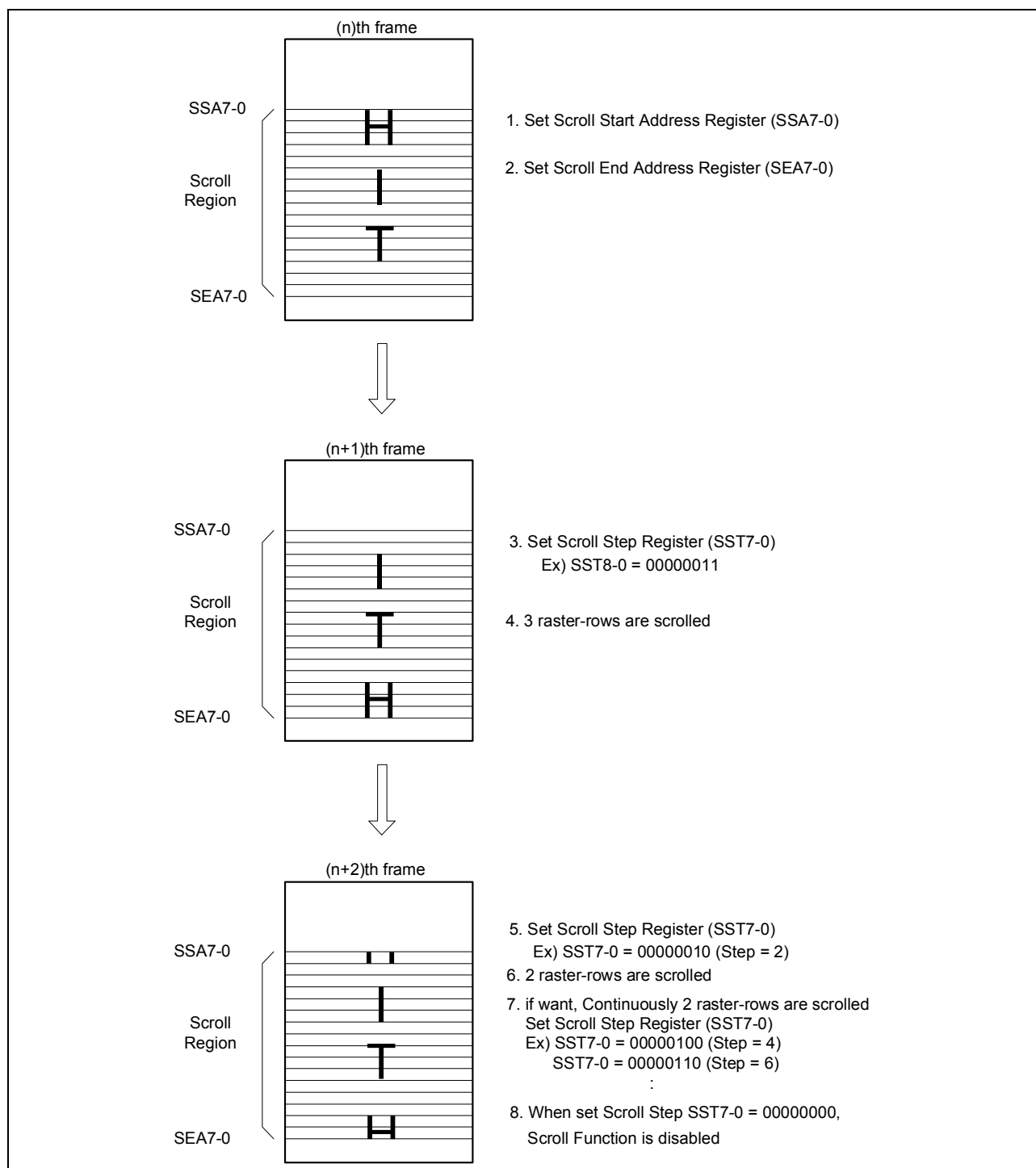


Figure 26. Vertical Scroll Display

8.3.24. Partial Screen Driving Position (R34h, R35h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	SE17	SE16	SE15	SE14	SE13	SE12	SE11	SE10
W	1	0	0	0	0	0	0	0	0	SS17	SS16	SS15	SS14	SS13	SS12	SS11	SS10

SE17–10: Specify the driving end position for the screen in a line unit. The LCD driving is performed to the 'set value + 1' gate driver. For instance, when SS17–10 = 019h and SE17–10 = 029h are set, the LCD driving is performed from G26 to G42, and non-display driving is performed for G1 to G25, G43, and others. Ensure that $SS18-10 \leq SE18-10 \leq DBh$.

SS17–10: Specify the drive starting position for the first screen in a line unit. The LCD driving starts from the 'set value +1' gate driver.

Note. Do not set the partial setting when the operation is in the normal display condition. Set this register only when in the partial display condition.

Ex) SS7-0=07h and SE7-0=10h are performed from G8 to G17.

The S6D0164 can select and drive partial screens at any position with the screen-driving position registers (R34H, R35H). Any partial screens required for display are selectively driven and reducing LCD-driving voltage and power consumption.

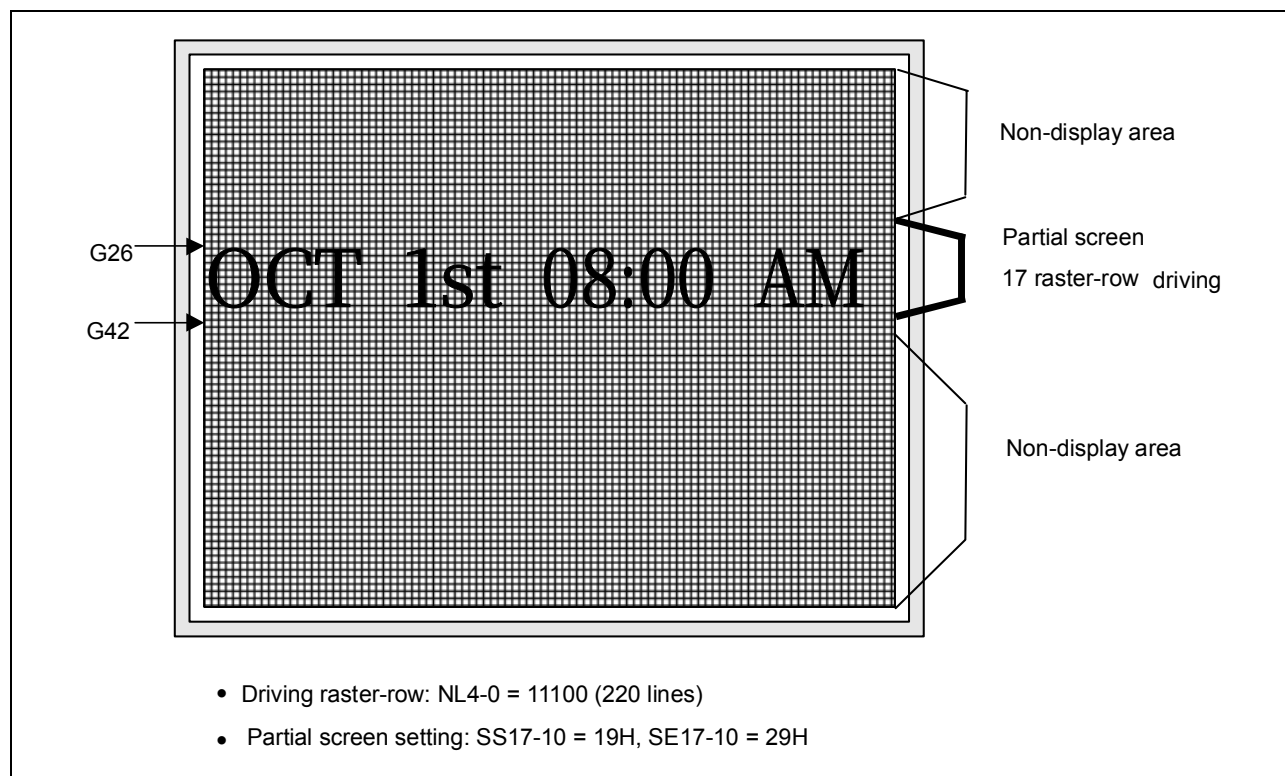


Figure 27. Driving On Partial Screen

8.3.25. Horizontal RAM Address Position (R36h, R37h)

8.3.26. Vertical RAM Address Position (R38h, R39h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	HEA 7	HEA 6	HEA 5	HEA 4	HEA 3	HEA 2	HEA 1	HEA 0
W	1	0	0	0	0	0	0	0	0	HSA 7	HSA 6	HSA 5	HSA 4	HSA 3	HSA 2	HSA 1	HSA 0
W	1	0	0	0	0	0	0	0	0	VEA 7	VEA 6	VEA 5	VEA 4	VEA 3	VEA 2	VEA 1	VEA 0
W	1	0	0	0	0	0	0	0	0	VSA 7	VSA 6	VSA 5	VSA 4	VSA 3	VSA 2	VSA 1	VSA 0

HSA7-0/HEA7-0: Specify the horizontal start/end positions of a window for access in memory. Data can be written to the GRAM from the address specified by HSA7-0 to the address specified by HEA7-0. Note that an address must be set before RAM is written.

VSA7-0/VEA7-0: Specify the vertical start/end positions of a window for access in memory. Data can be written to the GRAM from the address specified by VSA7-0 to the address specified by VEA7-0. Note that an address must be set before RAM is written.

When AM=0, $00h \leq HSA7-0 \leq HEA7-0 \leq AFh$ and $00h \leq VSA7-0 \leq VEA7-0 \leq DBh$

AM=1, $00h \leq HSA7-0 \leq HEA7-0 \leq DBh$ and $00h \leq VSA7-0 \leq VEA7-0 \leq AFh$

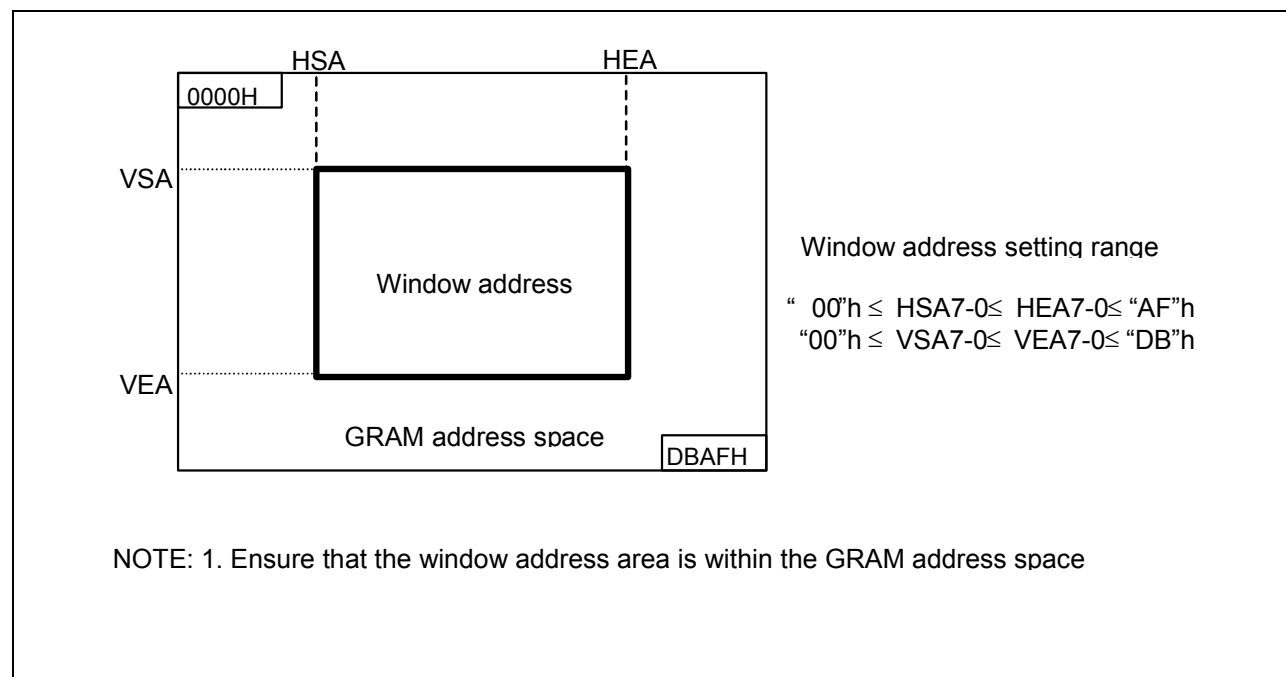


Figure 28. Window Address Setting Range (AM=0)

8.3.27. Gamma Control (R50h to R59h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	PKP 13	PKP 12	PKP 11	PKP 10	0	0	0	0	PKP 03	PKP 02	PKP 01	PKP 00
W	1	0	0	0	0	PKP 33	PKP 32	PKP 31	PKP 30	0	0	0	0	PKP 23	PKP 22	PKP 21	PKP 20
W	1	0	0	0	0	PKP 53	PKP 52	PKP 51	PKP 50	0	0	0	0	PKP 43	PKP 42	PKP 41	PKP 40
W	1	0	0	0	0	PRP 13	PRP 12	PRP 11	PRP 10	0	0	0	0	PRP 03	PRP 02	PRP 01	PRP 00
W	1	0	0	0	0	PKN 13	PKN 12	PKN 11	PKN 10	0	0	0	0	PKN 03	PKN 02	PKN 01	PKN 00
W	1	0	0	0	0	PKN 33	PKN 32	PKN 31	PKN 30	0	0	0	0	PKN 23	PKN 22	PKN 21	PKN 20
W	1	0	0	0	0	PKN 53	PKN 52	PKN 51	PKN 50	0	0	0	0	PKN 43	PKN 42	PKN 41	PKN 40
W	1	0	0	0	0	PRN 13	PRN 12	PRN 11	PRN 10	0	0	0	0	PRN 03	PRN 02	PRN 01	PRN 00
W	1	0	0	0	VRP 14	VRP 13	VRP 12	VRP 11	VRP 10	0	0	0	VRP 04	VRP 03	VRP 02	VRP 01	VRP 00
W	1	0	0	0	VRN 14	VRN 13	VRN 12	VRN 11	VRN 10	0	0	0	VRN 04	VRN 03	VRN 02	VRN 01	VRN 00

PKP53-00: The gamma fine adjustment registers for the positive polarity output

* Initial Value: PKP53-00 = 0000

PRP13-00: The gradient adjustment registers for the positive polarity output

* Initial Value: PRP13-00 = 0000

PKN53-00: The gamma fine adjustment registers for the negative polarity output

* Initial Value: PKN53-00 = 0000

PRN13-00: The gradient adjustment registers for the negative polarity output

* Initial Value: PRN13-00 = 0000

VRP14-00: The amplitude adjustment registers for the positive polarity output

* Initial Value: VRP14-00 = 00000

VRN14-00: The amplitude adjustment registers for the negative polarity output

* Initial Value: VRN14-00 = 00000

For details, see the GAMMA ADJUSTMENT FUNCTION.

8.3.28. TEST_KEY (R80h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	TEST_KEY[7:0]							

TEST_KEY

When you want to update MTP data, “8C” should be written to this register. And you should write different value for MTP data not to be corrupted.

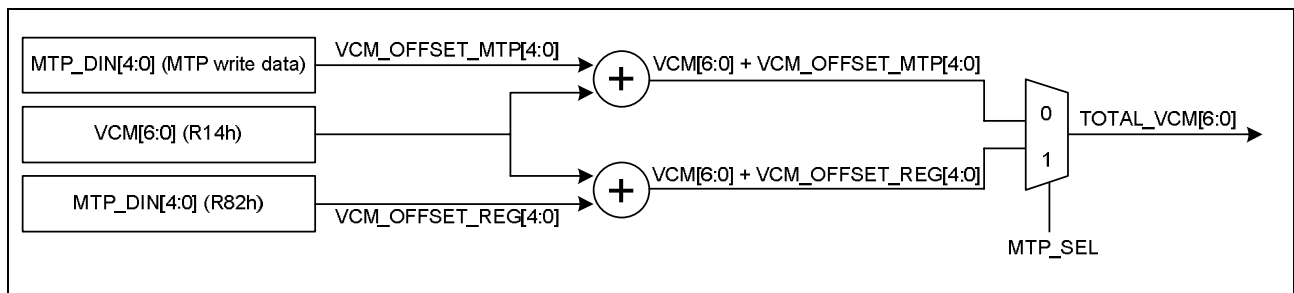
8.3.29. MTP Control (R81h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	MTP_ MODE	MTP_ _EX	0	MTP_ SEL	0	0	0	MTP_ ERB	0	0	0	MTP_ WRB	0	0	0	MTP_ LOAD

MTP_SEL: Select the VCOMH voltage setting register.

VCM [6:0]: Set the VCOMH voltage

MTP_SEL	VCOMH Control Data
0	VCM [6:0] Register + VCM_OFFSET_MTP[4:0]
1	VCM[6:0] Register + VCM_OFFSET_REG[4:0]



MTP_MODE: Set the 2nd booster operating condition.

MTP_MODE = 0: The 2nd booster operates as a user-specified condition. VGH/VGL voltages are generated as a designated level by BT2-0 setting.

MTP_MODE = 1: Available BT2-0 settings are limited only '010' & '101'.

MTP_MODE	MTP operation mode
0	All BT2-0 settings are available (Normal operating condition)
1	Setting of BT2-0 is limited. (An MTP-programming / erasing condition)

Note. Do not execute MTP programming / erasing operation when MTP_MODE = 0.

MTP_EX: Select MTP power supply source.

MTP_EX = 0: Internally generated VGH voltage is used as a MTP-programming/erasing potential.

MTP_EX = 1: External power should be applied for programming / erasing MTP via VGH pad.

MTP_EX	Erase / Initial / Program supply
0	Used internally generated VGH
1	Needed external power supply

Note. MTP_EX register is valid only in case that MTP_MODE = 1. Do not access MTP_EX register when MTP_MODE = 0.

MTP_ERB: Enable for MTP initial or erase. When MTP_ERB = 0, MTP initialization or erase is enabled.

MTP_WRB: MTP Write enable signal. If you want to write data to MTP cell, set MTP_WRB = 0

MTP_LOAD: When MTP_LOAD is High, MTP data is loaded into internal register.

8.3.30. MTP Data Write (R82h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	0	0	0	MTP_DIN4	MTP_DIN3	MTP_DIN2	MTP_DIN1	MTP_DIN0
R	1	X	X	X	X	X	X	X	X	X	X	MTP_DOUT5	MTP_DOUT4	MTP_DOUT3	MTP_DOUT2	MTP_DOUT1	MTP_DOUT0

MTP_DIN

MTP_DIN [5:0] contain VCM Offset data. This MTP data and VCM register determines VCOMH level.

TOTAL_VCM [6:0] = VCM [6:0] + VCM_OFFSET.

MTP_DIN[4:0]	VCM_OFFSET	MTP_DIN[4:0]	VCM_OFFSET
00000	0	10000	0
00001	+1	10001	-1
00010	+2	10010	-2
00011	+3	10011	-3
00100	+4	10100	-4
00101	+5	10101	-5
00110	+6	10110	-6
00111	+7	10111	-7
01000	+8	11000	-8
01001	+9	11001	-9
01010	+10	11010	-10
01011	+11	11011	-11
01100	+12	11100	-12
01101	+13	11101	-13
01110	+14	11110	-14
01111	+15	11111	-15

For example, if VCM [6:0] = 0001011 and MTP_DIN [4:0] = 10001 is selected, then MTP_OFFSET is “-1,” and therefore TOTAL_VCM is “0001010,” which results in VCOMH voltage = 2.697 from VCM6-0 table.

Note that TOTAL_VCM [6:0] cannot be set to the value above “1111111” or below “0000000,” that is, $127 \geq \text{VCM [6:0]} + \text{VCM_OFFSET} \geq 0$.

Note. TOTAL_VCM [6:0] is VCM [6:0] + VCM_OFFSET_MTP [4:0] when MTP_SEL=0 and is VCM [6:0] + VCM_OFFSET_REG [4:0] when MTP_SEL=1.

MTP_DOUT: Data of MTP Output

This command reads the VCOM MTP values, MTP_DOUT<5:0> bits. The MSB bit, MTP_DOUT5, protects re-write the MTP, when any bit of MTP is written, MSB bit is changed from low to high automatically.

Once MSB bit sets to be high, MTP cannot be written. To re-write the MTP, need initialization to set MTP_DOUT5 bit to be low.

MTP Control in Internal Mode

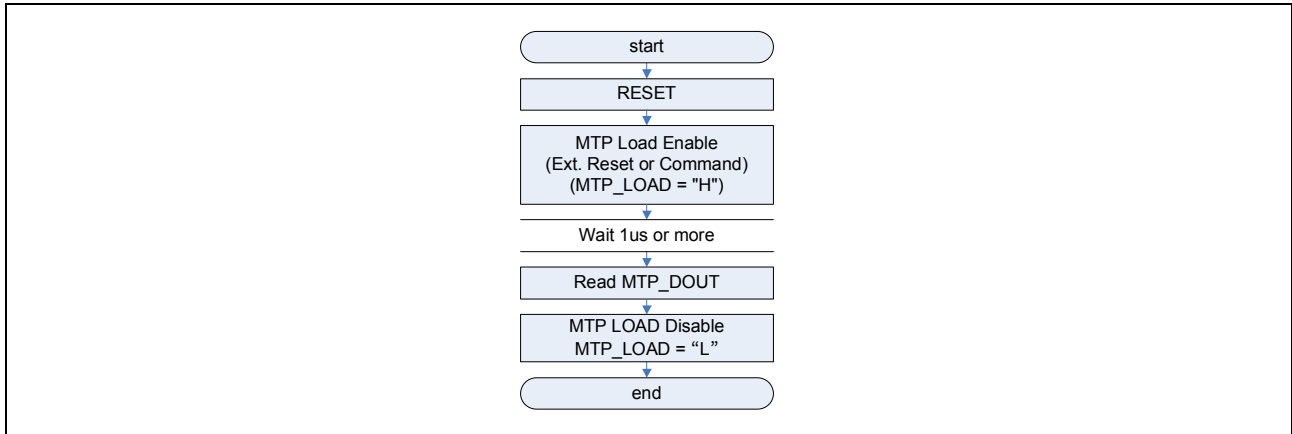


Figure 29. Flow of MTP Load / Read

a. Using VCI for MTP

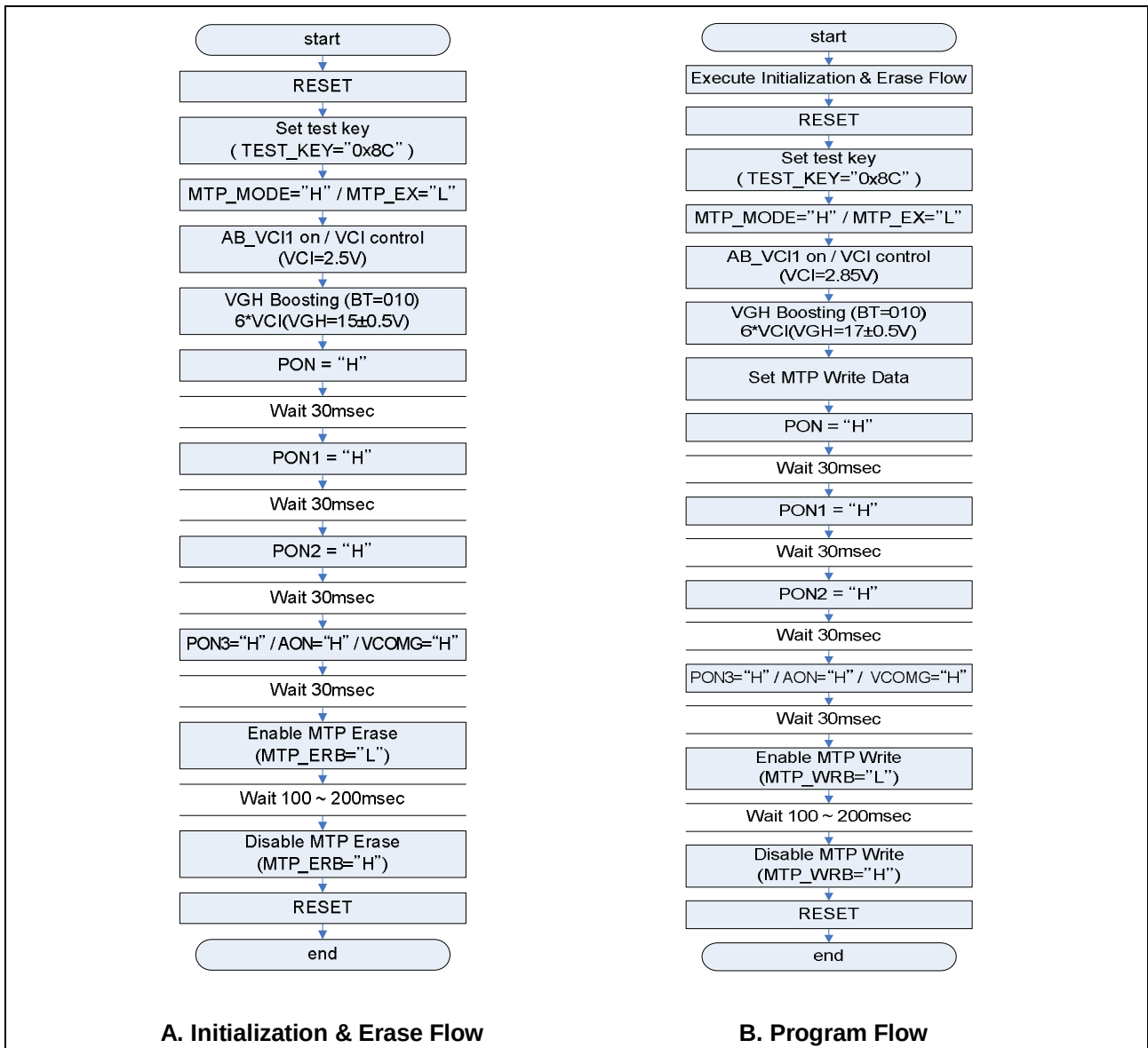


Figure 30. MTP Initialization, Erase and program

b. Using VCI1 for MTP

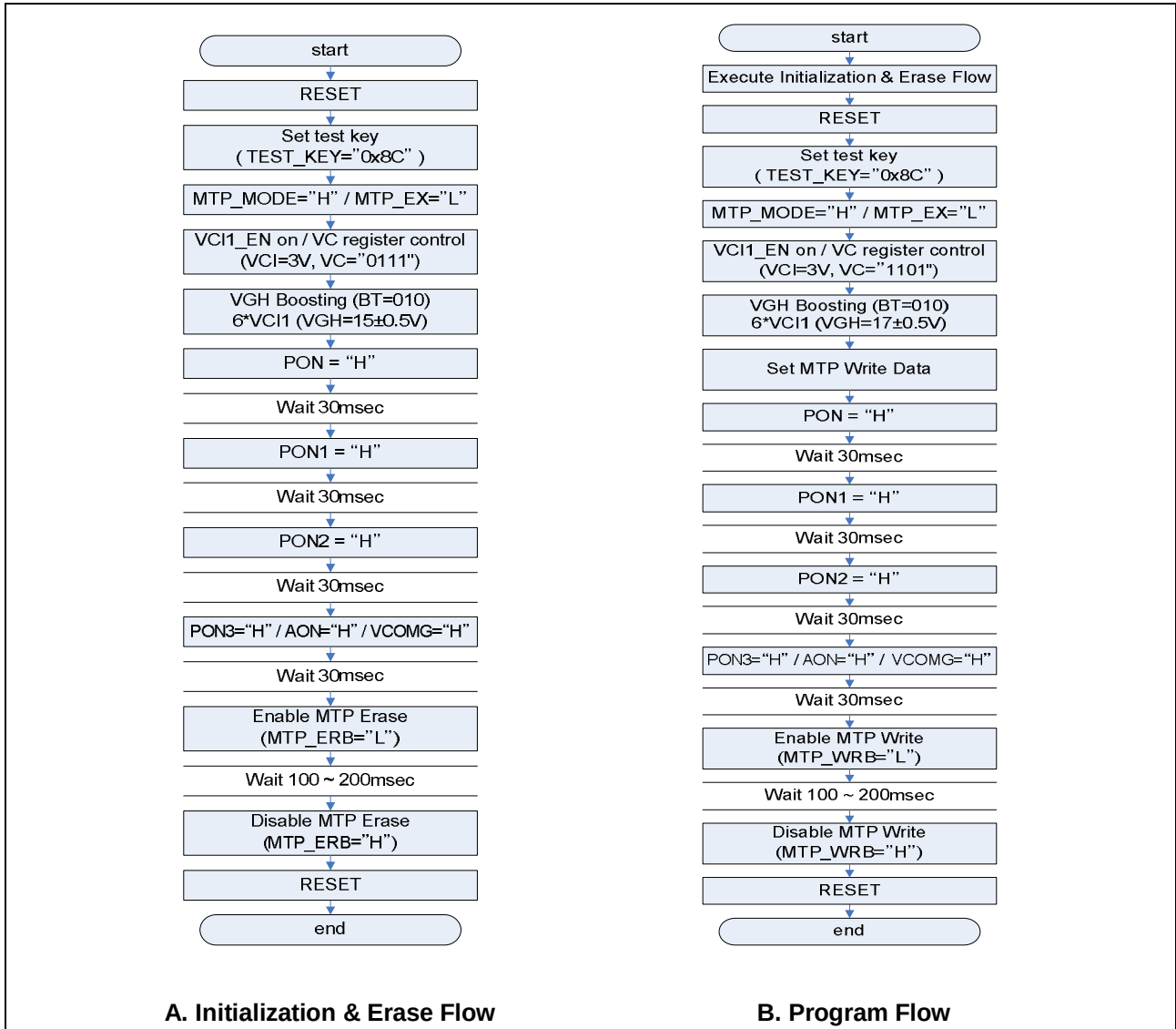


Figure 31. MTP Initialization, Erase and program

MTP Control in External Mode

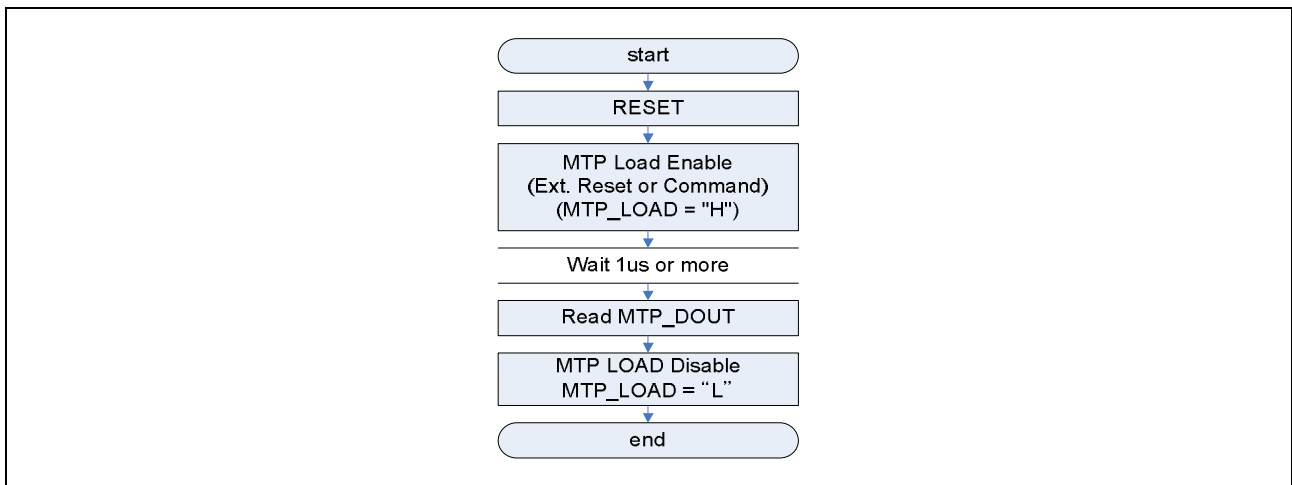


Figure 32. Flow of MTP Load / Read

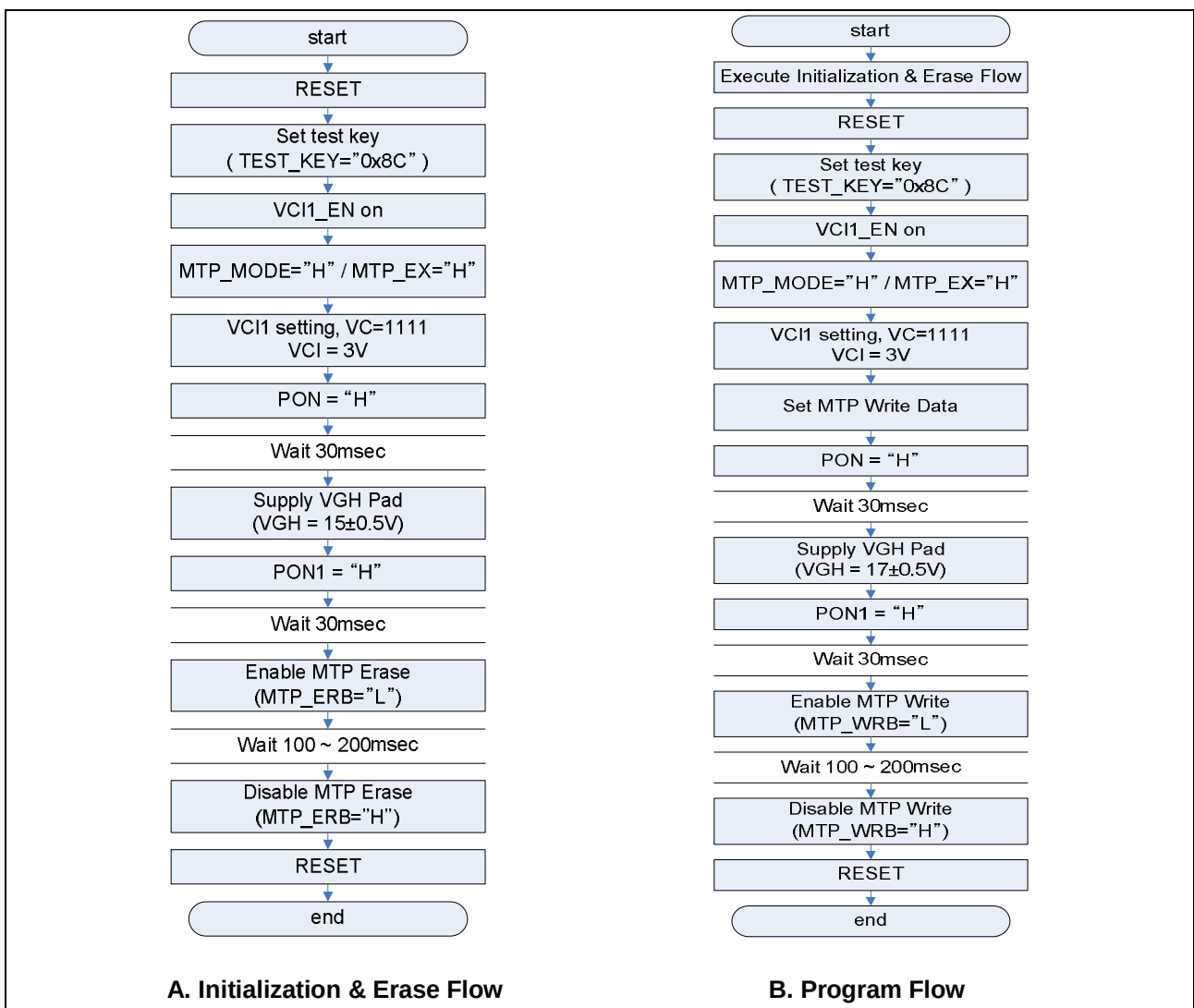


Figure 33. MTP Initialization, Erase and program

Timing of MTP Control

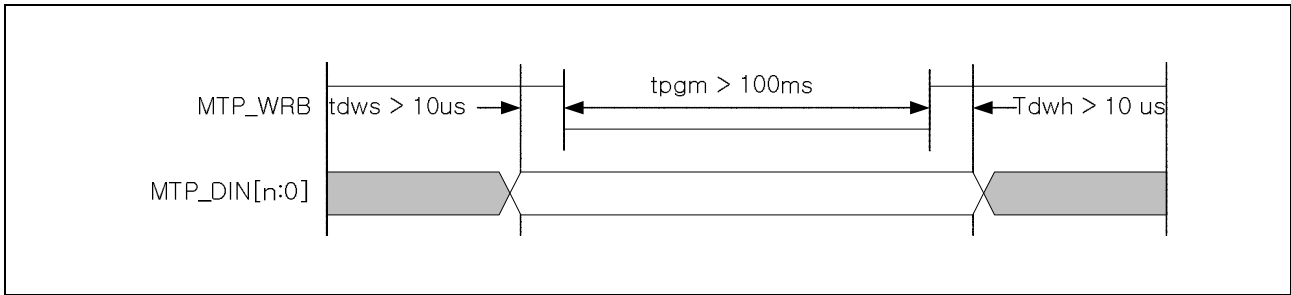


Figure 34. Timing of MTP Program

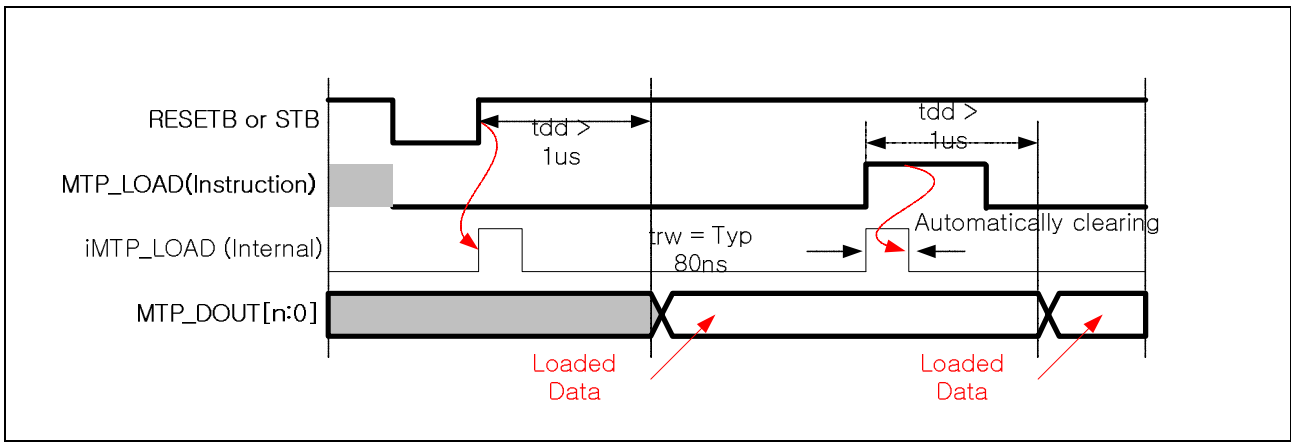


Figure 35. Timing of MTP Load

8.3.31. Panel ID (R93h)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
R	1	P_ID1	P_ID0	X	X	X	X	X	X	X	X	X	X	X	X	X	X

P_ID1-0: Panel ID can be read using TEST_DA and TEST_MODE [2:0] pads. Panel ID is used to multi-panel.

Pin Setting		P_ID[1:0]
TEST_DA	TEST_MODE[2:0]	
0	000	00
0	111	01
1	000	10
1	111	11

9. RESET FUNCTION

The S6D0164 is internally initialized by RESET input. The reset input must be held for at least 20us. Do not access the GRAM or initially set the instructions until the R-C oscillation frequency is stable after power has been supplied (10 ms).

Instruction Set Initialization

1. Driver output control (VSPL=0, HSPL=0, DPL=0, EPL=0, SM=0, GS=0, SS = 0, NL4-0 = 11100)
2. LCD driving AC control (INV1-0=01, FLD=0)
3. Entry mode set (BGR=0, MDT1-0 = 00, I/D1-0 = 11, AM=0)
4. Display control (FLM_MON = 0, GON = 0, CL = 0, REV = 0, D1-0 = 00: Display off)
5. Blank period control (FP3-0 = 1000, BP3-0 = 1000)
6. Frame cycle control (NO3-0 = 0001, SDT3-0 = 0001, RTN3-0 = 0000: 16 clock cycle in 1H period)
7. External display interface (RIM1-0=00:18-bit RGB interface, DM=0: operated by internal clock, RM=0: system interface)
8. Start oscillation (FOSC4-0=00101, OSCON=1)
9. Power control 1 (SAP3-0 = 0010, DSTB= 0: Deep Standby mode off, STB=0)
10. Power control 2 (APON=0, PON3-0=0000, AB_VCI1=0, AON = 0, VCI1_EN=0, VC3-0 =0000)
11. Power control 3 (BT2-0=000, DC11-0=00, DC21-0=00, DC31-0=00)
12. Power control 4 (DCR_EX=0, DCR2-0=000, GVD6-0 = 0000000)
13. Power control 5 (VCOMG=1, VCMR = 0, VCM6-0 = 0000000, VML6-0 = 0000000)
14. VCIR Recycling (VCIR2-0 = 000)
15. RAM Address data (AD7-0 = 00000000, AD15-8 = 00000000)
16. Gate scan position (SCN4-0=00000)
17. Vertical scroll control 1 (SSA7-0 = 0000000, SEA7-0 = 11011011)
18. Vertical scroll control 2 (SST7-0 = 0000000 : No vertical scroll)
19. Partial screen division (SE17-10 = 11011011, SS17-10 = 00000000)
20. Horizontal RAM address position (HEA7-0 = 10101111, HSA7-0 = 00000000)
21. Vertical RAM address position (VEA7-0 = 11011011, VSA7-0 = 00000000)
22. Gamma control
(PKP03-00 = 0000, PKP13-10 = 0000, PKP23-20 = 0000, PKP33-30 = 0000,
PKP43-40 = 0000, PKP53-50 = 0000, PRP03-00 = 0000, PRP13-10 = 0000)
(PKN03-00 = 0000, PKN13-10 = 0000, PKN23-20 = 0000, PKN33-30 = 0000,
PKN43-40 = 0000, PKN53-50 = 0000, PRN03-00 = 0000, PRN13-10 = 0000)
(VRP04-00 = 00000, VRP14-10 = 00000, VRN04-00 = 00000, VRN_14-10 = 00000)
23. MTP control (MTP_MODE=0, MTP_EX=0, MTP_SEL=0, MTP_ERB=1, MTP_WRB=1, MTP_LOAD=0)
24. Test key command (00000000)
25. MTP load data (MTP_DIN4-0 = 00000)

GRAM Data Initialization

GRAM is not automatically initialized by reset input but must be initialized by software while display is off (D1-0 = 00).

Output pin Initialization

LCD driver output pins (Source output): Output AVSS level

(Gate output): Output AVSS level

10. POWER SUPPLY

10.1. Power Supply Circuit

The following figure shows a configuration of the voltage generation circuit of S6D0164. The step-up circuits consist of step-up circuits 1, 2 and 3. Step-up circuit1 doubles input voltage supplied from VCI1 for AVDD level. Step-up circuit2 makes 2.5, 3 or 3.5 times AVDD voltage for VGH level, and makes -1.5, -2 or -2.5 times AVDD voltage for VGL level. Step-up circuit3 reverses the VCI1 voltage with respect to VSS to generate VCL level. These step-up circuits generate power supplies AVDD, VGH, VGL, and VCL. Reference voltage GVDD is generated with VREF from the voltage divide circuit. VCOMH and VCOML are generated with GVDD from the voltage adjustment circuit. Connect VCOM to the TFT panel.

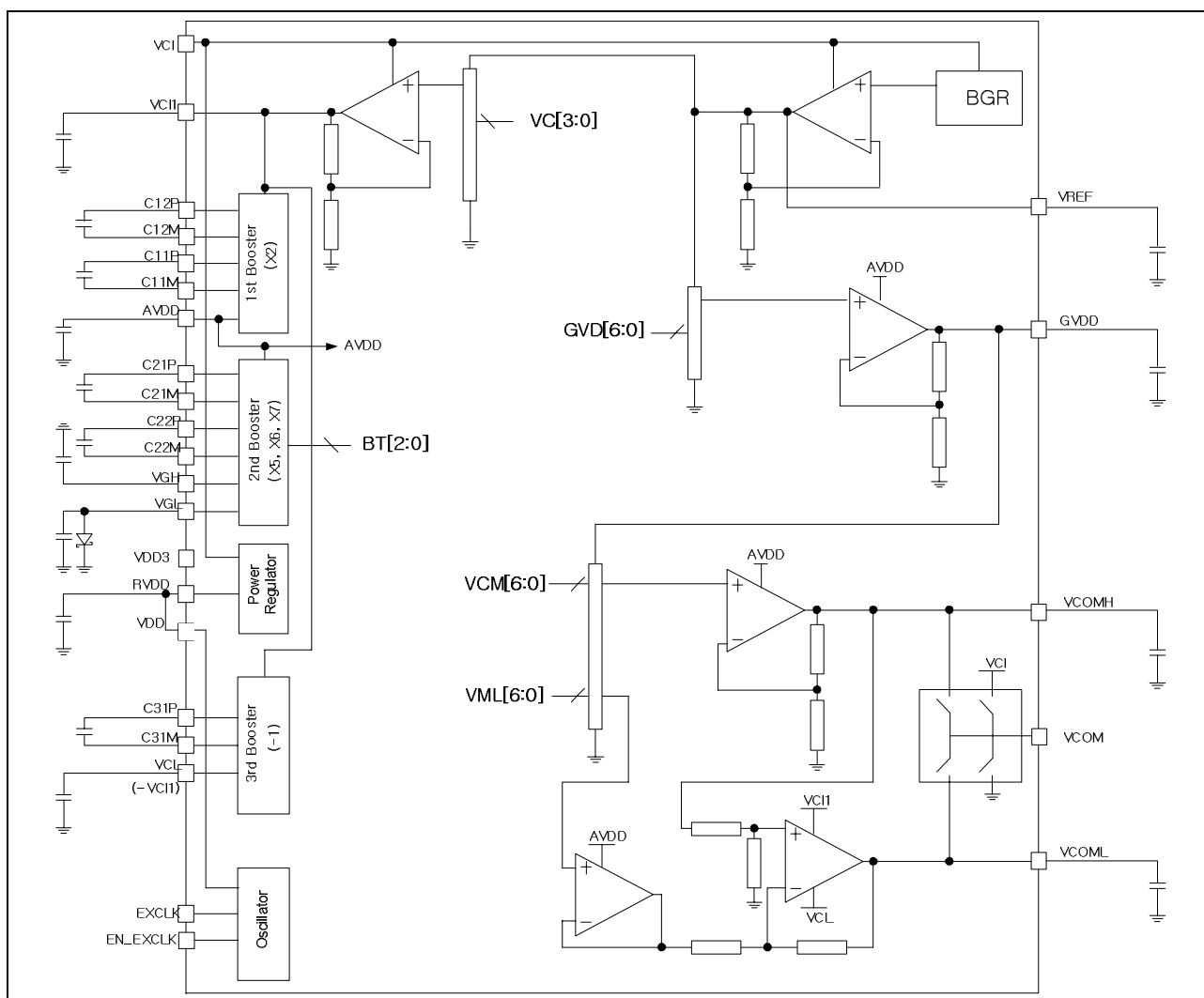


Figure 36. Configuration of the Internal Power-Supply Circuit

Note.

1. Use the 1uF capacitor.
2. Schottky diode between VGL and VSS is positively necessary for latch-up free.
3. The Capacitor between VREF and VSS may be used by case when VCOM swing level fluctuates.

10.2. Pattern Diagrams for Voltage Setting

The following figure shows a pattern diagram for the voltage setting and an example of waveforms.

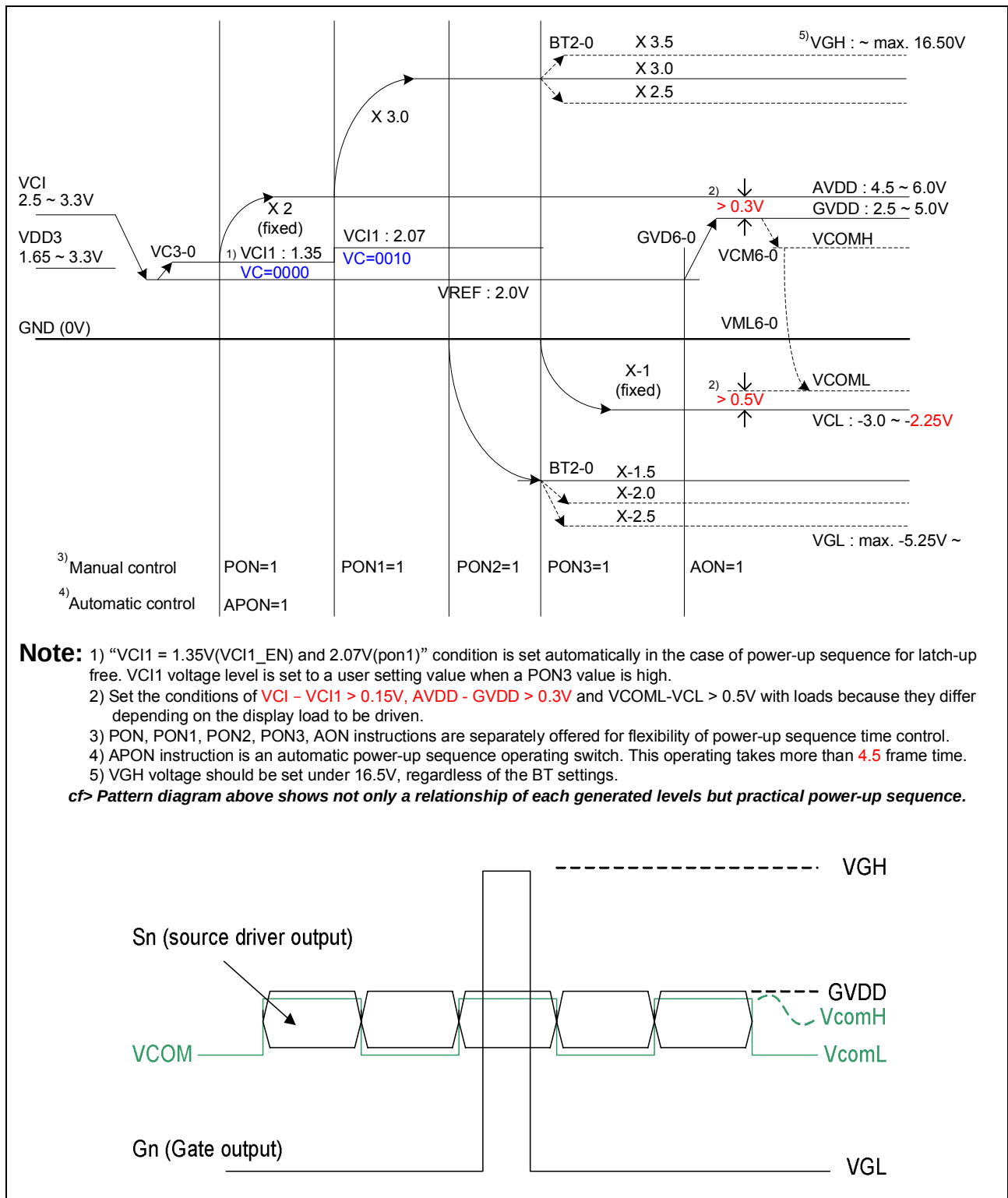


Figure 37. Power-Up Pattern Diagram & An Example Of Source/VCOM Waveforms

10.3. Set up Flow of Generated Power Supply

Apply the power in a sequential way as shown in the following figure. The settling time of the oscillation circuit, step-up1/2/3 circuits, and operational amplifier depend on the external resistance or capacitance value.

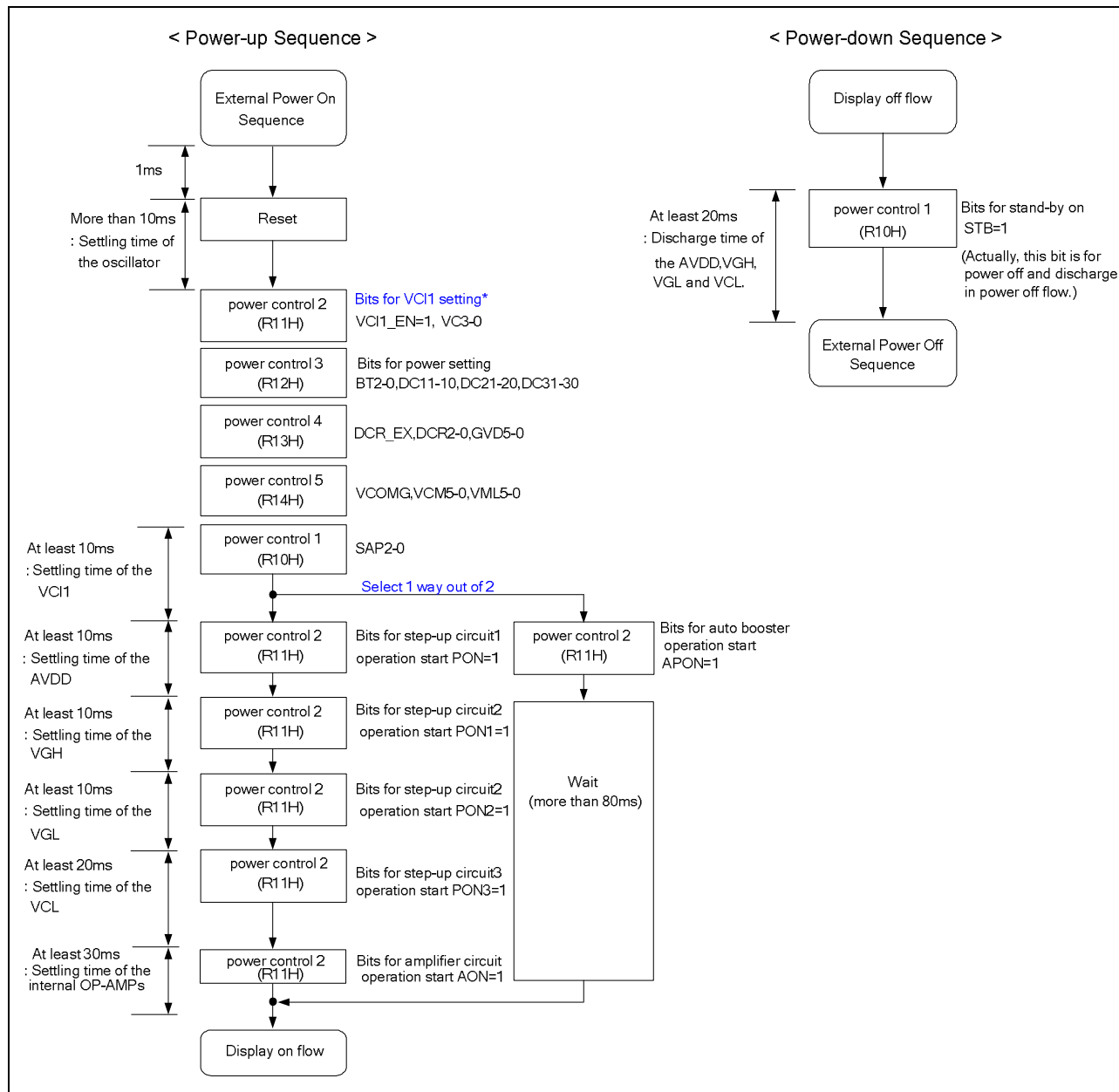


Figure 38. Setup Flow of Generated Power Supply

10.4. Voltage regulation function

The S6D0164 has the internal voltage regulator. By the use of this function, unexpected damages on internal logic circuit can be avoided. Furthermore, power consumption can also be obtained. Detailed function description and application configuration is described in the following diagram.

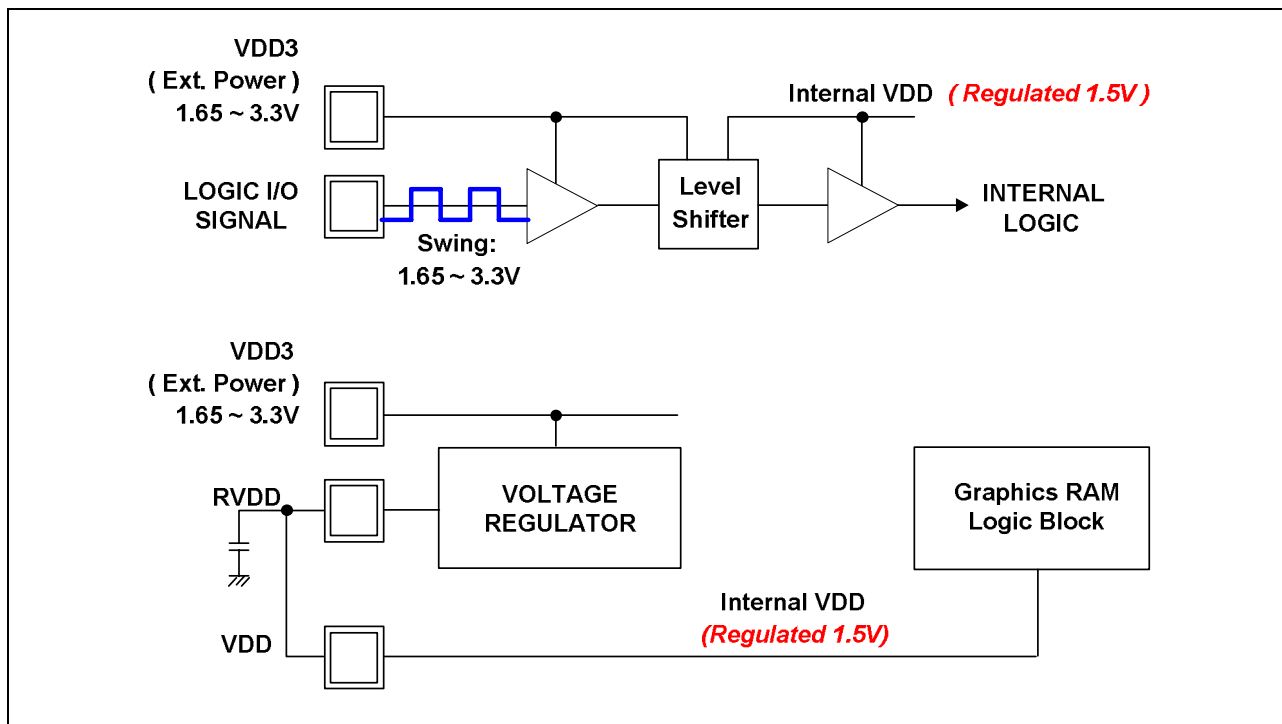


Figure 39. Voltage Regulation Function

11. INTERFACE SPECIFICATION

S6D0164 incorporates nine System Interfaces which are used to set instructions, and an RGB interface that is used to display motion pictures. Selecting one of these interfaces to match the screen data (motion picture or still picture) enables efficient transfer of data for display.

The External Clock Operation mode that uses RGB interface allows flicker-free screen update. In this mode, the synchronization signals (VSYNC, HSYNC, and DOTCLK) are available for display operation. The data for display (DB [17:0]) is written according to the status of ENABLE in synchronization with VSYNC, HSYNC, and DOTCLK. In addition, using Window Address function enables rewriting only to the internal GRAM area to display motion pictures. Using this function also enables simultaneously display of motion picture and the GRAM data that was written earlier.

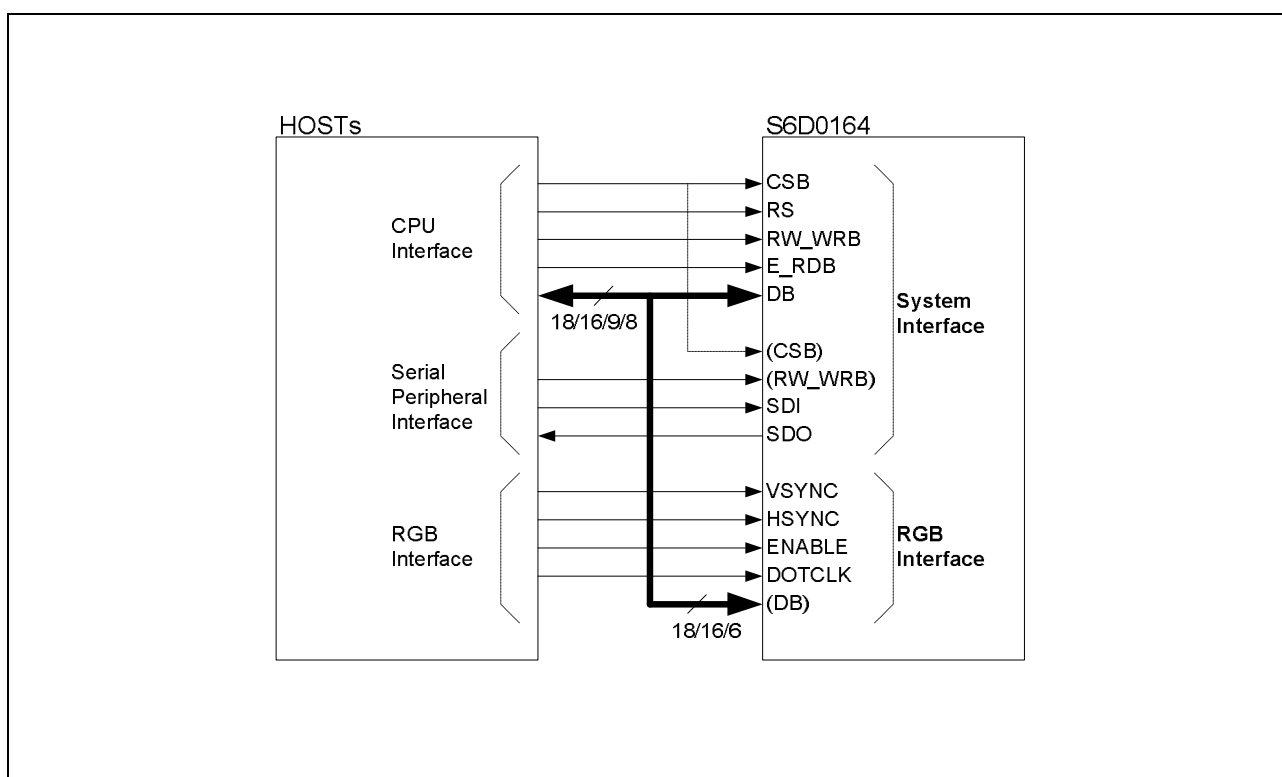


Figure 40. System Interface and RGB Interface

11.1. SYSTEM INTERFACE

S6D0164 has nine System Interfaces as show below.

Table 35. System Interfaces of S6D0164

IM[3:0]	System Interface
4'b0000	68-16bit CPU interface
4'b0001	68-8bit CPU interface
4'b0010	80-16bit CPU interface
4'b0011	80-8bit CPU interface
4'b010x	Serial peripheral interface (SPI)
4'b011x	Setting disabled
4'b1000	68-18bit CPU interface
4'b1001	68-9bit CPU interface
4'b1010	80-18bit CPU interface
4'b1011	80-9bit CPU interface
4'b11xx	Setting disabled

In order to select one of them you should set IM [3:0] properly. For detail, see "PIN DESCRIPTION" described earlier.

11.1.1. 68-18BIT CPU INTERFACE

Bit Assignment

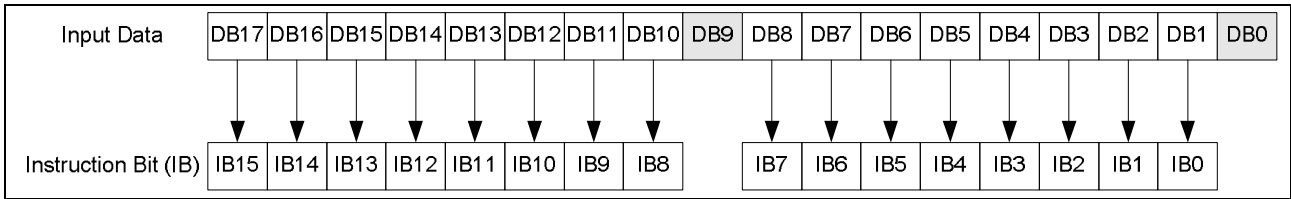


Figure 41. Bit Assignment of Instructions on 68-18bit CPU Interface

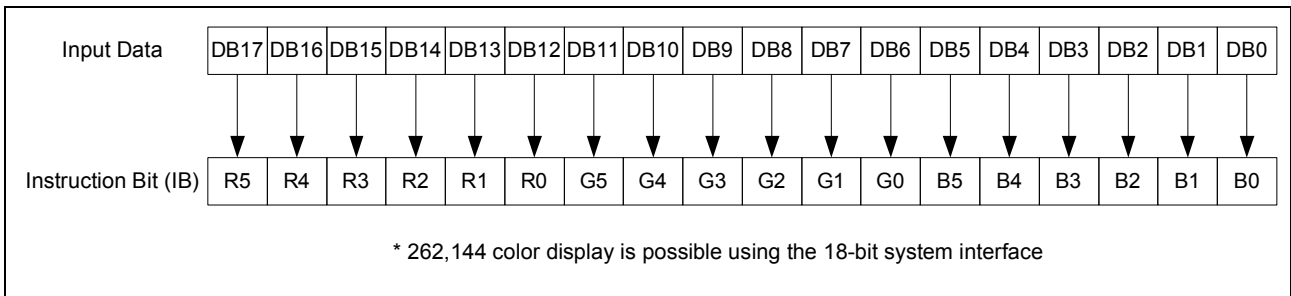


Figure 42. Bit Assignment of GRAM Data on 68-18bit CPU Interface

Timing Diagram

There are 4 timing conditions for 68 18-bit CPU interfaces, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

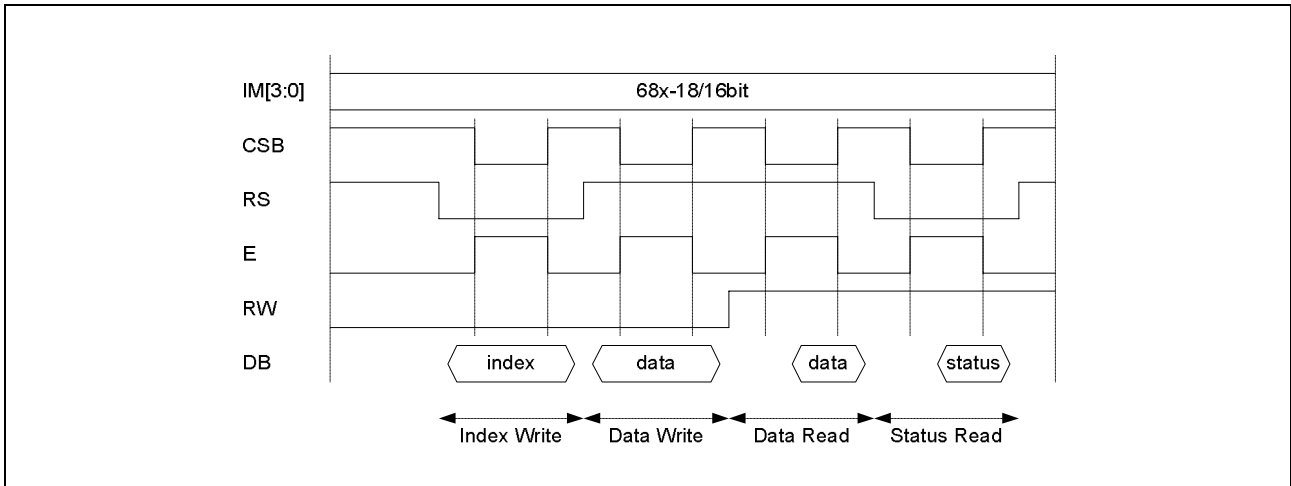


Figure 43. Timing Diagram of 68-18bit CPU Interface

11.1.2. 68-16BIT CPU INTERFACE

Bit Assignment

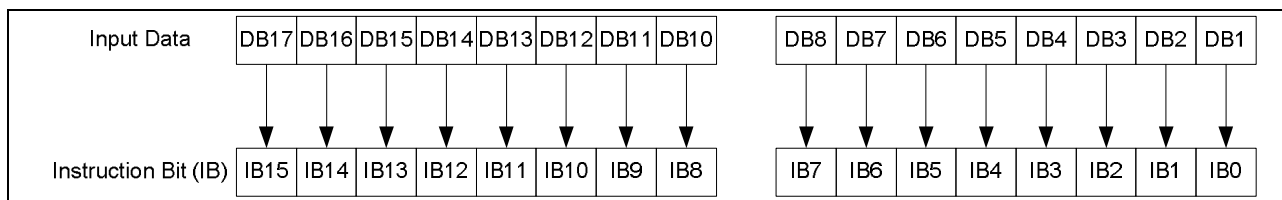


Figure 44. Bit Assignment of Instructions on 68-16bit CPU Interface

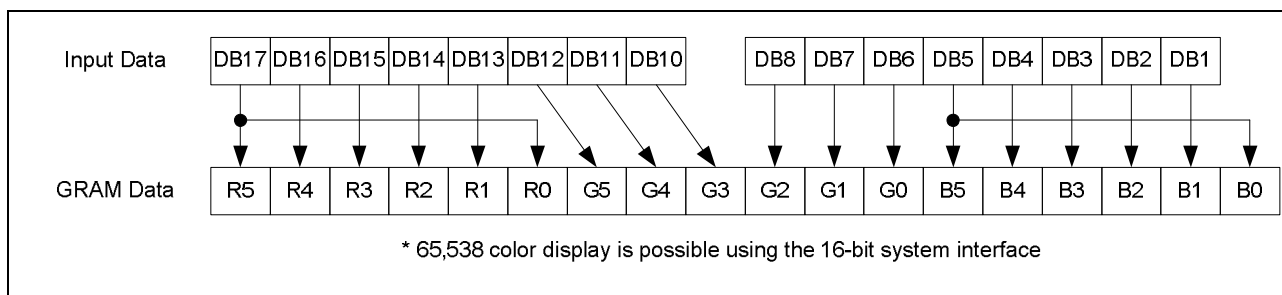


Figure 45. Bit Assignment of GRAM Data on 68-16bit CPU Interface

Timing Diagram

There are 4 timing conditions for 68-16bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

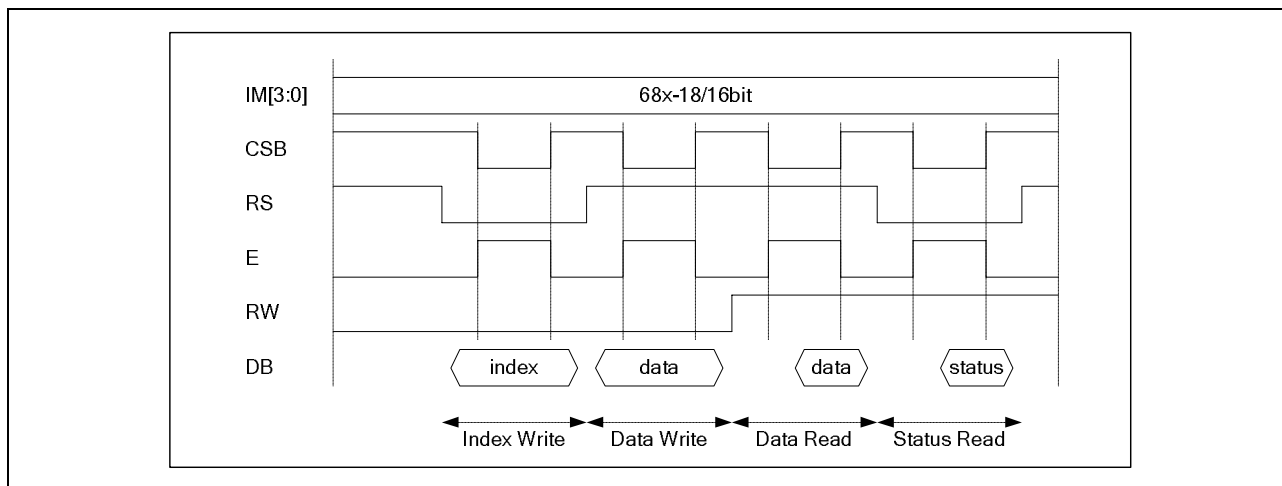


Figure 46. Timing Diagram of 68-16bit CPU Interface

11.1.3. 68-9BIT CPU INTERFACE

Bit Assignment

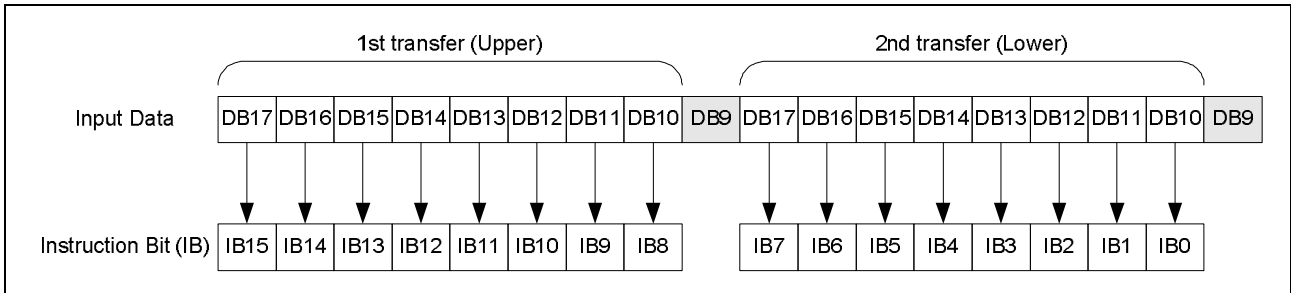


Figure 47. Bit Assignment of Instructions on 68-9bit CPU Interface

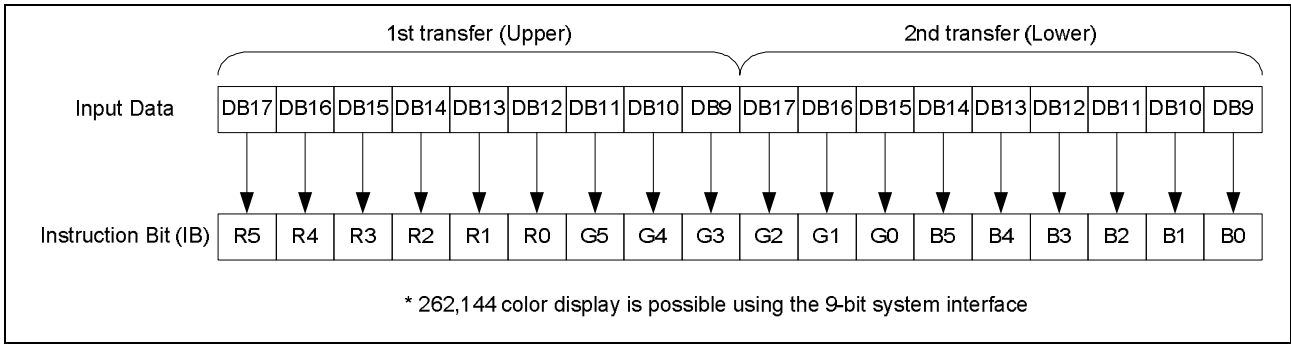


Figure 48. Bit Assignment of GRAM Data on 68-9bit CPU Interface

Timing Diagram

There are 4 timing conditions for 68-9bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

In this mode, 16-bit instructions and GRAM data are divided into two half words and the transfer starts from the upper half word.

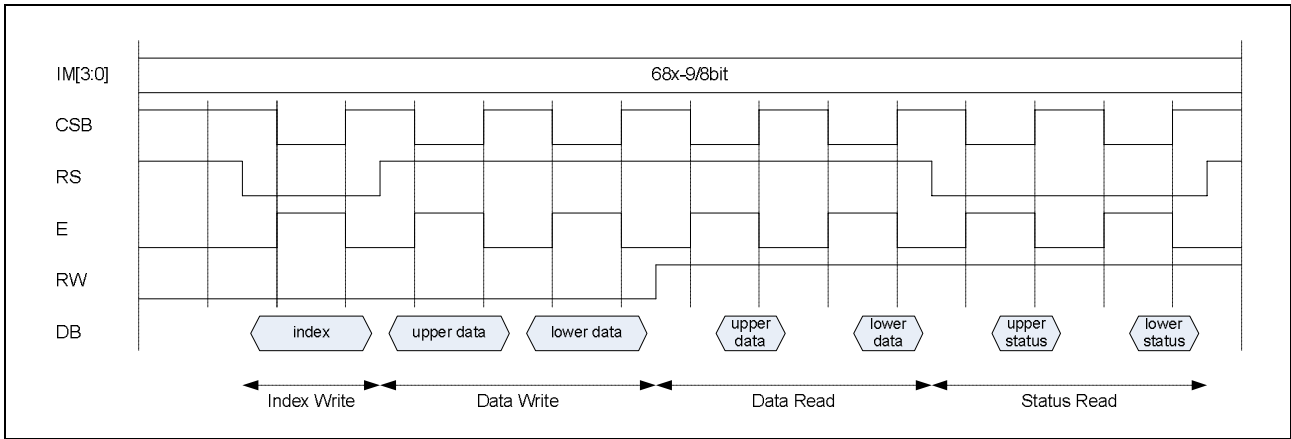


Figure 49. Timing Diagram of 68-9bit CPU Interface

11.1.4. 68-8BIT CPU INTERFACE

Bit Assignment

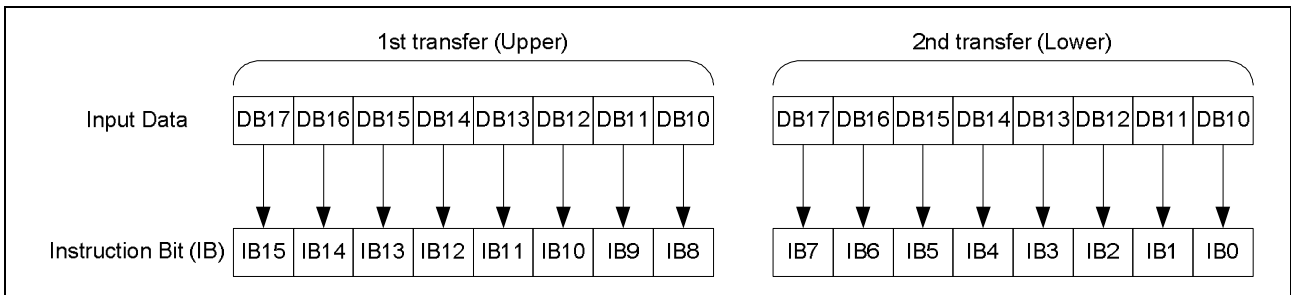


Figure 50. Bit Assignment of Instructions on 68-8bit CPU Interface

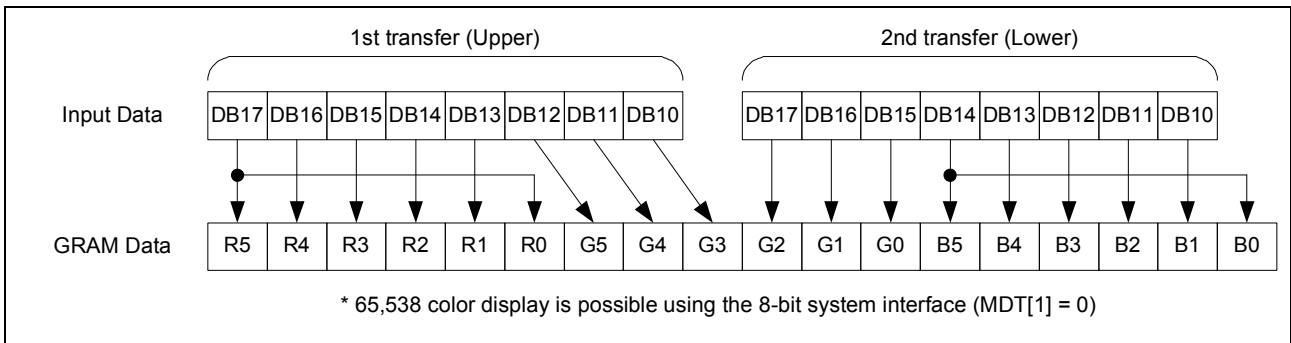


Figure 51. Bit Assignment of GRAM Data on 68-8bit CPU Interface

Timing Diagram

There are 4 timing conditions for 68-8bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

In this mode, 16-bit instructions and GRAM data are divided into two half words and the transfer starts from the upper half word.

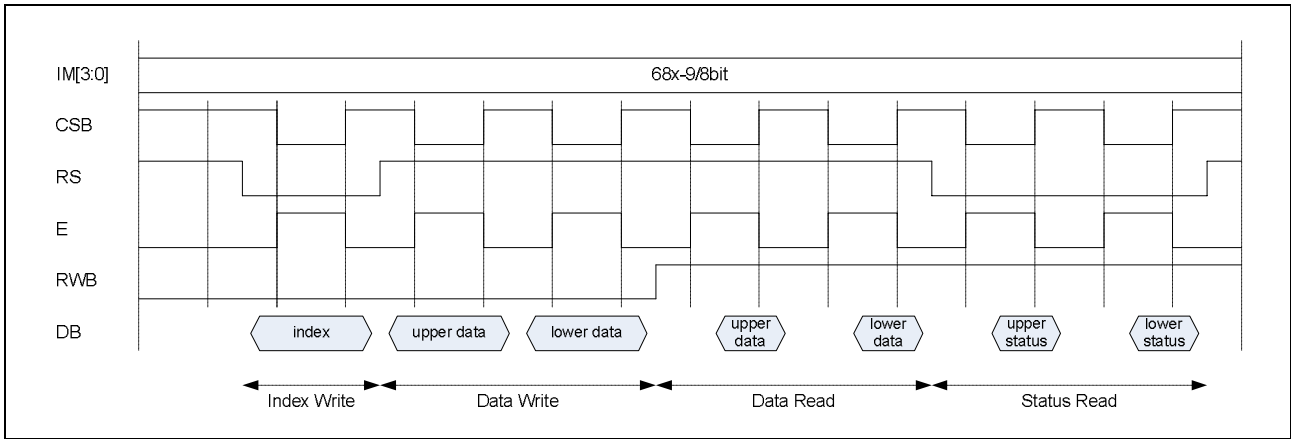


Figure 52. Timing Diagram of 68-8bit CPU Interface

11.1.5. 80-18BIT CPU INTERFACE

Bit Assignment

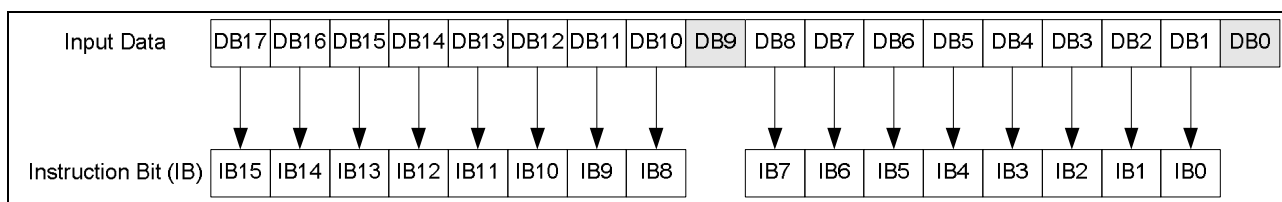


Figure 53. Bit Assignment of Instructions on 80-18bit CPU Interface

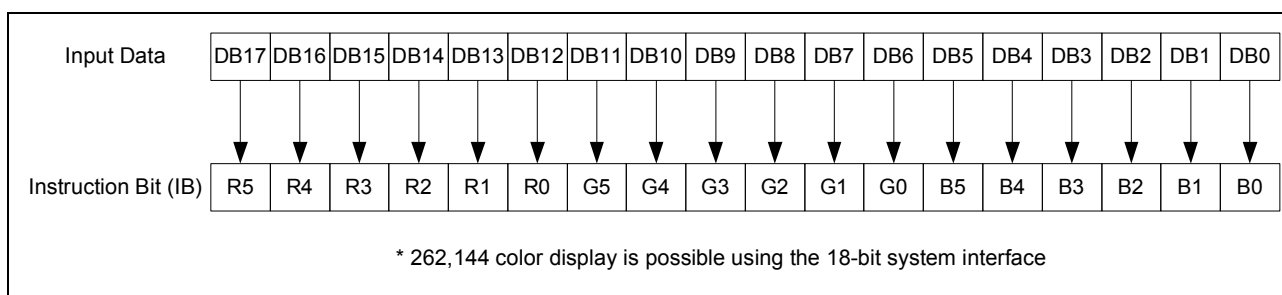


Figure 54. Bit Assignment of GRAM Data on 80-18bit CPU Interface

Timing Diagram

There are 4 timing conditions for 80 18-bit CPU interfaces, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

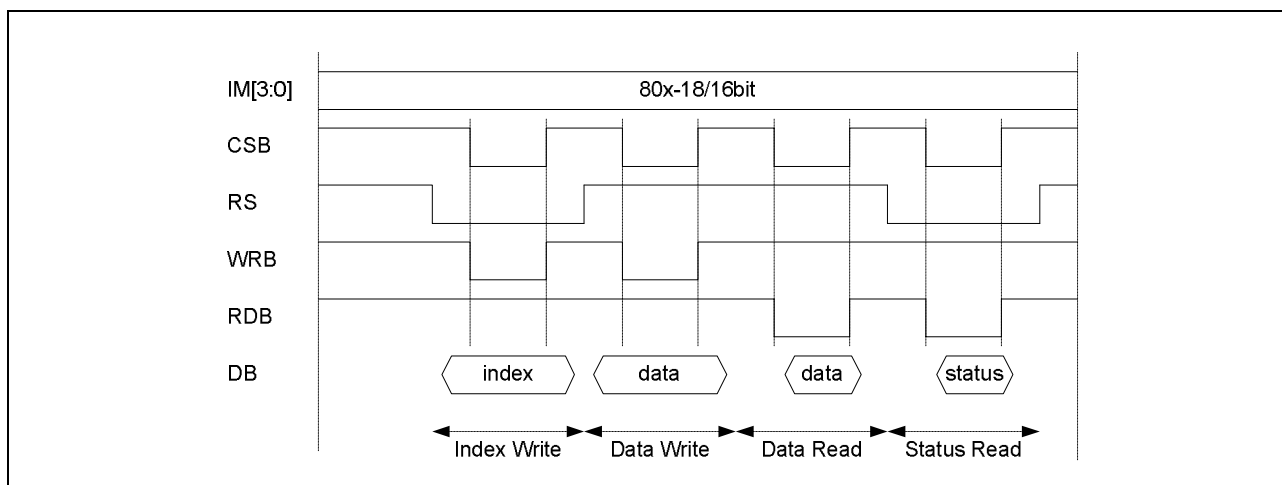


Figure 55. Timing Diagram of 80-18bit CPU Interface

11.1.6. 80-16BIT CPU INTERFACE

Bit Assignment

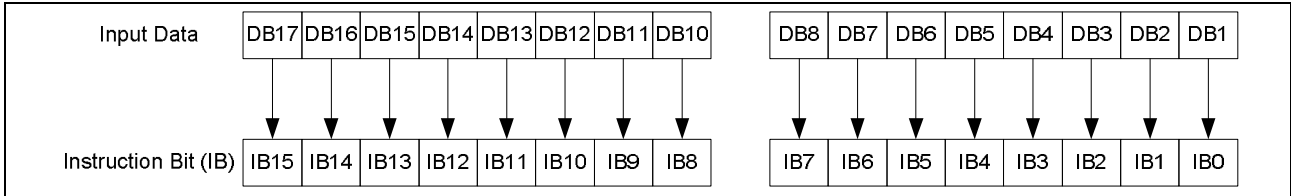


Figure 56. Bit Assignment of Instructions on 80-16bit CPU Interface

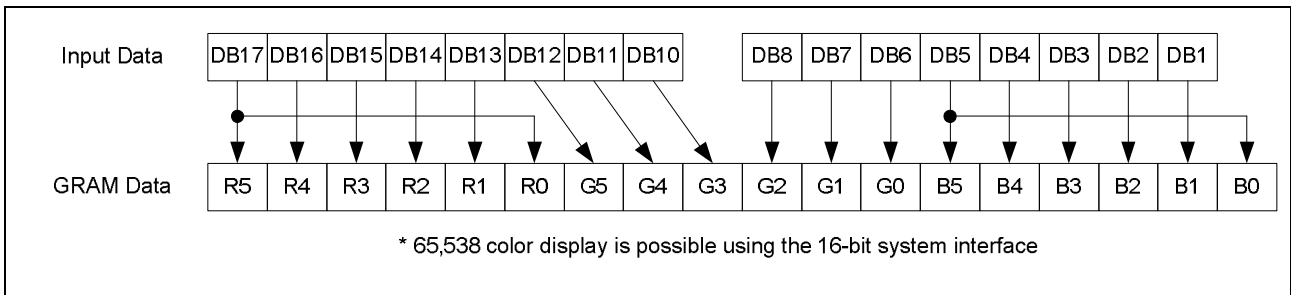


Figure 57. Bit Assignment of GRAM Data on 80-16bit CPU Interface

Timing Diagram

There are 4 timing conditions for 80-16bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

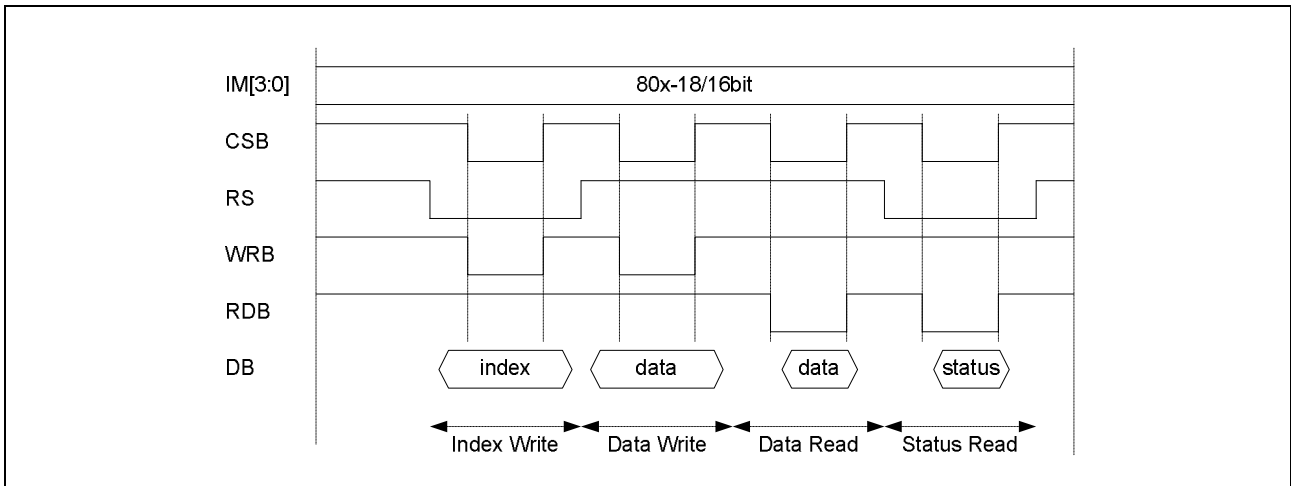


Figure 58. Timing Diagram of 80-16bit CPU Interface

11.1.7. 80-9BIT CPU INTERFACE

Bit Assignment

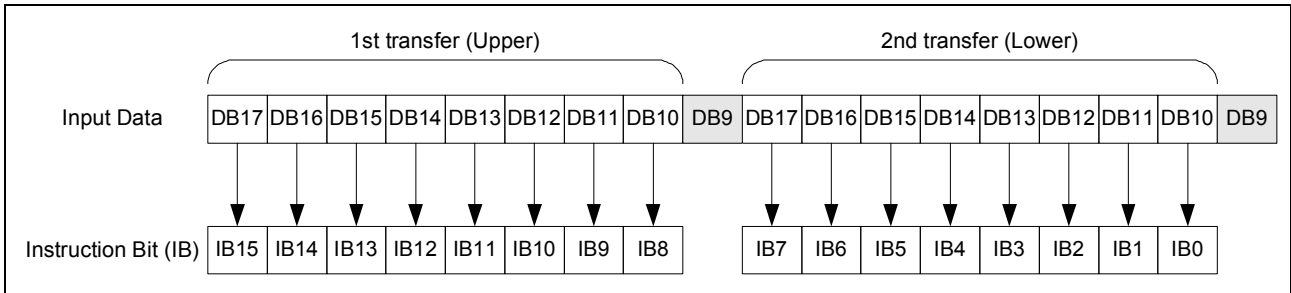


Figure 59. Bit Assignment of Instructions on 80-9bit CPU Interface

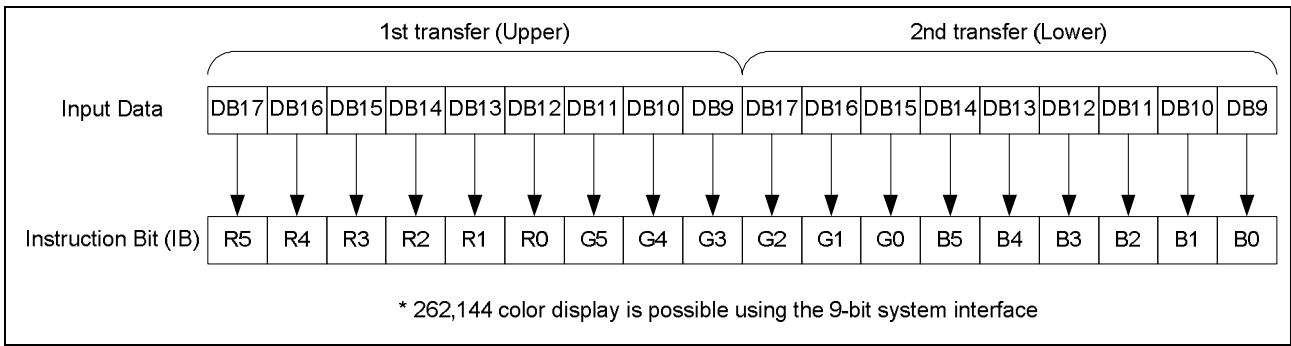


Figure 60. Bit Assignment of GRAM Data on 80-9bit CPU Interface

Timing Diagram

There are 4 timing conditions for 80-9bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

In this mode, 16-bit instructions and GRAM data are divided into two half words and the transfer starts from the upper half word.

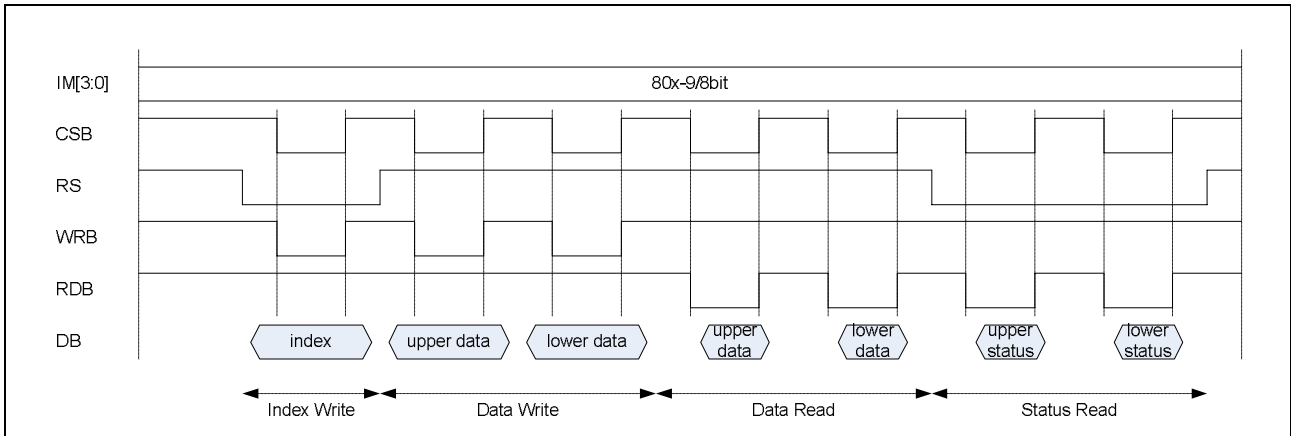


Figure 61. Timing Diagram of 80-9bit CPU Interface

11.1.8. 80-8BIT CPU INTERFACE

Bit Assignment

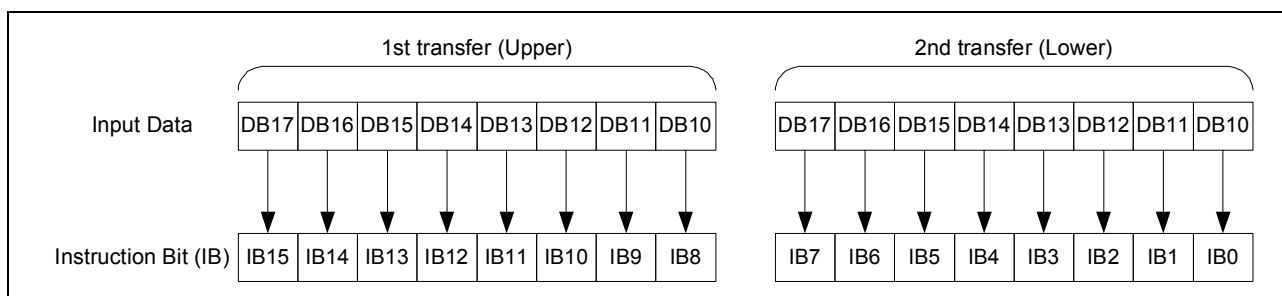


Figure 62. Bit Assignment of Instructions on 80-8bit CPU Interface

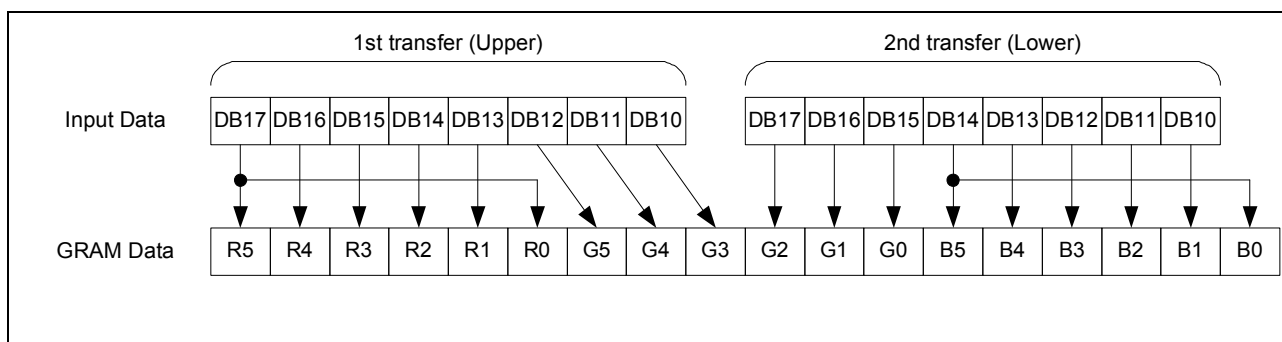


Figure 63. Bit Assignment of GRAM Data on 80-8bit CPU Interface

Timing Diagram

There are 4 timing conditions for 80-8bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

In this mode, 16-bit instructions and GRAM data are divided into two half words and the transfer starts from the upper half word.

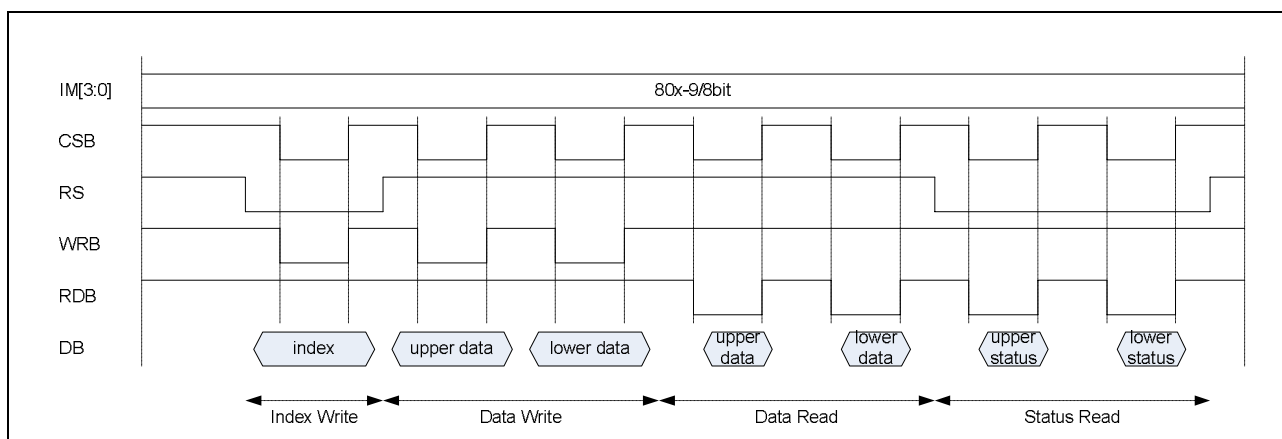


Figure 64. Timing Diagram of 80-8bit CPU Interface

11.1.9. SERIAL PERIPHERAL INTERFACE

Setting IM [3:0] properly allows standard clock-synchronized serial data transfer (SPI; Serial Peripheral Interface), using CSB (chip select), SCL (serial transfer clock), SDI (serial input data) and SDO (serial output data). For the serial interface, IM [0] is used as ID.

S6D0164 initiates serial data transfer by transferring the start byte at the falling edge of CSB input. It ends serial data transfer at the rising edge of CSB input.

S6D0164 is selected when the 6-bit chip address in the start byte transferred by the transmitting device matches the 6-bit device identification code assigned to S6D0164. ID is the least significant bit of the device identification code. S6D0164, when selected, receives the subsequent data string.

Two different chip addresses must be assigned to a single S6D0164 because the seventh bit of the start byte is used as a register select bit (RS): that is, when RS = "0", data can be written to the index register or status can be read, and when RS = "1", an instruction can be issued or data can be written to or read from GRAM. Read or write is determined according to the eighth bit of the start byte (R/WB bit). The data is written (receives) when the R/WB bit is "0", and is read (transmits) when the R/WB bit is "1".

After receiving the start byte, S6D0164 receives or transmits the subsequent data. The data is transferred with the MSB first. All S6D0164 instructions are 16 bits, so two bytes are received with the MSB first (DB15 to 0), and then the instruction is internally executed.

Five bytes of GRAM data read just after the start byte are invalid. S6D0164 starts to read correct GRAM data from the sixth byte. Likewise, it starts to read correct register/status from the second byte.

Table 36. Start Byte Format

Transfer Bit	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
Start byte format	Device Identification code						RS	R/WB
	0	1	1	1	0	ID		

Note. The IM [0] pin is used as ID

Table 37. RS and R/WB Bit Function

RS bit	R/WB bit	Function
0	0	Set index register
0	1	Read status
1	0	Writes instruction or RAM data
1	1	Reads instruction or RAM data

Bit Assignment

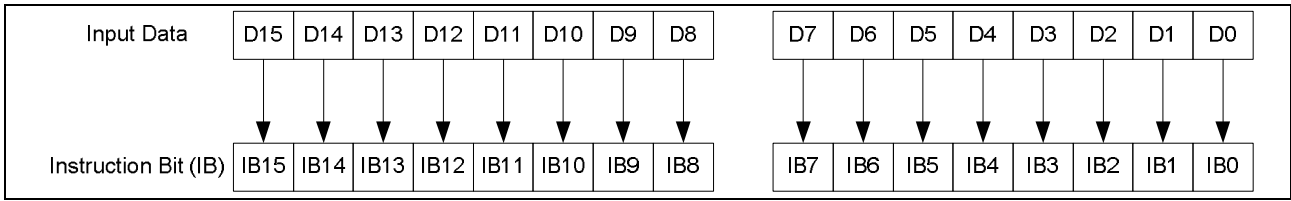


Figure 65. Bit Assignment of Instructions on SPI

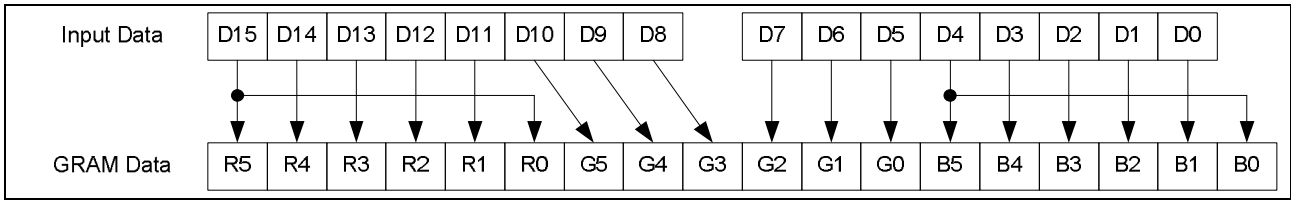


Figure 66. Bit Assignment of GRAM Data on SPI

Timing Diagrams

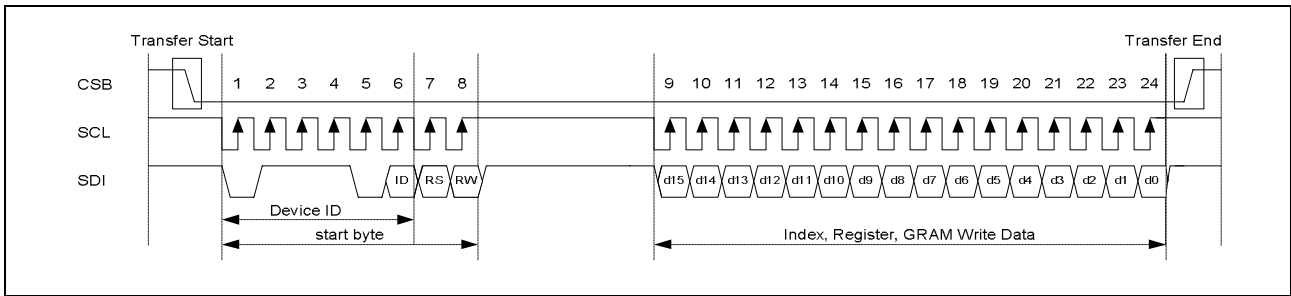


Figure 67. Basic Timing Diagram of Data Transfer through SPI

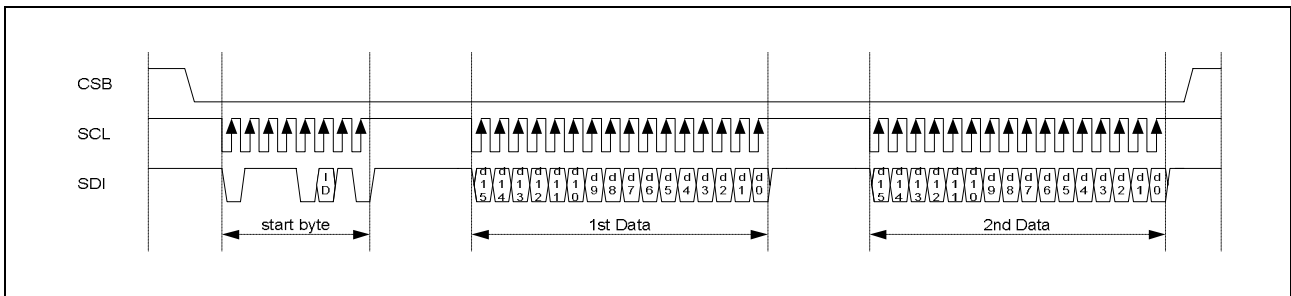


Figure 68. Timing Diagram of Consecutive Data-Write through SPI

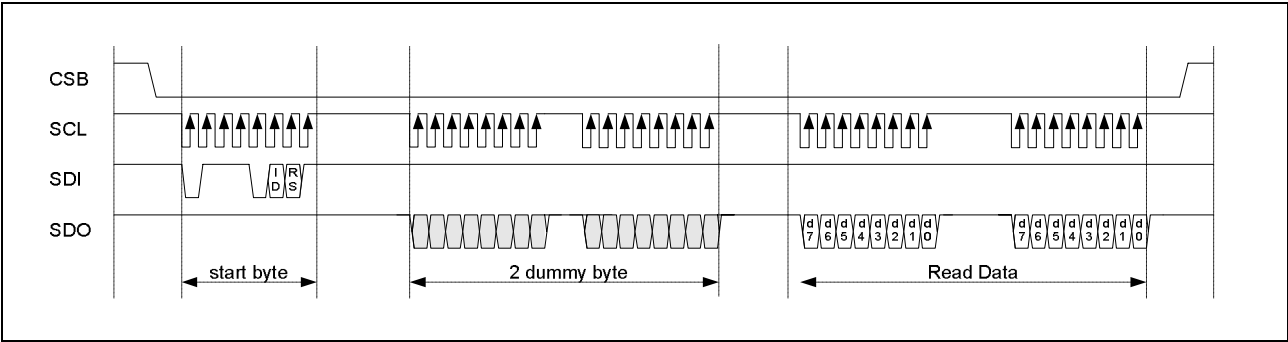


Figure 69. Timing Diagram of Register / Status Read through SPI

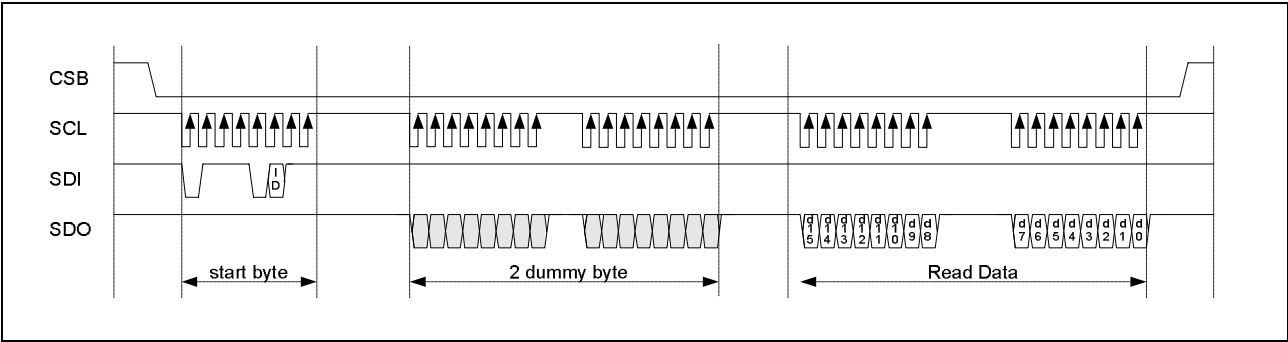


Figure 70. Timing Diagram of GRAM-Data Read through SPI

11.2. RGB INTERFACE

11.2.1. MOTION PICTURE DISPLAY

S6D0164 incorporates RGB interface to display motion pictures and GRAM to store data for display.

To display motion pictures, S6D0164 has the following features.

- Only motion picture area can be transferred by the Window Address function.
- Only motion picture area to be rewritten can be transferred selectively.
- Reducing the amount of data transferred enables reduce the power consumption of the whole system.
- Still picture area, such as an icon, can be updated while displaying motion pictures combining with the system interface (for details, refer to “GRAM ACCESS VIA RGB INTERFACE AND SPI” described later).

The RGB interface is performed in synchronization with VSYNC, HSYNC, and DOTCLK.

Window Address Function enables transfer only the screen to be updated and reduce the power consumption.

In the period between the completion of displaying one frame data and the next VSYNC signal, the display status will remain in front porch period.

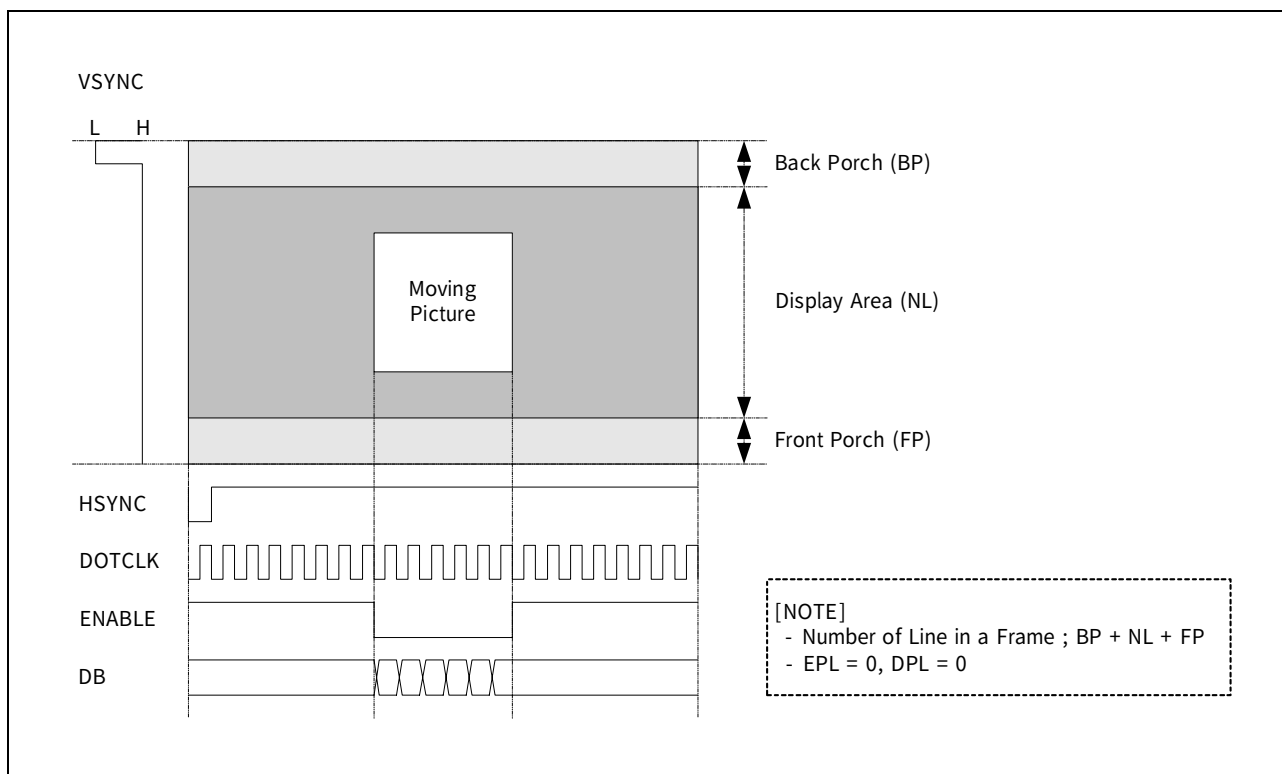


Figure 71. RGB Interface

Note. For RGB interface, VSYNC, HSYNC, DOTCLK should be supplied at much higher resolution than that of panel.

There are three timing conditions for RGB Interface that is determined according to RIM and each condition is described below.

11.2.2. 18BIT RGB INTERFACE

Bit Assignment

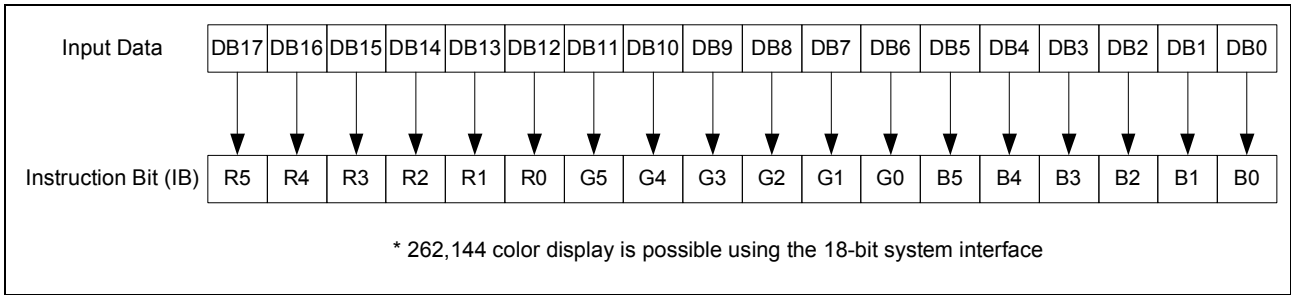


Figure 72. Bit Assignment of GRAM Data on 18bit RGB Interface

Timing Diagram

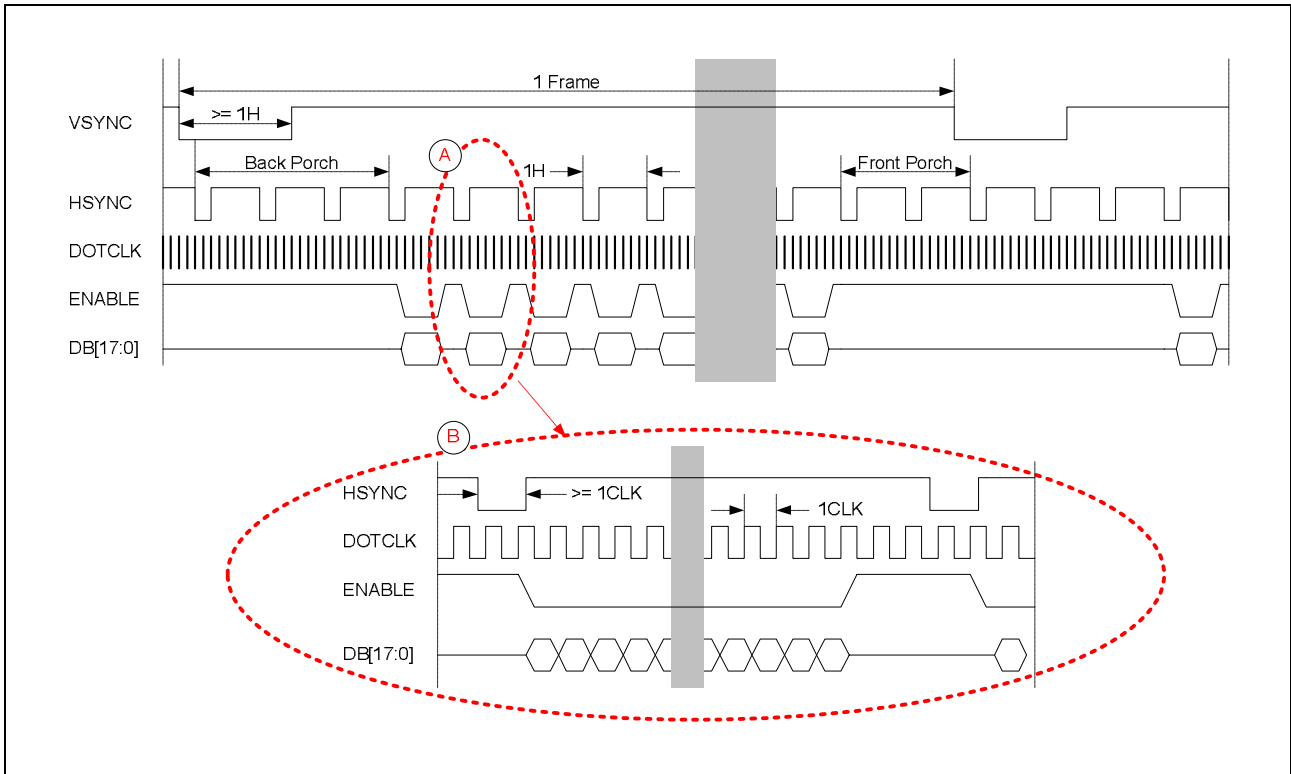


Figure 73. Timing Diagram of 18/16bit RGB Interface

11.2.3. 16BIT RGB INTERFACE

Bit Assignment

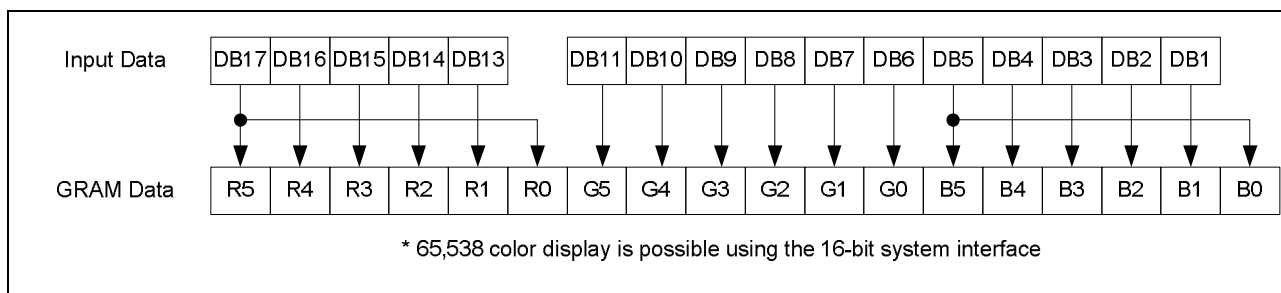


Figure 74. Bit Assignment of GRAM Data on 16bit RGB Interface

Timing Diagram

Time chart for RGB interface is shown below. (In case of EPL = 0)

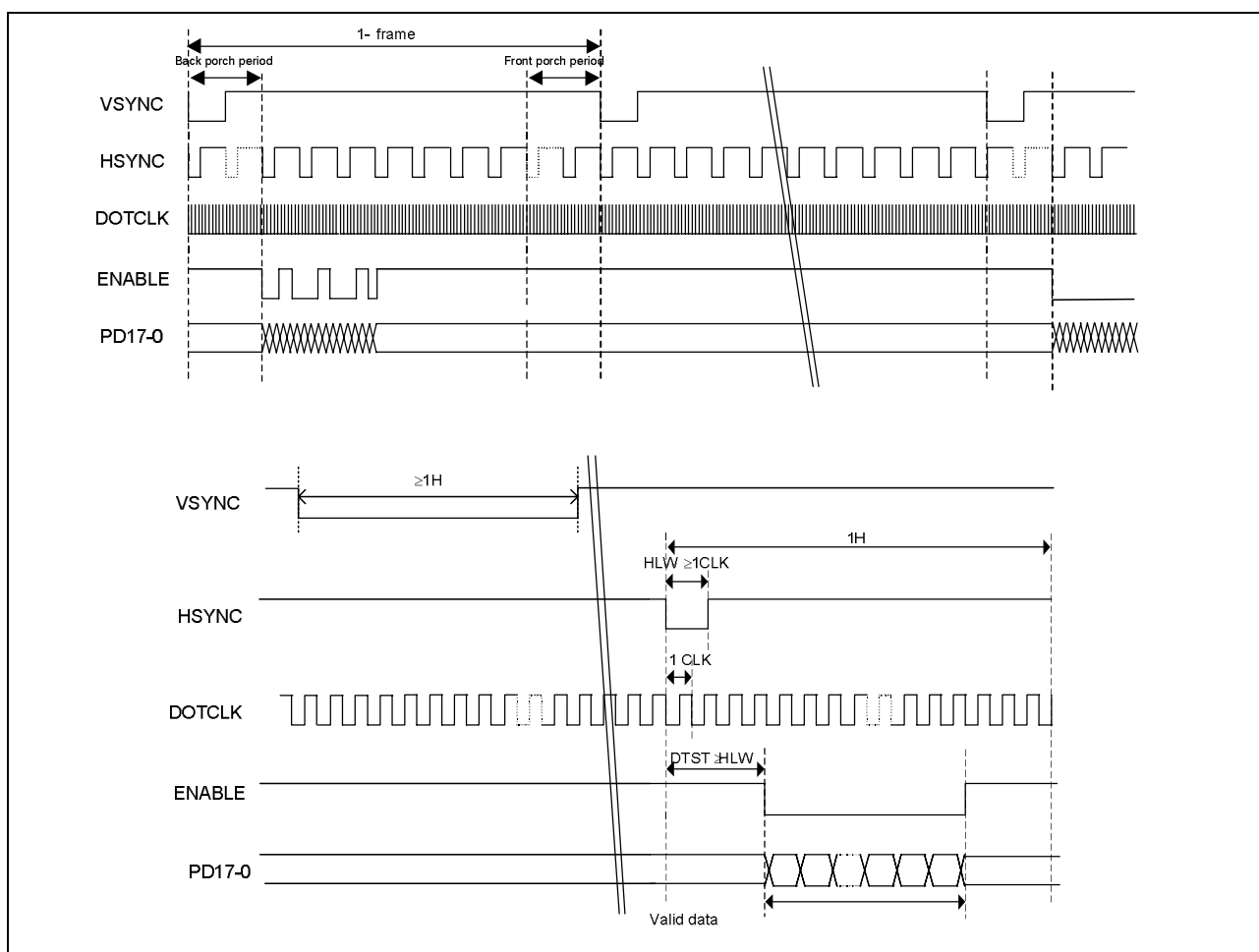


Figure 75. 16-/18-bit RGB Interface Timing (In case of EPL = 0, DPL = 0, VSPL = 0, HSPL = 0)

Note.

VLW: The period in which VSYNC is "Low" level

HLW: The period in which HSYNC is "Low" level

DTST: Set up time of data transfer

11.2.4. 6BIT RGB INTERFACE

In order to transfer data on 6bit RGB Interface there should be three transfers.

Bit Assignment

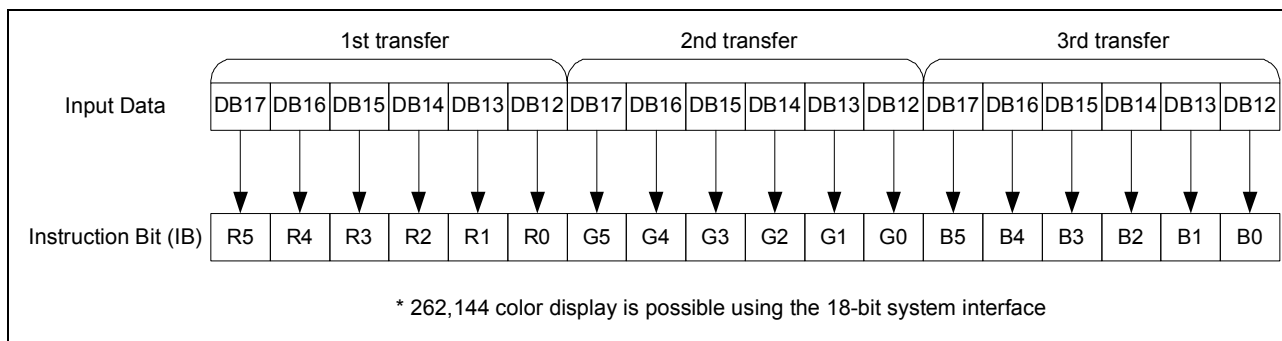


Figure 76. Bit Assignment of GRAM Data on 6bit RGB Interface

Timing Diagram

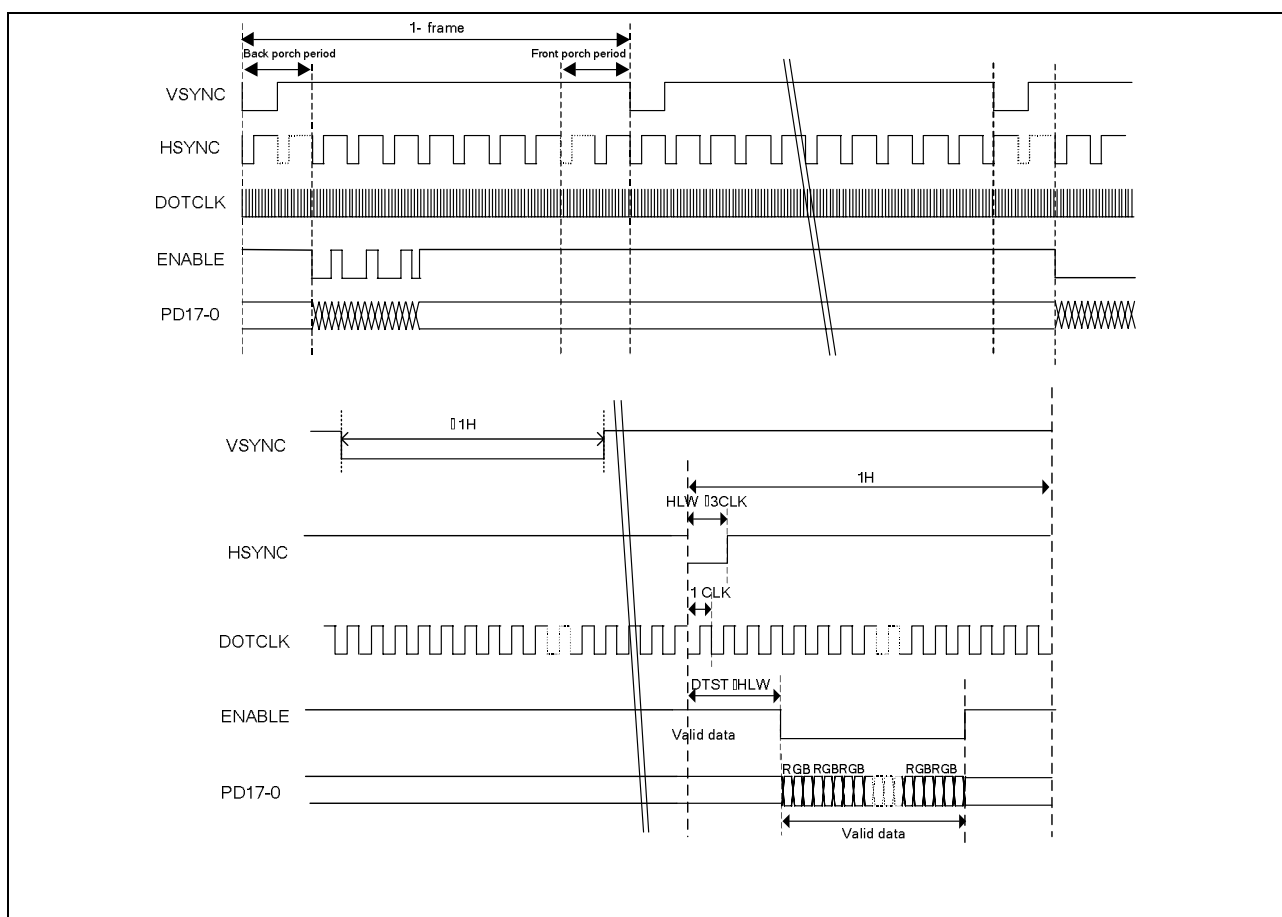


Figure 77. 6-bit RGB Interface Timing (In case of EPL = 0, DPL = 0, VSPL = 0, HSPL = 0)

Note.

Three clocks are regarded as one clock for transfer when data is transferred in 6-bit interface.

VSYNC, HSYNC, ENABLE, DOTCLK, and DB [17:12] should be transferred in units of three clocks.

Transfer Synchronization

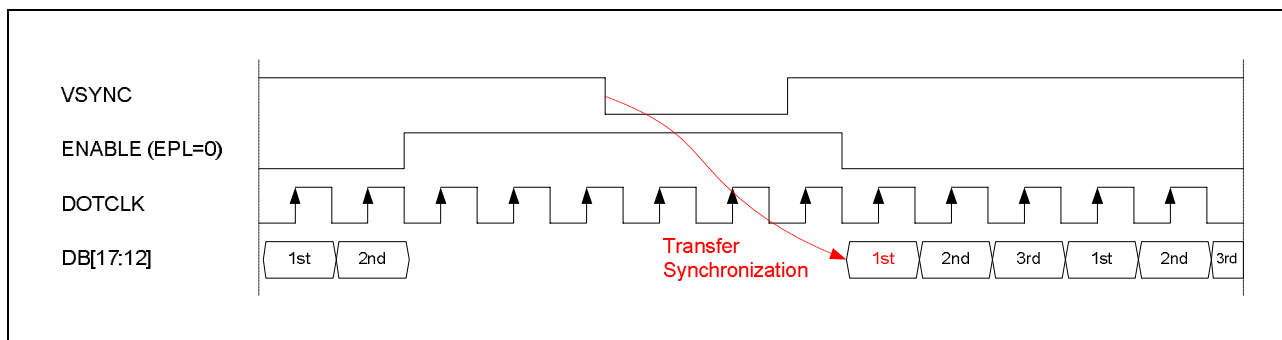


Figure 78. Transfer Synchronization Function in 6-bit RGB Interface mode

Note. The figure above shows Transfer Synchronization functions for 6bit RGB Interface. S6D0164 has a transfer counter internally to count 1st, 2nd and 3rd data transfer of 6bit RGB Interface. The transfer counter is reset on the falling edge of VSYNC and enters the 1st data transmission state. Transfer mismatch can be corrected at every VSYNC signal assertion. In this method, when data is consecutively transferred in for displaying motion pictures, the effect of transfer mismatch will be reduced and recovered by normal operation.

Note. The display is operated in units of three DOTCLKs. When DOTCLK is not input in units of pixels, clock mismatch occurs and the frame, which is operated, and the next frame are not displayed correctly.

11.3. INTERFACE SWAPPING FOR MEMORY ACCESS

11.3.1. DISPLAY MODES AND GRAM ACCESS CONTROL

Display mode and RAM Access is controlled as shown below. For each display status, display mode control and RAM Access control are combined properly.

Table 38. DISPLAY MODE & RAM ACCESS CONTROL

Display Status	GRAM Access (RM)	Display Mode (DM)
1. Still Picture Display	System Interface (RM = 0)	Internal Clock Operation (DM = 0)
2. Motion Picture Display	RGB Interface (RM = 1)	External Clock Operation (DM = 1)
3. Rewrite Still Picture while Motion Picture is being displayed	System Interface (RM = 0)	External Clock Operation (DM = 1)

Note.

Only system interface can set Instruction register.

When the RGB Interface is being operated, do not change the RGB Interface mode (RIM).

11.3.2. Internal Clock Operation mode with System Interface (1)

Every operation in Internal Clock Operation mode is done in synchronization with the internal clock which is generated by internal OSC. The signals input through RGB interface are all meaningless. Access to internal GRAM is done via system interface.

11.3.3. External Clock Operation mode with RGB Interface (2)

In External Clock Operation mode, frame sync signal (VSYNC), line sync signal (HSYNC) and DOTCLK are used for display operation. Display data is transferred in the unit of pixel through DB bus and saved to GRAM.

11.3.4. External Clock Operation mode with System Interface (3)

Write GRAM data via system interface even in External Clock Operation mode. There should not be any data transmission on RGB interface in this case. To restart data transmission on RGB interface, set RM to "1", set memory address properly and write index of 22h for GRAM write operation.

With the combination of Window Address function, motion picture and still picture may be saved in separated GRAM regions respectively. In this case motion picture and still picture are displayed simultaneously.

11.3.5. GRAM ACCESS VIA RGB INTERFACE AND SPI

All the data for display is written to the internal GRAM in S6D0164 when RGB interface is in use. In this method, data, including motion picture and screen update frame, can only be transferred via RGB interface.

With Window Address function, power consumption can be reduced and high-speed access can be achieved while motion pictures are being displayed. Data for display that is not in the motion picture area or the screen update frame can be written via System Interface.

GRAM can be accessed via SPI even when RGB interface is in use. To do that ENABLE should be inactive state to stop data writing via RGB interface, because the write operation to GRAM is always performed in synchronization with DOTCLK while ENABLE is active state. Then you may write any data through SPI. After this access to GRAM via SPI, a waiting time is needed for a write/read bus cycle before the next RAM access starts via RGB interface. When a RAM write conflict occurs, data writing is not guaranteed.

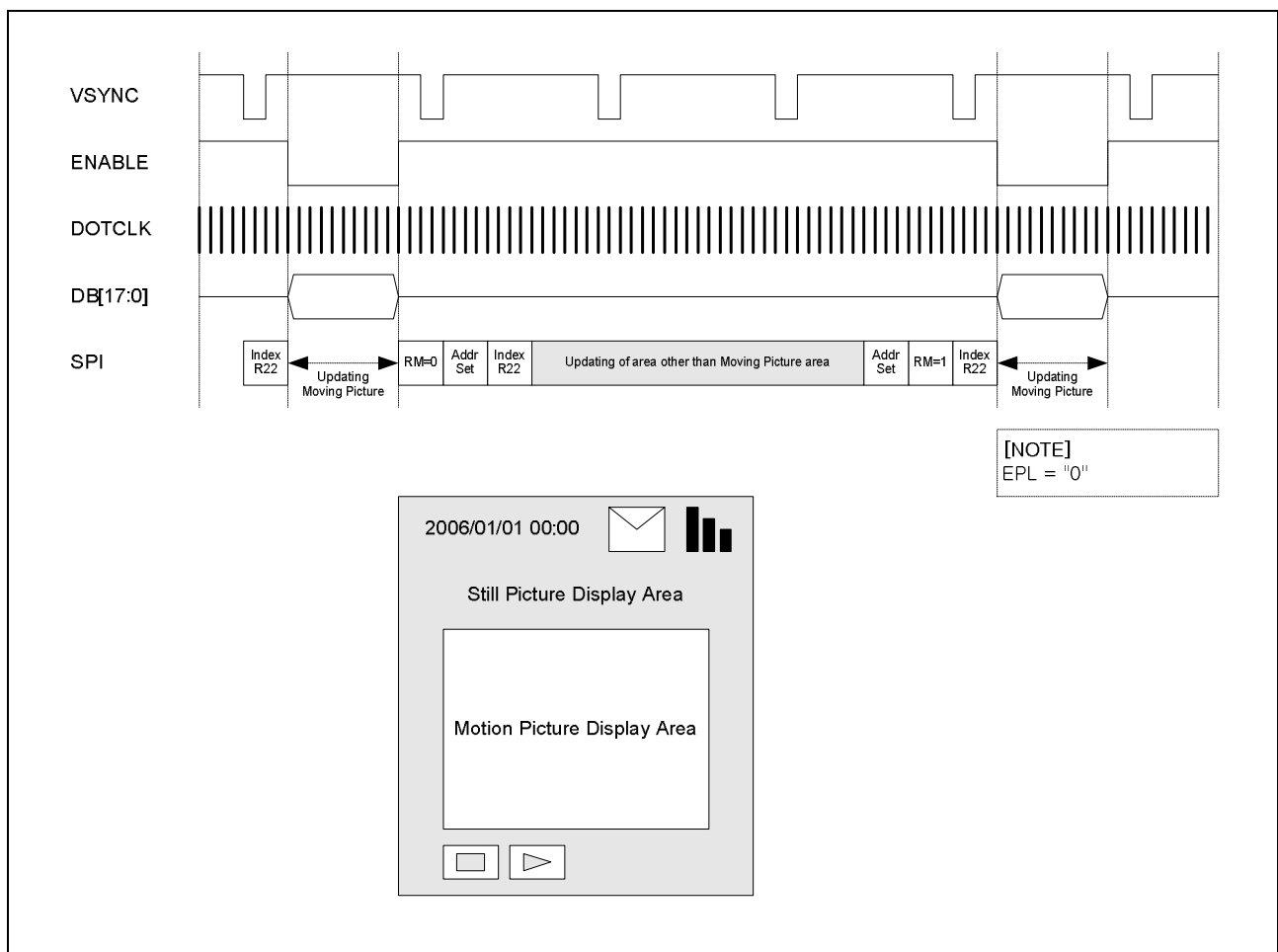


Figure 79. GRAM Access through RGB Interface and SPI

11.3.6. TRANSITION SEQUENCES BETWEEN DISPLAY MODES

Transitions between Internal Clock Operation mode and External Clock Operation mode should follow the mode transition sequence shown below.

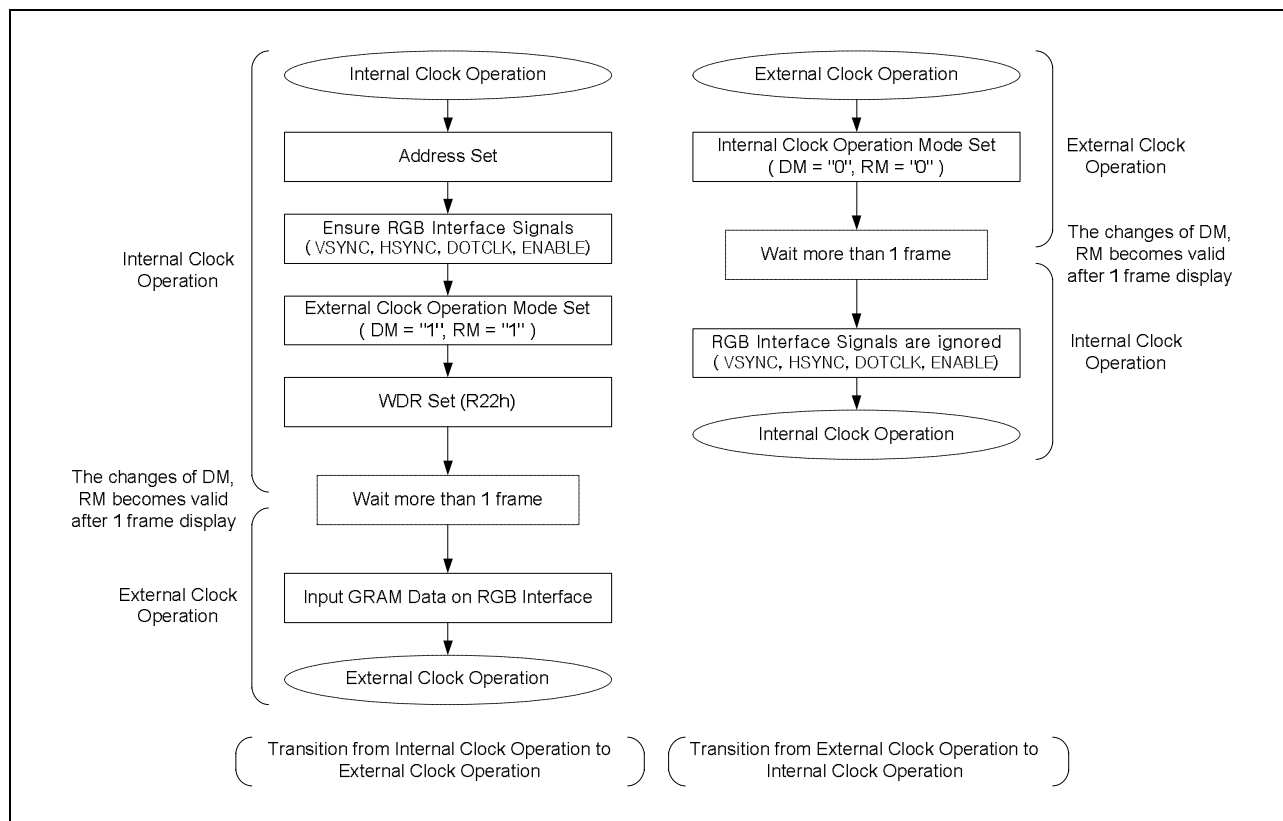


Figure 80. Transition between Internal Clock Operation Mode and External Clock Operation Mode

12. GAMMA ADJUSTMENT FUNCTION

S6D0164 provides the gamma adjustment function to display 262,144 colors simultaneously. The gamma adjustment is executed by the gradient adjustment register, the reference adjustment register, the amplitude adjustment register and the micro-adjustment register that determine the 8 grayscale levels. Furthermore, since these registers have the positive polarities and negative polarities, adjust them to match LCD panel respectively.

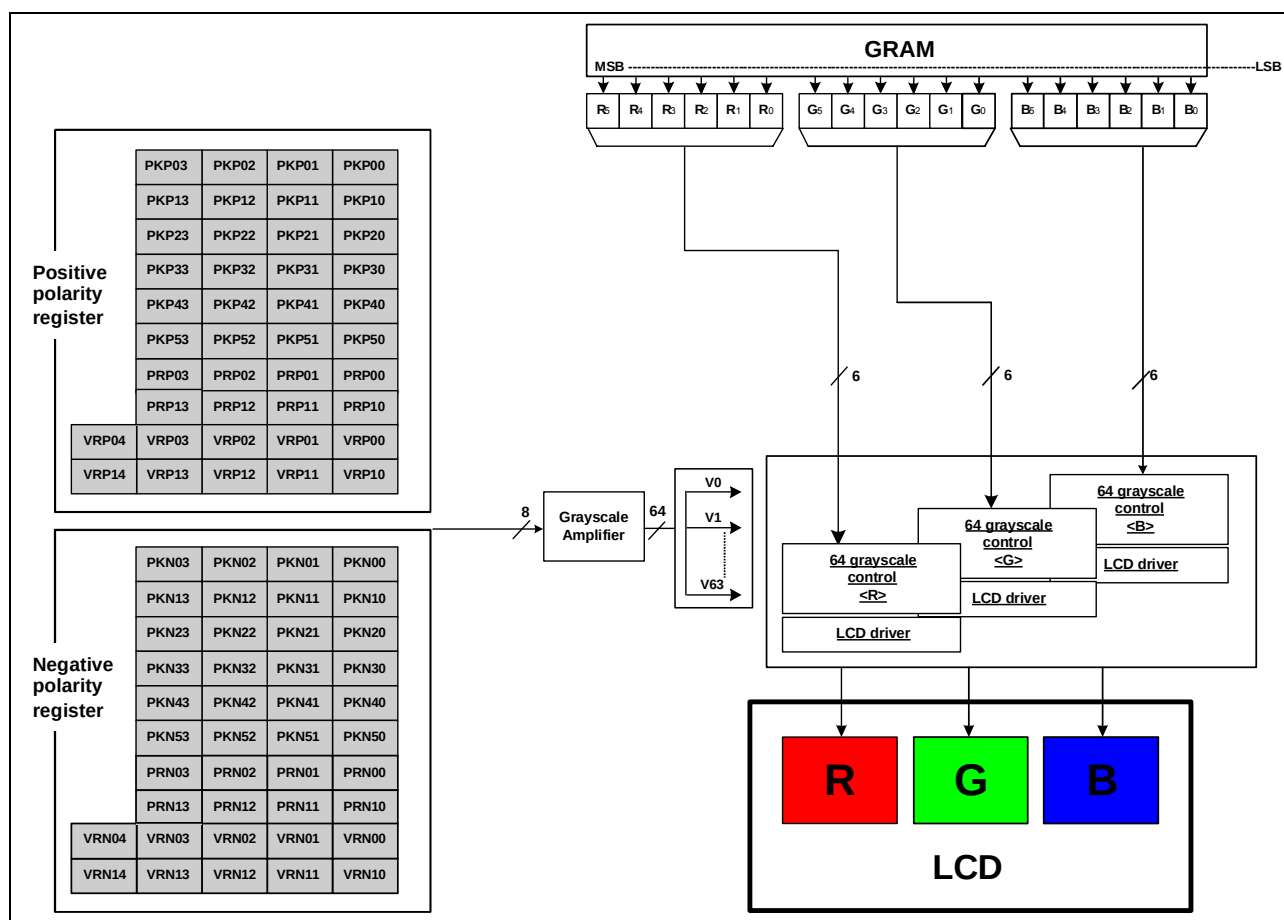


Figure 81. Grayscale Control

12.1. Structure of grayscale amplifier

The structure of grayscale amplifier is shown as below. The 8 voltage levels (VIN0-VIN7) between GVDD and VGS are determined by the gradient adjustment register, the reference adjustment register, the amplitude adjustment register and the micro-adjustment register. Each level is split into 8 levels again by the internal ladder resistor network. As a result, grayscale amplifier generates 64 voltage levels ranging from V0 to V63.

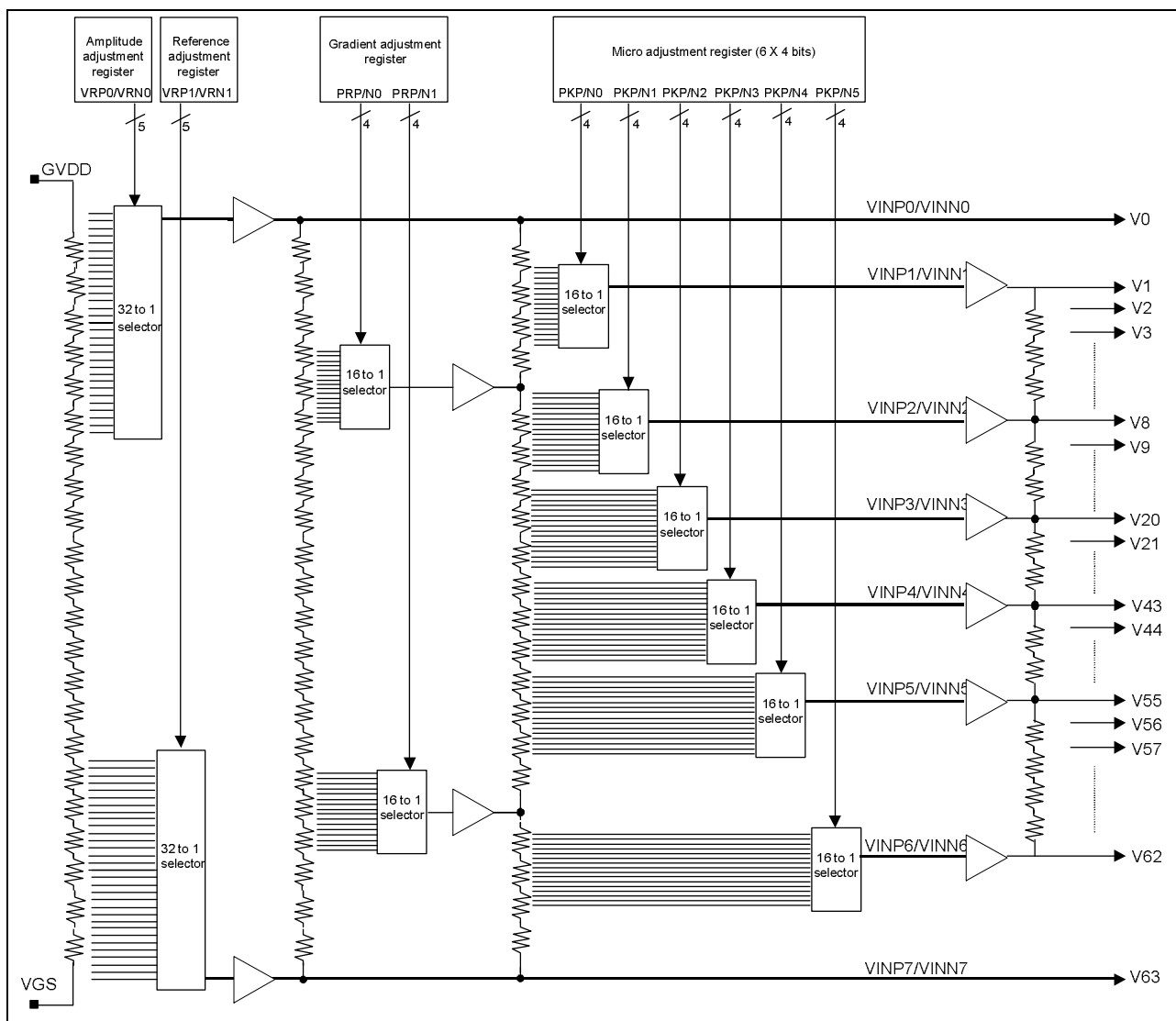


Figure 82. Structure of Grayscale Amplifier

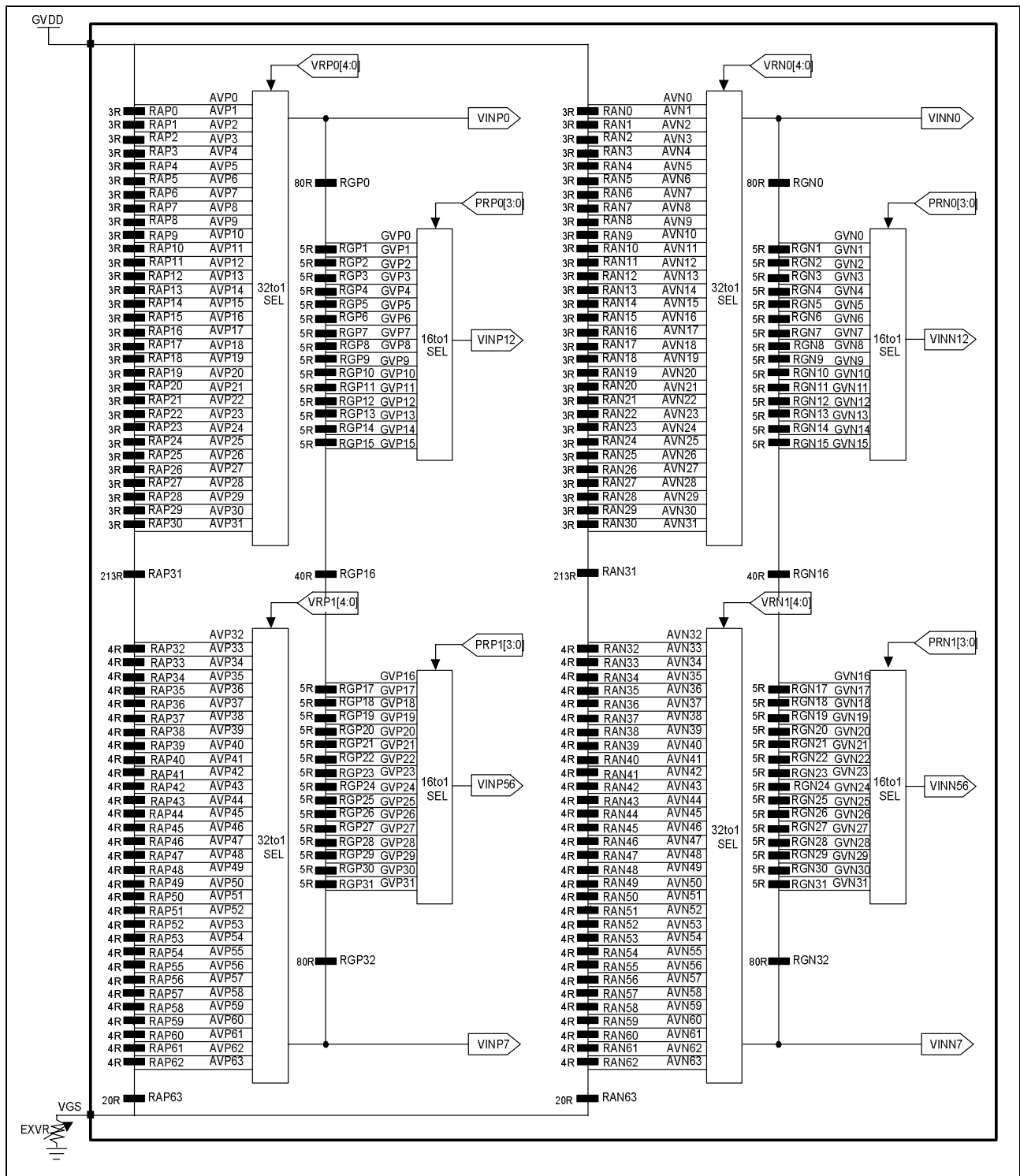


Figure 83. Structure of Resistor Ladder Network 1

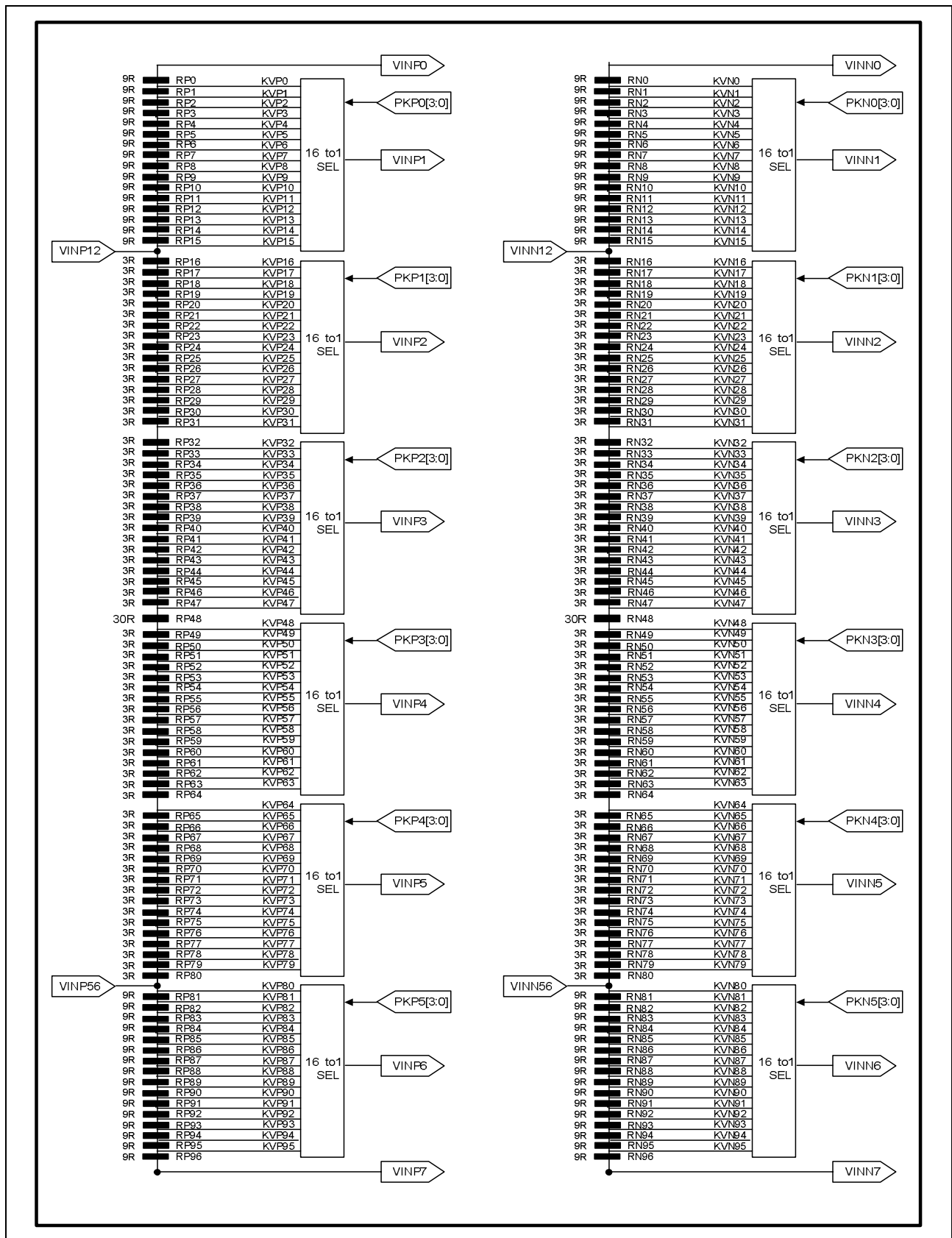


Figure 84. Structure of Resistor Ladder Network 2

12.2. Gamma adjustment register

This block has registers to set up the grayscale voltage adjusting to the gamma specification of the LCD panel. These registers can independently set up to positive/negative polarities and there are 4 types of register groups to adjust gradient and amplitude on number of the grayscale, characteristics of the grayscale voltage. The following figure indicates the operation of each adjustment register.

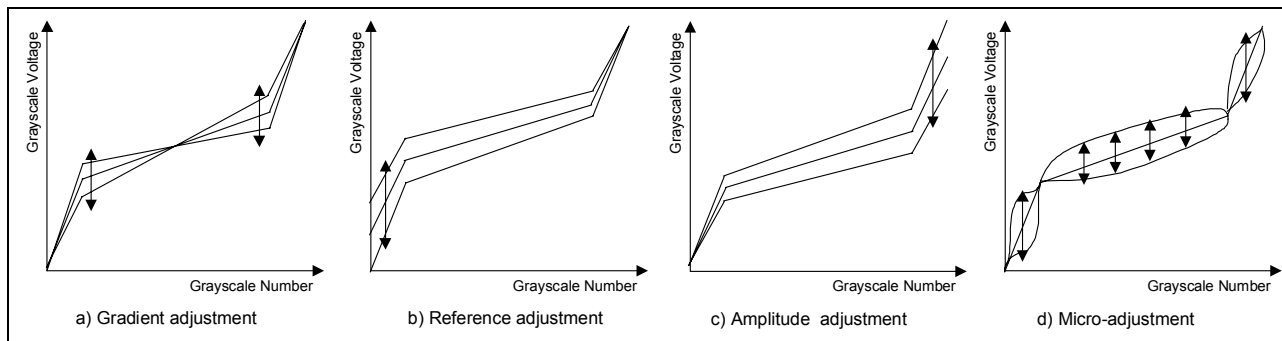


Figure 85. The Operation of Adjusting Register

Gradient adjusting register

The gradient adjustment register is to adjust the gradient in the middle of the grayscale characteristics for the voltage without changing the dynamic range. To accomplish the adjustment, it controls the VINP12/VINN12 and VINP56/VINN56 voltage level by the 16 to 1 selector towards the 16-leveled reference voltage generated from the resistor ladder between VINP0/VINN0 and VINP7/VINN7. Also, there is an independent register on the positive/negative polarities in order for corresponding to asymmetry drive.

Reference adjusting register

The Reference adjustment register is to adjust the reference of the grayscale voltage. To accomplish the adjustment, it controls the VINP7/VINN7 voltage level by 32 to 1 selector towards the 32-leveled reference voltage generated from the resistor ladder between GVDD and VGS.

Amplitude adjusting register

The Amplitude adjustment register is to adjust the amplitude of the grayscale voltage. To accomplish the adjustment, it controls the VINP0/VINN0 voltage level by 32 to 1 selector towards the 32-leveled reference voltage generated from the resistor ladder between GVDD and VGS.

Micro-adjusting register

The Micro adjustment register is to make subtle adjustment of the grayscale voltage level. To accomplish the adjustment, it controls the each reference voltage level by the 16 to 1 selector towards the 16-leveled reference voltage generated from the resistor ladder. Also, there is an independent register on the positive/negative polarities as well as other adjustment registers.

Table 39. Gamma Adjustment Register

Register	Positive polarity	Negative polarity	Set-up contents
Gradient adjustment	PRP0[3:0]	PRN0[3:0]	The voltage of VINP12/VINN12 is selected by the 16 to 1 selector
	PRP1[3:0]	PRN1[3:0]	The voltage of VINP56/VINN56 is selected by the 16 to 1 selector
Reference adjustment	VRP1[4:0]	VRN1[4:0]	The voltage of VINP7/VINN7 is selected by the 32 to 1 selector
Amplitude adjustment	VRP0[4:0]	VRN0[4:0]	The voltage of VINP0/VINN0 is selected by the 32 to 1 selector
Micro adjustment	PKP0[3:0]	PKN0[3:0]	The voltage of grayscale number 1 is selected by the 16 to 1 selector
	PKP1[3:0]	PKN1[3:0]	The voltage of grayscale number 8 is selected by the 16 to 1 selector
	PKP2[3:0]	PKN2[3:0]	The voltage of grayscale number 20 is selected by the 16 to 1 selector
	PKP3[3:0]	PKN3[3:0]	The voltage of grayscale number 43 is selected by the 16 to 1 selector
	PKP4[3:0]	PKN4[3:0]	The voltage of grayscale number 55 is selected by the 16 to 1 selector
	PKP5[3:0]	PKN5[3:0]	The voltage of grayscale number 62 is selected by the 16 to 1 selector

12.3. Resistor ladder network / selector

This block outputs the reference voltage of the grayscale voltage. There are four ladder resistors including the 8 to 1 selector selecting voltage generated by the ladder resistance voltage. Also, there are pins that connect to the external volume resistor. In addition, it allows compensating the dispersion of length between one panel and another.

Resistor ladder network 1 / selector

There are 4 adjustments that are for the gradient adjustment (VRHP (N) / VRLP (N)) and for the reference / amplitude adjustment (VRP (N) 1 / VRP (N) 0). The voltage level is set by the gradient adjustment register and the reference / amplitude adjustment registers as below.

Table 40. Amplitude Adjustment

Register value VRP(N)0 [4:0]	Selected voltage VINP(N)0	Formula of VINP(N)0
00000	AVP(N)0	$(450R/450R) * (GVDD-VGS) + VGS$
00001	AVP(N)1	$(447R/450R) * (GVDD-VGS) + VGS$
00010	AVP(N)2	$(444R/450R) * (GVDD-VGS) + VGS$
00011	AVP(N)3	$(441R/450R) * (GVDD-VGS) + VGS$
00100	AVP(N)4	$(438R/450R) * (GVDD-VGS) + VGS$
00101	AVP(N)5	$(435R/450R) * (GVDD-VGS) + VGS$
00110	AVP(N)6	$(432R/450R) * (GVDD-VGS) + VGS$
00111	AVP(N)7	$(429R/450R) * (GVDD-VGS) + VGS$
01000	AVP(N)8	$(426R/450R) * (GVDD-VGS) + VGS$
01001	AVP(N)9	$(423R/450R) * (GVDD-VGS) + VGS$
01010	AVP(N)10	$(420R/450R) * (GVDD-VGS) + VGS$
01011	AVP(N)11	$(417R/450R) * (GVDD-VGS) + VGS$
01100	AVP(N)12	$(414R/450R) * (GVDD-VGS) + VGS$
01101	AVP(N)13	$(411R/450R) * (GVDD-VGS) + VGS$
01110	AVP(N)14	$(408R/450R) * (GVDD-VGS) + VGS$
01111	AVP(N)15	$(405R/450R) * (GVDD-VGS) + VGS$
10000	AVP(N)16	$(402R/450R) * (GVDD-VGS) + VGS$
10001	AVP(N)17	$(399R/450R) * (GVDD-VGS) + VGS$
10010	AVP(N)18	$(396R/450R) * (GVDD-VGS) + VGS$
10011	AVP(N)19	$(393R/450R) * (GVDD-VGS) + VGS$
10100	AVP(N)20	$(390R/450R) * (GVDD-VGS) + VGS$
10101	AVP(N)21	$(387R/450R) * (GVDD-VGS) + VGS$
10110	AVP(N)22	$(384R/450R) * (GVDD-VGS) + VGS$
10111	AVP(N)23	$(381R/450R) * (GVDD-VGS) + VGS$
11000	AVP(N)24	$(378R/450R) * (GVDD-VGS) + VGS$
11001	AVP(N)25	$(375R/450R) * (GVDD-VGS) + VGS$
11010	AVP(N)26	$(372R/450R) * (GVDD-VGS) + VGS$
11011	AVP(N)27	$(369R/450R) * (GVDD-VGS) + VGS$
11100	AVP(N)28	$(366R/450R) * (GVDD-VGS) + VGS$
11101	AVP(N)29	$(363R/450R) * (GVDD-VGS) + VGS$
11110	AVP(N)30	$(360R/450R) * (GVDD-VGS) + VGS$
11111	AVP(N)31	$(357R/450R) * (GVDD-VGS) + VGS$

Table 41. Reference Adjustment

Register value VRP(N)1 [4:0]	Selected voltage VINP(N)7	Formula of VINP(N)7
00000	AVP(N)63	$(20R/450R) * (GVDD-VGS) + VGS$
00001	AVP(N)62	$(24R/450R) * (GVDD-VGS) + VGS$
00010	AVP(N)61	$(28R/450R) * (GVDD-VGS) + VGS$
00011	AVP(N)60	$(32R/450R) * (GVDD-VGS) + VGS$
00100	AVP(N)59	$(36R/450R) * (GVDD-VGS) + VGS$
00101	AVP(N)58	$(40R/450R) * (GVDD-VGS) + VGS$
00110	AVP(N)57	$(44R/450R) * (GVDD-VGS) + VGS$
00111	AVP(N)56	$(48R/450R) * (GVDD-VGS) + VGS$
01000	AVP(N)55	$(52R/450R) * (GVDD-VGS) + VGS$
01001	AVP(N)54	$(56R/450R) * (GVDD-VGS) + VGS$
01010	AVP(N)53	$(60R/450R) * (GVDD-VGS) + VGS$
01011	AVP(N)52	$(64R/450R) * (GVDD-VGS) + VGS$
01100	AVP(N)51	$(68R/450R) * (GVDD-VGS) + VGS$
01101	AVP(N)50	$(72R/450R) * (GVDD-VGS) + VGS$
01110	AVP(N)49	$(76R/450R) * (GVDD-VGS) + VGS$
01111	AVP(N)48	$(80R/450R) * (GVDD-VGS) + VGS$
10000	AVP(N)47	$(84R/450R) * (GVDD-VGS) + VGS$
10001	AVP(N)46	$(88R/450R) * (GVDD-VGS) + VGS$
10010	AVP(N)45	$(92R/450R) * (GVDD-VGS) + VGS$
10011	AVP(N)44	$(96R/450R) * (GVDD-VGS) + VGS$
10100	AVP(N)43	$(100R/450R) * (GVDD-VGS) + VGS$
10101	AVP(N)42	$(104R/450R) * (GVDD-VGS) + VGS$
10110	AVP(N)41	$(108R/450R) * (GVDD-VGS) + VGS$
10111	AVP(N)40	$(112R/450R) * (GVDD-VGS) + VGS$
11000	AVP(N)39	$(116R/450R) * (GVDD-VGS) + VGS$
11001	AVP(N)38	$(120R/450R) * (GVDD-VGS) + VGS$
11010	AVP(N)37	$(124R/450R) * (GVDD-VGS) + VGS$
11011	AVP(N)36	$(128R/450R) * (GVDD-VGS) + VGS$
11100	AVP(N)35	$(132R/450R) * (GVDD-VGS) + VGS$
11101	AVP(N)34	$(136R/450R) * (GVDD-VGS) + VGS$
11110	AVP(N)33	$(140R/450R) * (GVDD-VGS) + VGS$
11111	AVP(N)32	$(144R/450R) * (GVDD-VGS) + VGS$

Table 42. Gradient Adjustment (1)

Register value PRP(N)0[3:0]	Selected voltage VINP(N)12	Formula of VINP(N)12
0000	GVP(N)0	$(270R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0001	GVP(N)1	$(265R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0010	GVP(N)2	$(260R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0011	GVP(N)3	$(255R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0100	GVP(N)4	$(250R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0101	GVP(N)5	$(245R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0110	GVP(N)6	$(240R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0111	GVP(N)7	$(235R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1000	GVP(N)8	$(230R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1001	GVP(N)9	$(225R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1010	GVP(N)10	$(220R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1011	GVP(N)11	$(215R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1100	GVP(N)12	$(210R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1101	GVP(N)13	$(205R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1110	GVP(N)14	$(200R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1111	GVP(N)15	$(195R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$

Table 43. Gradient Adjustment (2)

Register value PRP(N)1 [3:0]	Selected voltage VINP(N)56	Formula of VINP(N)56
0000	GVP(N)31	$(80R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0001	GVP(N)30	$(85R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0010	GVP(N)29	$(90R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0011	GVP(N)28	$(95R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0100	GVP(N)27	$(100R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0101	GVP(N)26	$(105R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0110	GVP(N)25	$(110R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
0111	GVP(N)24	$(115R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1000	GVP(N)23	$(120R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1001	GVP(N)22	$(125R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1010	GVP(N)21	$(130R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1011	GVP(N)20	$(135R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1100	GVP(N)19	$(140R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1101	GVP(N)18	$(145R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1110	GVP(N)17	$(150R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$
1111	GVP(N)16	$(155R/350R) * (VINP(N)0 - VINP(N)7) + VINP(N)7$

Resistor ladder network 2 / selector

In the 16-to-1 selector, the voltage level must be selected by the given ladder resistance and the micro-adjustment register and output the six types of the reference voltage, VIN1 to VIN6.

Following figure explains the relationship between the micro-adjustment register and the selected voltage.

Table 44. Relationship between Micro-adjustment Register and Selected Voltage

Register value PKP(N) [3:0]	Selected voltage					
	VINP(N)1	VINP(N)2	VINP(N)3	VINP(N)4	VINP(N)5	VINP(N)6
0000	KVP(N)0	KVP(N)16	KVP(N)32	KVP(N)63	KVP(N)79	KVP(N)95
0001	KVP(N)1	KVP(N)17	KVP(N)33	KVP(N)62	KVP(N)78	KVP(N)94
0010	KVP(N)2	KVP(N)18	KVP(N)34	KVP(N)61	KVP(N)77	KVP(N)93
0011	KVP(N)3	KVP(N)19	KVP(N)35	KVP(N)60	KVP(N)76	KVP(N)92
0100	KVP(N)4	KVP(N)20	KVP(N)36	KVP(N)59	KVP(N)75	KVP(N)91
0101	KVP(N)5	KVP(N)21	KVP(N)37	KVP(N)58	KVP(N)74	KVP(N)90
0110	KVP(N)6	KVP(N)22	KVP(N)38	KVP(N)57	KVP(N)73	KVP(N)89
0111	KVP(N)7	KVP(N)23	KVP(N)39	KVP(N)56	KVP(N)72	KVP(N)88
1000	KVP(N)8	KVP(N)24	KVP(N)40	KVP(N)55	KVP(N)71	KVP(N)87
1001	KVP(N)9	KVP(N)25	KVP(N)41	KVP(N)54	KVP(N)70	KVP(N)86
1010	KVP(N)10	KVP(N)26	KVP(N)42	KVP(N)53	KVP(N)69	KVP(N)85
1011	KVP(N)11	KVP(N)27	KVP(N)43	KVP(N)52	KVP(N)68	KVP(N)84
1100	KVP(N)12	KVP(N)28	KVP(N)44	KVP(N)51	KVP(N)67	KVP(N)83
1101	KVP(N)13	KVP(N)29	KVP(N)45	KVP(N)50	KVP(N)66	KVP(N)82
1110	KVP(N)14	KVP(N)30	KVP(N)46	KVP(N)49	KVP(N)65	KVP(N)81
1111	KVP(N)15	KVP(N)31	KVP(N)47	KVP(N)48	KVP(N)64	KVP(N)80

Note. The grayscale levels are determined by the following formulas listed in the next pages.

Table 45. Formulas for Calculating Gamma Adjusting Voltage (Positive Polarity) 1

Pins	Formula	Micro-adjusting register value	Reference voltage
KVP0	$(45R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "0000"	VINP1
KVP1	$(42R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "0001"	
KVP2	$(39R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "0010"	
KVP3	$(36R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "0011"	
KVP4	$(33R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "0100"	
KVP5	$(30R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "0101"	
KVP6	$(27R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "0110"	
KVP7	$(24R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "0111"	
KVP8	$(21R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "1000"	
KVP9	$(18R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "1001"	
KVP10	$(15R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "1010"	
KVP11	$(12R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "1011"	
KVP12	$(9R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "1100"	
KVP13	$(6R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "1101"	
KVP14	$(3R/48R) * (VINP0 - VINP12) + VINP12$	PKP0[3:0] = "1110"	
KVP15	VINP12	PKP0[3:0] = "1111"	VINP2
KVP16	$(219R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "0000"	
KVP17	$(216R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "0001"	
KVP18	$(213R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "0010"	
KVP19	$(210R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "0011"	
KVP20	$(207R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "0100"	
KVP21	$(204R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "0101"	
KVP22	$(201R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "0110"	
KVP23	$(198R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "0111"	
KVP24	$(195R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "1000"	
KVP25	$(192R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "1001"	
KVP26	$(189R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "1010"	
KVP27	$(186R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "1011"	
KVP28	$(183R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "1100"	
KVP29	$(180R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "1101"	
KVP30	$(177R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "1110"	
KVP31	$(174R/222R) * (VINP12 - VINP56) + VINP56$	PKP1[3:0] = "1111"	VINP3
KVP32	$(171R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "0000"	
KVP33	$(168R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "0001"	
KVP34	$(165R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "0010"	
KVP35	$(162R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "0011"	
KVP36	$(159R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "0100"	
KVP37	$(156R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "0101"	
KVP38	$(153R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "0110"	
KVP39	$(150R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "0111"	
KVP40	$(147R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "1000"	
KVP41	$(144R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "1001"	
KVP42	$(141R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "1010"	
KVP43	$(138R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "1011"	
KVP44	$(135R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "1100"	
KVP45	$(132R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "1101"	
KVP46	$(129R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "1110"	
KVP47	$(126R/222R) * (VINP12 - VINP56) + VINP56$	PKP2[3:0] = "1111"	

Pins	Formula	Micro-adjusting register value	Reference voltage
KVP48	$(96R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "1111"	VINP4
KVP49	$(93R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "1110"	
KVP50	$(90R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "1101"	
KVP51	$(87R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "1100"	
KVP52	$(84R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "1011"	
KVP53	$(81R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "1010"	
KVP54	$(78R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "1001"	
KVP55	$(75R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "1000"	
KVP56	$(72R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "0111"	
KVP57	$(69R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "0110"	
KVP58	$(66R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "0101"	
KVP59	$(63R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "0100"	
KVP60	$(60R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "0011"	
KVP61	$(57R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "0010"	
KVP62	$(54R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "0001"	
KVP63	$(51R/222R) * (VINP12 - VINP56) + VINP56$	PKP3[3:0] = "0000"	
KVP64	$(48R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "1111"	VINP5
KVP65	$(45R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "1110"	
KVP66	$(42R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "1101"	
KVP67	$(39R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "1100"	
KVP68	$(36R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "1011"	
KVP69	$(33R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "1010"	
KVP70	$(30R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "1001"	
KVP71	$(27R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "1000"	
KVP72	$(24R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "0111"	
KVP73	$(21R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "0110"	
KVP74	$(18R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "0101"	
KVP75	$(15R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "0100"	
KVP76	$(12R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "0011"	
KVP77	$(9R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "0010"	
KVP78	$(6R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "0001"	
KVP79	$(3R/222R) * (VINP12 - VINP56) + VINP56$	PKP4[3:0] = "0000"	
KVP80	VINP56	PKP5[3:0] = "1111"	VINP6
KVP81	$(45R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "1110"	
KVP82	$(42R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "1101"	
KVP83	$(39R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "1100"	
KVP84	$(36R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "1011"	
KVP85	$(33R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "1010"	
KVP86	$(30R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "1001"	
KVP87	$(27R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "1000"	
KVP88	$(24R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "0111"	
KVP89	$(21R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "0110"	
KVP90	$(18R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "0101"	
KVP91	$(15R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "0100"	
KVP92	$(12R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "0011"	
KVP93	$(9R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "0010"	
KVP94	$(6R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "0001"	
KVP95	$(3R/48R) * (VINP56 - VINP7) + VINP7$	PKP5[3:0] = "0000"	

Table 46. Formulas for Calculating Gamma Adjusting Voltage (Positive Polarity) 2

Grayscale voltage	Formula	Grayscale voltage	Formula
V0	VINP0	V32	$V20 - (V20 - V43) * (12/23)$
V1	VINP1	V33	$V20 - (V20 - V43) * (13/23)$
V2	$V1 - (V1 - V8) * (28/96)$	V34	$V20 - (V20 - V43) * (14/23)$
V3	$V1 - (V1 - V8) * (42/96)$	V35	$V20 - (V20 - V43) * (15/23)$
V4	$V1 - (V1 - V8) * (60/96)$	V36	$V20 - (V20 - V43) * (16/23)$
V5	$V1 - (V1 - V8) * (69/96)$	V37	$V20 - (V20 - V43) * (17/23)$
V6	$V1 - (V1 - V8) * (78/96)$	V38	$V20 - (V20 - V43) * (18/23)$
V7	$V1 - (V1 - V8) * (87/96)$	V39	$V20 - (V20 - V43) * (19/23)$
V8	VINP2	V40	$V20 - (V20 - V43) * (20/23)$
V9	$V8 - (V8 - V20) * (2/24)$	V41	$V20 - (V20 - V43) * (21/23)$
V10	$V8 - (V8 - V20) * (4/24)$	V42	$V20 - (V20 - V43) * (22/23)$
V11	$V8 - (V8 - V20) * (6/24)$	V43	VINP4
V12	$V8 - (V8 - V20) * (8/24)$	V44	$V43 - (V43 - V55) * (2/24)$
V13	$V8 - (V8 - V20) * (10/24)$	V45	$V43 - (V43 - V55) * (4/24)$
V14	$V8 - (V8 - V20) * (12/24)$	V46	$V43 - (V43 - V55) * (6/24)$
V15	$V8 - (V8 - V20) * (14/24)$	V47	$V43 - (V43 - V55) * (8/24)$
V16	$V8 - (V8 - V20) * (16/24)$	V48	$V43 - (V43 - V55) * (10/24)$
V17	$V8 - (V8 - V20) * (18/24)$	V49	$V43 - (V43 - V55) * (12/24)$
V18	$V8 - (V8 - V20) * (20/24)$	V50	$V43 - (V43 - V55) * (14/24)$
V19	$V8 - (V8 - V20) * (22/24)$	V51	$V43 - (V43 - V55) * (16/24)$
V20	VINP3	V52	$V43 - (V43 - V55) * (18/24)$
V21	$V20 - (V20 - V43) * (1/23)$	V53	$V43 - (V43 - V55) * (20/24)$
V22	$V20 - (V20 - V43) * (2/23)$	V54	$V43 - (V43 - V55) * (22/24)$
V23	$V20 - (V20 - V43) * (3/23)$	V55	VINP5
V24	$V20 - (V20 - V43) * (4/23)$	V56	$V55 - (V55 - V62) * (9/96)$
V25	$V20 - (V20 - V43) * (5/23)$	V57	$V55 - (V55 - V62) * (18/96)$
V26	$V20 - (V20 - V43) * (6/23)$	V58	$V55 - (V55 - V62) * (27/96)$
V27	$V20 - (V20 - V43) * (7/23)$	V59	$V55 - (V55 - V62) * (36/96)$
V28	$V20 - (V20 - V43) * (8/23)$	V60	$V55 - (V55 - V62) * (54/96)$
V29	$V20 - (V20 - V43) * (9/23)$	V61	$V55 - (V55 - V62) * (68/96)$
V30	$V20 - (V20 - V43) * (10/23)$	V62	VINP6
V31	$V20 - (V20 - V43) * (11/23)$	V63	VINP7

Table 47. Formulas for Calculating Gamma Adjusting Voltage (Negative Polarity) 1

Pins	Formula	Micro-adjusting register value	Reference voltage
KVN0	$(45R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "0000"	VINN1
KVN1	$(42R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "0001"	
KVN2	$(39R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "0010"	
KVN3	$(36R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "0011"	
KVN4	$(33R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "0100"	
KVN5	$(30R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "0101"	
KVN6	$(27R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "0110"	
KVN7	$(24R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "0111"	
KVN8	$(21R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "1000"	
KVN9	$(18R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "1001"	
KVN10	$(15R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "1010"	
KVN11	$(12R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "1011"	
KVN12	$(9R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "1100"	
KVN13	$(6R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "1101"	
KVN14	$(3R/48R) * (VINN0 - VINN12) + VINN12$	PKN0[3:0] = "1110"	
KVN15	VINN12	PKN0[3:0] = "1111"	
KVN16	$(219R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "0000"	VINN2
KVN17	$(216R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "0001"	
KVN18	$(213R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "0010"	
KVN19	$(210R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "0011"	
KVN20	$(207R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "0100"	
KVN21	$(204R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "0101"	
KVN22	$(201R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "0110"	
KVN23	$(198R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "0111"	
KVN24	$(195R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "1000"	
KVN25	$(192R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "1001"	
KVN26	$(189R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "1010"	
KVN27	$(186R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "1011"	
KVN28	$(183R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "1100"	
KVN29	$(180R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "1101"	
KVN30	$(177R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "1110"	
KVN31	$(174R/222R) * (VINN12 - VINN56) + VINN56$	PKN1[3:0] = "1111"	
KVN32	$(171R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "0000"	VINN3
KVN33	$(168R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "0001"	
KVN34	$(165R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "0010"	
KVN35	$(162R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "0011"	
KVN36	$(159R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "0100"	
KVN37	$(156R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "0101"	
KVN38	$(153R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "0110"	
KVN39	$(150R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "0111"	
KVN40	$(147R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "1000"	
KVN41	$(144R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "1001"	
KVN42	$(141R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "1010"	
KVN43	$(138R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "1011"	
KVN44	$(135R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "1100"	
KVN45	$(132R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "1101"	
KVN46	$(129R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "1110"	
KVN47	$(126R/222R) * (VINN12 - VINN56) + VINN56$	PKN2[3:0] = "1111"	

Pins	Formula	Micro-adjusting register value	Reference voltage
KVN48	$(96R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "1111"	VINN4
KVN49	$(93R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "1110"	
KVN50	$(90R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "1101"	
KVN51	$(87R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "1100"	
KVN52	$(84R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "1011"	
KVN53	$(81R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "1010"	
KVN54	$(78R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "1001"	
KVN55	$(75R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "1000"	
KVN56	$(72R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "0111"	
KVN57	$(69R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "0110"	
KVN58	$(66R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "0101"	
KVN59	$(63R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "0100"	
KVN60	$(60R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "0011"	
KVN61	$(57R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "0010"	
KVN62	$(54R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "0001"	
KVN63	$(51R/222R) * (VINN12 - VINN56) + VINN56$	PKN3[3:0] = "0000"	VINN5
KVN64	$(48R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "1111"	
KVN65	$(45R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "1110"	
KVN66	$(42R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "1101"	
KVN67	$(39R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "1100"	
KVN68	$(36R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "1011"	
KVN69	$(33R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "1010"	
KVN70	$(30R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "1001"	
KVN71	$(27R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "1000"	
KVN72	$(24R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "0111"	
KVN73	$(21R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "0110"	
KVN74	$(18R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "0101"	
KVN75	$(15R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "0100"	
KVN76	$(12R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "0011"	
KVN77	$(9R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "0010"	VINN6
KVN78	$(6R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "0001"	
KVN79	$(3R/222R) * (VINN12 - VINN56) + VINN56$	PKN4[3:0] = "0000"	
KVN80	VINN56	PKN5[3:0] = "1111"	
KVN81	$(45R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "1110"	
KVN82	$(42R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "1101"	
KVN83	$(39R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "1100"	
KVN84	$(36R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "1011"	
KVN85	$(33R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "1010"	
KVN86	$(30R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "1001"	
KVN87	$(27R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "1000"	
KVN88	$(24R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "0111"	
KVN89	$(21R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "0110"	
KVN90	$(18R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "0101"	
KVN91	$(15R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "0100"	
KVN92	$(12R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "0011"	
KVN93	$(9R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "0010"	
KVN94	$(6R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "0001"	
KVN95	$(3R/48R) * (VINN56 - VINN7) + VINN7$	PKN5[3:0] = "0000"	

Table 48. Formulas for Calculating Gamma Adjusting Voltage (Negative Polarity) 2

Grayscale voltage	Formula	Grayscale voltage	Formula
V0	VINN0	V32	$V20 - (V20 - V43) * (12/23)$
V1	VINN1	V33	$V20 - (V20 - V43) * (13/23)$
V2	$V1 - (V1 - V8) * (28/96)$	V34	$V20 - (V20 - V43) * (14/23)$
V3	$V1 - (V1 - V8) * (42/96)$	V35	$V20 - (V20 - V43) * (15/23)$
V4	$V1 - (V1 - V8) * (60/96)$	V36	$V20 - (V20 - V43) * (16/23)$
V5	$V1 - (V1 - V8) * (69/96)$	V37	$V20 - (V20 - V43) * (17/23)$
V6	$V1 - (V1 - V8) * (78/96)$	V38	$V20 - (V20 - V43) * (18/23)$
V7	$V1 - (V1 - V8) * (87/96)$	V39	$V20 - (V20 - V43) * (19/23)$
V8	VINN2	V40	$V20 - (V20 - V43) * (20/23)$
V9	$V8 - (V8 - V20) * (2/24)$	V41	$V20 - (V20 - V43) * (21/23)$
V10	$V8 - (V8 - V20) * (4/24)$	V42	$V20 - (V20 - V43) * (22/23)$
V11	$V8 - (V8 - V20) * (6/24)$	V43	VINN4
V12	$V8 - (V8 - V20) * (8/24)$	V44	$V43 - (V43 - V55) * (2/24)$
V13	$V8 - (V8 - V20) * (10/24)$	V45	$V43 - (V43 - V55) * (4/24)$
V14	$V8 - (V8 - V20) * (12/24)$	V46	$V43 - (V43 - V55) * (6/24)$
V15	$V8 - (V8 - V20) * (14/24)$	V47	$V43 - (V43 - V55) * (8/24)$
V16	$V8 - (V8 - V20) * (16/24)$	V48	$V43 - (V43 - V55) * (10/24)$
V17	$V8 - (V8 - V20) * (18/24)$	V49	$V43 - (V43 - V55) * (12/24)$
V18	$V8 - (V8 - V20) * (20/24)$	V50	$V43 - (V43 - V55) * (14/24)$
V19	$V8 - (V8 - V20) * (22/24)$	V51	$V43 - (V43 - V55) * (16/24)$
V20	VINN3	V52	$V43 - (V43 - V55) * (18/24)$
V21	$V20 - (V20 - V43) * (1/23)$	V53	$V43 - (V43 - V55) * (20/24)$
V22	$V20 - (V20 - V43) * (2/23)$	V54	$V43 - (V43 - V55) * (22/24)$
V23	$V20 - (V20 - V43) * (3/23)$	V55	VINN5
V24	$V20 - (V20 - V43) * (4/23)$	V56	$V55 - (V55 - V62) * (9/96)$
V25	$V20 - (V20 - V43) * (5/23)$	V57	$V55 - (V55 - V62) * (18/96)$
V26	$V20 - (V20 - V43) * (6/23)$	V58	$V55 - (V55 - V62) * (27/96)$
V27	$V20 - (V20 - V43) * (7/23)$	V59	$V55 - (V55 - V62) * (36/96)$
V28	$V20 - (V20 - V43) * (8/23)$	V60	$V55 - (V55 - V62) * (54/96)$
V29	$V20 - (V20 - V43) * (9/23)$	V61	$V55 - (V55 - V62) * (68/96)$
V30	$V20 - (V20 - V43) * (10/23)$	V62	VINN6
V31	$V20 - (V20 - V43) * (11/23)$	V63	VINN7

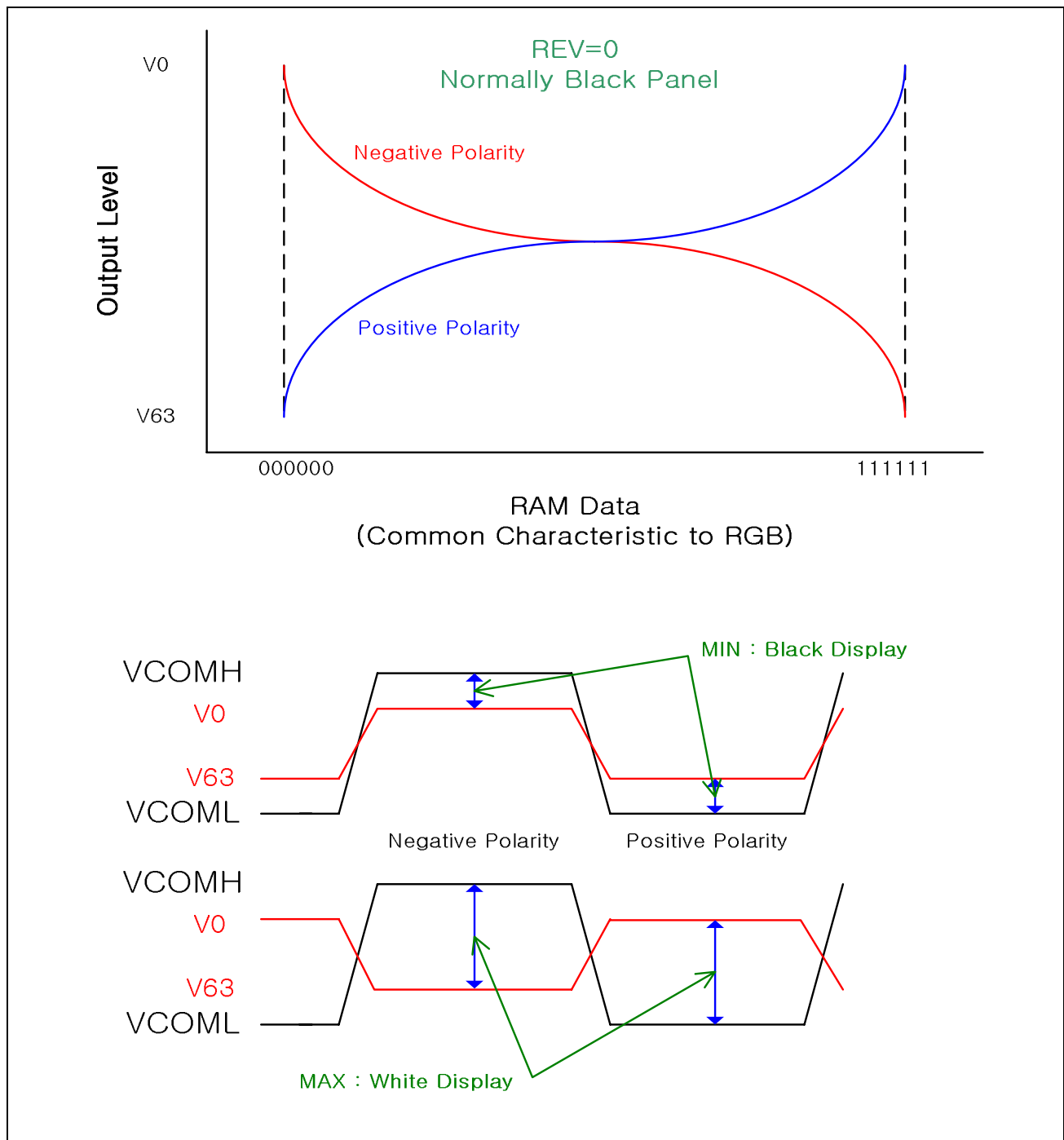


Figure 86. Relationship Between RAM Data, Source Output Voltage and VCOM (REV=0)

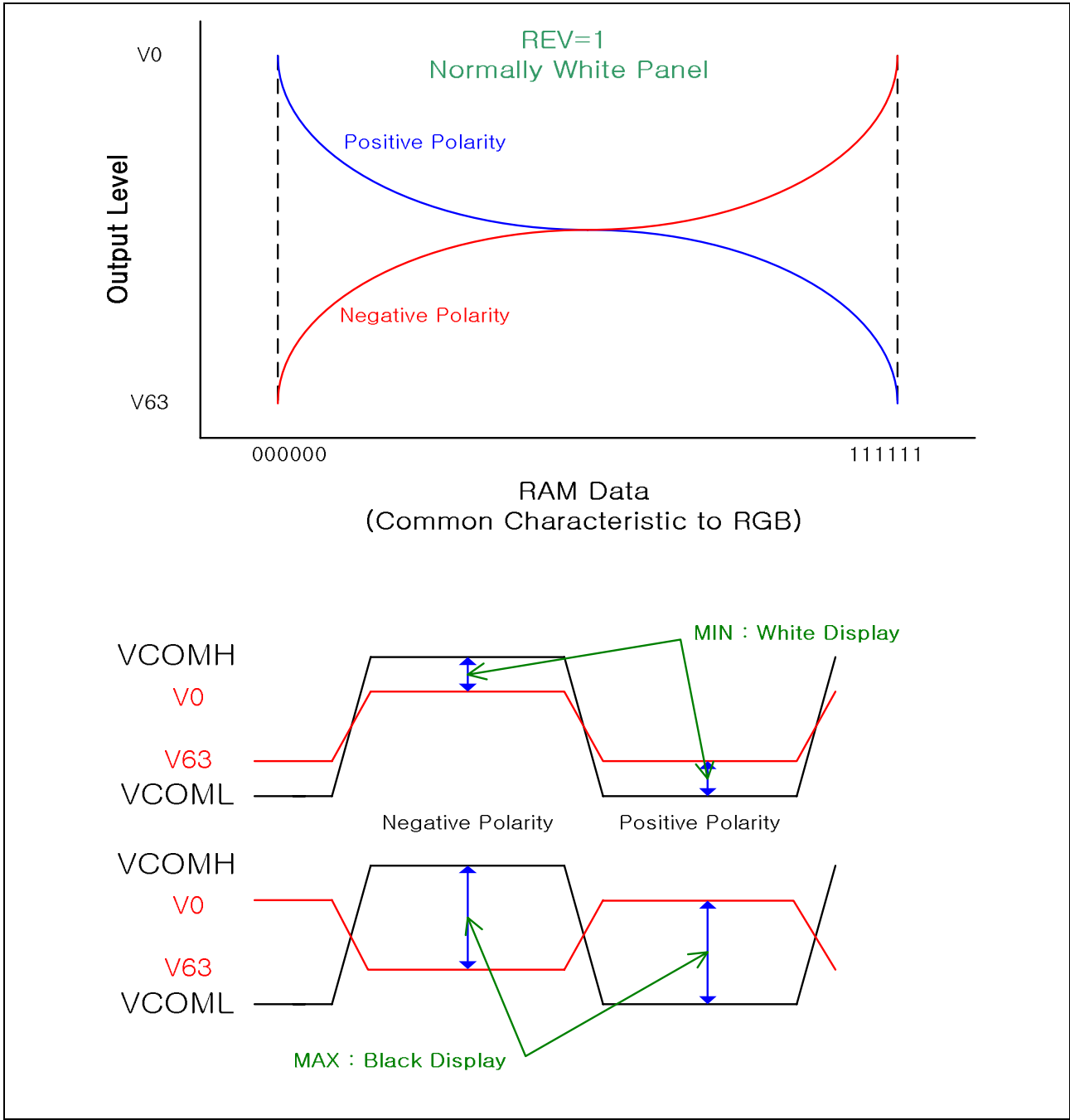


Figure 87. Relationship Between RAM Data , Source Output Voltage and VCOM (REV=1)

13. THE 8-COLOR DISPLAY MODE

The S6D0164 incorporates 8-color display mode. The grayscale levels to be used are V0 (GVDD) and V63 (AVSS) and all the other levels (V1~V62) are halt. So that it attempts to lower power consumption.

During the 8-color mode, the Gamma micro adjustment register, PKP00-PKP52 and PKN00-PKN52 are invalid.

The grayscale levels (V1-V62) are in OFF condition in order to select V0/V63.

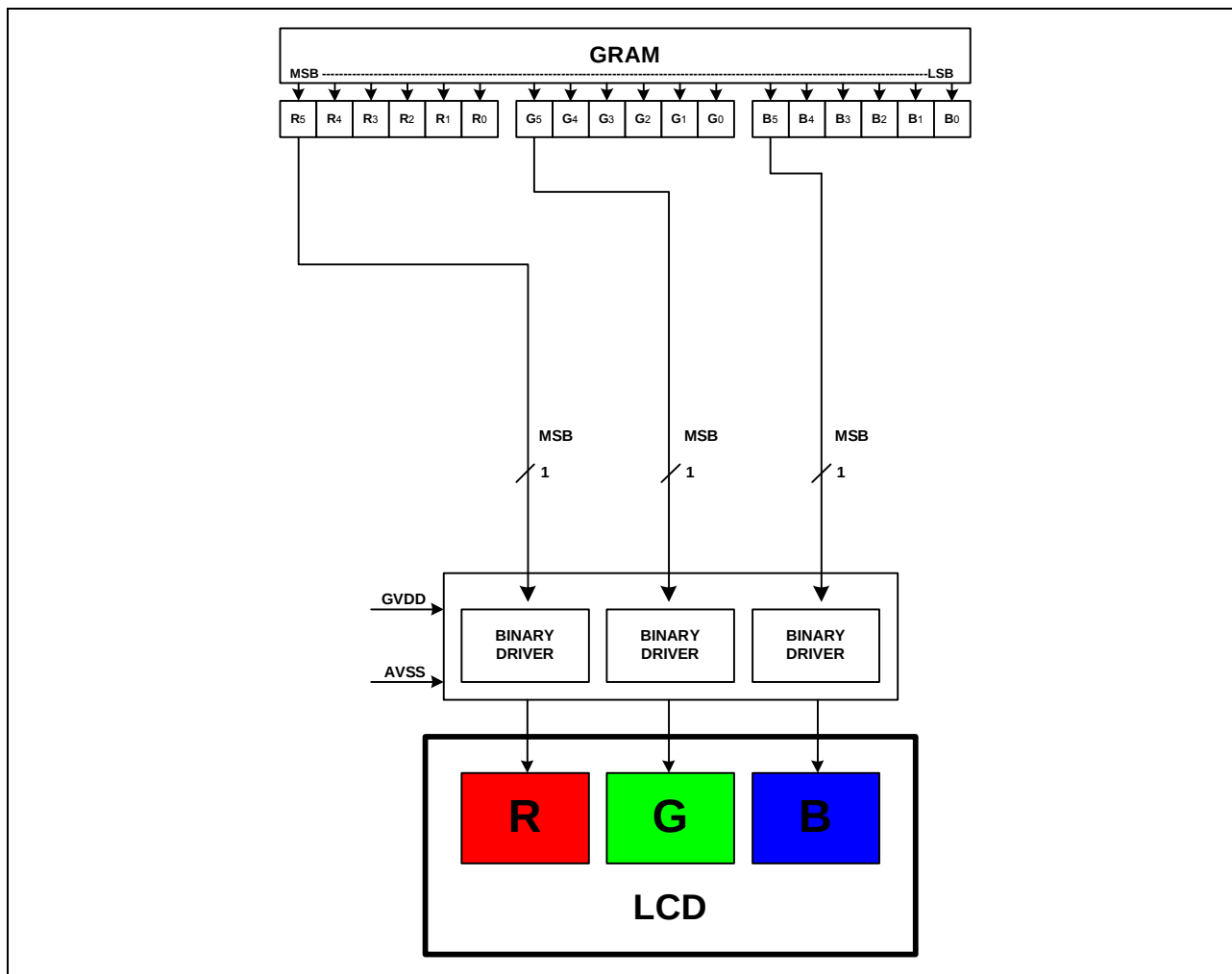


Figure 88. 8-Color Display Control

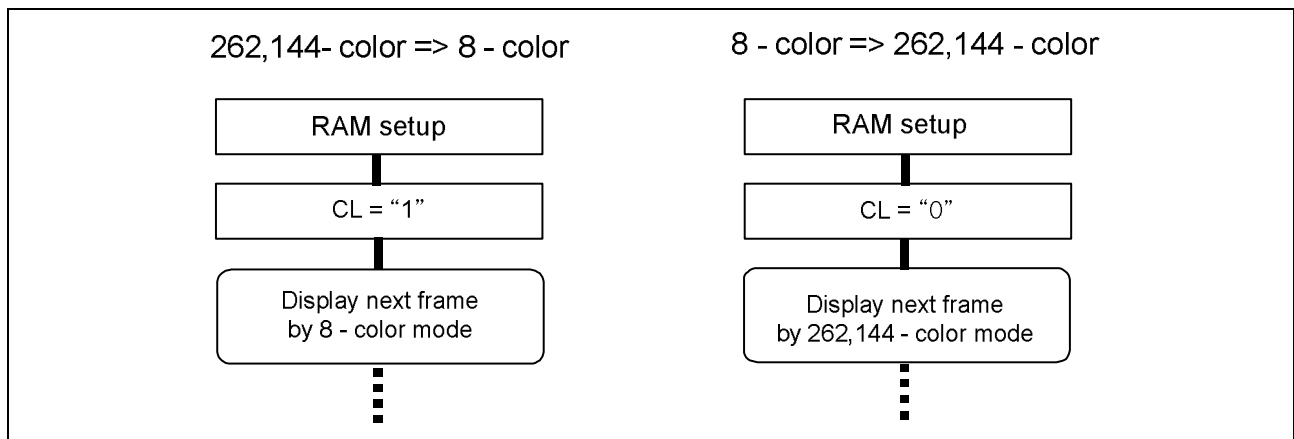


Figure 89. Setup Procedure For The 8-Color Mode

14. INSTRUCTION SET UP FLOW

14.1. DISPLAY ON / OFF SEQUENCE

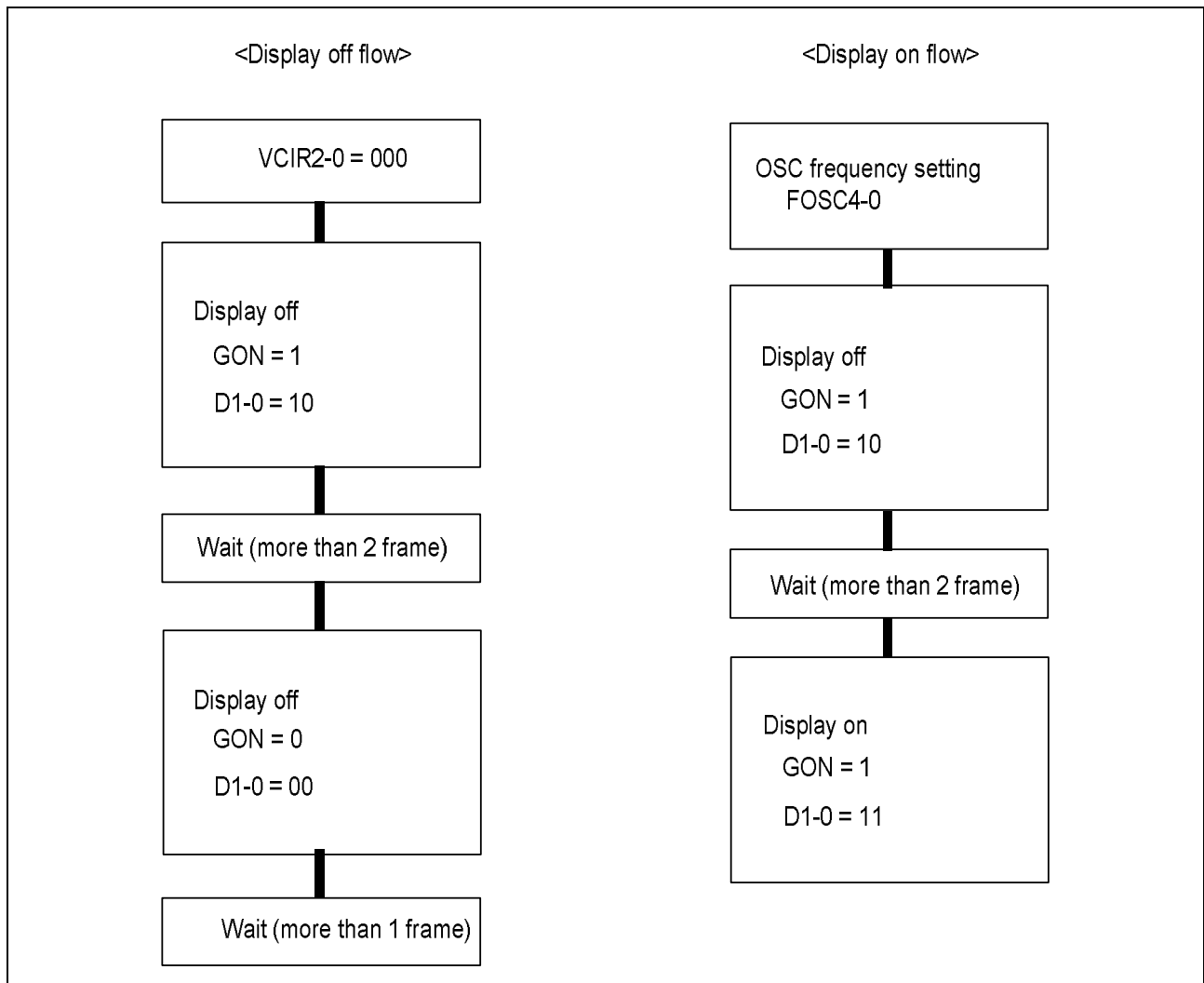


Figure 90. DISPLAY ON / OFF SEQUENCE

14.2. D-STAND-BY / STAND-BY SEQUENCE

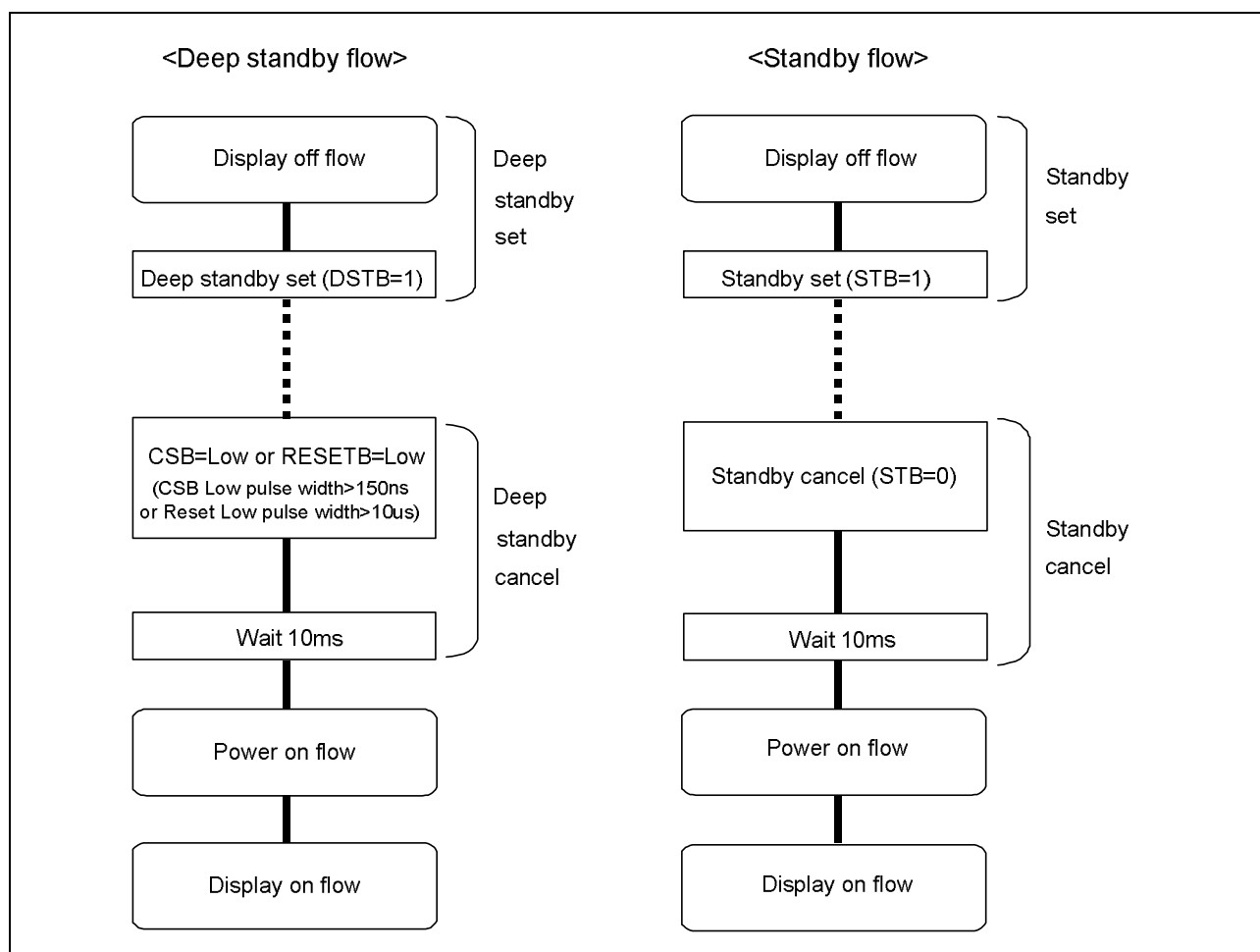


Figure 91. D-STAND-BY/STAND-BY SEQUENCE

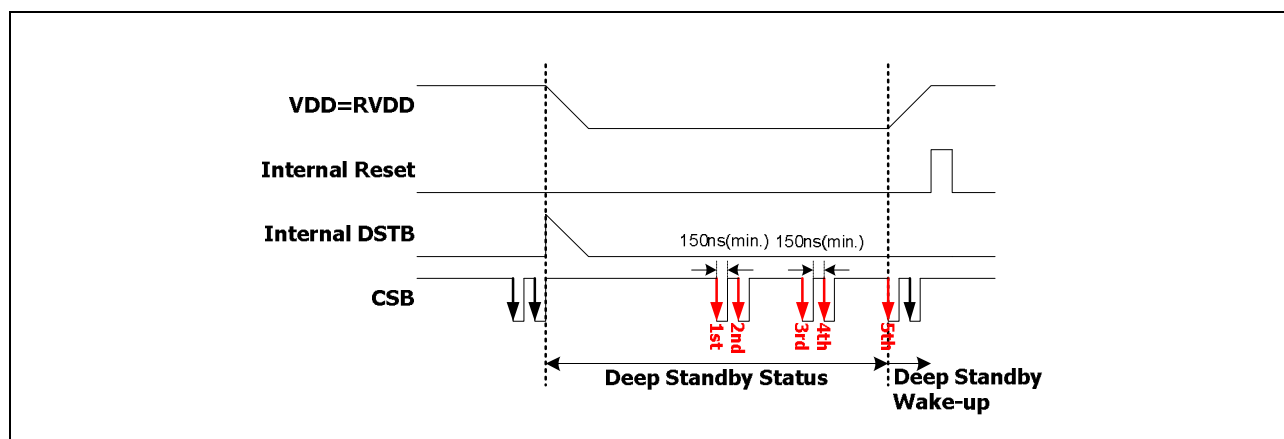


Figure 92. DEEP STAND-BY EXIT FLOW

15. OSCILLATION CIRCUIT

The S6D0164 can provide R-C oscillation. S6D0164 internal oscillator does not need to attach the external resistor. The appropriate oscillation frequency for operating voltage, display size, and frame frequency can be obtained by adjusting the oscillator frequency control register setting. Since R-C oscillation stops during the standby mode, power consumption can be reduced.

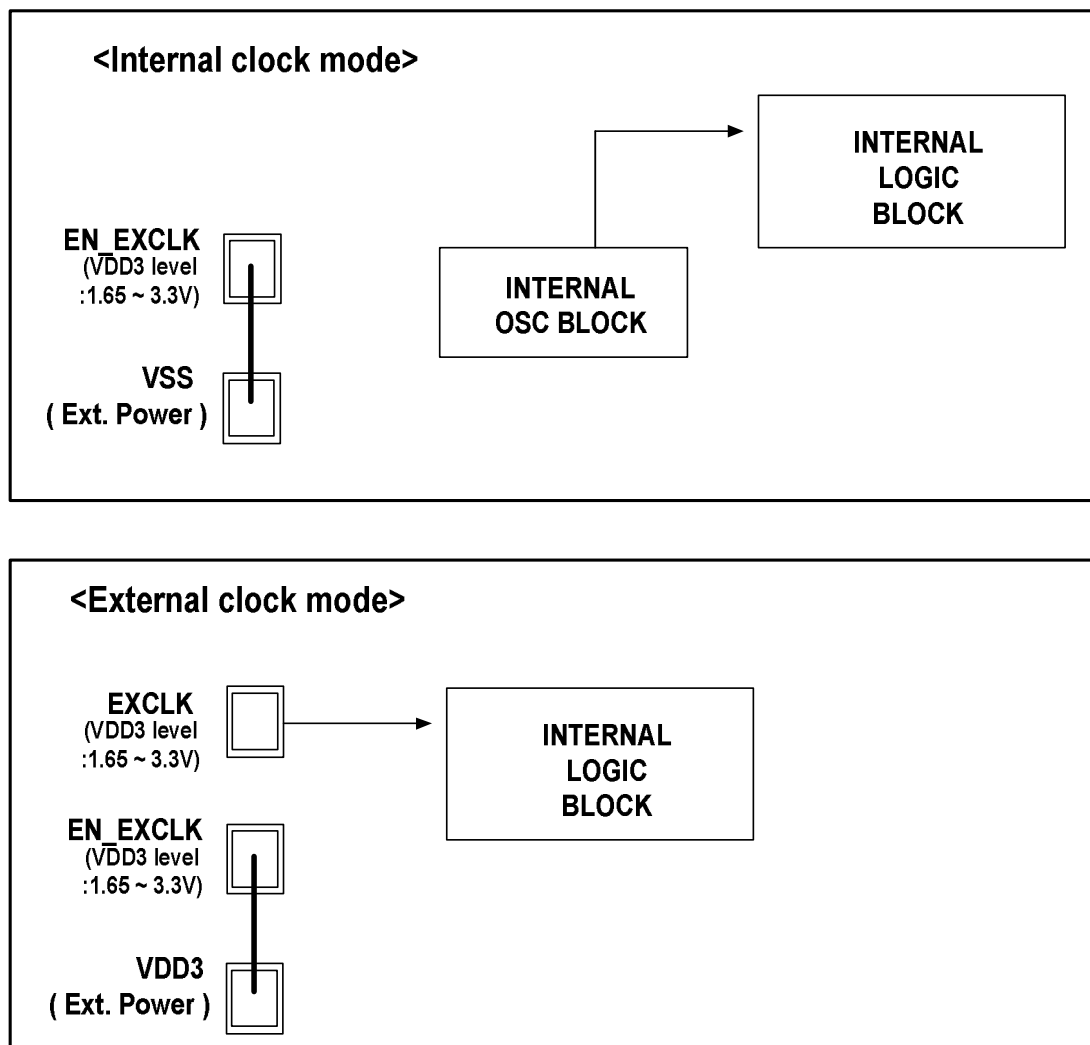


Figure 93. Oscillation Circuit

16. FRAME FREQUENCY ADJUSTING FUNCTION

The S6D0164 has an on-chip frame-frequency adjustment function. The frame frequency can be adjusted by the instruction setting (RTN) during the LCD driver as the oscillation frequency is always same.

If the oscillation frequency is set to high, animation or a static image can be displayed in suitable ways by changing the frame frequency. When a static image is displayed, the frame frequency can be set low and the low-power consumption mode can be entered. When high-speed screen switching for an animated display, etc. is required, the frame frequency can be set high.

16.1. Relationship between LCD drive duty and frame frequency

The relationships between the LCD drive duty and the frame frequency is calculated by the following expression. The frame frequency can be adjusted in the 1H period adjusting bit (RTN).

$$\text{Frame Frequency} = \frac{f_{\text{osc}}}{\text{Clock cycles per raster-row} \times (\text{Line} + \text{B})} \quad [\text{Hz}]$$

f_{osc} : R-C oscillation frequency
 Line: Number of raster-rows (NL bit)
 Clock cycles per raster-row: RTN bit
 B: Blank period(Back porch + Front Porch)

Figure 94. Formula for the Frame Frequency

Calculation Example:

Driver raster-rows: 220

1H period: 16 clock (RTN2 to 0 = 000)

B: Blank period (BP + FP): 16

$$f_{\text{osc}} = 60\text{Hz} \times 16 \text{ clock} \times (220 + 16) \text{ lines} = 227 \text{ [kHz]}$$

The calculation result says that the required oscillation frequency in the display condition listed above is 227 kHz. So the external resistance value of the R-C oscillator should be selected to make oscillation frequency 227 kHz.

17. Appendices

17.1. PAD COORDINATES

Table 49. Pad Center Coordinates

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
1	DUMMY1	-6695	-335	51	C11P	-3695	-335	101	VCI	-695	-335
2	DUMMY2	-6635	-335	52	C11P	-3635	-335	102	VCI	-635	-335
3	VCOM	-6575	-335	53	C11P	-3575	-335	103	VCI	-575	-335
4	VCOM	-6515	-335	54	C11P	-3515	-335	104	VCI	-515	-335
5	VCOM	-6455	-335	55	C11P	-3455	-335	105	VCI	-455	-335
6	VCOM	-6395	-335	56	C11P	-3395	-335	106	VCL	-395	-335
7	DUMMY3	-6335	-335	57	C11P	-3335	-335	107	VCL	-335	-335
8	VGH	-6275	-335	58	C11M	-3275	-335	108	VCL	-275	-335
9	VGH	-6215	-335	59	C11M	-3215	-335	109	VCL	-215	-335
10	VGH	-6155	-335	60	C11M	-3155	-335	110	VCL	-155	-335
11	VGH	-6095	-335	61	C11M	-3095	-335	111	DUMMY8	-95	-335
12	VGH	-6035	-335	62	C11M	-3035	-335	112	RS	-35	-335
13	DUMMY4	-5975	-335	63	C11M	-2975	-335	113	CSB	25	-335
14	VGL	-5915	-335	64	C11M	-2915	-335	114	VSNC	85	-335
15	VGL	-5855	-335	65	C11M	-2855	-335	115	HSNC	145	-335
16	VGL	-5795	-335	66	C12P	-2795	-335	116	DOTCLK	205	-335
17	VGL	-5735	-335	67	C12P	-2735	-335	117	ENABLE	265	-335
18	VGL	-5675	-335	68	C12P	-2675	-335	118	RESETB	325	-335
19	DUMMY5	-5615	-335	69	C12P	-2615	-335	119	SDI	385	-335
20	C22P	-5555	-335	70	C12P	-2555	-335	120	E_RDB	445	-335
21	C22P	-5495	-335	71	C12P	-2495	-335	121	RW_WRB	505	-335
22	C22P	-5435	-335	72	C12M	-2435	-335	122	DB<17>	565	-335
23	C22M	-5375	-335	73	C12M	-2375	-335	123	DB<16>	625	-335
24	C22M	-5315	-335	74	C12M	-2315	-335	124	DB<15>	685	-335
25	C22M	-5255	-335	75	C12M	-2255	-335	125	DB<14>	745	-335
26	C21P	-5195	-335	76	C12M	-2195	-335	126	DB<13>	805	-335
27	C21P	-5135	-335	77	C12M	-2135	-335	127	DB<12>	865	-335
28	C21P	-5075	-335	78	C31P	-2075	-335	128	DB<11>	925	-335
29	C21M	-5015	-335	79	C31P	-2015	-335	129	DB<10>	985	-335
30	C21M	-4955	-335	80	C31P	-1955	-335	130	DB<9>	1045	-335
31	C21M	-4895	-335	81	C31P	-1895	-335	131	DB<8>	1105	-335
32	DUMMY6	-4835	-335	82	C31P	-1835	-335	132	DB<7>	1165	-335
33	DUMMY7	-4775	-335	83	C31M	-1775	-335	133	DB<6>	1225	-335
34	VSSC	-4715	-335	84	C31M	-1715	-335	134	DB<5>	1285	-335
35	VSSC	-4655	-335	85	C31M	-1655	-335	135	DB<4>	1345	-335
36	VSSC	-4595	-335	86	C31M	-1595	-335	136	DB<3>	1405	-335
37	VSSC	-4535	-335	87	C31M	-1535	-335	137	DB<2>	1465	-335
38	VSSC	-4475	-335	88	AVDD	-1475	-335	138	DB<1>	1525	-335
39	VSSC	-4415	-335	89	AVDD	-1415	-335	139	DB<0>	1585	-335
40	VSSC	-4355	-335	90	AVDD	-1355	-335	140	IM<3>	1645	-335
41	VSSC	-4295	-335	91	AVDD	-1295	-335	141	IM<2>	1705	-335
42	VSSC	-4235	-335	92	AVDD	-1235	-335	142	IM<1>	1765	-335
43	VSSC	-4175	-335	93	AVDD	-1175	-335	143	IM<0>	1825	-335
44	VCI1	-4115	-335	94	AVDD	-1115	-335	144	SDO	1885	-335
45	VCI1	-4055	-335	95	AVDD	-1055	-335	145	M	1945	-335
46	VCI1	-3995	-335	96	VCI	-995	-335	146	FLM	2005	-335
47	VCI1	-3935	-335	97	VCI	-935	-335	147	CL1	2065	-335
48	VCI1	-3875	-335	98	VCI	-875	-335	148	TEST_MODE<2>	2125	-335
49	VCI1	-3815	-335	99	VCI	-815	-335	149	TEST_MODE<1>	2185	-335
50	C11P	-3755	-335	100	VCI	-755	-335	150	TEST_MODE<0>	2245	-335

Table 50. Pad Center Coordinates

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
151	TEST_MUX<2>	2855	-335	201	GVDD	5855	-335	251	G<64>	6212	151
152	TEST_MUX<1>	2915	-335	202	VCOMH	5915	-335	252	G<66>	6196	304
153	TEST_MUX<0>	2975	-335	203	VCOMH	5975	-335	253	G<68>	6180	151
154	TEST_DA	3035	-335	204	VCOML	6035	-335	254	G<70>	6164	304
155	EN_EXCLK	3095	-335	205	VCOML	6095	-335	255	G<72>	6148	151
156	EXCLK	3155	-335	206	VCOMR	6155	-335	256	G<74>	6132	304
157	AVSS	3215	-335	207	CONTACT	6215	-335	257	G<76>	6116	151
158	AVSS	3275	-335	208	CONTACT	6275	-335	258	G<78>	6100	304
159	AVSS	3335	-335	209	DUMMY10	6335	-335	259	G<80>	6084	151
160	AVSS	3395	-335	210	VCOM	6395	-335	260	G<82>	6068	304
161	AVSS	3455	-335	211	VCOM	6455	-335	261	G<84>	6052	151
162	AVSS	3515	-335	212	VCOM	6515	-335	262	G<86>	6036	304
163	AVSS	3575	-335	213	VCOM	6575	-335	263	G<88>	6020	151
164	AVSS	3635	-335	214	DUMMY11	6635	-335	264	G<90>	6004	304
165	AVSS	3695	-335	215	DUMMY12	6695	-335	265	G<92>	5988	151
166	VSS	3755	-335	216	DUMMY13	6772	304	266	G<94>	5972	304
167	VSS	3815	-335	217	DUMMY14	6756	151	267	G<96>	5956	151
168	VSS	3875	-335	218	DUMMY15	6740	304	268	G<98>	5940	304
169	VSS	3935	-335	219	DUMMY16	6724	151	269	G<100>	5924	151
170	VSS	3995	-335	220	G<2>	6708	304	270	G<102>	5908	304
171	VSS	4055	-335	221	G<4>	6692	151	271	G<104>	5892	151
172	VSS	4115	-335	222	G<6>	6676	304	272	G<106>	5876	304
173	VSS	4175	-335	223	G<8>	6660	151	273	G<108>	5860	151
174	VSS	4235	-335	224	G<10>	6644	304	274	G<110>	5844	304
175	VSS	4295	-335	225	G<12>	6628	151	275	G<112>	5828	151
176	VGS	4355	-335	226	G<14>	6612	304	276	G<114>	5812	304
177	VGS	4415	-335	227	G<16>	6596	151	277	G<116>	5796	151
178	RVDD	4475	-335	228	G<18>	6580	304	278	G<118>	5780	304
179	RVDD	4535	-335	229	G<20>	6564	151	279	G<120>	5764	151
180	RVDD	4595	-335	230	G<22>	6548	304	280	G<122>	5748	304
181	RVDD	4655	-335	231	G<24>	6532	151	281	G<124>	5732	151
182	RVDD	4715	-335	232	G<26>	6516	304	282	G<126>	5716	304
183	RVDD	4775	-335	233	G<28>	6500	151	283	G<128>	5700	151
184	VDD	4835	-335	234	G<30>	6484	304	284	G<130>	5684	304
185	VDD	4895	-335	235	G<32>	6468	151	285	G<132>	5668	151
186	VDD	4955	-335	236	G<34>	6452	304	286	G<134>	5652	304
187	VDD	5015	-335	237	G<36>	6436	151	287	G<136>	5636	151
188	VDD	5075	-335	238	G<38>	6420	304	288	G<138>	5620	304
189	VDD	5135	-335	239	G<40>	6404	151	289	G<140>	5604	151
190	VDD3	5195	-335	240	G<42>	6388	304	290	G<142>	5588	304
191	VDD3	5255	-335	241	G<44>	6372	151	291	G<144>	5572	151
192	VDD3	5315	-335	242	G<46>	6356	304	292	G<146>	5556	304
193	VDD3	5375	-335	243	G<48>	6340	151	293	G<148>	5540	151
194	VDD3	5435	-335	244	G<50>	6324	304	294	G<150>	5524	304
195	VDD3	5495	-335	245	G<52>	6308	151	295	G<152>	5508	151
196	DUMMY9	5555	-335	246	G<54>	6292	304	296	G<154>	5492	304
197	VREF	5615	-335	247	G<56>	6276	151	297	G<156>	5476	151
198	GVDD	5675	-335	248	G<58>	6260	304	298	G<158>	5460	304
199	GVDD	5735	-335	249	G<60>	6244	151	299	G<160>	5444	151
200	GVDD	5795	-335	250	G<62>	6228	304	300	G<162>	5428	304

Table 51. Pad Center Coordinates

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
301	G<164>	5412	151	351	S<516>	4612	151	401	S<466>	3812	151
302	G<166>	5396	304	352	S<515>	4596	304	402	S<465>	3796	304
303	G<168>	5380	151	353	S<514>	4580	151	403	S<464>	3780	151
304	G<170>	5364	304	354	S<513>	4564	304	404	S<463>	3764	304
305	G<172>	5348	151	355	S<512>	4548	151	405	S<462>	3748	151
306	G<174>	5332	304	356	S<511>	4532	304	406	S<461>	3732	304
307	G<176>	5316	151	357	S<510>	4516	151	407	S<460>	3716	151
308	G<178>	5300	304	358	S<509>	4500	304	408	S<459>	3700	304
309	G<180>	5284	151	359	S<508>	4484	151	409	S<458>	3684	151
310	G<182>	5268	304	360	S<507>	4468	304	410	S<457>	3668	304
311	G<184>	5252	151	361	S<506>	4452	151	411	S<456>	3652	151
312	G<186>	5236	304	362	S<505>	4436	304	412	S<455>	3636	304
313	G<188>	5220	151	363	S<504>	4420	151	413	S<454>	3620	151
314	G<190>	5204	304	364	S<503>	4404	304	414	S<453>	3604	304
315	G<192>	5188	151	365	S<502>	4388	151	415	S<452>	3588	151
316	G<194>	5172	304	366	S<501>	4372	304	416	S<451>	3572	304
317	G<196>	5156	151	367	S<500>	4356	151	417	S<450>	3556	151
318	G<198>	5140	304	368	S<499>	4340	304	418	S<449>	3540	304
319	G<200>	5124	151	369	S<498>	4324	151	419	S<448>	3524	151
320	G<202>	5108	304	370	S<497>	4308	304	420	S<447>	3508	304
321	G<204>	5092	151	371	S<496>	4292	151	421	S<446>	3492	151
322	G<206>	5076	304	372	S<495>	4276	304	422	S<445>	3476	304
323	G<208>	5060	151	373	S<494>	4260	151	423	S<444>	3460	151
324	G<210>	5044	304	374	S<493>	4244	304	424	S<443>	3444	304
325	G<212>	5028	151	375	S<492>	4228	151	425	S<442>	3428	151
326	G<214>	5012	304	376	S<491>	4212	304	426	S<441>	3412	304
327	G<216>	4996	151	377	S<490>	4196	151	427	S<440>	3396	151
328	G<218>	4980	304	378	S<489>	4180	304	428	S<439>	3380	304
329	G<220>	4964	151	379	S<488>	4164	151	429	S<438>	3364	151
330	DUMMY17	4948	304	380	S<487>	4148	304	430	S<437>	3348	304
331	DUMMY18	4932	151	381	S<486>	4132	151	431	S<436>	3332	151
332	DUMMY19	4916	304	382	S<485>	4116	304	432	S<435>	3316	304
333	DUMMY20	4900	151	383	S<484>	4100	151	433	S<434>	3300	151
334	DUMMY21	4884	304	384	S<483>	4084	304	434	S<433>	3284	304
335	DUMMY22	4868	151	385	S<482>	4068	151	435	S<432>	3268	151
336	DUMMY23	4852	304	386	S<481>	4052	304	436	S<431>	3252	304
337	DUMMY24	4836	151	387	S<480>	4036	151	437	S<430>	3236	151
338	DUMMY25	4820	304	388	S<479>	4020	304	438	S<429>	3220	304
339	S<528>	4804	151	389	S<478>	4004	151	439	S<428>	3204	151
340	S<527>	4788	304	390	S<477>	3988	304	440	S<427>	3188	304
341	S<526>	4772	151	391	S<476>	3972	151	441	S<426>	3172	151
342	S<525>	4756	304	392	S<475>	3956	304	442	S<425>	3156	304
343	S<524>	4740	151	393	S<474>	3940	151	443	S<424>	3140	151
344	S<523>	4724	304	394	S<473>	3924	304	444	S<423>	3124	304
345	S<522>	4708	151	395	S<472>	3908	151	445	S<422>	3108	151
346	S<521>	4692	304	396	S<471>	3892	304	446	S<421>	3092	304
347	S<520>	4676	151	397	S<470>	3876	151	447	S<420>	3076	151
348	S<519>	4660	304	398	S<469>	3860	304	448	S<419>	3060	304
349	S<518>	4644	151	399	S<468>	3844	151	449	S<418>	3044	151
350	S<517>	4628	304	400	S<467>	3828	304	450	S<417>	3028	304

Table 52. Pad Center Coordinates

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
451	S<416>	3012	151	501	S<366>	2162	151	551	S<316>	1362	151
452	S<415>	2996	304	502	S<365>	2146	304	552	S<315>	1346	304
453	S<414>	2980	151	503	S<364>	2130	151	553	S<314>	1330	151
454	S<413>	2964	304	504	S<363>	2114	304	554	S<313>	1314	304
455	S<412>	2948	151	505	S<362>	2098	151	555	S<312>	1298	151
456	S<411>	2932	304	506	S<361>	2082	304	556	S<311>	1282	304
457	S<410>	2916	151	507	S<360>	2066	151	557	S<310>	1266	151
458	S<409>	2900	304	508	S<359>	2050	304	558	S<309>	1250	304
459	S<408>	2884	151	509	S<358>	2034	151	559	S<308>	1234	151
460	S<407>	2868	304	510	S<357>	2018	304	560	S<307>	1218	304
461	S<406>	2852	151	511	S<356>	2002	151	561	S<306>	1202	151
462	S<405>	2836	304	512	S<355>	1986	304	562	S<305>	1186	304
463	S<404>	2820	151	513	S<354>	1970	151	563	S<304>	1170	151
464	S<403>	2804	304	514	S<353>	1954	304	564	S<303>	1154	304
465	S<402>	2788	151	515	S<352>	1938	151	565	S<302>	1138	151
466	S<401>	2772	304	516	S<351>	1922	304	566	S<301>	1122	304
467	S<400>	2756	151	517	S<350>	1906	151	567	S<300>	1106	151
468	S<399>	2740	304	518	S<349>	1890	304	568	S<299>	1090	304
469	S<398>	2724	151	519	S<348>	1874	151	569	S<298>	1074	151
470	S<397>	2708	304	520	S<347>	1858	304	570	S<297>	1058	304
471	S<396>	2642	151	521	S<346>	1842	151	571	S<296>	1042	151
472	S<395>	2626	304	522	S<345>	1826	304	572	S<295>	1026	304
473	S<394>	2610	151	523	S<344>	1810	151	573	S<294>	1010	151
474	S<393>	2594	304	524	S<343>	1794	304	574	S<293>	994	304
475	S<392>	2578	151	525	S<342>	1778	151	575	S<292>	978	151
476	S<391>	2562	304	526	S<341>	1762	304	576	S<291>	962	304
477	S<390>	2546	151	527	S<340>	1746	151	577	S<290>	946	151
478	S<389>	2530	304	528	S<339>	1730	304	578	S<289>	930	304
479	S<388>	2514	151	529	S<338>	1714	151	579	S<288>	914	151
480	S<387>	2498	304	530	S<337>	1698	304	580	S<287>	898	304
481	S<386>	2482	151	531	S<336>	1682	151	581	S<286>	882	151
482	S<385>	2466	304	532	S<335>	1666	304	582	S<285>	866	304
483	S<384>	2450	151	533	S<334>	1650	151	583	S<284>	850	151
484	S<383>	2434	304	534	S<333>	1634	304	584	S<283>	834	304
485	S<382>	2418	151	535	S<332>	1618	151	585	S<282>	818	151
486	S<381>	2402	304	536	S<331>	1602	304	586	S<281>	802	304
487	S<380>	2386	151	537	S<330>	1586	151	587	S<280>	786	151
488	S<379>	2370	304	538	S<329>	1570	304	588	S<279>	770	304
489	S<378>	2354	151	539	S<328>	1554	151	589	S<278>	754	151
490	S<377>	2338	304	540	S<327>	1538	304	590	S<277>	738	304
491	S<376>	2322	151	541	S<326>	1522	151	591	S<276>	722	151
492	S<375>	2306	304	542	S<325>	1506	304	592	S<275>	706	304
493	S<374>	2290	151	543	S<324>	1490	151	593	S<274>	690	151
494	S<373>	2274	304	544	S<323>	1474	304	594	S<273>	674	304
495	S<372>	2258	151	545	S<322>	1458	151	595	S<272>	658	151
496	S<371>	2242	304	546	S<321>	1442	304	596	S<271>	642	304
497	S<370>	2226	151	547	S<320>	1426	151	597	S<270>	626	151
498	S<369>	2210	304	548	S<319>	1410	304	598	S<269>	610	304
499	S<368>	2194	151	549	S<318>	1394	151	599	S<268>	594	151
500	S<367>	2178	304	550	S<317>	1378	304	600	S<267>	578	304

Table 53. Pad Center Coordinates

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
601	S<266>	562	151	651	S<216>	-1322	151	701	S<166>	-2122	151
602	S<265>	546	304	652	S<215>	-1338	304	702	S<165>	-2138	304
603	S<264>	-554	151	653	S<214>	-1354	151	703	S<164>	-2154	151
604	S<263>	-570	304	654	S<213>	-1370	304	704	S<163>	-2170	304
605	S<262>	-586	151	655	S<212>	-1386	151	705	S<162>	-2186	151
606	S<261>	-602	304	656	S<211>	-1402	304	706	S<161>	-2202	304
607	S<260>	-618	151	657	S<210>	-1418	151	707	S<160>	-2218	151
608	S<259>	-634	304	658	S<209>	-1434	304	708	S<159>	-2234	304
609	S<258>	-650	151	659	S<208>	-1450	151	709	S<158>	-2250	151
610	S<257>	-666	304	660	S<207>	-1466	304	710	S<157>	-2266	304
611	S<256>	-682	151	661	S<206>	-1482	151	711	S<156>	-2282	151
612	S<255>	-698	304	662	S<205>	-1498	304	712	S<155>	-2298	304
613	S<254>	-714	151	663	S<204>	-1514	151	713	S<154>	-2314	151
614	S<253>	-730	304	664	S<203>	-1530	304	714	S<153>	-2330	304
615	S<252>	-746	151	665	S<202>	-1546	151	715	S<152>	-2346	151
616	S<251>	-762	304	666	S<201>	-1562	304	716	S<151>	-2362	304
617	S<250>	-778	151	667	S<200>	-1578	151	717	S<150>	-2378	151
618	S<249>	-794	304	668	S<199>	-1594	304	718	S<149>	-2394	304
619	S<248>	-810	151	669	S<198>	-1610	151	719	S<148>	-2410	151
620	S<247>	-826	304	670	S<197>	-1626	304	720	S<147>	-2426	304
621	S<246>	-842	151	671	S<196>	-1642	151	721	S<146>	-2442	151
622	S<245>	-858	304	672	S<195>	-1658	304	722	S<145>	-2458	304
623	S<244>	-874	151	673	S<194>	-1674	151	723	S<144>	-2474	151
624	S<243>	-890	304	674	S<193>	-1690	304	724	S<143>	-2490	304
625	S<242>	-906	151	675	S<192>	-1706	151	725	S<142>	-2506	151
626	S<241>	-922	304	676	S<191>	-1722	304	726	S<141>	-2522	304
627	S<240>	-938	151	677	S<190>	-1738	151	727	S<140>	-2538	151
628	S<239>	-954	304	678	S<189>	-1754	304	728	S<139>	-2554	304
629	S<238>	-970	151	679	S<188>	-1770	151	729	S<138>	-2570	151
630	S<237>	-986	304	680	S<187>	-1786	304	730	S<137>	-2586	304
631	S<236>	-1002	151	681	S<186>	-1802	151	731	S<136>	-2602	151
632	S<235>	-1018	304	682	S<185>	-1818	304	732	S<135>	-2618	304
633	S<234>	-1034	151	683	S<184>	-1834	151	733	S<134>	-2634	151
634	S<233>	-1050	304	684	S<183>	-1850	304	734	S<133>	-2650	304
635	S<232>	-1066	151	685	S<182>	-1866	151	735	S<132>	-2716	151
636	S<231>	-1082	304	686	S<181>	-1882	304	736	S<131>	-2732	304
637	S<230>	-1098	151	687	S<180>	-1898	151	737	S<130>	-2748	151
638	S<229>	-1114	304	688	S<179>	-1914	304	738	S<129>	-2764	304
639	S<228>	-1130	151	689	S<178>	-1930	151	739	S<128>	-2780	151
640	S<227>	-1146	304	690	S<177>	-1946	304	740	S<127>	-2796	304
641	S<226>	-1162	151	691	S<176>	-1962	151	741	S<126>	-2812	151
642	S<225>	-1178	304	692	S<175>	-1978	304	742	S<125>	-2828	304
643	S<224>	-1194	151	693	S<174>	-1994	151	743	S<124>	-2844	151
644	S<223>	-1210	304	694	S<173>	-2010	304	744	S<123>	-2860	304
645	S<222>	-1226	151	695	S<172>	-2026	151	745	S<122>	-2876	151
646	S<221>	-1242	304	696	S<171>	-2042	304	746	S<121>	-2892	304
647	S<220>	-1258	151	697	S<170>	-2058	151	747	S<120>	-2908	151
648	S<219>	-1274	304	698	S<169>	-2074	304	748	S<119>	-2924	304
649	S<218>	-1290	151	699	S<168>	-2090	151	749	S<118>	-2940	151
650	S<217>	-1306	304	700	S<167>	-2106	304	750	S<117>	-2956	304

Table 54. Pad Center Coordinates

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
751	S<116>	-2972	151	801	S<66>	-3772	151	851	S<16>	-4572	151
752	S<115>	-2988	304	802	S<65>	-3788	304	852	S<15>	-4588	304
753	S<114>	-3004	151	803	S<64>	-3804	151	853	S<14>	-4604	151
754	S<113>	-3020	304	804	S<63>	-3820	304	854	S<13>	-4620	304
755	S<112>	-3036	151	805	S<62>	-3836	151	855	S<12>	-4636	151
756	S<111>	-3052	304	806	S<61>	-3852	304	856	S<11>	-4652	304
757	S<110>	-3068	151	807	S<60>	-3868	151	857	S<10>	-4668	151
758	S<109>	-3084	304	808	S<59>	-3884	304	858	S<9>	-4684	304
759	S<108>	-3100	151	809	S<58>	-3900	151	859	S<8>	-4700	151
760	S<107>	-3116	304	810	S<57>	-3916	304	860	S<7>	-4716	304
761	S<106>	-3132	151	811	S<56>	-3932	151	861	S<6>	-4732	151
762	S<105>	-3148	304	812	S<55>	-3948	304	862	S<5>	-4748	304
763	S<104>	-3164	151	813	S<54>	-3964	151	863	S<4>	-4764	151
764	S<103>	-3180	304	814	S<53>	-3980	304	864	S<3>	-4780	304
765	S<102>	-3196	151	815	S<52>	-3996	151	865	S<2>	-4796	151
766	S<101>	-3212	304	816	S<51>	-4012	304	866	S<1>	-4812	304
767	S<100>	-3228	151	817	S<50>	-4028	151	867	DUMMY26	-4828	151
768	S<99>	-3244	304	818	S<49>	-4044	304	868	DUMMY27	-4844	304
769	S<98>	-3260	151	819	S<48>	-4060	151	869	DUMMY28	-4860	151
770	S<97>	-3276	304	820	S<47>	-4076	304	870	DUMMY29	-4876	304
771	S<96>	-3292	151	821	S<46>	-4092	151	871	DUMMY30	-4892	151
772	S<95>	-3308	304	822	S<45>	-4108	304	872	DUMMY31	-4908	304
773	S<94>	-3324	151	823	S<44>	-4124	151	873	DUMMY32	-4924	151
774	S<93>	-3340	304	824	S<43>	-4140	304	874	DUMMY33	-4940	304
775	S<92>	-3356	151	825	S<42>	-4156	151	875	G<219>	-4956	151
776	S<91>	-3372	304	826	S<41>	-4172	304	876	G<217>	-4972	304
777	S<90>	-3388	151	827	S<40>	-4188	151	877	G<215>	-4988	151
778	S<89>	-3404	304	828	S<39>	-4204	304	878	G<213>	-5004	304
779	S<88>	-3420	151	829	S<38>	-4220	151	879	G<211>	-5020	151
780	S<87>	-3436	304	830	S<37>	-4236	304	880	G<209>	-5036	304
781	S<86>	-3452	151	831	S<36>	-4252	151	881	G<207>	-5052	151
782	S<85>	-3468	304	832	S<35>	-4268	304	882	G<205>	-5068	304
783	S<84>	-3484	151	833	S<34>	-4284	151	883	G<203>	-5084	151
784	S<83>	-3500	304	834	S<33>	-4300	304	884	G<201>	-5100	304
785	S<82>	-3516	151	835	S<32>	-4316	151	885	G<199>	-5116	151
786	S<81>	-3532	304	836	S<31>	-4332	304	886	G<197>	-5132	304
787	S<80>	-3548	151	837	S<30>	-4348	151	887	G<195>	-5148	151
788	S<79>	-3564	304	838	S<29>	-4364	304	888	G<193>	-5164	304
789	S<78>	-3580	151	839	S<28>	-4380	151	889	G<191>	-5180	151
790	S<77>	-3596	304	840	S<27>	-4396	304	890	G<189>	-5196	304
791	S<76>	-3612	151	841	S<26>	-4412	151	891	G<187>	-5212	151
792	S<75>	-3628	304	842	S<25>	-4428	304	892	G<185>	-5228	304
793	S<74>	-3644	151	843	S<24>	-4444	151	893	G<183>	-5244	151
794	S<73>	-3660	304	844	S<23>	-4460	304	894	G<181>	-5260	304
795	S<72>	-3676	151	845	S<22>	-4476	151	895	G<179>	-5276	151
796	S<71>	-3692	304	846	S<21>	-4492	304	896	G<177>	-5292	304
797	S<70>	-3708	151	847	S<20>	-4508	151	897	G<175>	-5308	151
798	S<69>	-3724	304	848	S<19>	-4524	304	898	G<173>	-5324	304
799	S<68>	-3740	151	849	S<18>	-4540	151	899	G<171>	-5340	151
800	S<67>	-3756	304	850	S<17>	-4556	304	900	G<169>	-5356	304

Table 55. Pad Center Coordinates

NO	NAME	X	Y	NO	NAME	X	Y
901	G<167>	-5372	151	951	G<67>	-6172	151
902	G<165>	-5388	304	952	G<65>	-6188	304
903	G<163>	-5404	151	953	G<63>	-6204	151
904	G<161>	-5420	304	954	G<61>	-6220	304
905	G<159>	-5436	151	955	G<59>	-6236	151
906	G<157>	-5452	304	956	G<57>	-6252	304
907	G<155>	-5468	151	957	G<55>	-6268	151
908	G<153>	-5484	304	958	G<53>	-6284	304
909	G<151>	-5500	151	959	G<51>	-6300	151
910	G<149>	-5516	304	960	G<49>	-6316	304
911	G<147>	-5532	151	961	G<47>	-6332	151
912	G<145>	-5548	304	962	G<45>	-6348	304
913	G<143>	-5564	151	963	G<43>	-6364	151
914	G<141>	-5580	304	964	G<41>	-6380	304
915	G<139>	-5596	151	965	G<39>	-6396	151
916	G<137>	-5612	304	966	G<37>	-6412	304
917	G<135>	-5628	151	967	G<35>	-6428	151
918	G<133>	-5644	304	968	G<33>	-6444	304
919	G<131>	-5660	151	969	G<31>	-6460	151
920	G<129>	-5676	304	970	G<29>	-6476	304
921	G<127>	-5692	151	971	G<27>	-6492	151
922	G<125>	-5708	304	972	G<25>	-6508	304
923	G<123>	-5724	151	973	G<23>	-6524	151
924	G<121>	-5740	304	974	G<21>	-6540	304
925	G<119>	-5756	151	975	G<19>	-6556	151
926	G<117>	-5772	304	976	G<17>	-6572	304
927	G<115>	-5788	151	977	G<15>	-6588	151
928	G<113>	-5804	304	978	G<13>	-6604	304
929	G<111>	-5820	151	979	G<11>	-6620	151
930	G<109>	-5836	304	980	G<9>	-6636	304
931	G<107>	-5852	151	981	G<7>	-6652	151
932	G<105>	-5868	304	982	G<5>	-6668	304
933	G<103>	-5884	151	983	G<3>	-6684	151
934	G<101>	-5900	304	984	G<1>	-6700	304
935	G<99>	-5916	151	985	DUMMY34	-6716	151
936	G<97>	-5932	304	986	DUMMY35	-6732	304
937	G<95>	-5948	151	987	DUMMY36	-6748	151
938	G<93>	-5964	304	988	DUMMY37	-6764	304
939	G<91>	-5980	151				
940	G<89>	-5996	304				
941	G<87>	-6012	151				
942	G<85>	-6028	304				
943	G<83>	-6044	151				
944	G<81>	-6060	304				
945	G<79>	-6076	151				
946	G<77>	-6092	304				
947	G<75>	-6108	151				
948	G<73>	-6124	304				
949	G<71>	-6140	151				
950	G<69>	-6156	304				

The following figure indicates a schematic diagram of application circuit for S6D0164.



17.3. EXTERNAL COMPONENT

Table 56. External Component

Name	Device	Value	Connection	Note	
C1	Capacitor	1uF	VCI1 - GND	Maximum Ratings Voltage	3V
C2	Capacitor	1uF	AVDD - GND		10V
C3	Capacitor	1uF	VGH - GND		18V
C4	Capacitor	1uF	VGL - GND		18V
C5	Capacitor	1uF	GVDD - GND		6V
C6	Capacitor	1uF	VCOMH - GND		6V
C7	Capacitor	1uF	VCONL - GND		6V
C8	Capacitor	1uF	C11M - C11P		6V
C9	Capacitor	1uF	C12M - C12P		6V
C10	Capacitor	1uF	C21M - C21P		18V
C11	Capacitor	1uF	C22M - C22P		18V
C12	Capacitor	1uF	C31M - C31P		6V
C13	Capacitor	1uF	VCL - GND		6V
C14	Capacitor	1uF	RVDD - GND		3V
C15	Capacitor	1uF	VREF - GND		3V
D1	Diode	-	(+)VGL - GND(-)	VF ≤ 0.3V (@ IF = 20mA, Ta = 25°C) VR ≥ max.20V	

Cf> Component C15 (described in gray-color) is optional for this application.